



Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

May/June 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **20** pages. Blank pages are indicated.

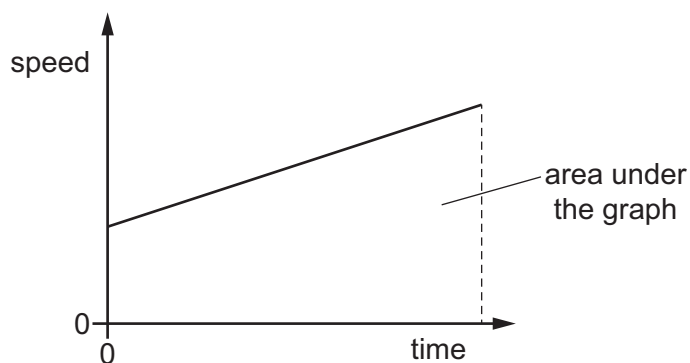


- 1 A pendulum makes 50 complete swings in 2 min 40 s.

What is the time period for 1 complete swing?

- A 1.6 s B 3.2 s C 4.8 s D 6.4 s

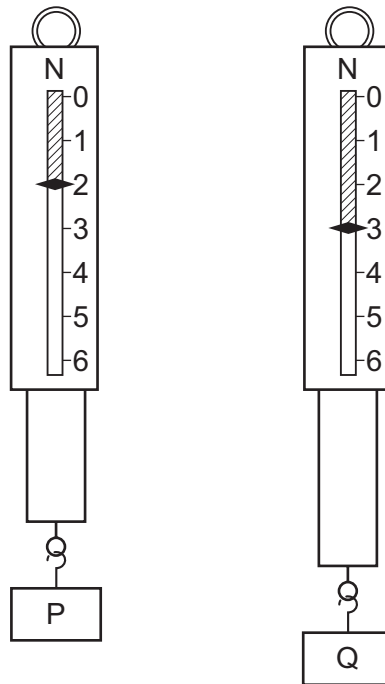
- 2 The motion of an object is represented by the speed–time graph shown.



Which quantity is equal to the area under the graph?

- A acceleration
B average speed
C distance travelled
D kinetic energy
- 3 Which statement about acceleration is correct?
- A It is related to the changing speed of an object.
B It is the distance an object travels in one second.
C It is the force acting on an object divided by the distance it travels in one second.
D It is the force acting on an object when it is near to the Earth.

- 4 Two metal blocks P and Q have identical dimensions. They hang on identical spring balances.



Which statement about P and Q is correct?

- A They have different volumes and different weights.
 - B They have different volumes and equal weights.
 - C They have equal volumes and equal weights.
 - D They have equal volumes and different weights.
- 5 A mass of 6.0 kg rests on the surface of a planet.

On this planet, $g = 20 \text{ N/kg}$.

What is the weight of the object?

- A 0.30 N B 0.60 N C 60 N D 120 N

- 6 The mass of a measuring cylinder is 190 g.

400 cm³ of liquid is put into the measuring cylinder.

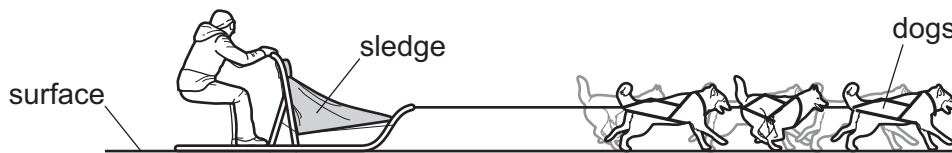
The total mass of the measuring cylinder and the liquid is 560 g.

Four solid objects are lowered in turn into the liquid. The densities of the objects are shown.

- 1 0.40 g/cm³
- 2 0.90 g/cm³
- 3 1.2 g/cm³
- 4 2.7 g/cm³

Which objects will float in the liquid?

- A** 1 only **B** 1 and 2 only **C** 1, 2 and 3 **D** 3 and 4 only
- 7 A sledge is pulled in a straight line by dogs, as shown.



The dogs produce a total horizontal driving force of 600 N.

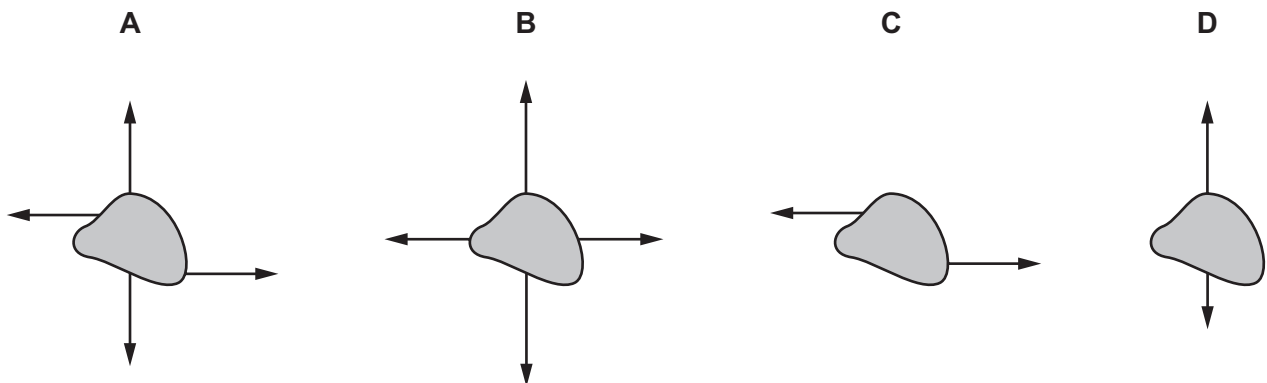
The frictional force between the sledge and the surface is 150 N and the air resistance on the sledge is 450 N.

What is the resultant force acting on the sledge?

- A** 0 N **B** 300 N **C** 900 N **D** 1200 N
- 8 Forces are applied to four identical objects.

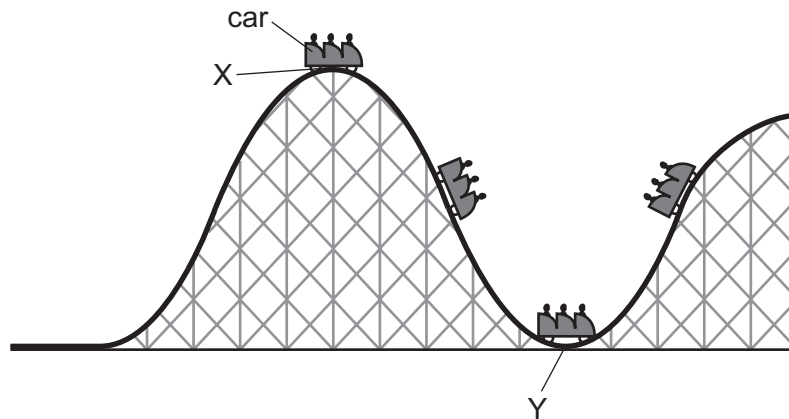
The lengths of the arrows indicate the magnitude of each force.

Which object is in equilibrium?



- 9 The diagram shows part of a rollercoaster ride with the car at different positions.

The car runs freely down from position X to position Y and up the hill on the other side.



What happens to the kinetic energy and to the gravitational potential energy of the car as it moves from position X to position Y?

	kinetic energy	gravitational potential energy
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

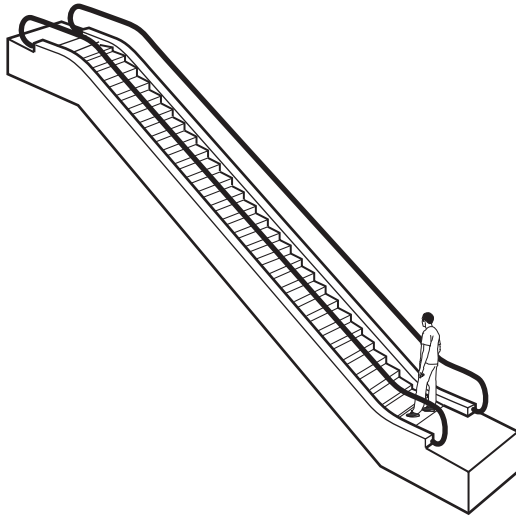
- 10 Student P uses a force of 35 N to push a box 3.0 m across the floor.

Student Q uses a force of 22 N to push another box 1.8 m across the floor.

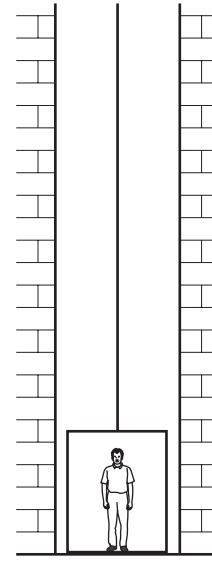
Which statement gives a full explanation why student P uses more energy than student Q?

- A** Student P pushes his box a greater distance than student Q.
- B** Student P pushes his box a greater distance and uses a bigger force than student Q.
- C** Student P uses a bigger force than student Q.
- D** Student P pushes a heavier box than student Q.

- 11 A man can either take an escalator or a lift to travel up between two floors in a hotel.



escalator



lift

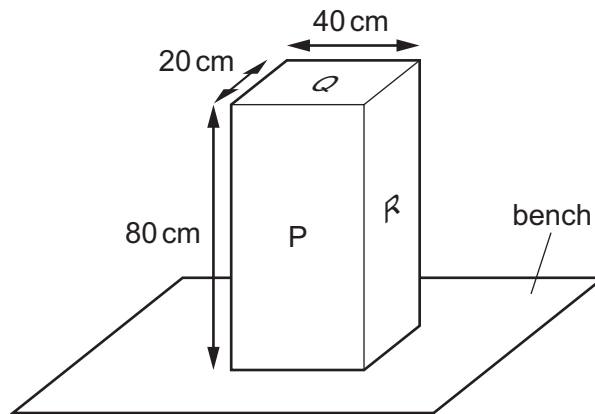
The escalator takes 20 seconds to carry the man between the two floors. The useful work done against gravity is W . The useful power developed is P .

The lift takes 30 seconds to carry the same man between the same two floors.

How much useful work against gravity is done by the lift, and how much useful power is developed by the lift?

	useful work done against gravity by lift	useful power developed by lift
A	more than W	less than P
B	more than W	P
C	W	less than P
D	W	P

12 The diagram shows a solid block resting on a bench. The dimensions of the block are shown.

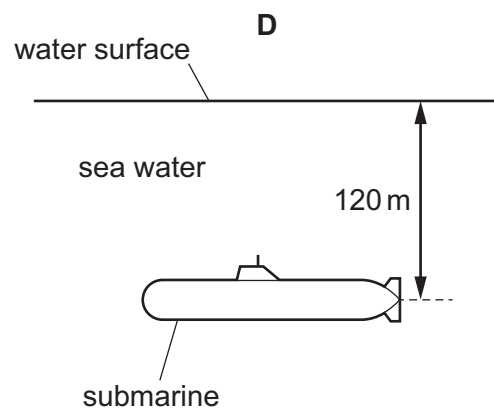
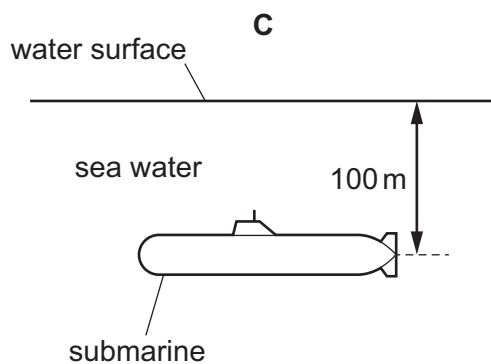
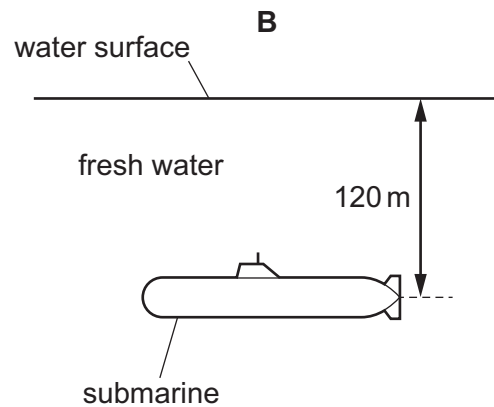
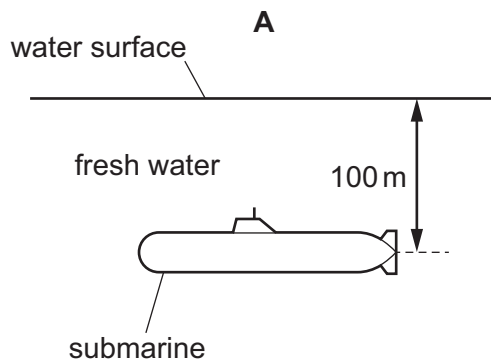


On which labelled surface should the block rest to produce the smallest pressure on the bench?

- A P
- B Q
- C R
- D P, Q and R produce the same pressure

- 13** Four submarines are submerged. The density of fresh water is 1000 kg/m^3 and the density of sea water is 1020 kg/m^3 .

Which submarine experiences the greatest pressure due to the water?



- 14** A gas is heated in a sealed container.

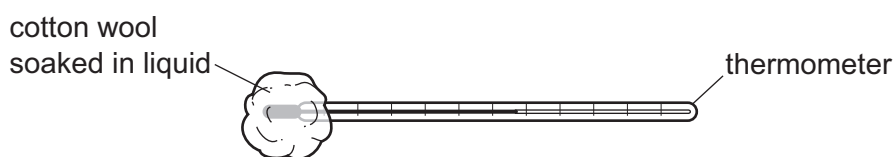
The volume of the container does not change.

What happens to the molecules of the gas?

- A** The average distance between molecules increases.
- B** The average kinetic energy of the molecules increases.
- C** The mass of each molecule increases.
- D** The volume of each molecule increases.

- 15** A student is investigating evaporation. She soaks a piece of cotton wool in a liquid and attaches this to the bulb of a thermometer.

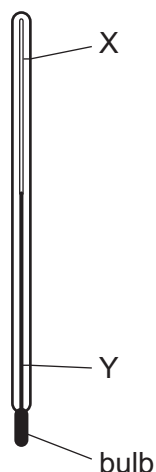
As the liquid evaporates the temperature reading on the thermometer changes.



Which statement is correct?

- A** The more energetic molecules leave the liquid and the temperature reading decreases.
 - B** The more energetic molecules leave the liquid and the temperature reading increases.
 - C** The less energetic molecules leave the liquid and the temperature reading decreases.
 - D** The less energetic molecules leave the liquid and the temperature reading increases.
- 16** The thermometer in the diagram has no scale.

Before this thermometer can be used to measure temperature, two standard temperatures known as fixed points are needed. These are labelled X and Y.



Which row describes these fixed points on the Celsius scale?

	X	Y
A	temperature of pure boiling water	normal body temperature
B	temperature of pure boiling water	temperature of pure melting ice
C	normal body temperature	temperature of pure boiling water
D	temperature of pure melting ice	temperature of pure boiling water

- 17 When a hot gas is left to cool, its internal energy decreases.

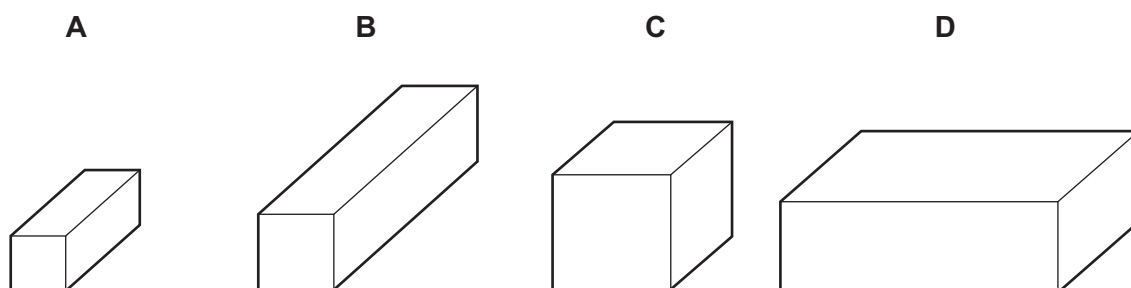
What causes this?

- A a decrease in the kinetic energy of the gas particles
- B a decrease in the gravitational potential energy of the gas particles
- C an increase in the average speed of the gas particles
- D an increase in the average distance of separation of the gas particles

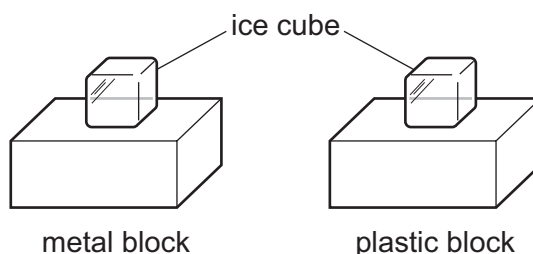
- 18 The diagrams show four blocks of steel. The blocks are all drawn to the same scale.

The same quantity of thermal energy is given to each block.

Which block shows the greatest rise in temperature?



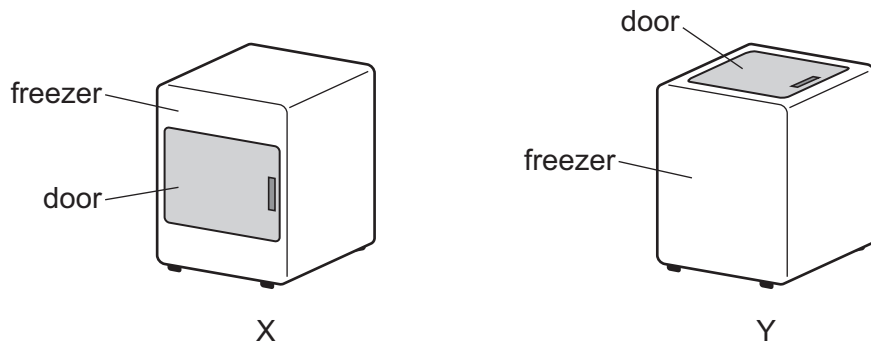
- 19 One ice cube is placed on a metal block. An identical ice cube is placed on a plastic block. The blocks are left next to each other on a table in a laboratory.



Which ice cube melts first and why?

- A The ice cube on the plastic block melts first because plastic is a good insulator of thermal energy.
- B The ice cube on the plastic block melts first because plastic is a good conductor of thermal energy.
- C The ice cube on the metal block melts first because metal is a good conductor of thermal energy.
- D The ice cube on the metal block melts first because metal is a good insulator of thermal energy.

- 20 Two freezers X and Y are identical except that one has a door opening at the front and the other has a door opening at the top.

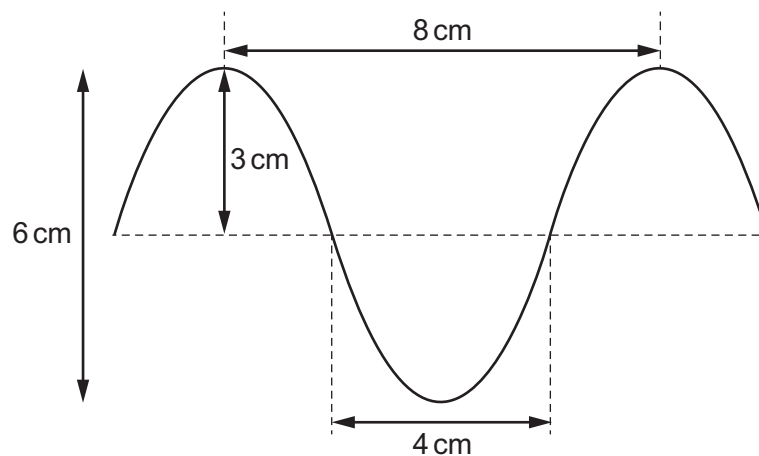


Both doors are the same size and are opened for the same amount of time.

Which freezer gains the least amount of thermal energy in this time and why?

	freezer gaining the least thermal energy	reason
A	X	cold air falls
B	X	warm air falls
C	Y	cold air falls
D	Y	warm air falls

- 21 The diagram shows a wave.



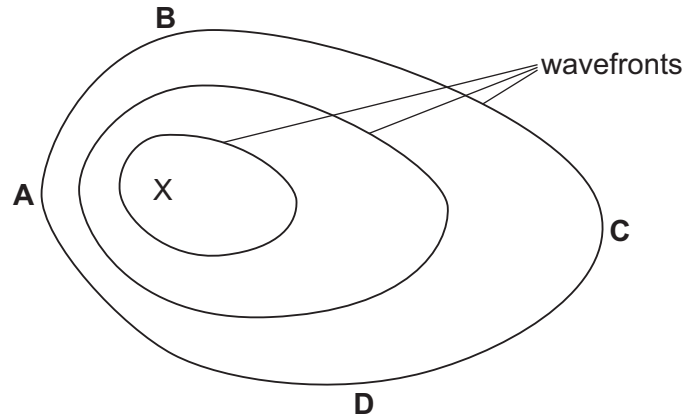
What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

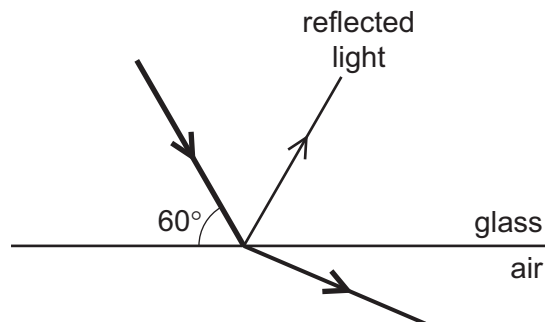
- 22** Waves travel more quickly on the surface of water when the water is deep.

A stone is dropped at point X into a pool of varying depth. The diagram shows the first three wavefronts on the surface of the pool.

The region between X and which labelled point is likely to be the deepest?



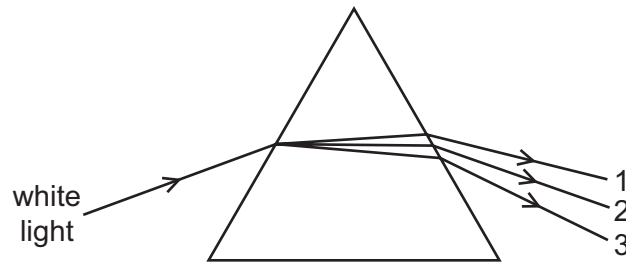
- 23** The diagram shows a beam of light travelling through glass and meeting a glass-air interface.



Which row correctly describes what is happening at the glass-air interface?

	angle of incidence at the interface	observation
A	30°	some internal reflection
B	30°	total internal reflection
C	60°	some internal reflection
D	60°	total internal reflection

- 24** A narrow beam of white light passes through a prism and is dispersed into a spectrum.



Which row is correct?

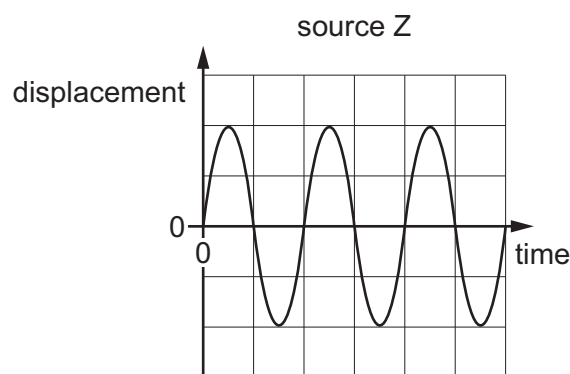
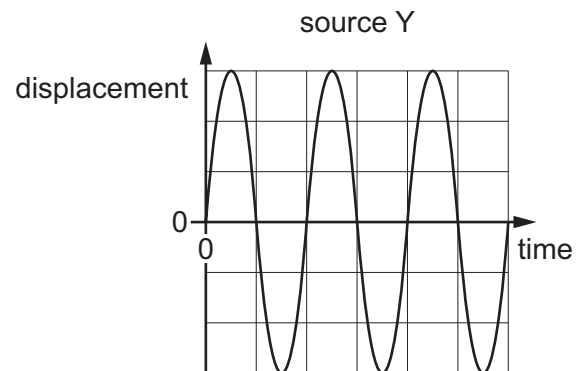
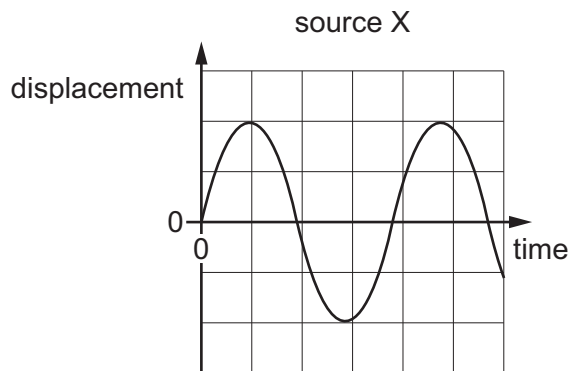
	colour 1	colour 2	colour 3
A	blue	yellow	red
B	red	blue	yellow
C	red	yellow	blue
D	yellow	blue	red

- 25** An intruder alarm sensor detects that a person is warmer than his surroundings.

Which type of electromagnetic wave does the sensor detect?

- A** infrared
- B** radio
- C** ultraviolet
- D** visible light

- 26** The graphs show the displacement of particles in sound waves from three sources X, Y and Z. The scales on the graphs are all identical.



Which sources are producing sound waves with the same pitch?

- A** X and Y only **B** Y and Z only **C** X and Z only **D** X, Y and Z

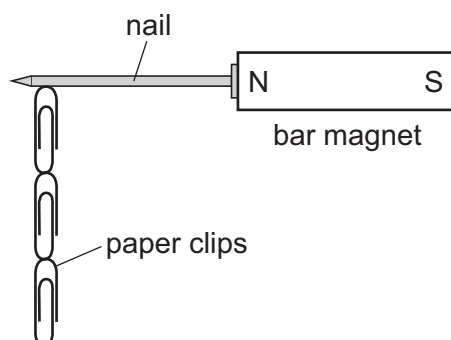
- 27** A permanent magnet is to be made.

Which row gives the correct material to use and the correct reason for choosing this material?

	material	reason
A	soft iron	easy to demagnetise
B	soft iron	hard to demagnetise
C	steel	easy to demagnetise
D	steel	hard to demagnetise

28 Four nails **A**, **B**, **C** and **D** are tested to find which makes the strongest permanent magnet.

One of the nails is placed against a bar magnet and the number of paper clips which the nail can support is recorded.



The bar magnet is then removed and the number of paper clips remaining attached to the nail is recorded. Each nail is tested individually.

Which nail becomes the strongest permanent magnet?

	number of paper clips attached to the nail	
	bar magnet present	bar magnet removed
A	2	0
B	2	1
C	4	3
D	5	2

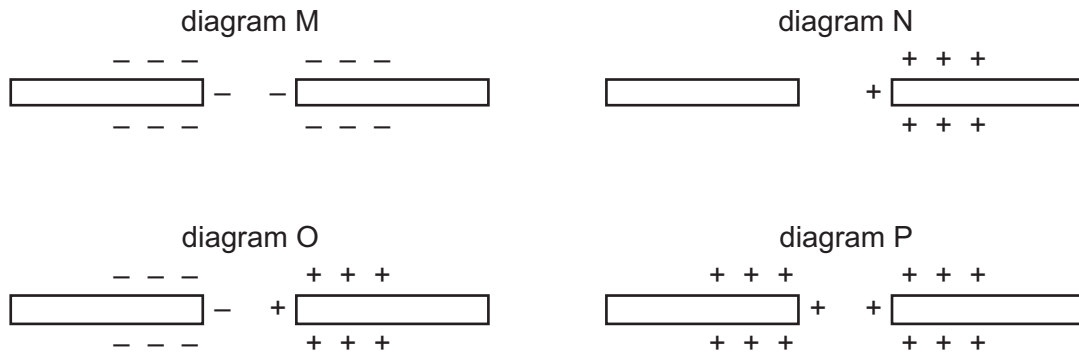
29 Three statements about electric charge are given.

- 1 An ammeter directly measures how much electric charge is in an object.
- 2 A moving electric charge can be detected by an ammeter.
- 3 A flow of electric charge is an electric current.

Which statements are correct?

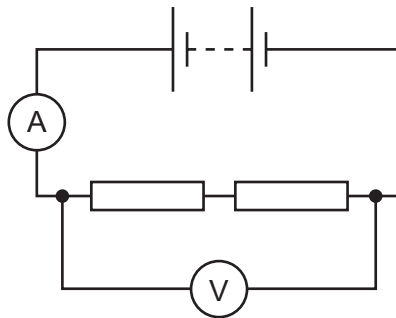
- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 30 Each of the four diagrams M, N, O and P shows a separate pair of insulating rods. Each rod is charged as shown.



In which **two** arrangements do the pairs of rods experience a force of repulsion?

- A** M and N **B** O and P **C** M and P **D** N and O
- 31 A student uses the circuit shown to determine the resistance of two identical resistors.

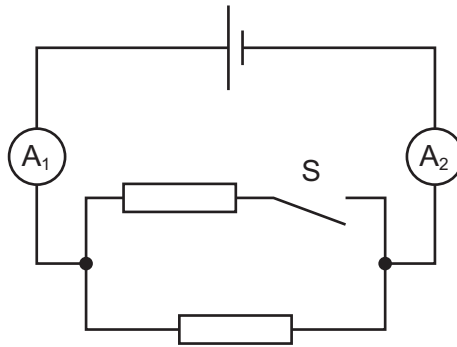


The voltmeter reading is 2.2 V and the ammeter reading is 0.25 A.

What is the resistance of each resistor?

- A** $0.275\ \Omega$ **B** $0.55\ \Omega$ **C** $4.4\ \Omega$ **D** $8.8\ \Omega$

32 In the circuit shown, A_1 and A_2 are ammeters.



Switch S is closed.

Which row is correct?

	the resistance of the whole circuit	reading of A_1	reading of A_2
A	decreases	stays the same	increases
B	decreases	increases	increases
C	increases	stays the same	stays the same
D	increases	decreases	decreases

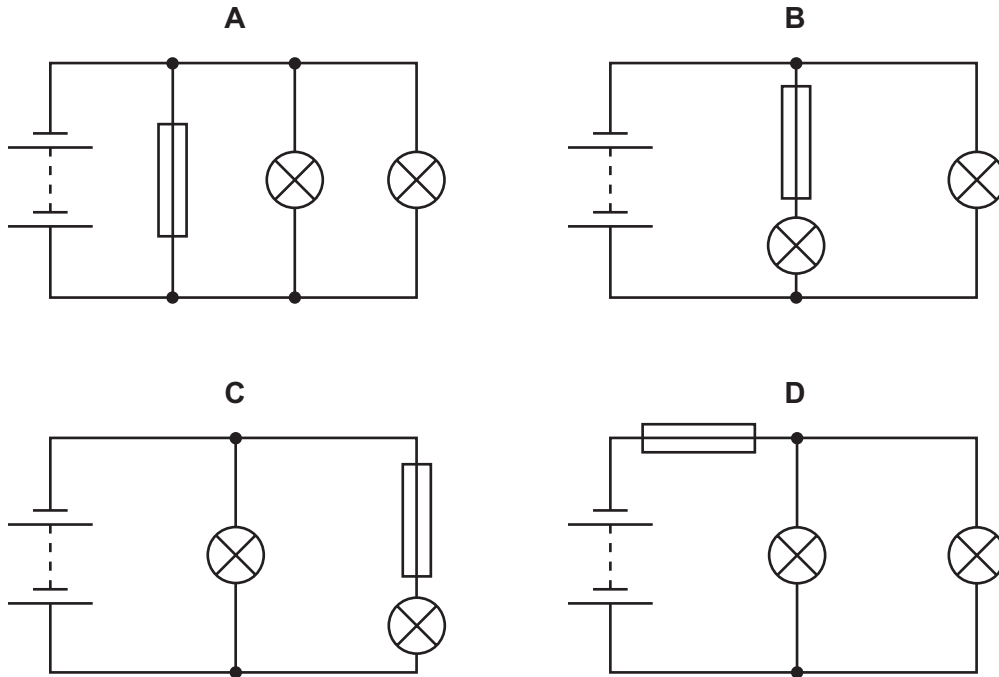
33 What happens to the resistance of an LDR when the brightness of light falling on it increases?

- A** Its resistance decreases.
- B** Its resistance increases.
- C** Its resistance increases then decreases.
- D** Its resistance stays the same.

- 34** A student constructs four circuits, each containing a fuse.

The fuse blows in one circuit and both lamps in the circuit go out.

In which circuit does the fuse blow and both lamps go out?



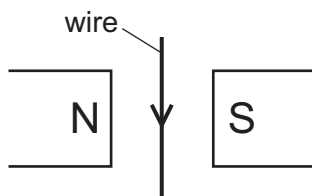
- 35** A transformer has 200 turns on its primary coil and is connected to a 240 V a.c. supply.

The output voltage of the transformer is 60 V a.c..

How many turns are on the secondary coil of the transformer?

- A** 20 **B** 50 **C** 72 **D** 800

- 36** A conducting wire is placed between the poles of a magnet. When an electric current in the wire is in the direction shown, then the force on the wire acts out of the page.



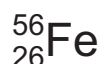
Three statements of different conditions and how the wire is affected are given.

- 1 The current is towards the top of the page and the direction of the magnetic field is unchanged then the force produced acts into the page.
- 2 The current is towards the bottom of the page and the magnetic field is reversed then the force produced acts into the page.
- 3 The current in the wire is alternating and the wire vibrates into and out of the page.

Which statements are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 37** A nuclide of the element iron has the symbol shown.



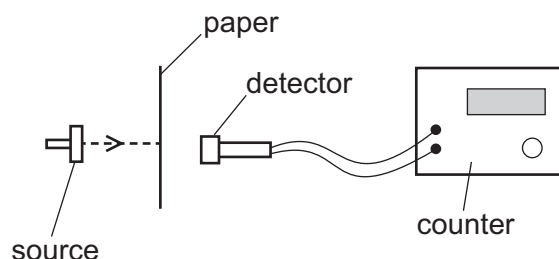
What does a neutral atom of this nuclide contain?

	protons	neutrons	electrons
A	26	30	26
B	26	56	30
C	30	26	56
D	56	26	30

- 38** What is the nucleon number of a nuclide?

- A** the number of neutrons in the nucleus
B the number of protons in the nucleus
C the number of protons minus the number of neutrons in the nucleus
D the number of protons plus the number of neutrons in the nucleus

- 39 A thin sheet of paper is placed between a radioactive source and a radiation detector. The count rate falls to a very low reading.



From this result, which type of radiation is the source emitting?

- A α -particles
 - B β -particles
 - C γ -rays
 - D X-rays
- 40 In 1986 the Chernobyl nuclear power station in Ukraine suffered a meltdown.

This caused background radiation in many countries, thousands of kilometres from Chernobyl, to increase.

What was transported in the atmosphere to these countries to cause this rise in background radiation?

- A α -particles
- B β -particles
- C γ -rays
- D radioactive isotopes

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May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	C	1
3	A	1
4	D	1
5	D	1
6	B	1
7	A	1
8	B	1
9	C	1
10	B	1
11	C	1
12	A	1
13	D	1
14	B	1
15	A	1
16	B	1
17	A	1
18	A	1
19	C	1
20	C	1
21	B	1
22	C	1
23	A	1
24	C	1
25	A	1
26	B	1
27	D	1
28	C	1

Question	Answer	Marks
29	C	1
30	C	1
31	C	1
32	B	1
33	A	1
34	D	1
35	B	1
36	D	1
37	A	1
38	D	1
39	A	1
40	D	1



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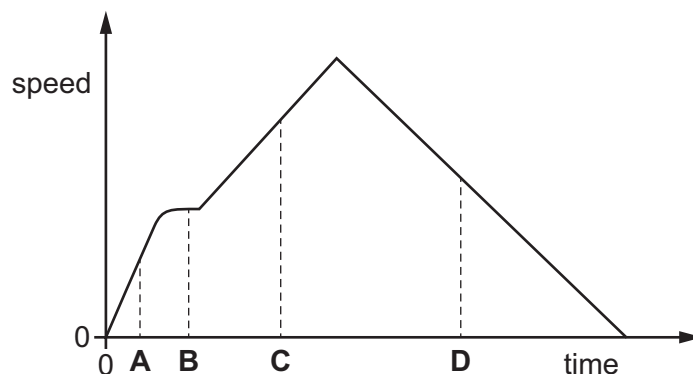
- 1 Five athletes P, Q, R, S and T compete in a race. The table shows the finishing times for the athletes.

athlete	P	Q	R	S	T
finishing time / s	22.50	24.40	25.20	26.50	23.20

Which statement is correct?

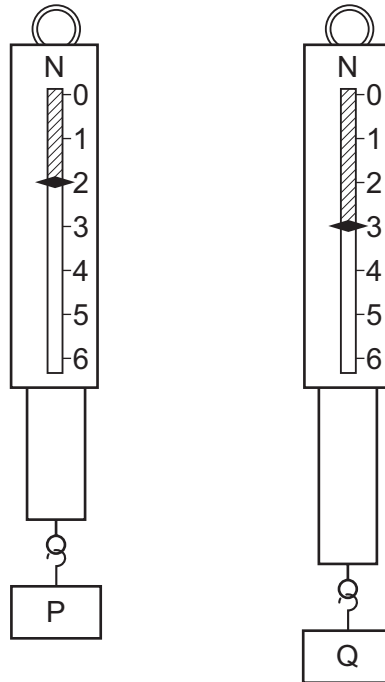
- A** Athlete P won the race and was 0.70 s ahead of the athlete in second place.
B Athlete P won the race and was 1.90 s ahead of the athlete in second place.
C Athlete S won the race and was 1.30 s ahead of the athlete in second place.
D Athlete S won the race and was 2.10 s ahead of the athlete in second place.
- 2 The graph shows how the speed of an object varies with time.

At which labelled time is the acceleration greatest?



- 3 Which statement about acceleration is correct?
- A** It is related to the changing speed of an object.
B It is the distance an object travels in one second.
C It is the force acting on an object divided by the distance it travels in one second.
D It is the force acting on an object when it is near to the Earth.

- 4 Two metal blocks P and Q have identical dimensions. They hang on identical spring balances.



Which statement about P and Q is correct?

- A They have different volumes and different weights.
 - B They have different volumes and equal weights.
 - C They have equal volumes and equal weights.
 - D They have equal volumes and different weights.
- 5 A space probe is taken from the Earth to Mars.

The force of gravity on the surface of Mars is less than the force of gravity on the surface of the Earth.

How do the weight and the mass of a space probe on the surface of Mars compare to their values when the probe is on the surface of the Earth?

	weight on Mars	mass on Mars
A	decreased	decreased
B	decreased	unchanged
C	unchanged	decreased
D	unchanged	unchanged

- 6 Water has a density of 1000 kg/m^3 .

A rectangular swimming pool has an average depth of 1.6 m.

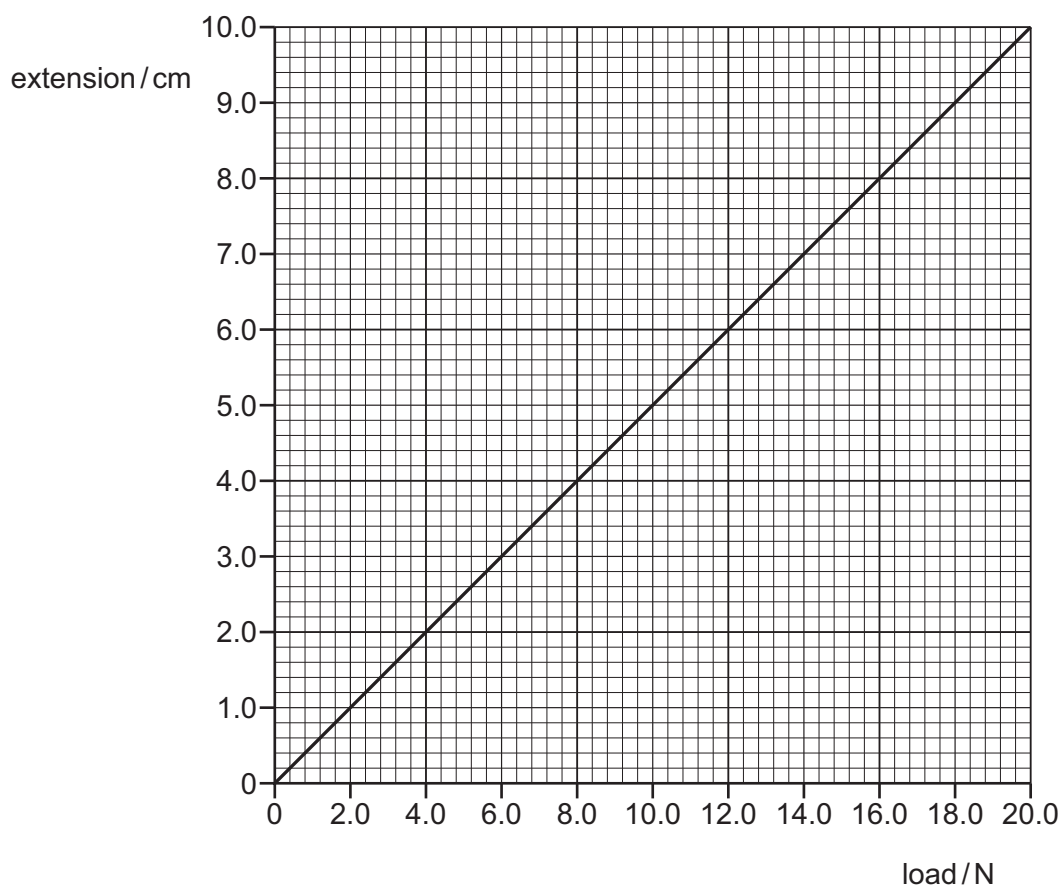
The length of the pool is 25 m.

The width of the pool is 10 m.

What is the mass of the water in the swimming pool?

- A** 2.5 kg **B** 400 kg **C** 400 000 kg **D** 800 000 kg

- 7 The diagram shows an extension–load graph for a spring.



An empty can of weight 3.0 N is suspended from the spring.

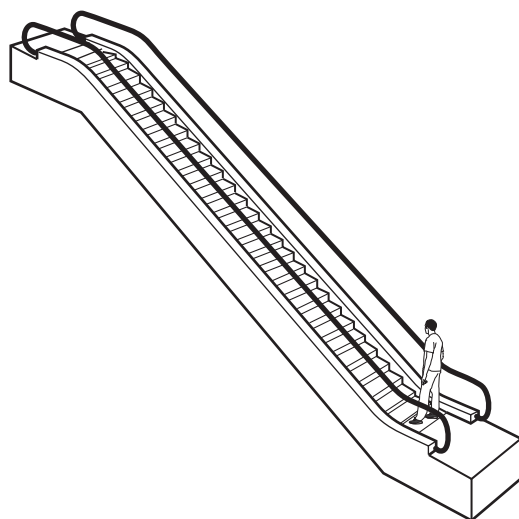
Liquid is poured into the can until the extension is 8.0 cm.

What is the weight of the liquid?

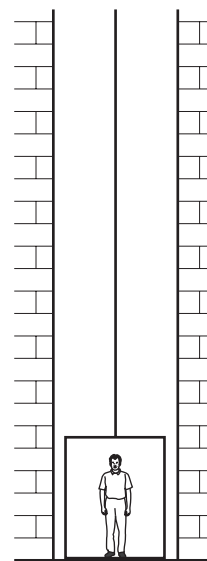
- A** 4.0 N **B** 10.0 N **C** 13.0 N **D** 16.0 N

- 8 What is meant by the moment of a force on an object?
- A the magnitude of the force on the object
 - B the direction of the force on the object
 - C the time for which the force acts on the object
 - D the turning effect of the force on the object
- 9 Which energy resource is **not** renewable?
- A geothermal
 - B nuclear fission
 - C solar
 - D wind
- 10 Two motors X and Y lift loads of the same weight through the same vertical distance.
- Motor X is more efficient than motor Y.
- Which statement about the motors is correct?
- A The useful energy output of motor X is larger than that of motor Y.
 - B The useful energy output of motor X is smaller than that of motor Y.
 - C The energy input of motor X is larger than that of motor Y.
 - D The energy input of motor X is smaller than that of motor Y.

- 11 A man can either take an escalator or a lift to travel up between two floors in a hotel.



escalator



lift

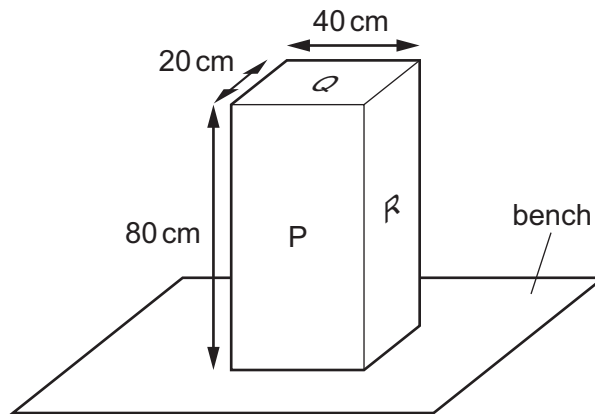
The escalator takes 20 seconds to carry the man between the two floors. The useful work done against gravity is W . The useful power developed is P .

The lift takes 30 seconds to carry the same man between the same two floors.

How much useful work against gravity is done by the lift, and how much useful power is developed by the lift?

	useful work done against gravity by lift	useful power developed by lift
A	more than W	less than P
B	more than W	P
C	W	less than P
D	W	P

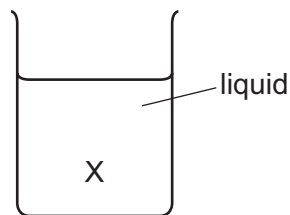
- 12 The diagram shows a solid block resting on a bench. The dimensions of the block are shown.



On which labelled surface should the block rest to produce the smallest pressure on the bench?

- A P
- B Q
- C R
- D P, Q and R produce the same pressure

- 13 A beaker contains a liquid.



On what does the liquid pressure at position X depend?

- A both the density of the liquid and the depth of X below the surface
 - B both the surface area of the liquid and the depth of X below the surface
 - C both the surface area of the liquid and the volume of the liquid
 - D the depth of X below the surface only
- 14 Evaporation occurs from the surface of a pool of water.

Which statement describes this change of state?

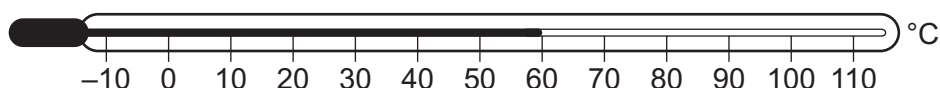
- A Electrons move from the liquid and become a gas.
- B Molecules that move from the liquid have the same energy as those that stay in the liquid.
- C The more energetic molecules escape the liquid.
- D The more energetic molecules remain in the liquid.

- 15** A gas, in a sealed container, is compressed slowly so that its temperature does not change.

What happens to the molecules of the gas as a result of this compression?

- A** The average speed of the molecules increases.
- B** The average force in a collision between a molecule and the container increases.
- C** There are more frequent collisions between molecules and the container.
- D** The volume of each molecule decreases.

- 16** The diagram shows a liquid-in-glass thermometer.



What is the temperature difference between the two fixed points on the Celsius scale?

- A** 10 °C
- B** 100 °C
- C** 110 °C
- D** 120 °C

- 17** A metal block is left overnight in a cool, shady room. In the morning, the metal block is moved into warm surroundings.

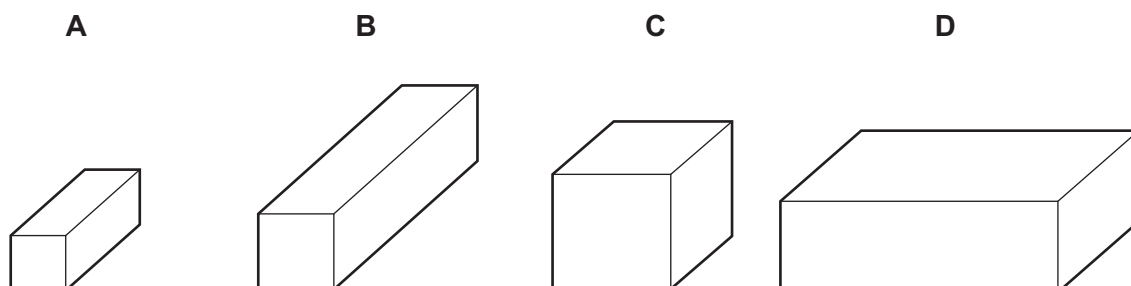
Which statement about the metal block is correct in the morning?

- A** The internal energy of the metal block increases.
- B** The temperature of the metal block decreases.
- C** Convection transfers energy throughout the metal block.
- D** The metal contracts slightly.

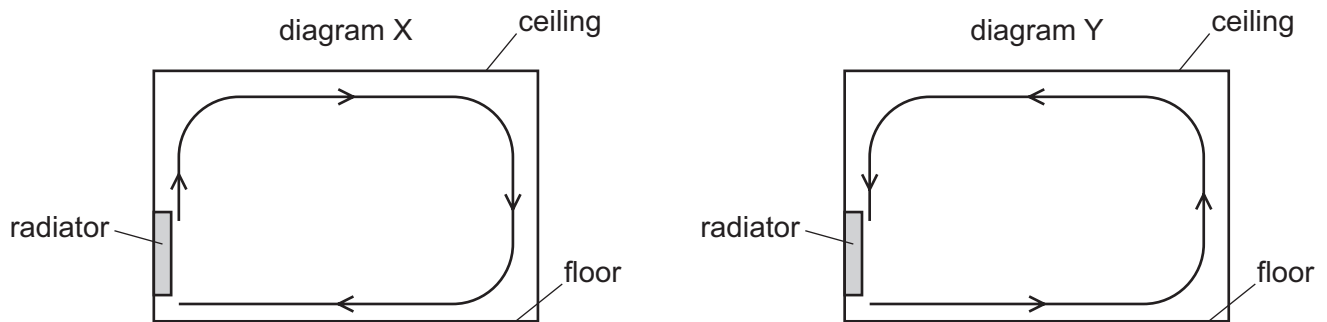
- 18** The diagrams show four blocks of steel. The blocks are all drawn to the same scale.

The same quantity of thermal energy is given to each block.

Which block shows the greatest rise in temperature?



- 19** A room is heated by a radiator. The diagrams X and Y show two possible circulations of hot air, which heat the room.



Which diagram and reason explain the heating of the room by convection?

	diagram	reason
A	X	air density decreases when air is heated
B	X	air density increases when air is heated
C	Y	air density decreases when air is heated
D	Y	air density increases when air is heated

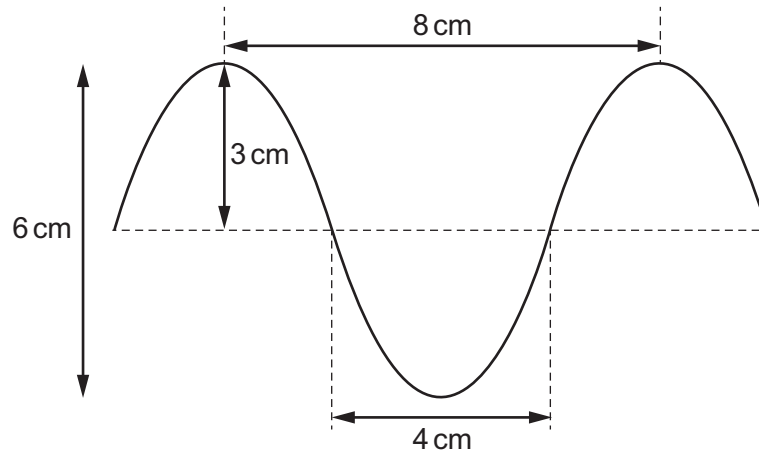
- 20** Two similar liquid-in-glass thermometers P and Q are placed in direct sunlight.

The bulb of thermometer P is painted white. The bulb of thermometer Q is painted black.

How and why would the thermometer readings differ?

- A** P would read higher than Q because black is a good absorber of radiation.
- B** P would read higher than Q because black is a poor absorber of radiation.
- C** P would read lower than Q because black is a good absorber of radiation.
- D** P would read lower than Q because black is a poor absorber of radiation.

21 The diagram shows a wave.



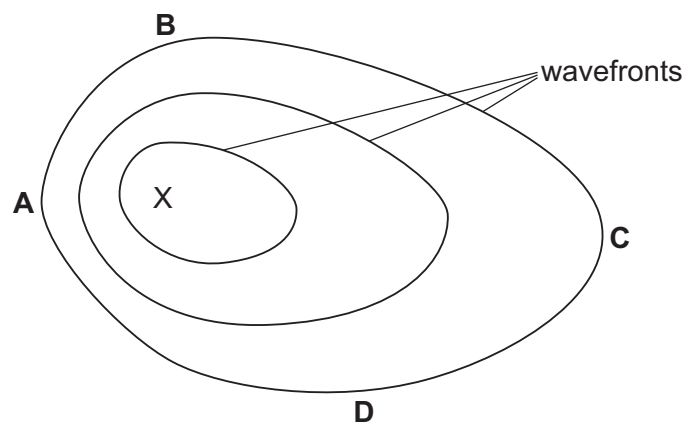
What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

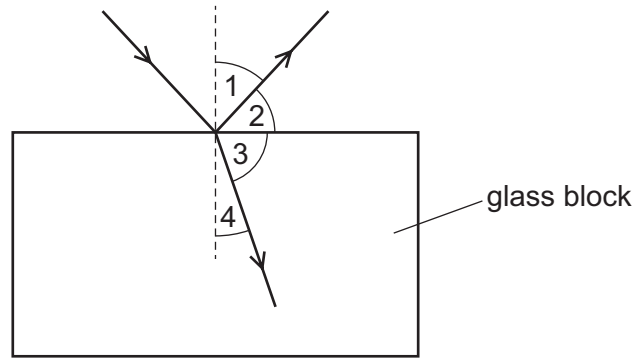
22 Waves travel more quickly on the surface of water when the water is deep.

A stone is dropped at point X into a pool of varying depth. The diagram shows the first three wavefronts on the surface of the pool.

The region between X and which labelled point is likely to be the deepest?



- 23 The diagram shows a ray of light incident on the surface of a glass block.



The ray of light is partially reflected back into the air and partially refracted into the glass block.

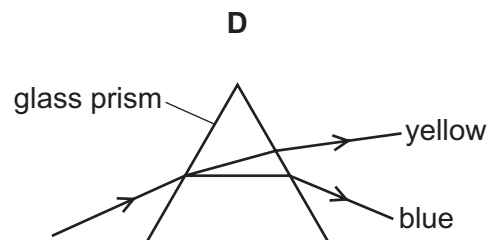
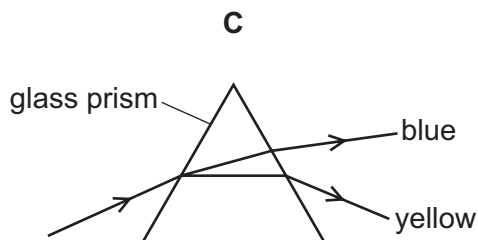
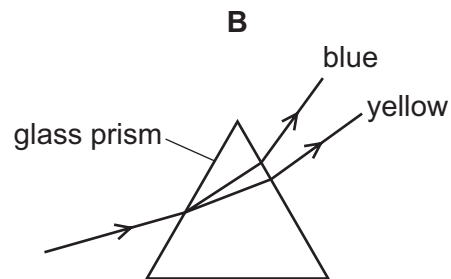
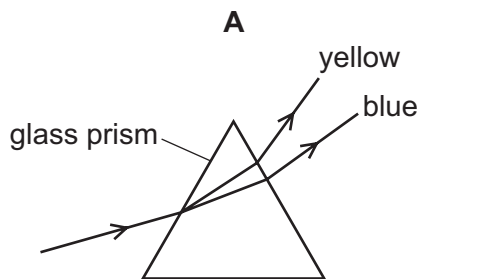
Which row correctly identifies the angle of reflection and the angle of refraction?

	angle of reflection	angle of refraction
A	1	3
B	1	4
C	2	3
D	2	4

- 24 A beam of light consists of yellow and blue light.

The beam of light is incident on a glass prism.

Which diagram is correct?



- 25** An intruder alarm sensor detects that a person is warmer than his surroundings.

Which type of electromagnetic wave does the sensor detect?

- A** infrared
- B** radio
- C** ultraviolet
- D** visible light

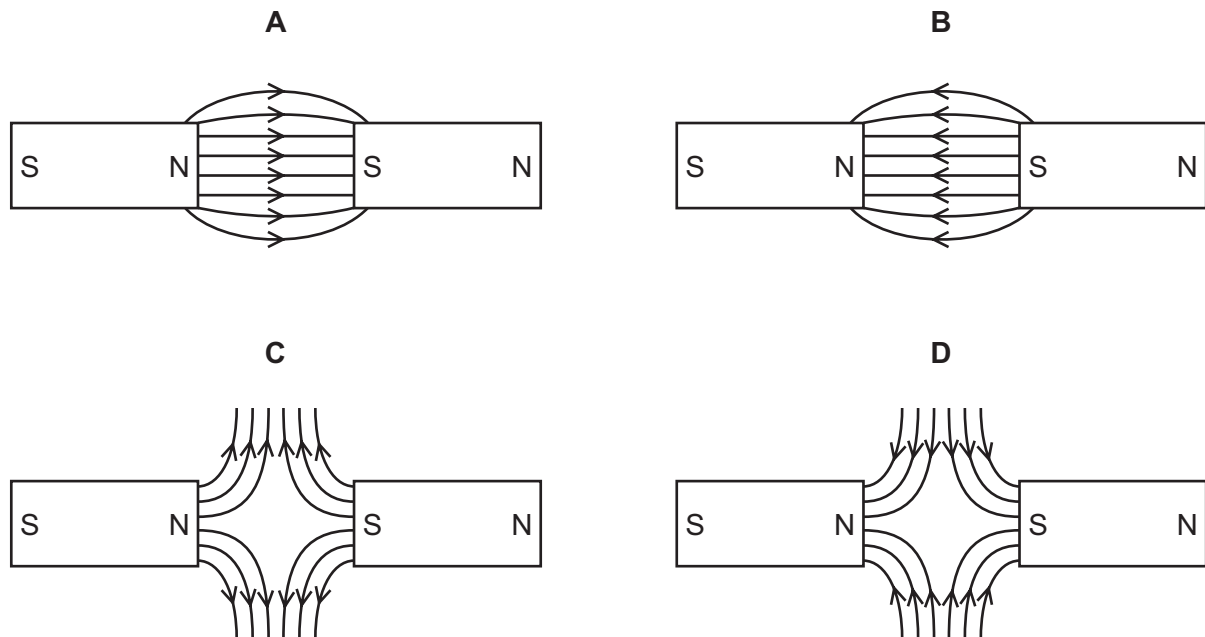
- 26** A dolphin sends out a sound wave. An echo returns 0.010 s later from a fish which is 7.5 m from the dolphin.

What is the speed of the sound wave in water?

- A** 0.075 m/s **B** 0.15 m/s **C** 750 m/s **D** 1500 m/s

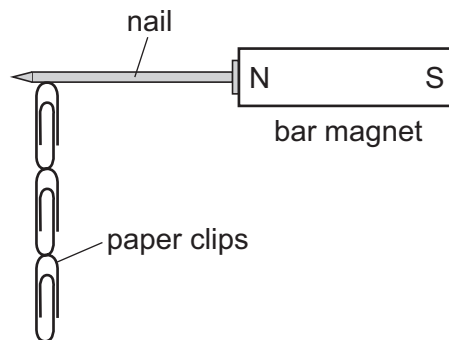
- 27** The diagrams show two bar magnets which are attracting each other.

Which diagram shows the magnetic field pattern between the poles?



28 Four nails **A**, **B**, **C** and **D** are tested to find which makes the strongest permanent magnet.

One of the nails is placed against a bar magnet and the number of paper clips which the nail can support is recorded.



The bar magnet is then removed and the number of paper clips remaining attached to the nail is recorded. Each nail is tested individually.

Which nail becomes the strongest permanent magnet?

	number of paper clips attached to the nail	
	bar magnet present	bar magnet removed
A	2	0
B	2	1
C	4	3
D	5	2

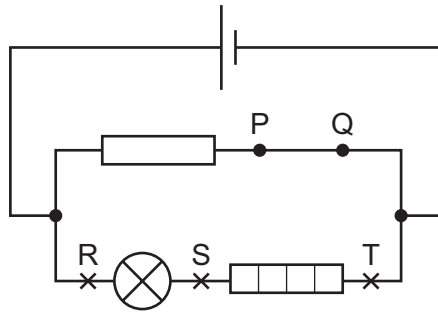
29 A cloth is used to rub an uncharged plastic rod.

Both the rod and the cloth become charged.

Why does the plastic rod become negatively charged and the cloth become positively charged?

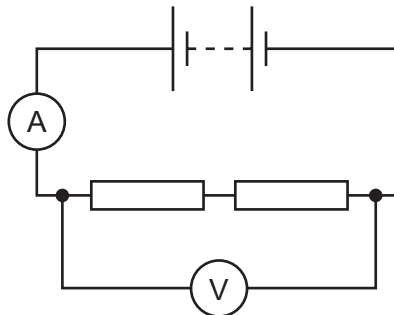
- A** The rod gains electrons and the cloth gains positive charges.
- B** The rod gains electrons and the cloth loses electrons.
- C** The rod loses electrons and the cloth gains electrons.
- D** The rod loses electrons and the cloth loses positive charges.

- 30 The diagram shows a circuit. The wire between P and Q can be removed and replaced by a circuit component.



Where should a voltmeter be connected to measure the potential difference (p.d.) across the lamp?

- A between P and Q in place of the wire
 - B in parallel with R and S
 - C in parallel with R and T
 - D in parallel with S and T
- 31 A student uses the circuit shown to determine the resistance of two identical resistors.

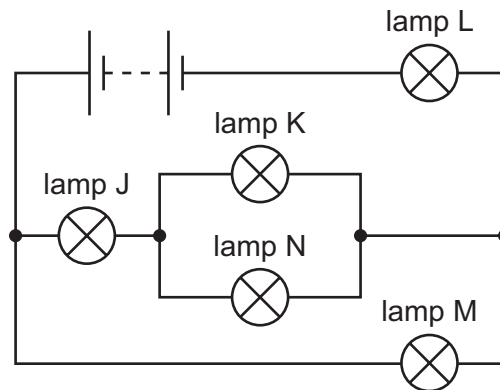


The voltmeter reading is 2.2 V and the ammeter reading is 0.25 A.

What is the resistance of each resistor?

- A $0.275\ \Omega$
- B $0.55\ \Omega$
- C $4.4\ \Omega$
- D $8.8\ \Omega$

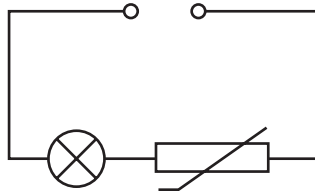
- 32 The circuit shown contains five lamps J, K, L, M and N. All the lamps are glowing.



One lamp is removed and two other lamps go out.

Which lamp is removed?

- A lamp J
 - B lamp K
 - C lamp L
 - D lamp M
- 33 The diagram shows a control circuit. The lamp is lit.



The temperature of the surroundings increases.

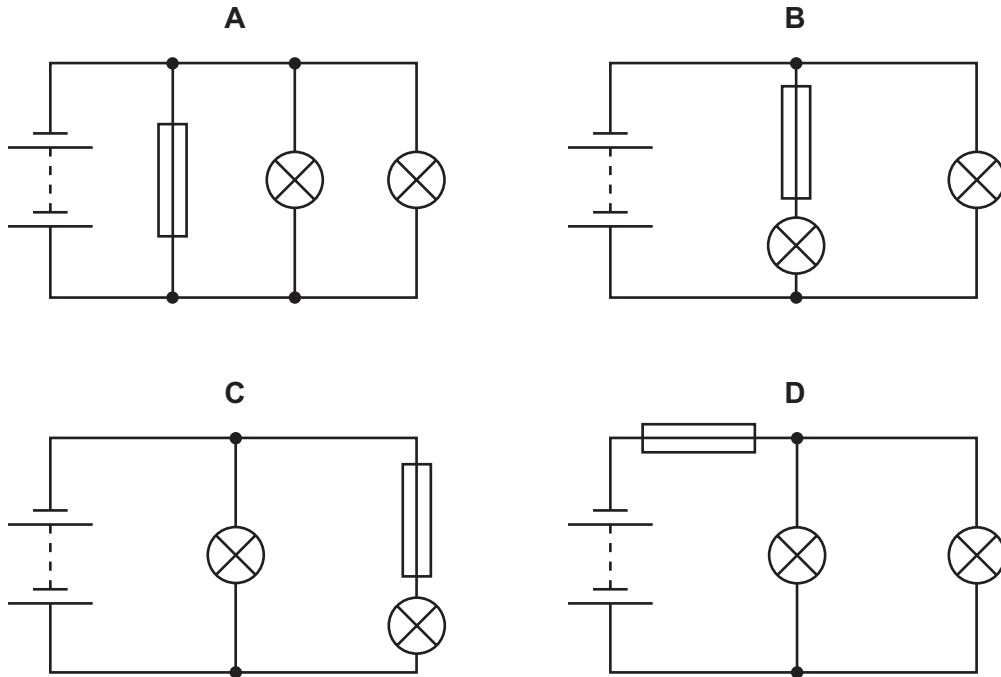
What will happen to the brightness of the lamp?

- A It will be brighter.
- B It will be less bright.
- C It will not change.
- D It will become brighter and then less bright.

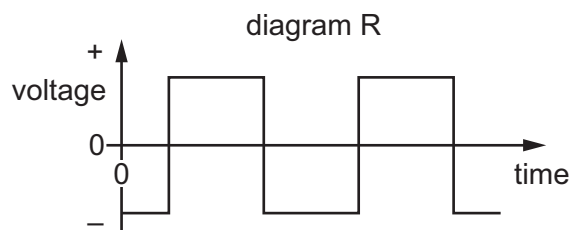
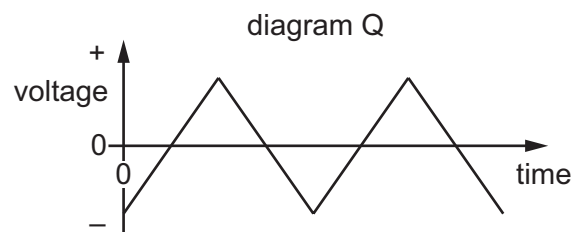
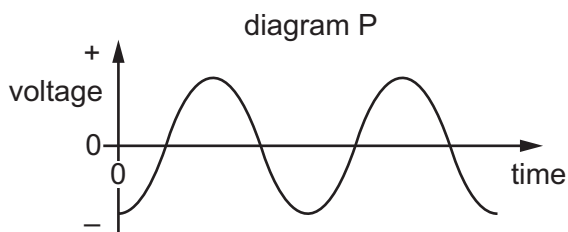
34 A student constructs four circuits, each containing a fuse.

The fuse blows in one circuit and both lamps in the circuit go out.

In which circuit does the fuse blow and both lamps go out?



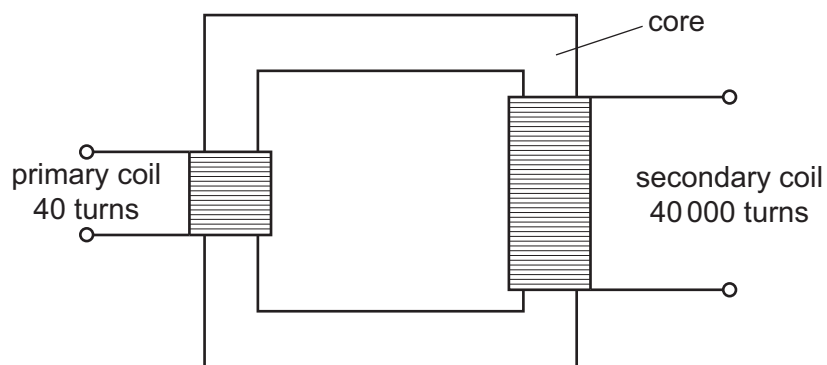
35 The diagrams P, Q and R show three voltage–time graphs.



Which graphs show an alternating voltage?

A P and Q only **B** P and R only **C** Q and R only **D** P, Q and R

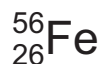
36 The diagram shows a transformer.



Which statement about this transformer is correct?

- A It can operate from a 12 V battery.
- B It has a core which is made of steel.
- C It steps the input voltage up by a factor of 1000.
- D It steps the input voltage down by a factor of 1000.

37 A nuclide of the element iron has the symbol shown.



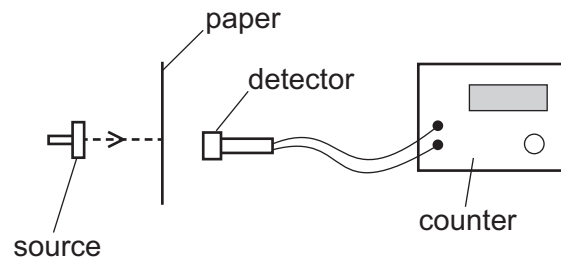
What does a neutral atom of this nuclide contain?

	protons	neutrons	electrons
A	26	30	26
B	26	56	30
C	30	26	56
D	56	26	30

38 Which statement about the nuclei of all atoms is correct?

- A They all contain electrons.
- B They are all always stable.
- C They all contain protons and electrons.
- D They all have a positive charge.

- 39** A thin sheet of paper is placed between a radioactive source and a radiation detector. The count rate falls to a very low reading.



From this result, which type of radiation is the source emitting?

- A** α -particles
 - B** β -particles
 - C** γ -rays
 - D** X-rays
- 40** A radioactive isotope has a half-life of 120 minutes.
It emits radiation at a rate of 100 particles per second.
How long does it take for the rate of emission to fall to 25 particles per second?
- A** 30 minutes **B** 45 minutes **C** 90 minutes **D** 240 minutes

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/12

Paper 1 Multiple Choice (Core)

May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	A	1
3	A	1
4	D	1
5	B	1
6	C	1
7	C	1
8	D	1
9	B	1
10	D	1
11	C	1
12	A	1
13	A	1
14	C	1
15	C	1
16	B	1
17	A	1
18	A	1
19	A	1
20	C	1
21	B	1
22	C	1
23	B	1
24	D	1
25	A	1
26	D	1
27	A	1
28	C	1

Question	Answer	Marks
29	B	1
30	B	1
31	C	1
32	A	1
33	A	1
34	D	1
35	D	1
36	C	1
37	A	1
38	D	1
39	A	1
40	D	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

October/November 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

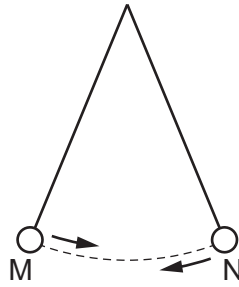
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **16** pages. Blank pages are indicated.



- 1 The diagram shows a pendulum. The pendulum bob swings repeatedly between points M and N.



A student starts a stop-watch when the bob reaches point M.

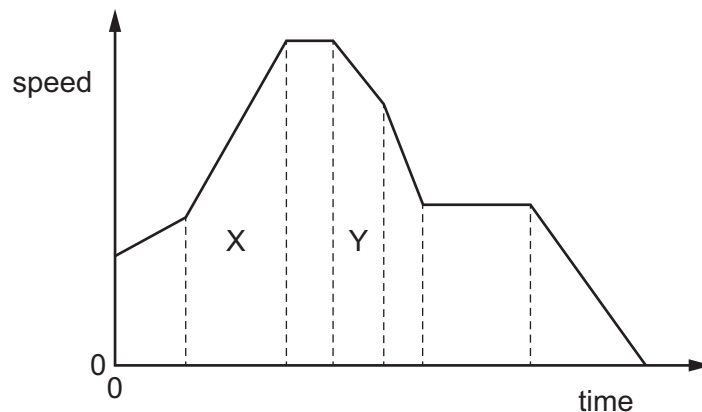
He counts each time the bob changes direction and stops the watch on the tenth change in direction.

The watch shows a time of 12.0 seconds.

What is the period of the pendulum?

- A** 0.60 s **B** 1.2 s **C** 2.4 s **D** 12.0 s

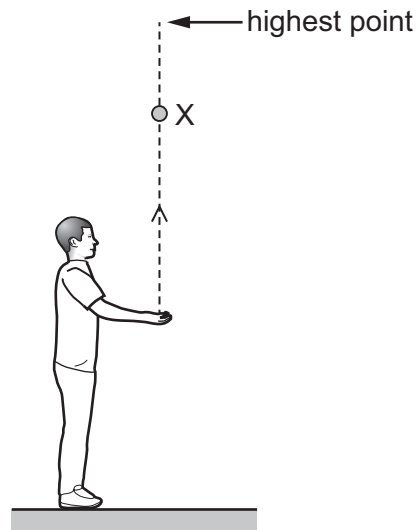
- 2 The speed–time graph represents a journey.



How does the graph show that the distance travelled in section X of the journey is greater than the distance travelled in section Y?

- A** The area below section X of the graph is greater than the area below section Y.
B The gradient of section X of the graph is greater than the gradient of section Y.
C The speed at the end of section X of the journey is greater than the speed at the end of section Y.
D The time for section X of the journey is greater than the time for section Y.

- 3 A boy throws a ball vertically upwards with a speed v .



Which row describes the speed and the acceleration of the ball at point X on the way upwards?

	speed	acceleration
A	decreasing	upwards
B	decreasing	downwards
C	increasing	upwards
D	increasing	downwards

- 4 A student compares the weights of different objects.

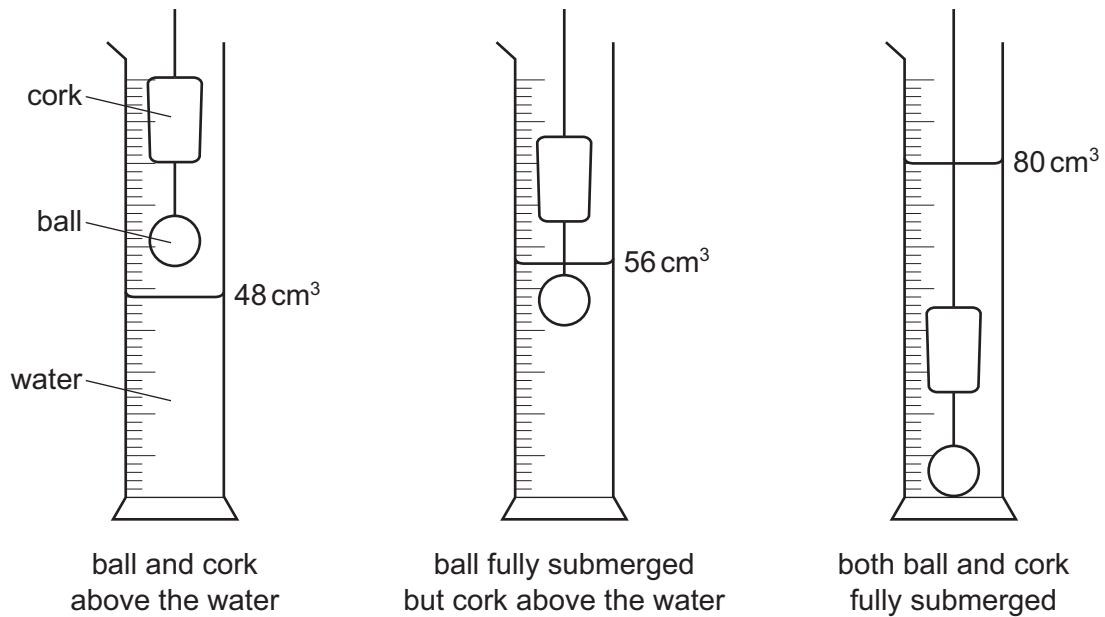
Which apparatus does he use?

- A** balance
- B** measuring cylinder
- C** stop-watch
- D** thermometer

- 5 Which quantity is weight an example of?

- A** acceleration
- B** force
- C** mass
- D** pressure

- 6 A metal ball is attached to a cork and is lowered into a measuring cylinder, pulling the cork into the water, as shown.

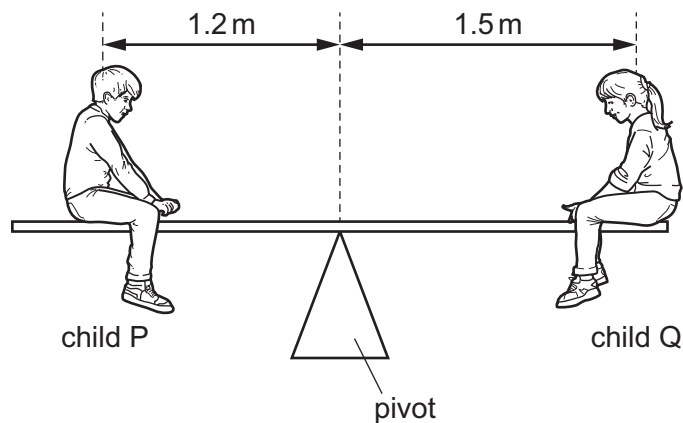


The mass of the cork is 4.8 g.

What is the density of the cork?

- A** 0.15 g/cm^3 **B** 0.20 g/cm^3 **C** 0.60 g/cm^3 **D** 5.0 g/cm^3
- 7 A uniform plank rests on a pivot at its centre.

Two children P and Q sit on the plank in the positions shown.



The mass of child P is 25 kg.

The plank is balanced.

What is the mass of child Q?

- A** 20 kg **B** 25 kg **C** 31 kg **D** 45 kg

- 8 An object is in equilibrium on the Earth.

Which statement is correct?

- A All the forces acting on the object are in the same direction.
- B All the forces acting on the object have the same value.
- C The object is weightless.
- D The resultant force acting on the object is zero.

- 9 A man jumps from a stationary balloon. After falling several hundred metres, he opens his parachute.

At which position is his kinetic energy greatest?

- A just after he jumps from the balloon
- B just before he opens his parachute
- C just after his parachute opens
- D just before he lands

- 10 The arrows show an outline of the processes in an oil-fired power station.

oil is burnt → turns turbine → turns generator → output transformer

What are the processes for the transfer of energy between the turbine and the generator and between the generator and the output transformer?

	turbine to generator	generator to output transformer
A	electrical working	electrical working
B	mechanical working	electrical working
C	mechanical working	transfer of thermal energy
D	transfer of thermal energy	mechanical working

- 11 To calculate the power produced by a force, the size of the force must be known.

What else needs to be known to calculate the power?

	the distance that the force moves the object	the time for which the force acts on the object
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = needed

✗ = not needed

- 12** A rectangular marble block has dimensions 1 m by 1 m by 5 m and weighs 125 000 N.

The marble block is stored with the long side resting on the ground, as in diagram 1.

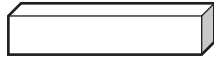


diagram 1

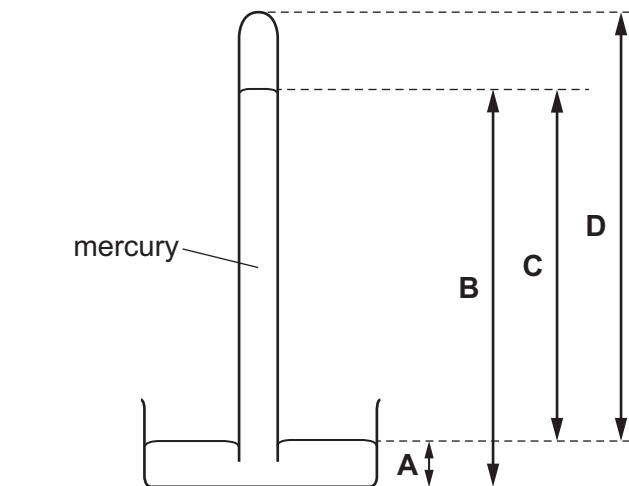


diagram 2

What is the change in the pressure on the ground due to the block when the block is stored as in diagram 2 rather than diagram 1?

- A** a decrease of 25 000 Pa
 - B** an increase of 100 000 Pa
 - C** an increase of 125 000 Pa
 - D** no change
- 13** The diagram shows a mercury barometer.

Which height is used as a measurement of atmospheric pressure?

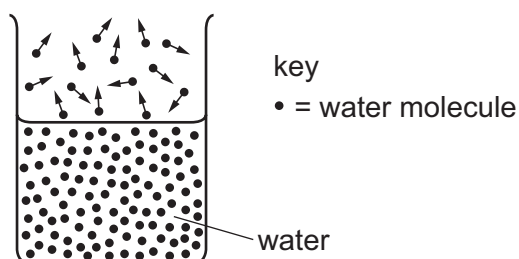


- 14** On a warm day, a driver checks the air pressure in a car tyre. At night, the temperature drops and the air pressure in the tyre decreases. There are no air leaks in the tyre.

Why does the pressure decrease?

- A** The air molecules in the tyre move more slowly.
- B** The air molecules in the tyre stop moving.
- C** The volume of the air in the tyre decreases.
- D** The volume of the air in the tyre increases.

- 15** The diagram shows the more energetic water molecules escaping from the surface of liquid water.



What is this process called?

- A** Brownian motion
 - B** condensation
 - C** evaporation
 - D** conduction
- 16** When a bridge is built, a gap is left between each concrete slab.

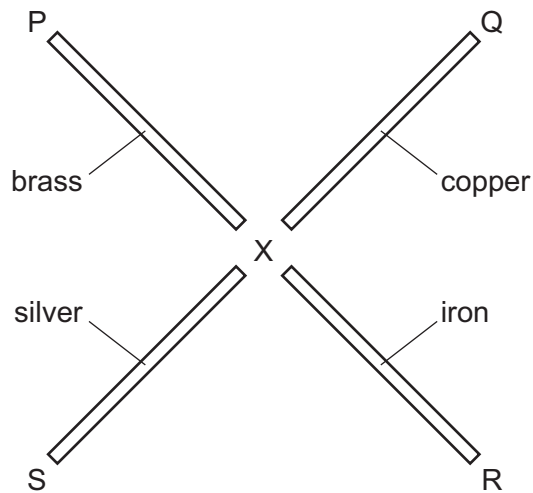
Why are these gaps left?

- A** Concrete expands on warm days.
 - B** Concrete contracts on warm days.
 - C** The gaps expand on warm days.
 - D** The gaps contract on cold days.
- 17** The table shows the melting and boiling points of four different substances.

Which substance becomes a liquid when placed in a beaker of boiling water?

	melting point / °C	boiling point / °C
A	10	90
B	40	150
C	105	230
D	110	320

18 The diagram shows four rods. Each rod is made of a different metal.



Wax is used to attach small metal balls at the rod ends P, Q, R and S.

Each rod is the same size. They are heated uniformly by a Bunsen burner at point X.

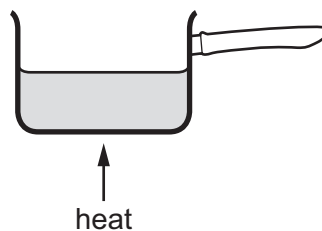
As the rods warm up, the wax melts and the balls fall off.

Why does the ball on the silver rod fall first?

- A** Silver is the best conductor of heat.
- B** Silver is the worst conductor of heat.
- C** Silver is the best radiator of heat.
- D** Silver is the worst radiator of heat.

19 The diagram shows a pan of water being heated.

After a short time, all the water in the pan begins to boil.



What is the main process by which thermal energy is transferred through the water?

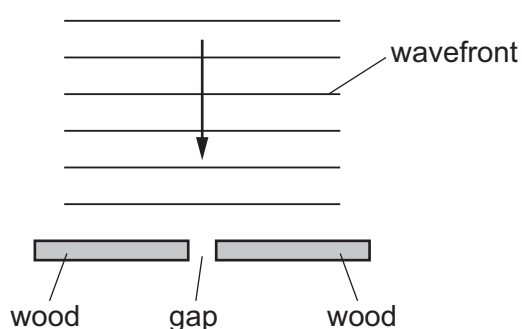
- A** conduction
- B** convection
- C** evaporation
- D** radiation

20 Which row correctly describes light waves?

	wave type	direction of vibrations
A	longitudinal	parallel to direction of wave travel
B	longitudinal	perpendicular to direction of wave travel
C	transverse	parallel to direction of wave travel
D	transverse	perpendicular to direction of wave travel

21 The diagram shows two pieces of wood resting in shallow water of constant depth.

Straight, parallel wavefronts approach the pieces of wood as indicated.



The gap between the pieces of wood is 2.0 cm wide.

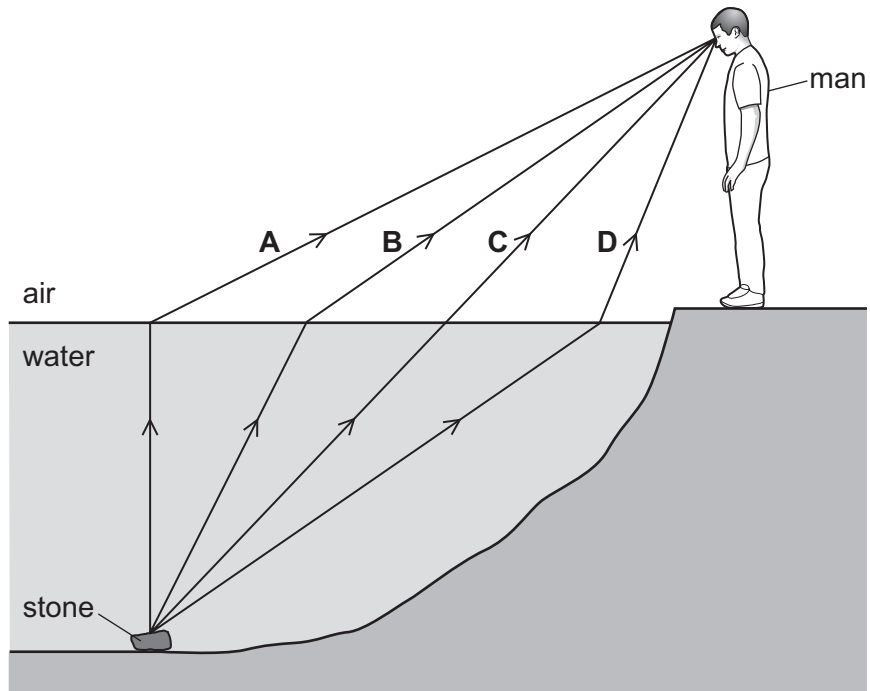
The wavefronts are 3.0 cm apart.

What is the appearance of the wavefronts after they pass through the gap?

- A** semicircular and 2.0 cm apart
- B** semicircular and 3.0 cm apart
- C** straight and 2.0 cm apart
- D** straight and 3.0 cm apart

22 A man sees a stone at the bottom of a pool of water.

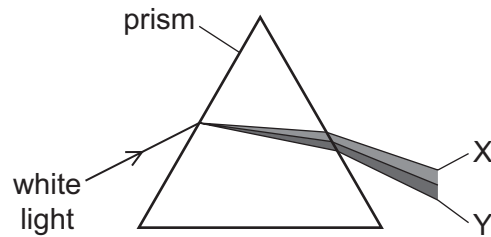
Which path could be taken by light from the stone to the man?



23 Which statement about a thin converging lens is correct?

- A** All rays of light refracted by the lens pass through the principal focus.
- B** All rays initially parallel to the principal axis of the lens are refracted through the principal focus.
- C** The focal length of the lens is the distance between the image and the principal focus.
- D** The focal length of the lens is the distance between the object and the image.

- 24** The diagram shows a beam of white light passing through a triangular prism. A spectrum is produced.



Which row correctly shows a wave property involved in producing the colours at X and Y?

	wave property	X	Y
A	diffraction	red	violet
B	dispersion	red	violet
C	reflection	violet	red
D	refraction	violet	red

- 25** The diagram shows the electromagnetic spectrum.

γ -rays	E	ultraviolet	F	infrared	microwaves	G
----------------	---	-------------	---	----------	------------	---

Which types of wave are E, F and G?

	E	F	G
A	radio	visible light	X-rays
B	radio	X-rays	ultrasound
C	X-rays	radio	ultrasound
D	X-rays	visible light	radio

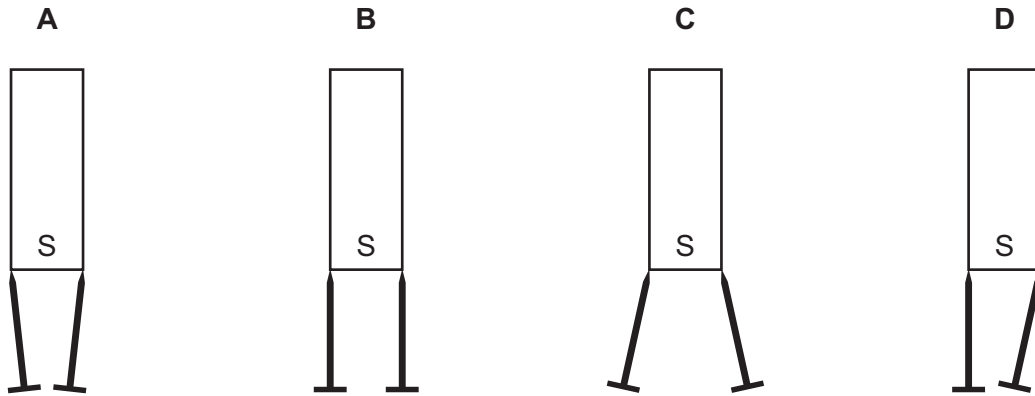
- 26** A police car with its siren sounding is stationary in heavy traffic. A pedestrian notices that, although the loudness of the sound produced does not change, the pitch varies.

Which row describes the amplitude and the frequency of the sound?

	amplitude	frequency
A	constant	constant
B	constant	varying
C	varying	constant
D	varying	varying

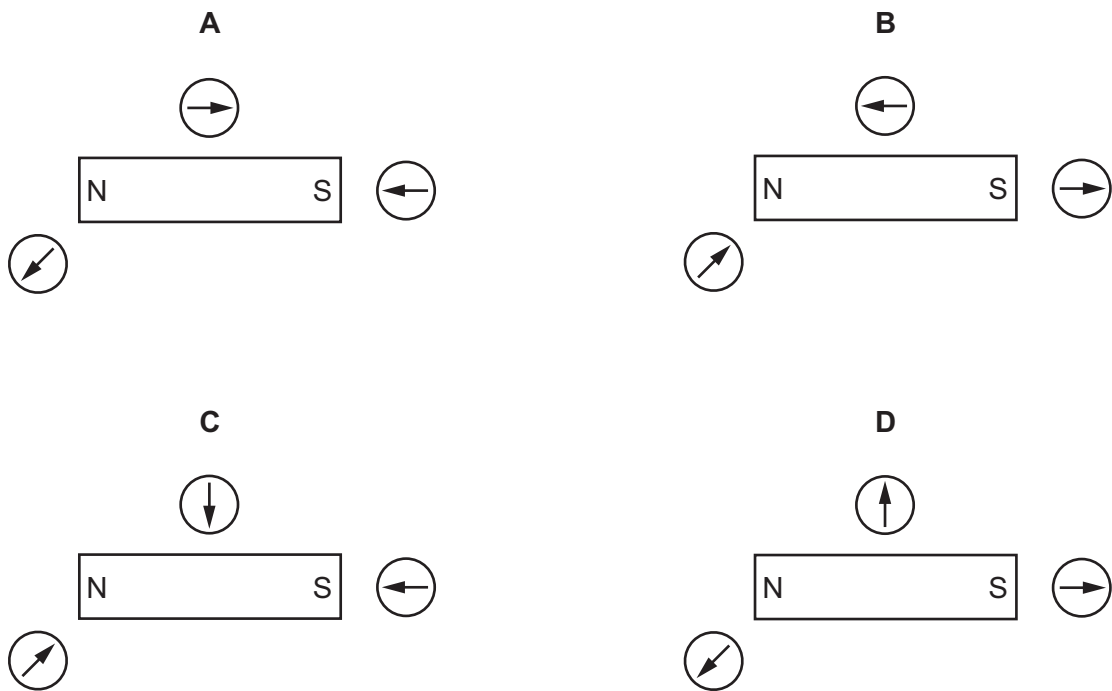
- 27** Two soft-iron pins are suspended from the S pole of a bar magnet.

Which diagram shows how the pins are deflected?



- 28** A student uses three small plotting compasses to investigate the magnetic field around a bar magnet.

Which diagram shows the directions in which the compass needles point?



- 29** What is an electric current in a metal wire?

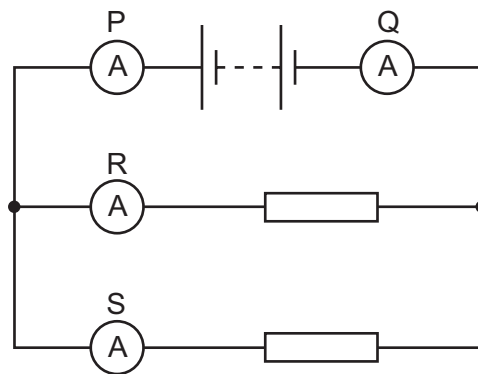
- A** a flow of electrons
- B** a flow of neutrons
- C** a flow of nucleons
- D** a flow of protons

- 30 Four wires are made of the same metal.

Which wire has the greatest resistance?

- A a 100 cm long wire with a diameter of 3.0 mm
- B a 100 cm long wire with a diameter of 6.0 mm
- C a 10 cm long wire with a diameter of 3.0 mm
- D a 10 cm long wire with a diameter of 6.0 mm

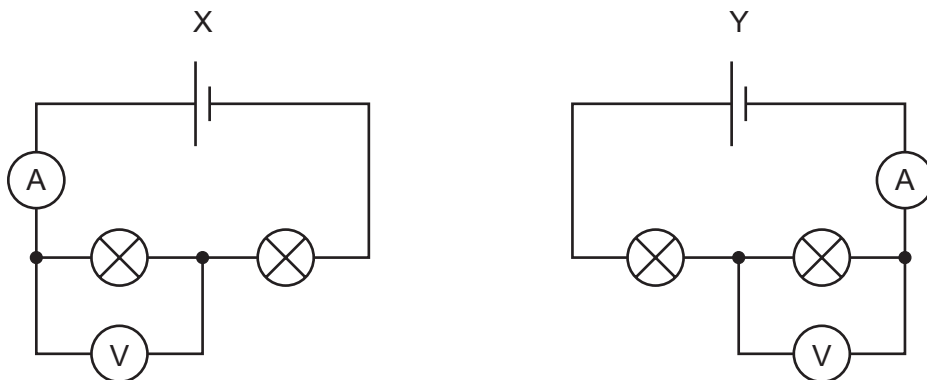
- 31 A student uses four ammeters P, Q, R and S to measure the current in different parts of the circuit shown.



Which two ammeters read the largest current?

- A P and Q
- B P and R
- C R and Q
- D R and S

- 32 A circuit X is set up with two identical lamps. Circuit Y is then set up, changing the positions of the meters.



On which meters do the readings change?

- A both the ammeter and the voltmeter
- B the ammeter only
- C the voltmeter only
- D neither the ammeter nor the voltmeter

- 33** An electrical appliance is powered from a mains supply.

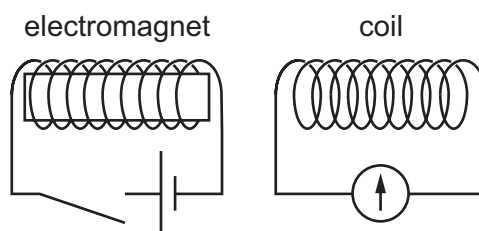
The appliance normally uses a current of 3 A, but the current briefly rises to 4 A at the instant the appliance is switched on. The cable to the appliance is designed for currents up to 6 A.

A fuse is used to protect the circuit.

What should be the rating of the fuse?

- A** 1 A **B** 3 A **C** 5 A **D** 13 A

- 34** An electromagnet is positioned close to a coil of wire.



The electromagnet is switched on, remains on for a short time, and is then switched off.

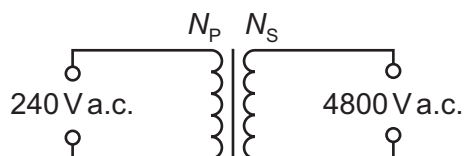
Three statements about the pointer on the galvanometer during this sequence are given.

- 1 The pointer kicks to one side as the electromagnet is switched on.
- 2 The pointer records a steady non-zero value while the electromagnet remains switched on.
- 3 The pointer kicks to the other side as the electromagnet is switched off.

Which statements are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 35** A transformer is needed to convert a supply of 240 V a.c. into 4800 V a.c..

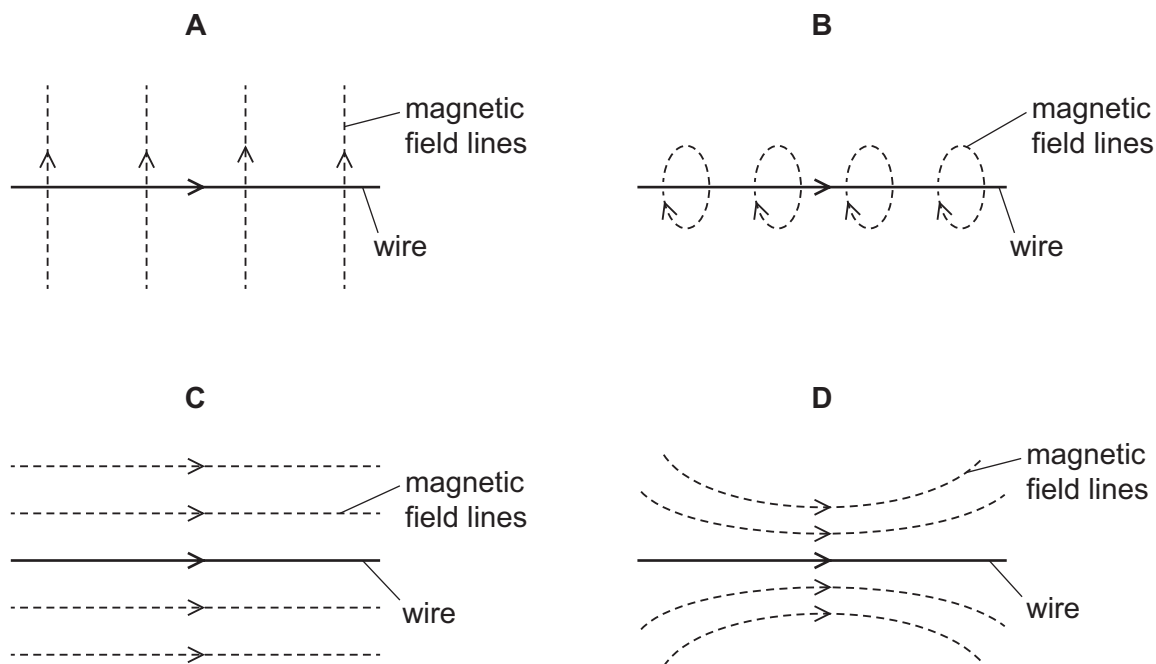


Which pair of coils would be suitable for this transformer?

	number of turns on primary coil N_p	number of turns on secondary coil N_s
A	50	1 000
B	240	48 000
C	480	24
D	2000	100

36 The diagrams show a current-carrying wire with an arrow in the direction of the current.

Which diagram shows the magnetic field produced by the current?



37 Which statement is correct for the nucleus of **any** atom?

- A The nucleus contains electrons, neutrons and protons.
- B The nucleus contains the same number of protons as neutrons.
- C The nucleus has a total charge of zero.
- D The nucleus is very small compared with the size of the atom.

38 The nucleus of an atom X is represented by the notation shown.



How many protons and how many neutrons are in this nucleus?

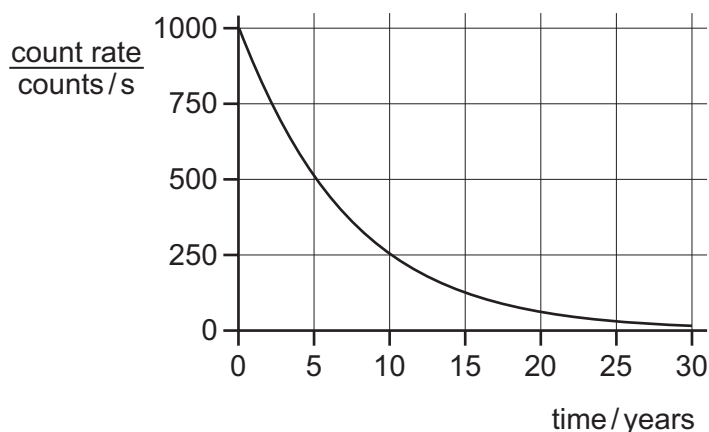
	number of protons	number of neutrons
A	P	Q
B	P	Q – P
C	Q	P
D	Q	P – Q

- 39 A radiation detector records a low reading even when no radioactive source is close. This is due to background radiation.

What does **not** contribute to this background radiation?

- A rocks on Earth
- B cosmic rays from the Sun
- C satellite TV signals
- D waste from nuclear power stations

- 40 The graph shows the radioactive decay curve of a substance.



What is the half-life of this substance?

- A 0.5 years
- B 5 years
- C 15 years
- D 30 years

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Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

October/November 2020

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	A	1
3	B	1
4	A	1
5	B	1
6	B	1
7	A	1
8	D	1
9	B	1
10	B	1
11	A	1
12	B	1
13	C	1
14	A	1
15	C	1
16	A	1
17	B	1
18	A	1
19	B	1
20	D	1
21	B	1
22	B	1
23	B	1
24	B	1
25	D	1
26	B	1
27	C	1
28	A	1

Question	Answer	Marks
29	A	1
30	A	1
31	A	1
32	D	1
33	C	1
34	B	1
35	A	1
36	B	1
37	D	1
38	D	1
39	C	1
40	B	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

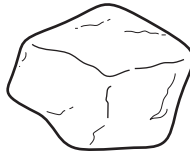
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 The diagram shows a stone of irregular shape.

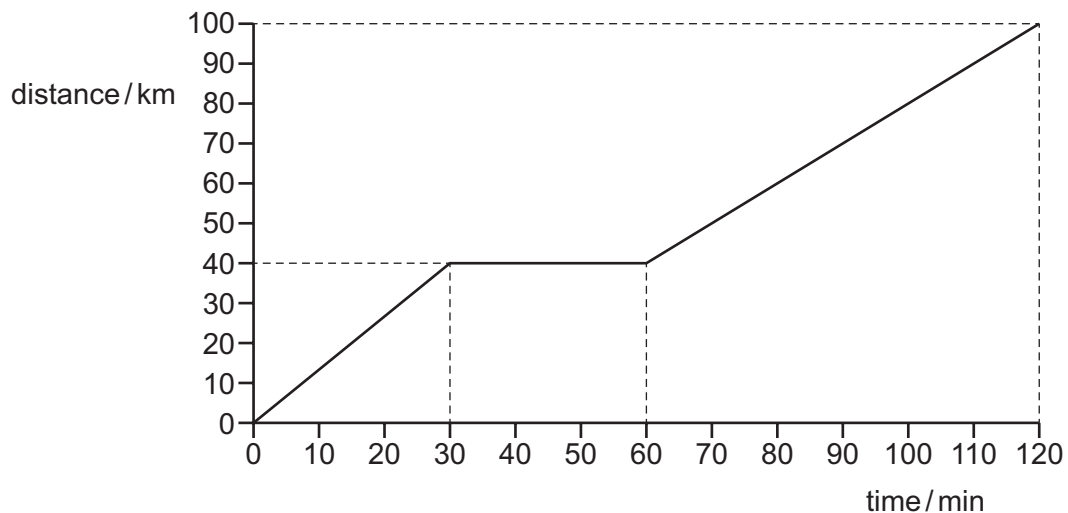


Which property of the stone can be found by lowering it into a measuring cylinder half-filled with water?

- A** length
B mass
C volume
D weight
- 2 Two stones of different weights fall at the same time from a table. Air resistance may be ignored.
 What will happen and why?

	what will happen	why
A	Both stones hit the floor at the same time.	Acceleration of free fall is constant.
B	Both stones hit the floor at the same time.	They fall at constant speed.
C	The heavier stone hits the floor first.	Acceleration increases with weight.
D	The heavier stone hits the floor first.	Speed increases with weight.

- 3 The distance–time graph for a motorway journey is shown.



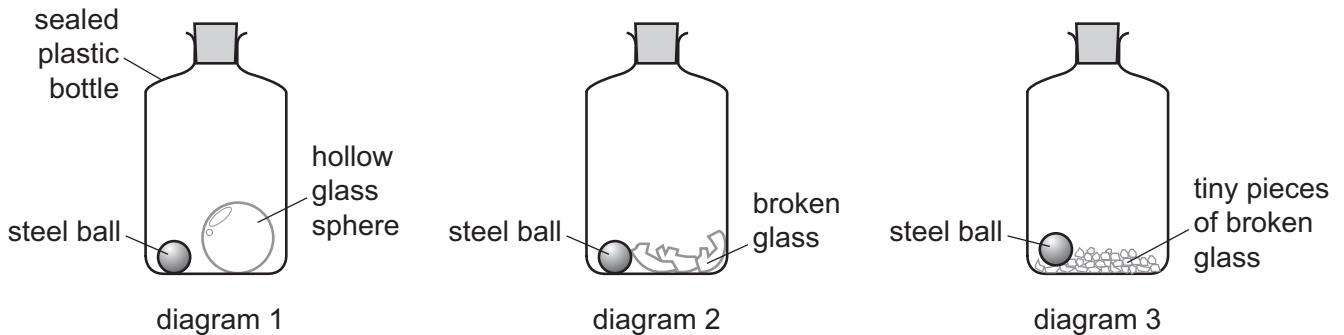
What is the average speed for the journey?

- A** 50 km/h **B** 67 km/h **C** 70 km/h **D** 83 km/h

- 4 Diagram 1 shows a sealed plastic bottle containing a hollow glass sphere and a steel ball.

Diagram 2 shows the same bottle after it has been shaken.

Diagram 3 shows the same bottle after it has been shaken again until the broken glass is in tiny pieces.



The mass of the bottle and contents in diagram 1 is m_1 .

The mass of the bottle and contents in diagram 2 is m_2 .

The mass of the bottle and contents in diagram 3 is m_3 .

Which statement gives the correct relation between m_1 , m_2 and m_3 ?

- A m_1 is equal to m_2 and m_2 is equal to m_3 .
 - B m_1 is greater than m_2 and m_2 is greater than m_3 .
 - C m_1 is less than m_2 and m_2 is greater than m_3 .
 - D m_1 is less than m_2 and m_2 is less than m_3 .
- 5 On Mars, the acceleration of free fall g is 3.7 m/s^2 .

What is the weight of a 2.0 kg mass on Mars?

- A 0.54 N B 1.9 N C 7.4 N D 20 N

- 6 The mass of an empty flask is 34 g .

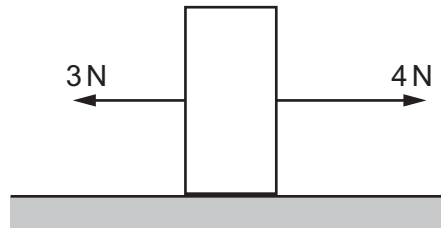
The volume of liquid added to the flask is 20 cm^3 .

The total mass of the flask and the liquid is 50 g .

What is the density of the liquid?

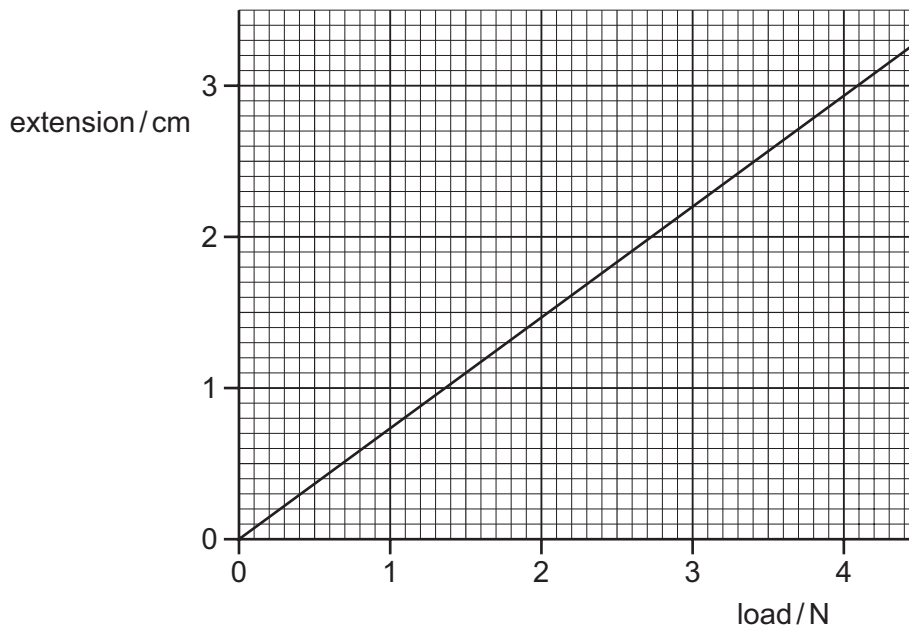
- A 0.80 g/cm^3 B 1.25 g/cm^3 C 2.50 g/cm^3 D 4.20 g/cm^3

- 7 The diagram shows a solid object on a flat surface, with two forces acting on the object.



What is the resultant force on the object?

- A 1 N to the left
 - B 1 N to the right
 - C 7 N to the left
 - D 7 N to the right
- 8 The extension–load graph for a spring is shown. The unstretched length of the spring is 17.0 cm.



When an object is suspended from the spring, the length of the spring is 19.2 cm.

What is the weight of the object?

- A 1.4 N
 - B 1.6 N
 - C 2.6 N
 - D 3.0 N
- 9 Which situation involves no work being done and no energy being transferred?
- A a car skidding to a stop on a road
 - B a crane lifting a load
 - C a heavy load hanging from a strong bar
 - D a student dragging a big box over a rough floor

- 10** A student suggests that there are several ways of transferring energy to a small, stationary block of iron on a smooth table. He makes the following suggestions.

- 1 Heat it.
- 2 Shine light on it.
- 3 Pass a current through it.

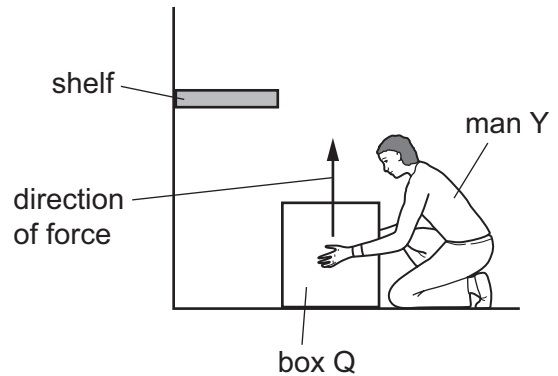
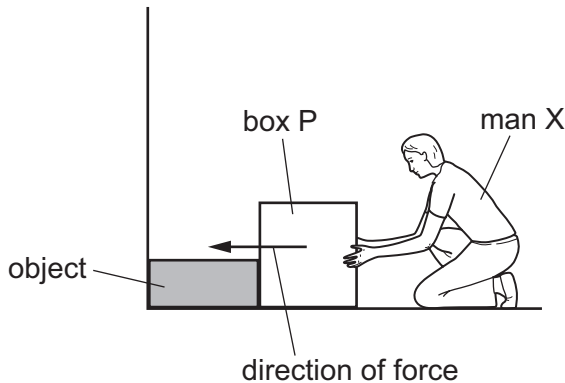
Which suggestions are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 11** Two men, X and Y, try to move identical heavy boxes, P and Q.

Man X tries to push box P along the floor. The box does not move because an object is in the way.

Man Y lifts box Q from the floor onto a shelf.



Which man does the most work on their box, and which box gains the most energy?

	man doing most work	box gaining most energy
A	X	P
B	X	Q
C	Y	P
D	Y	Q

- 12** A book has a mass of 400 g.

The surface of the book in contact with a table has dimensions $0.10 \text{ m} \times 0.20 \text{ m}$.

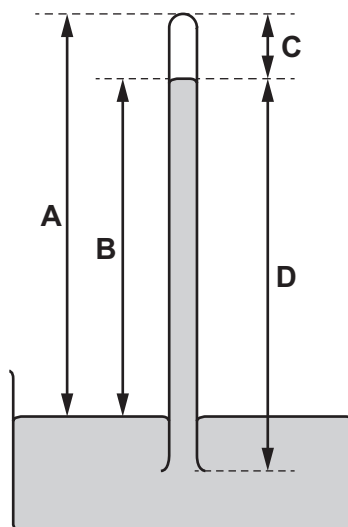
The gravitational field strength g is 10 N/kg .

What is the pressure exerted on the table due to the book?

- A** 0.08 N/m^2 **B** 8.0 N/m^2 **C** 20 N/m^2 **D** 200 N/m^2

13 A mercury barometer is used to measure atmospheric pressure.

Which distance gives a measure of the atmospheric pressure?



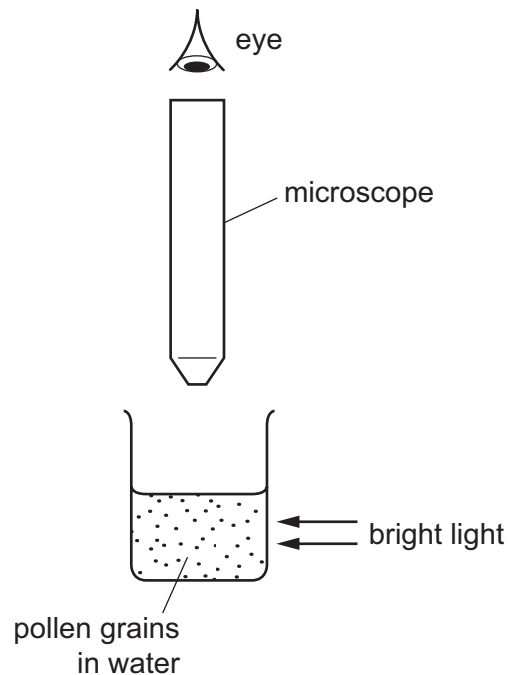
14 Air trapped in a sealed bottle increases in temperature.

Which row shows the changes that occur to the trapped air?

	pressure of the air	average speed of molecules in the air
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 15 Very small pollen grains are suspended in water. A bright light shines from the side.

When looked at through a microscope, small specks of light are seen to be moving in a random, jerky manner.



What are the moving specks of light?

- A pollen grains being hit by other pollen grains
 - B pollen grains being hit by water molecules
 - C water molecules being hit by other water molecules
 - D water molecules being hit by pollen grains
- 16 Why are small gaps left between the metal rails of a railway track?
- A to allow for expansion of the rails on a hot day
 - B to allow for contraction of the rails on a hot day
 - C to allow for expansion of the rails on a cold day
 - D to allow for contraction of the rails on a cold day
- 17 All thermometers require a physical property that changes with temperature.

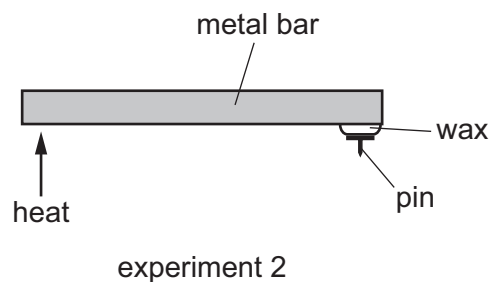
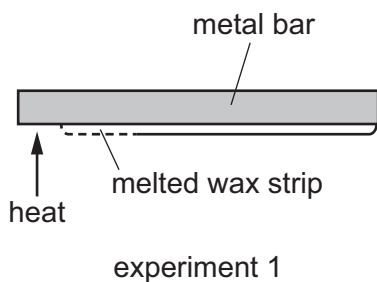
Which property would **not** be suitable for use in a thermometer?

- A pressure
- B volume
- C electrical resistance
- D mass

- 18 The thermal capacity of object Y is greater than that of object Z.

What is a consequence of this?

- A** Object Y needs less thermal energy to melt it than object Z.
- B** Object Y needs less thermal energy to raise its temperature by 1°C than object Z.
- C** Object Y needs more thermal energy to melt it than object Z.
- D** Object Y needs more thermal energy to raise its temperature by 1°C than object Z.
- 19 Two students carry out different experiments to compare the abilities of different metals to conduct thermal energy.



In experiment 1, the bar is heated for one minute and the length of wax strip that melts is measured.

In experiment 2, the bar is heated and the time taken for the pin to drop off is measured.

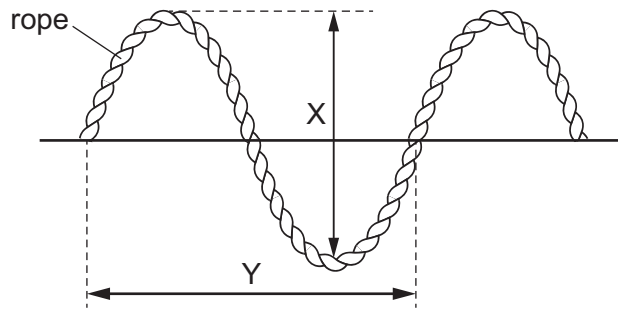
What happens to each of these measurements when a better conductor of thermal energy is tested?

	length of melted wax strip	time taken for the pin to drop
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 20 In which substances is convection a method of thermal energy transfer?

- A** air and water only
- B** air only
- C** air, water and wood
- D** water only

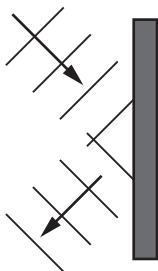
21 The diagram represents a wave on a rope.



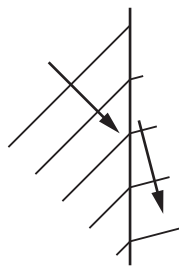
Which type of wave is shown and which labelled arrow shows the wavelength of the wave?

	type of wave	wavelength
A	longitudinal	X
B	longitudinal	Y
C	transverse	X
D	transverse	Y

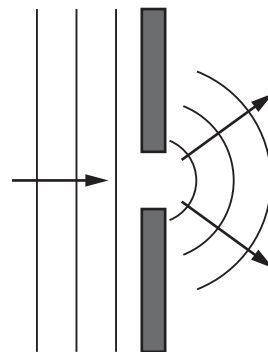
22 The diagrams show water waves in three different situations. The arrows show the direction of travel of the waves.



1



2

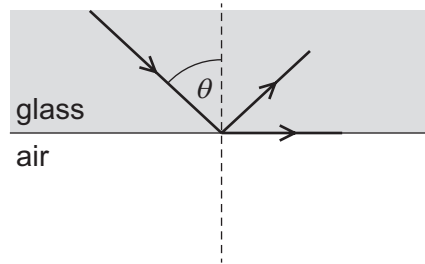


3

What does each diagram show?

	1	2	3
A	diffraction	reflection	refraction
B	diffraction	refraction	reflection
C	reflection	diffraction	refraction
D	reflection	refraction	diffraction

- 23** The diagram shows a narrow beam of light incident on a glass-air boundary. Some of the light emerges along the surface of the glass and some is reflected back into the glass.

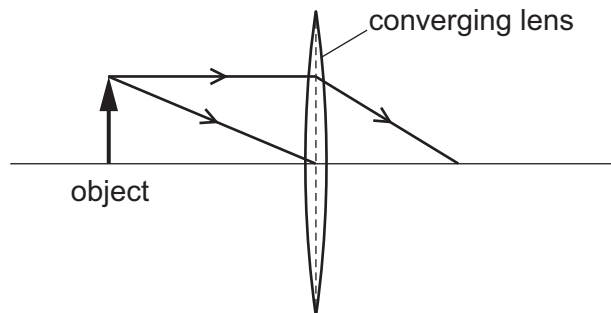


Which row is correct?

	this is an example of total internal reflection	angle θ is the critical angle
A	no	yes
B	no	no
C	yes	no
D	yes	yes

- 24** An object is placed in front of a thin converging lens.

The diagram shows the paths of two rays from the top of the object.



An image of the object is formed on a screen to the right of the lens.

How does this image compare with the object?

- A** It is larger and inverted.
- B** It is larger and the same way up.
- C** It is smaller and inverted.
- D** It is smaller and the same way up.

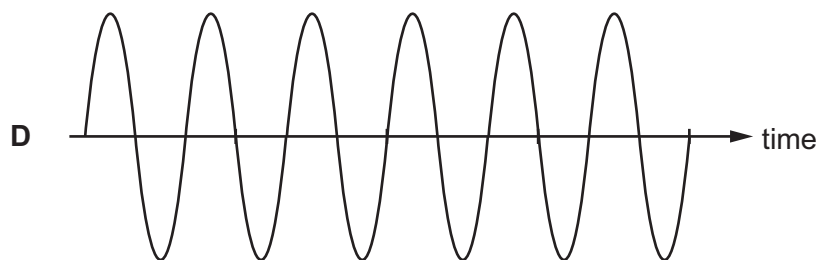
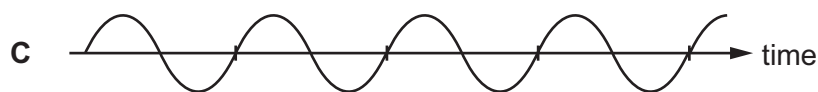
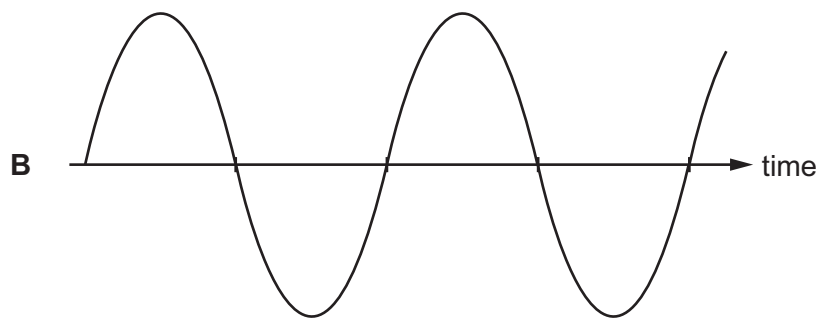
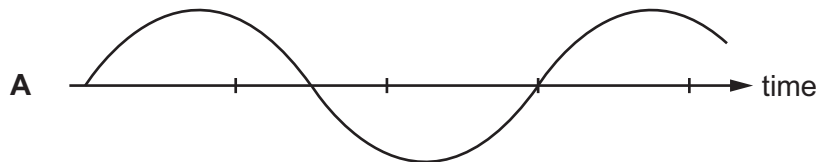
25 A student is asked to give two uses of four different types of electromagnetic radiation.

In which row are both the uses correct?

	radiation	use 1	use 2
A	radio	sterilising medical equipment	mobile phone masts
B	microwaves	mobile phones	sterilising medical equipment
C	infrared	remote controllers	intruder alarms
D	X-rays	security in airports	intruder alarms

26 The diagrams represent the waves produced by four sources of sound. The scales are the same for all the diagrams.

Which sound has the highest frequency?



- 27** Diagram 1 shows a small compass needle with its poles marked. It is not near any magnetic materials.

Diagram 2 shows a bar magnet with its poles marked. The compass needle is placed at point P.

diagram 1

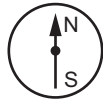


diagram 2

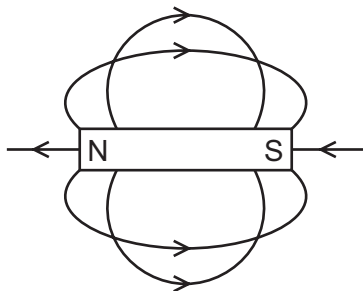


• P

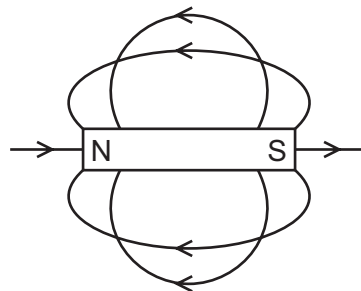
In which direction will the N pole of the compass needle point?

- A** to the left
 - B** to the right
 - C** down the page
 - D** up the page
- 28** Which diagram shows the pattern and direction of the magnetic field lines around a bar magnet?

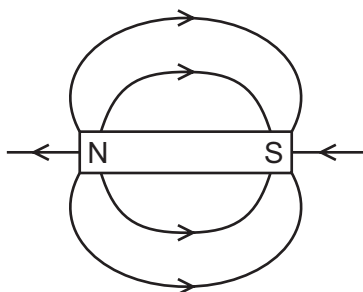
A



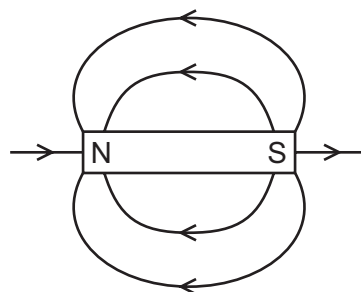
B



C



D

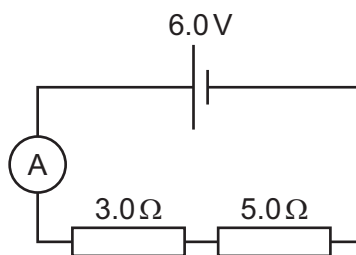


- 29 A student rubs a plastic rod with a cloth.

The rod becomes positively charged.

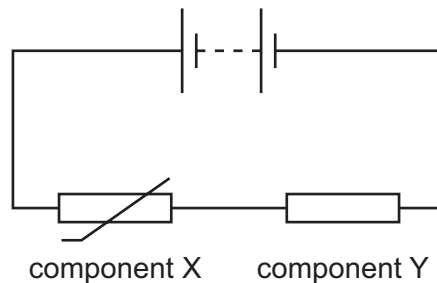
What has happened to the rod?

- A** It has gained electrons.
B It has gained protons.
C It has lost electrons.
D It has lost protons.
- 30 The circuit diagram shows a cell connected to an ammeter and two resistors.



What is the current in the circuit?

- A** 0.75 A **B** 1.3 A **C** 12 A **D** 48 A
- 31 The diagram shows an electrical circuit.



Which row describes what happens when the temperature rises?

	resistance of component X	potential difference (p.d.) across component Y
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 32 Diagram 1 shows a resistor connected in a circuit. Diagram 2 shows an identical resistor connected in parallel with the first one.

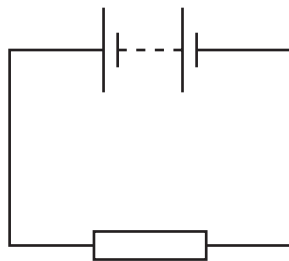


diagram 1

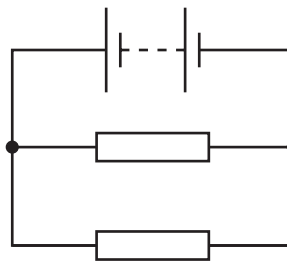


diagram 2

What is the combined resistance of the two resistors?

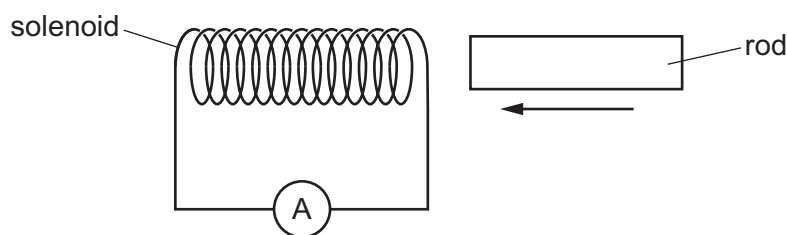
- A greater than in the circuit of diagram 1
 - B less than in the circuit of diagram 1
 - C the same as in the circuit of diagram 1
 - D zero
- 33 The current in a lamp connected on its own to the mains supply is 0.60 A.

A table decoration has three of these lamps connected in parallel.

Which rating of fuse is suitable to protect this circuit?

- A 0.2 A
- B 0.6 A
- C 1.0 A
- D 5.0 A

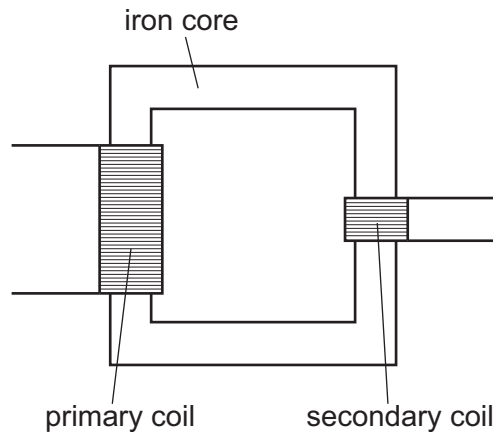
- 34 A solenoid is connected to a very sensitive ammeter. A rod is inserted into one end of the solenoid. The ammeter shows that there is a small electric current in the solenoid while the rod is moving.



Which rod is being inserted?

- A a heated copper rod
- B a magnetised steel rod
- C an uncharged nylon rod
- D a radioactive uranium rod

- 35** The diagram shows a transformer. There are 460 turns on the primary coil and 24 turns on the secondary coil. The primary voltage is 230 V.



What is the secondary voltage?

- A** 2.0 V **B** 12 V **C** 48 V **D** 4400 V
- 36** A class is designing a d.c. motor. To achieve a greater turning effect, three suggestions are made.
- 1 Have a larger current in the coil of the motor.
 - 2 Have a stronger magnet in the motor.
 - 3 Put a larger number of turns on the coil.

Which suggestions will help to increase the turning effect?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only
- 37** The charge on a proton is e .

What is the charge on an electron and what is the charge on a neutron?

	electron	neutron
A	e	e
B	e	0
C	$-e$	$-e$
D	$-e$	0

38 The nuclide notation of the isotope strontium-90 is $^{90}_{38}\text{Sr}$.

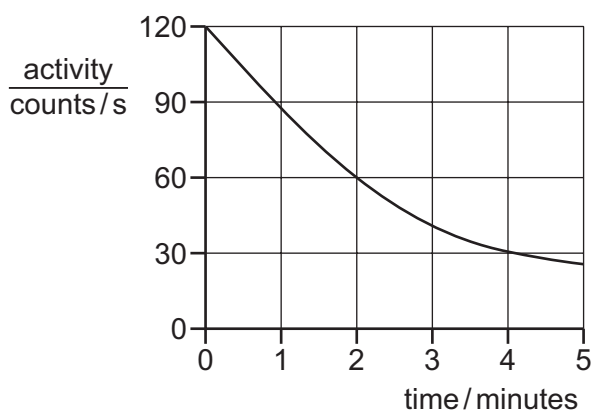
Which statement is correct?

- A A nucleus of strontium-90 has 38 neutrons.
- B A nucleus of strontium-90 has 52 neutrons.
- C A nucleus of strontium-90 has 90 electrons.
- D A nucleus of strontium-90 has 90 neutrons.

39 Which statement about α -particles and γ -rays is correct?

- A α -particles are a form of electromagnetic radiation.
- B α -particles penetrate materials more easily than γ -rays.
- C The emission of an α -particle produces a nucleus of a different element.
- D γ -rays are more ionising than α -particles.

40 The graph shows the activity of a radioactive source over a period of time.



What is the half-life of the source?

- A 1.0 minute
- B 2.0 minutes
- C 2.5 minutes
- D 4.0 minutes

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/11

Paper 1 Multiple Choice (Core)

May/June 2021

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	A	1
3	A	1
4	A	1
5	C	1
6	A	1
7	B	1
8	D	1
9	C	1
10	D	1
11	D	1
12	D	1
13	B	1
14	D	1
15	B	1
16	A	1
17	D	1
18	D	1
19	C	1
20	A	1
21	D	1
22	D	1
23	A	1
24	A	1
25	C	1
26	D	1
27	B	1
28	C	1

Question	Answer	Marks
29	C	1
30	A	1
31	B	1
32	B	1
33	D	1
34	B	1
35	B	1
36	A	1
37	D	1
38	B	1
39	C	1
40	B	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

October/November 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.



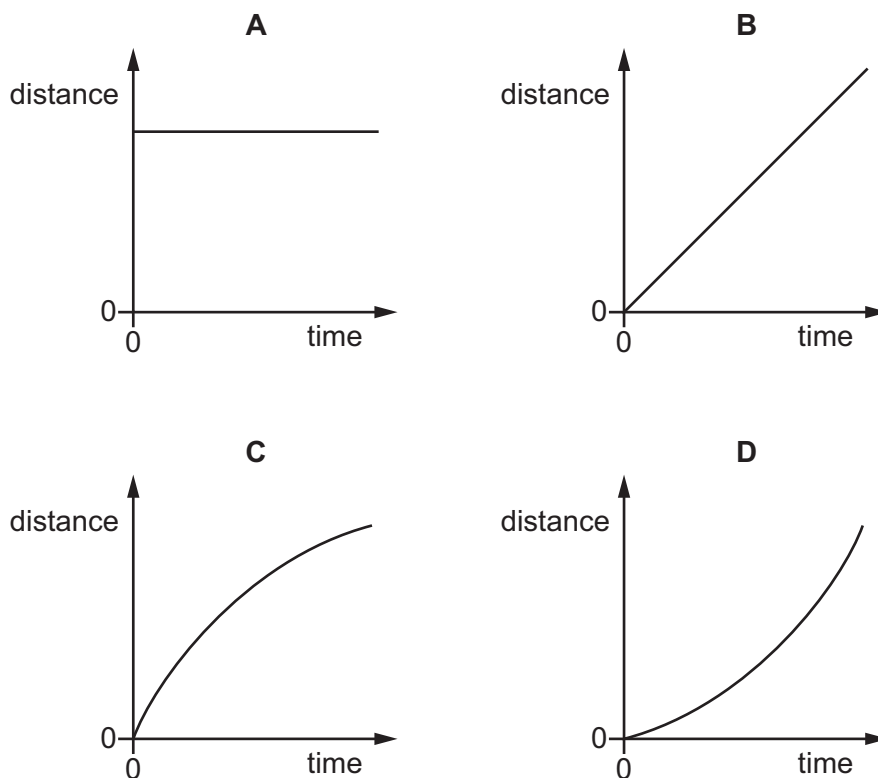
- 1 A student measures the volume of a quantity of water.

Which apparatus is suitable?

- A a balance
- B a measuring cylinder
- C a ruler
- D a thermometer

- 2 The diagrams show distance–time graphs for four objects.

Which graph represents an object moving with an increasing speed?



- 3 The gravitational field strength on the Moon is smaller than that on the Earth.

A scientist examines a rock which has been brought back from the Moon.

He measures three quantities.

- 1 the density of the rock
- 2 the mass of the rock
- 3 the weight of the rock

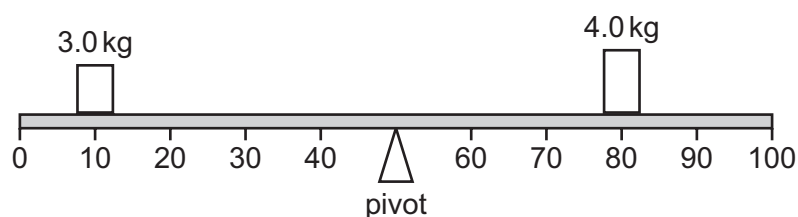
Which quantities are the same size on the surface of the Earth and on the surface of the Moon?

- A 1 and 2 only
- B 1 and 3 only
- C 2 and 3 only
- D 1, 2 and 3

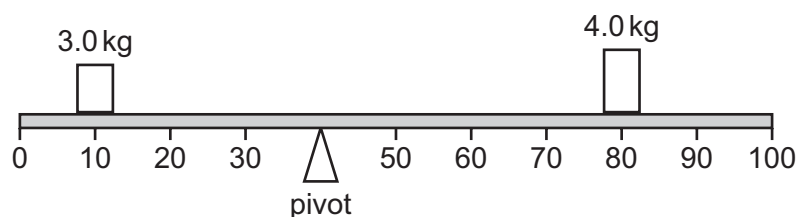
- 4 Which substance in the table has the lowest density?

	substance	mass / g	volume / cm ³
A	nylon	1.2	1.0
B	cotton	1.5	1.0
C	olive oil	1.8	2.0
D	water	2.0	2.0

- 5 A 100 cm beam balances as shown.



The pivot is moved 10 cm to the left.



What will be the effect of this change on the anticlockwise and clockwise moments about the pivot?

	anticlockwise moment	clockwise moment
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

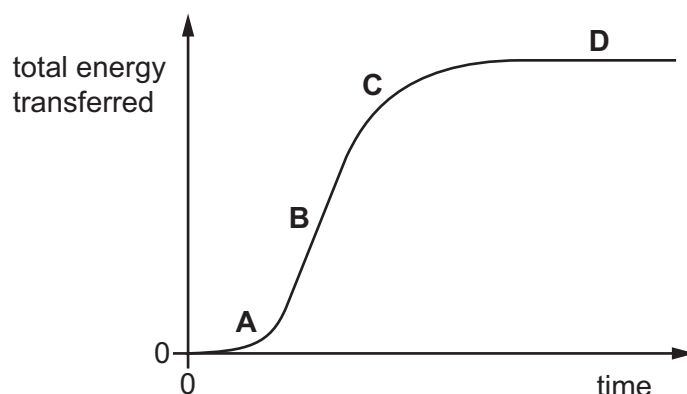
- 6 A spacecraft is travelling in space with no resultant force and no resultant moment acting on it.

Which statement about the spacecraft is correct?

- A** Its direction is changing.
- B** It is in equilibrium.
- C** Its speed is decreasing.
- D** Its speed is increasing.

- 7 The graph shows the total energy transferred by an electric motor over a period of time.

In which region of the graph is the greatest power being developed by the motor?

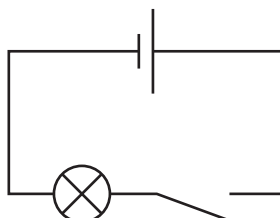


- 8 Electrical energy may be obtained from nuclear fission.

In which order is the energy transferred in this process?

- A** nuclear fuel → generator → reactor and boiler → turbines
B nuclear fuel → generator → turbines → reactor and boiler
C nuclear fuel → reactor and boiler → generator → turbines
D nuclear fuel → reactor and boiler → turbines → generator

- 9 The diagram shows an electric circuit. When the switch is closed, the lamp is lit.



Which row states the type of energy stored in the cell and how this energy is usefully transferred to the lamp?

	type of energy stored in the cell	how this energy is usefully transferred to the lamp
A	chemical	by electric current
B	chemical	by light
C	electrical	by electric current
D	electrical	by light

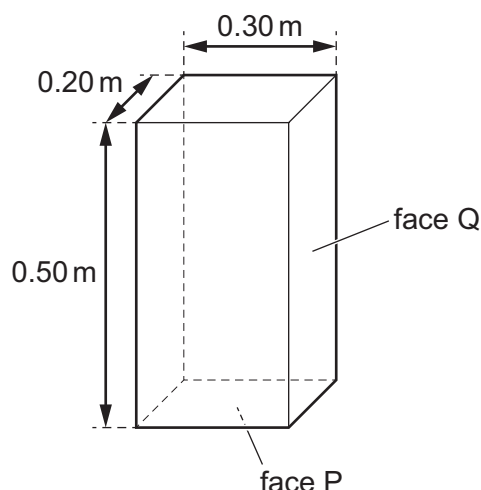
- 10** A scientist uses an electric motor to lift a load through a vertical distance of 2.0 m.

He then increases the input power to the motor and repeats the experiment. The efficiency of the motor does not change.

Which row correctly describes the effect that this has on the useful work done lifting the load and the time taken to lift it?

	work done	time taken
A	decreases	decreases
B	stays the same	decreases
C	decreases	stays the same
D	stays the same	stays the same

- 11** The box shown has a weight of 15 N.



The box is resting on a horizontal surface with face P in contact with the surface.

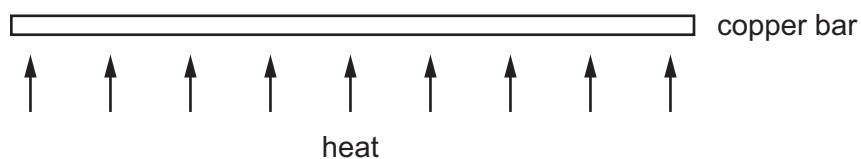
What is the change in pressure on the surface if the box falls over onto face Q?

- A** $0.0040 \text{ m}^2/\text{N}$ **B** $0.0067 \text{ m}^2/\text{N}$ **C** 100 N/m^2 **D** 250 N/m^2
- 12** A liquid is evaporating. The liquid is not boiling.
- Which statement about the liquid is correct at an instant in time?
- A** Any molecule can escape, and from any part of the liquid.
B Any molecule can escape, but only from the liquid's surface.
C Only molecules with enough energy can escape, and only from the liquid's surface.
D Only molecules with enough energy can escape, but from any part of the liquid.

13 Which row correctly describes the movement of particles in solids and liquids?

	solids	liquids
A	no movement	move around each other
B	no movement	vibration only
C	vibration only	move around each other
D	vibration only	vibration only

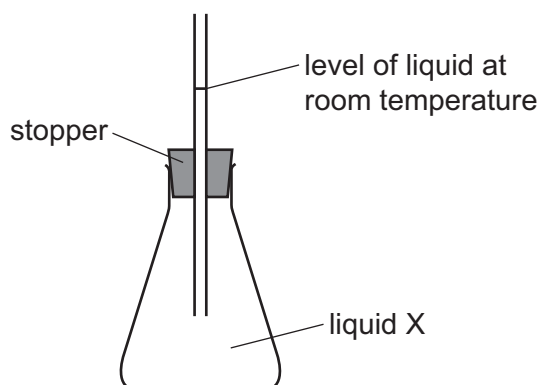
14 A long, thin bar of copper is heated gently and evenly along its length.



What happens to the bar?

- A** It becomes less heavy.
- B** It becomes longer.
- C** It becomes shorter.
- D** It bends at the ends.

- 15 The diagram shows a flask which has been filled with liquid X at room temperature.



When the flask is placed in warm water, the liquid rises higher up the tube. When the flask is put in cold water, the liquid drops below the original level in the tube.

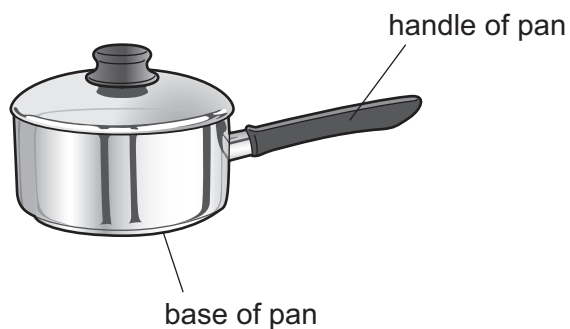
The experiment is repeated using an identical flask but a different liquid Y. The initial level of the liquid in the tube is the same as that in the original experiment.

Liquid Y expands more, per degree increase in temperature, than liquid X.

Which row is correct for the level of the liquid?

	rises most in warm water	falls most in cold water
A	X	X
B	X	Y
C	Y	X
D	Y	Y

16 The diagram shows a pan used for cooking food.



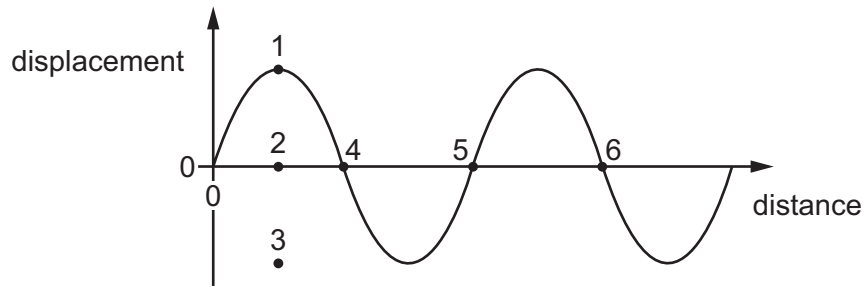
Which row is correct for the materials used to make the base and the handle of the pan?

	base of pan	handle of pan
A	good thermal conductor	good thermal conductor
B	good thermal conductor	poor thermal conductor
C	poor thermal conductor	good thermal conductor
D	poor thermal conductor	poor thermal conductor

17 Which row gives the correct description of a sound wave?

	action of wave	type of wave
A	transfers energy without transferring matter	longitudinal
B	transfers energy without transferring matter	transverse
C	transfers matter without transferring energy	longitudinal
D	transfers matter without transferring energy	transverse

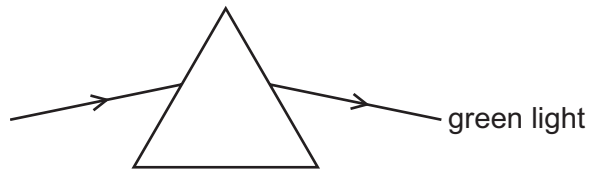
18 The diagram shows a wave.



Which row correctly indicates the amplitude and the wavelength of the wave?

	amplitude	wavelength
A	the distance between 1 and 2	the distance between 4 and 5
B	the distance between 1 and 2	the distance between 4 and 6
C	the distance between 1 and 3	the distance between 4 and 5
D	the distance between 1 and 3	the distance between 4 and 6

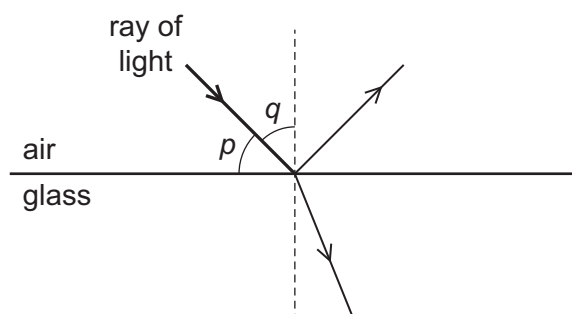
19 A ray of green light passes through a glass prism as shown.



Which colours of light refract as shown in the table?

	refract more than green	refract less than green
A	red	blue
B	red	yellow
C	violet	blue
D	violet	yellow

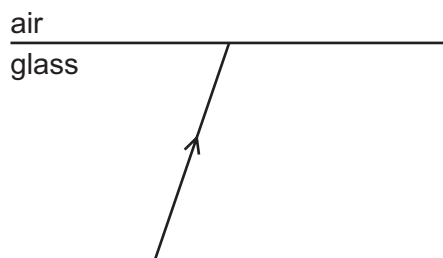
- 20 The diagram shows a ray of light in air incident on a glass block. Some of the light is refracted and some of the light is reflected. Two angles, p and q , are marked on the diagram.



Which row gives the angle of incidence and states whether total internal reflection occurs?

	angle of incidence	total internal reflection
A	p	no
B	p	yes
C	q	no
D	q	yes

- 21 The diagram shows a ray of light in glass incident on the surface between the glass and air.



What happens if the angle of incidence is made larger than the critical angle for the glass?

- A** The angle of refraction becomes equal to 90° .
 - B** There is a refracted ray and a ray reflected inside the glass.
 - C** There is a refracted ray only.
 - D** There is only a ray reflected inside the glass.
- 22 Radiation from which part of the electromagnetic spectrum is used in the remote controller for a television?
- A** infrared waves
 - B** microwaves
 - C** radio waves
 - D** ultraviolet waves

23 Two rays of light are different colours.

Which row is correct?

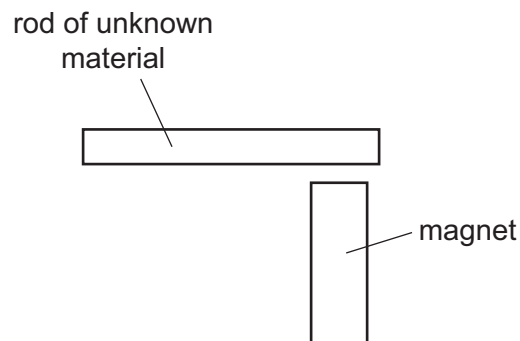
	speed of the two colours in a vacuum	wavelengths of the two colours in a vacuum
A	different	different
B	different	the same
C	the same	different
D	the same	the same

24 What is ultrasound?

- A** sound waves that are so loud that they damage human hearing
- B** sound waves that are too high-pitched for humans to hear
- C** sound waves that are too low-pitched for humans to hear
- D** sound waves that are too quiet for humans to hear

25 A bar magnet is brought near to a rod of unknown material.

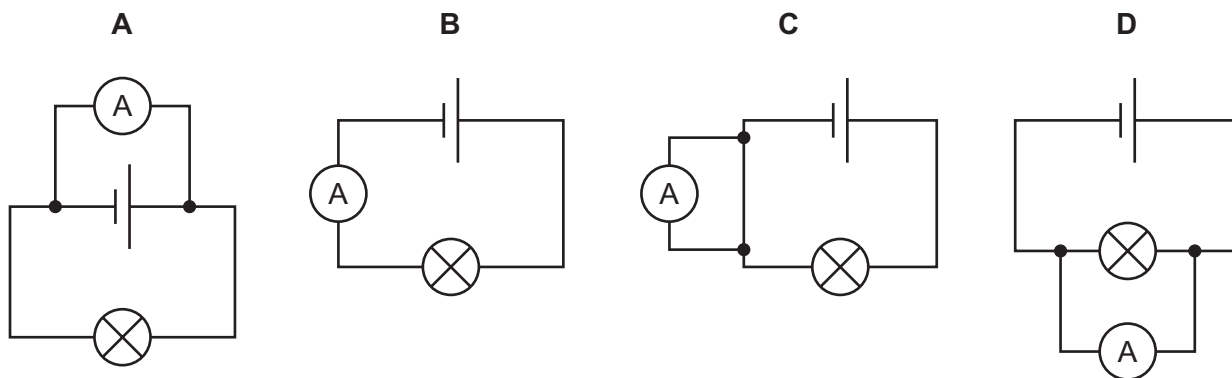
The unknown rod is attracted to both ends of the bar magnet.



Which material is the bar made from?

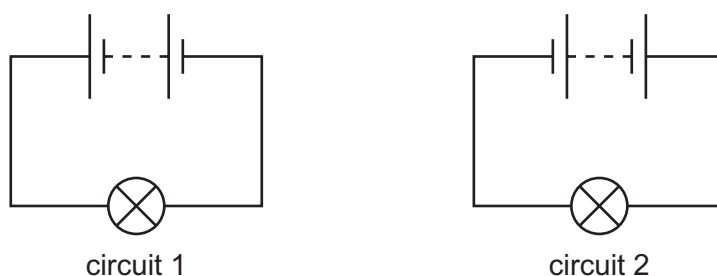
- A** aluminium
- B** magnetised steel
- C** unmagnetised iron
- D** plastic

26 In which circuit is the ammeter measuring the flow of charge through the lamp?



27 In circuit 1, a negative charge flows in a clockwise direction. The bulb is bright.

In circuit 2, the battery is reversed as shown. The bulb is equally bright.



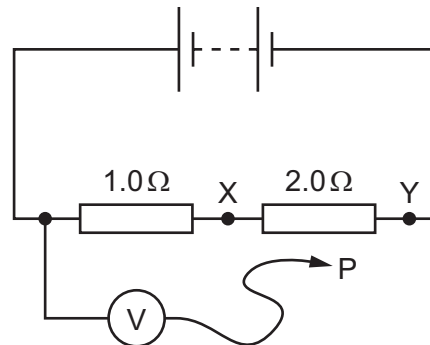
Which charge flows in circuit 2 and in which direction?

	charge	direction
A	negative	anticlockwise
B	negative	clockwise
C	positive	anticlockwise
D	positive	clockwise

- 28** The diagram shows a circuit containing two resistors of resistance $1.0\ \Omega$ and $2.0\ \Omega$.

A voltmeter is connected across the $1.0\ \Omega$ resistor by connecting P to X.

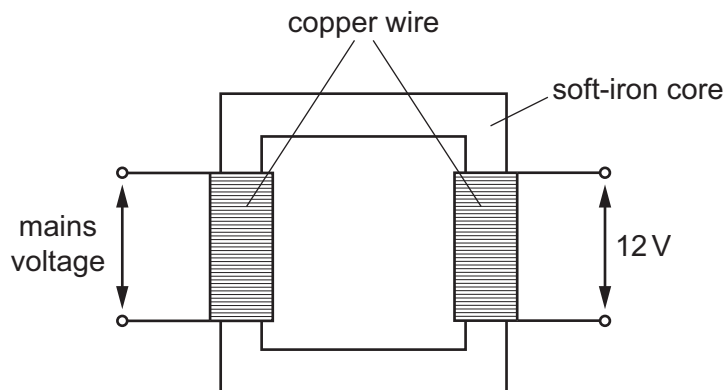
The reading on the voltmeter is 6.0 V .



P is moved to point Y in the circuit.

What is the new reading on the voltmeter?

- A** 3.0 V **B** 6.0 V **C** 12 V **D** 18 V
- 29** The step-down transformer shown reduces mains voltage to 12 V .

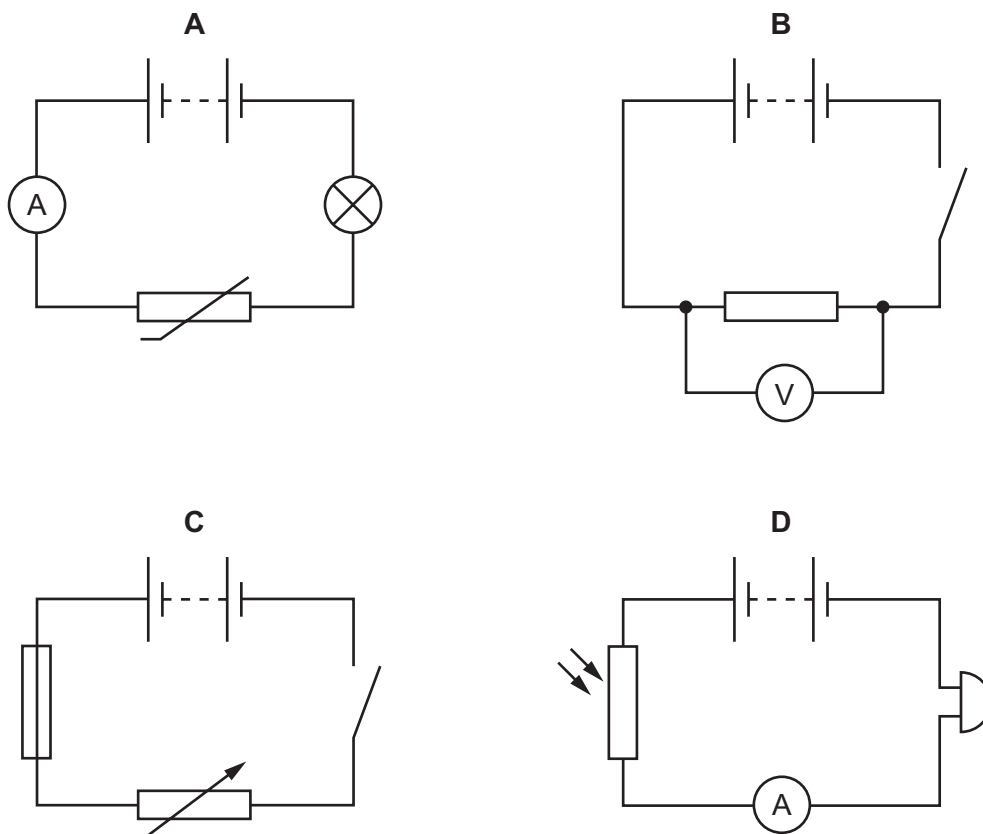


When the transformer is used, some energy is transferred to the surroundings.

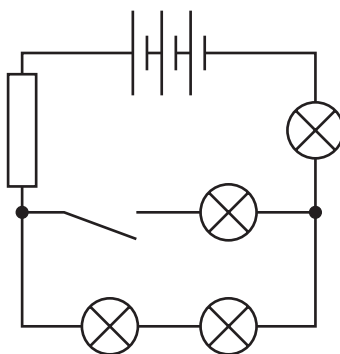
Which type of energy is transferred to the surroundings?

- A** chemical energy
B light energy
C thermal energy
D elastic energy

30 Which circuit contains a fuse?



31 The diagram shows an electric circuit.

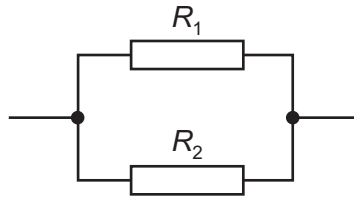


Which row correctly shows the numbers of different components in the circuit?

	cells	lamps	switches
A	1	1	4
B	1	4	1
C	3	1	4
D	3	4	1

- 32 Two resistors, with resistances R_1 and R_2 , are connected in parallel.

The resistance R_1 is greater than the resistance R_2 .



What is the resistance of the parallel combination?

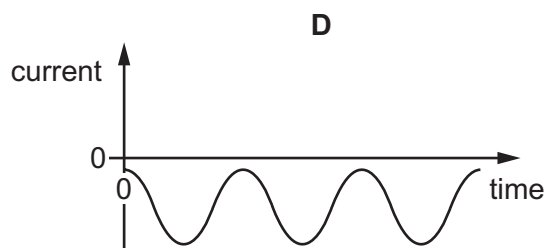
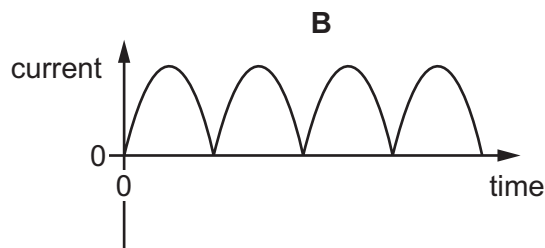
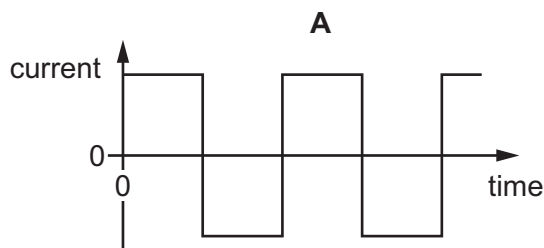
- A less than either R_1 or R_2
 - B equal to R_1
 - C equal to R_2
 - D the average of R_1 and R_2
- 33 The metal cases of electrical appliances are connected to an earth wire.
- Which statement is **not** correct?
- A The live wire may become loose and touch the metal case.
 - B If the metal case becomes live, the earth wire conducts current to the ground.
 - C The earth wire needs to have a high resistance.
 - D Earthing metal cases helps prevent a person from receiving an electric shock.
- 34 A teacher asks, 'Why do we put a fuse in a mains circuit?'

Student 1 says, 'It protects the wiring from overheating.' Student 2 says, 'It protects us from getting a shock if we touch the live wire.'

Who is correct?

- A both students
- B neither student
- C student 1 only
- D student 2 only

35 Which graph represents an alternating current (a.c.)?



36 The current in a small electric heater is 4.0 A.

The cable connected to the heater is able to carry currents up to 10 A.

Fuses rated 1 A, 3 A, 5 A and 13 A are available.

Which fuse should be used?

- A** 1 A **B** 3 A **C** 5 A **D** 13 A

37 A very important experiment improved scientists' understanding of the structure of matter.

The experiment involved α -particles being fired at a thin, gold foil.

What happened?

- A** All the α -particles were absorbed by the nuclei of the gold atoms.
B All the α -particles were unaffected by the gold atoms.
C Some of the α -particles were attracted by the neutrons in the nuclei of the gold atoms.
D Some of the α -particles were repelled by the protons in the nuclei of the gold atoms.

38 Some sources of background radiation are natural and others are due to human activity.

Which source is natural?

- A** medical X-rays
B nuclear weapons testing
C radioactive waste from power stations
D radon gas from rocks

- 39** A radioactive material is placed near a detector.

The detector shows a count rate of 28 000 counts/min.

When a piece of card is put between the material and the counter, the reading decreases to 25 000 counts/min.

When an aluminium sheet is put between the material and the counter, the reading remains at 25 000 counts/min.

When a sheet of lead is put between the material and the counter, the reading decreases to 19 000 counts/min.

What is being emitted by the radioactive material?

- A** α , β and γ -radiation
 - B** α and β -radiation only
 - C** α and γ -radiation only
 - D** β and γ -radiation only
- 40** A radioactive isotope has a half-life of 3 years.

A sample gives a count rate of 100 counts/min on a detector.

Which calculation is used to predict the count rate after 12 years?

- A** $100 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$
- B** $100 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$
- C** $100 \times \frac{3}{12}$
- D** $100 \times \frac{12}{3} \times \frac{1}{2}$

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Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

October/November 2021

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	D	1
3	A	1
4	C	1
5	B	1
6	B	1
7	B	1
8	D	1
9	A	1
10	B	1
11	C	1
12	C	1
13	C	1
14	B	1
15	D	1
16	B	1
17	A	1
18	B	1
19	D	1
20	C	1
21	D	1
22	A	1
23	C	1
24	B	1
25	C	1
26	B	1
27	A	1
28	D	1

Question	Answer	Marks
29	C	1
30	C	1
31	D	1
32	A	1
33	C	1
34	C	1
35	A	1
36	C	1
37	D	1
38	D	1
39	C	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

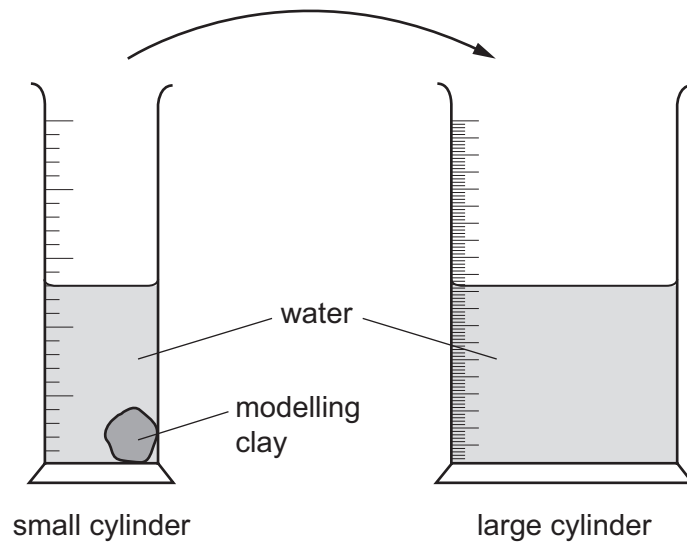
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.



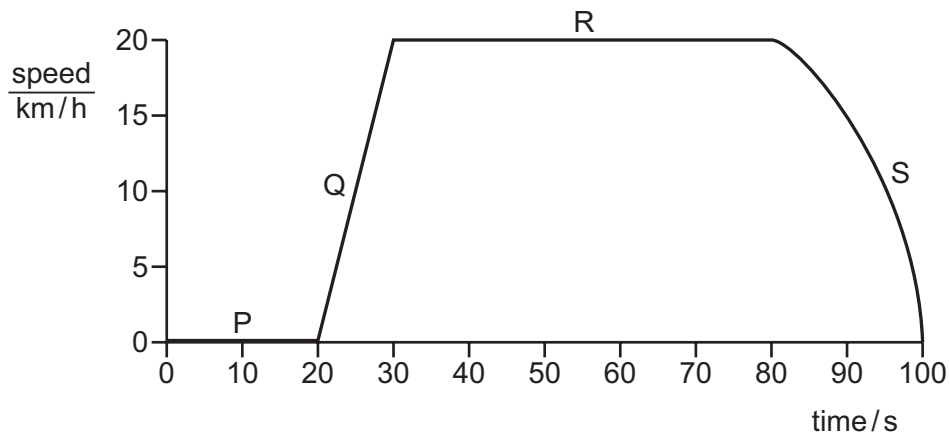
- 1 A lump of modelling clay is moved from a small measuring cylinder to a large measuring cylinder that has twice the diameter.



The reading on the small measuring cylinder goes down by 20 cm^3 .

By how much does the reading on the large cylinder go up?

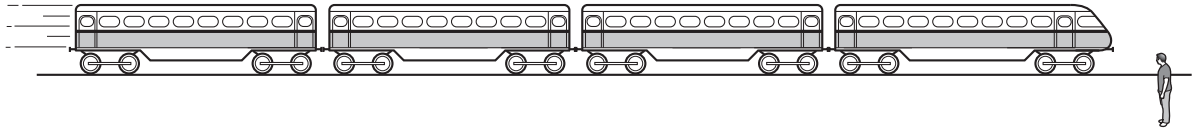
- A** 10 cm^3 **B** 20 cm^3 **C** 40 cm^3 **D** 80 cm^3
- 2 The speed–time graph for a train is shown.



Which regions of the graph show the train moving?

- A** P, Q, R and S
B Q, R and S only
C Q and S only
D R only

- 3 A man stands next to a railway track.



A train travelling at 40 m/s takes 2.0 s to pass the man.

What is the length of the train?

- A** 20 m **B** 38 m **C** 40 m **D** 80 m
- 4 A spring balance operates by the compression of a spring. It has been calibrated on the Earth in grams.

A beam balance operates by balancing standard masses against the unknown mass to be measured.

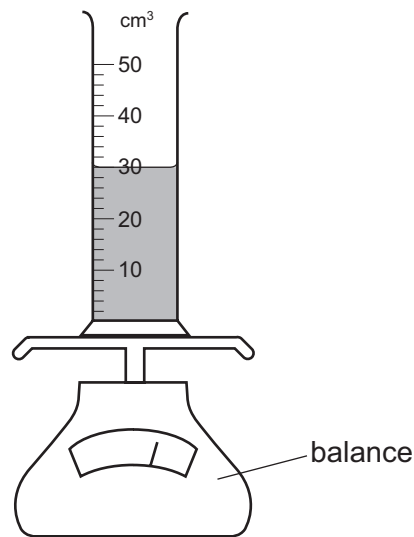
The same unknown mass is measured with each balance on the Earth and on the Moon.

The gravitational field strength on the Earth is greater than that on the Moon.

How would the measurements on the Earth compare with those on the Moon?

	spring balance measurements	beam balance measurements
A	larger on the Earth than on the Moon	larger on the Earth than on the Moon
B	larger on the Earth than on the Moon	same on the Earth as on the Moon
C	same on the Earth as on the Moon	larger on the Earth than on the Moon
D	same on the Earth as on the Moon	same on the Earth as on the Moon

- 5 A measuring cylinder contains 30 cm^3 of a liquid.



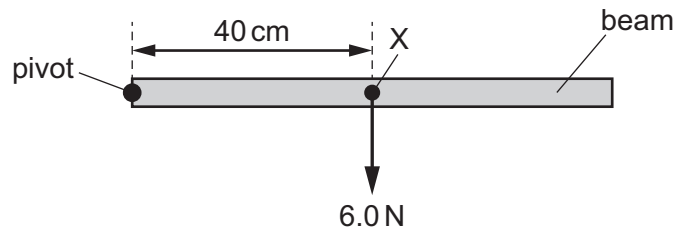
Some more of the liquid is added until the liquid level reaches the 50 cm^3 mark.

The reading on the balance increases by 30 g.

What is the density of the liquid?

- A** 0.60 g/cm^3 **B** 0.67 g/cm^3 **C** 1.5 g/cm^3 **D** 1.7 g/cm^3
- 6 Which pieces of apparatus are the most appropriate for an experiment to plot an extension–load graph of a spring?
- A** ruler, newton meter, clamp and stand
B balance, stop-watch, ruler
C light gate, ruler, newton meter
D stop-watch, balance, measuring cylinder

- 7 A beam is pivoted at one end, as shown.



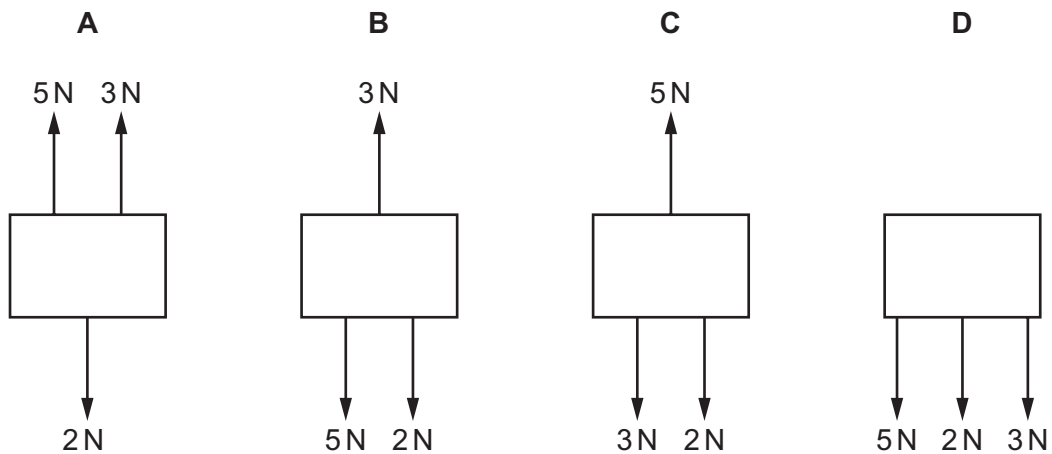
The beam weighs 6.0 N and its weight acts at a point X 40 cm from the pivot.

A force of 4.0 N is applied to the beam causing it to balance horizontally.

In which direction and where is the 4.0 N force applied?

- A** vertically downwards at 20 cm to the left of X
 - B** vertically downwards at 20 cm to the right of X
 - C** vertically upwards at 20 cm to the left of X
 - D** vertically upwards at 20 cm to the right of X
- 8 The diagrams show the forces acting on four moving objects.

Which object is moving at a constant speed?



- 9 A mass hangs vertically from a spring.

The mass is raised to a point P and is then released.

The mass oscillates repeatedly between point P and a lower point Q.

Which energies alternately increase and decrease throughout the oscillations?

- A** gravitational potential energy, kinetic energy and elastic energy
B gravitational potential energy and kinetic energy only
C gravitational potential energy, kinetic energy and internal energy
D internal energy and elastic energy
- 10 A force F acts on an object and the object moves a distance d in the direction of the force.

What is the work done on the object?

- A** $\frac{F}{d}$ **B** $\frac{d}{F}$ **C** $F \times d$ **D** $\frac{1}{(F \times d)}$

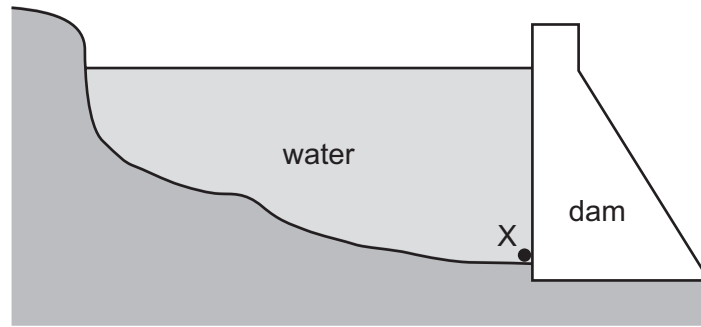
- 11 Liquid of mass 92 kg is contained in a rectangular tank.

The area of the base of the tank is 0.23 m^2 .

What is the pressure exerted by the liquid on the base of the tank?

- A** $2.5 \times 10^{-4} \text{ N/m}^2$
B $2.5 \times 10^{-3} \text{ N/m}^2$
C 400 N/m^2
D 4000 N/m^2

- 12 The diagram shows a deep reservoir formed by a dam.



On what does the pressure at X depend?

- A the depth of the water at X
 - B the length of the reservoir
 - C the surface area of the water
 - D the thickness of the dam wall
- 13 The list gives three properties of different states of matter.

- 1 They cannot be compressed significantly.
- 2 They can flow.
- 3 They always completely fill their container.

Which properties are correct for liquids?

- A 1 and 2 only
 - B 2 and 3 only
 - C 1 and 3 only
 - D 1, 2 and 3
- 14 Two thermometers, P and Q, give the same reading at room temperature.

The bulb of thermometer Q is wrapped in gauze and dipped in a beaker of water at room temperature.

Air at room temperature is blown over the two thermometer bulbs.

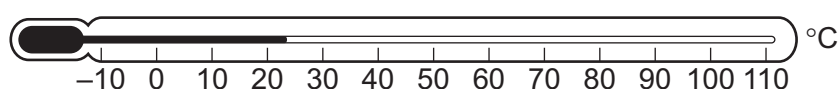
Which statement correctly describes and explains what happens?

- A P reads higher than Q because bulb P absorbs energy from the air.
- B P reads higher than Q because water evaporates from the gauze and cools bulb Q.
- C P reads lower than Q because the wet gauze insulates bulb Q.
- D P reads lower than Q because water evaporates from the gauze and warms bulb Q.

- 15 What happens to the volumes of liquid metal and of solid metal when heated at constant pressure?

	volume of liquid metal	volume of solid metal
A	decreases	increases
B	decreases	no change
C	increases	increases
D	increases	no change

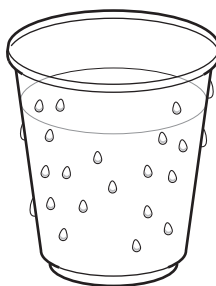
- 16 The diagram shows a liquid-in-glass thermometer.



A student wishes to check the marking of the upper fixed point on this thermometer.

What should she do?

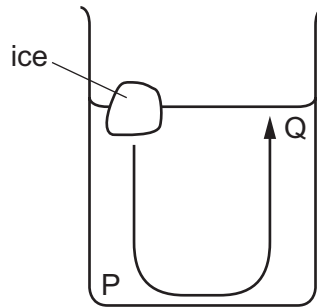
- A** Put the bulb in a beaker of boiling sea water.
 - B** Put the bulb in a beaker of boiling pure water.
 - C** Put the bulb in a beaker of ice and salt.
 - D** Put the bulb in a beaker of pure melting ice.
- 17 A glass contains an iced drink on a warm and humid day. Water starts to form on the outside of the glass.



What is the name of the effect by which the water forms?

- A** condensation
- B** conduction
- C** convection
- D** evaporation

- 18 The diagram shows a convection current caused by a piece of ice placed in a beaker of water at room temperature.



Which row correctly compares the temperatures and densities at water points P and Q?

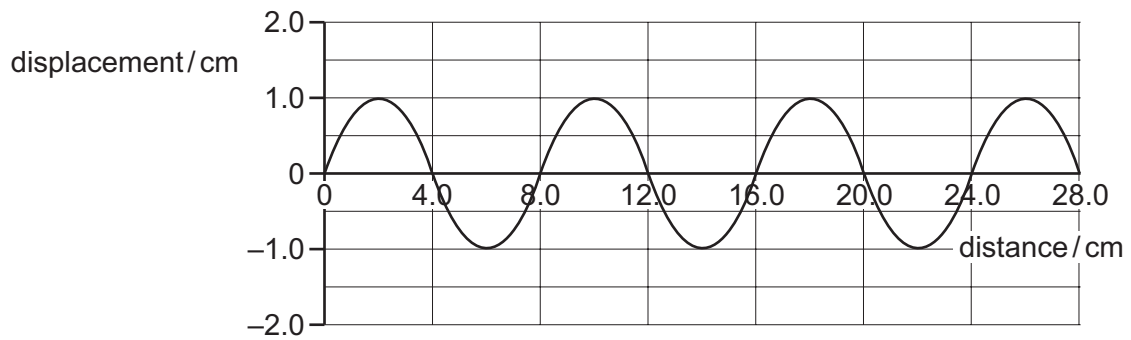
	temperature at P	density at P
A	higher than at Q	higher than at Q
B	higher than at Q	lower than at Q
C	lower than at Q	higher than at Q
D	lower than at Q	lower than at Q

- 19 The outside of one of two identical shiny metal containers is painted dull black. The containers are filled with equal masses of hot water at the same temperature.

Why does the dull black container cool more quickly?

- A** Black surfaces are better conductors than shiny surfaces.
- B** Black surfaces are better emitters of radiation than shiny surfaces.
- C** Black surfaces are better reflectors of radiation than shiny surfaces.
- D** Black surfaces are worse absorbers of radiation than shiny surfaces.

20 The diagram shows a wave.



Which row is correct?

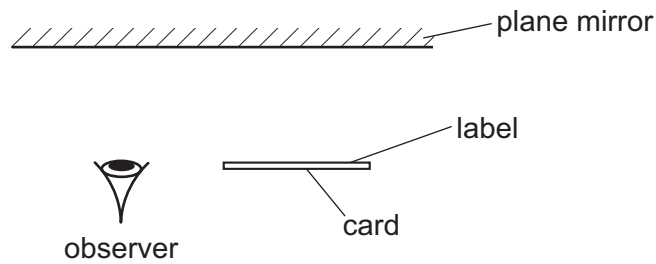
	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

21 A girl is sitting on a rock in the sea looking at passing waves. She notices that five complete wavelengths pass her in 20 s.

What is the frequency of this wave?

- A** 0.25 Hz **B** 4.0 Hz **C** 15 Hz **D** 100 Hz

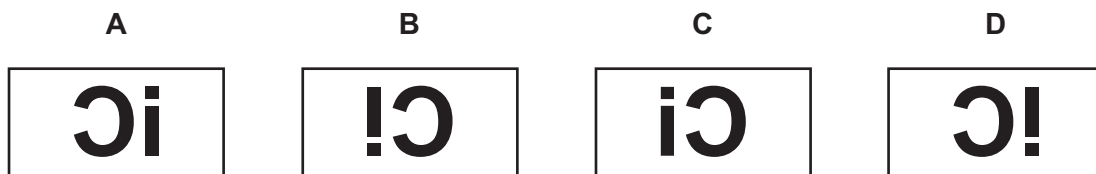
- 22 A card is placed in front of a plane mirror so that its label is facing the mirror, as shown.



The label is shown.

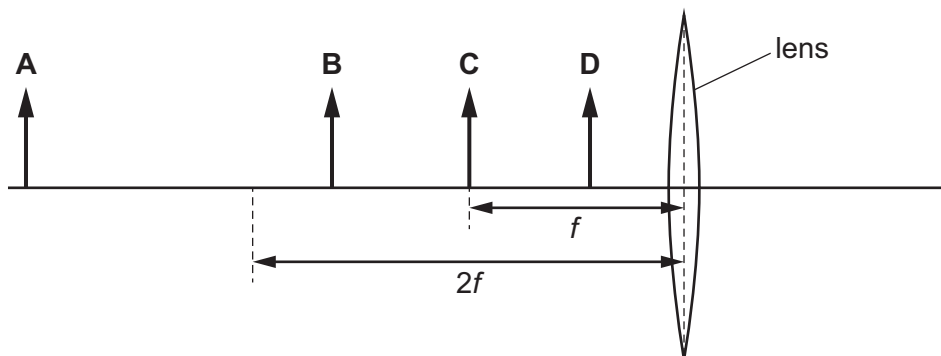


How does the image of the label formed by the mirror appear to the observer?



- 23 An object is placed in front of a converging lens. The lens has a focal length f .

In which labelled position should the object be placed in order to produce a real image that is smaller than the object?



- 24 Which part of the electromagnetic spectrum is used by a remote controller for a television?

- A infrared waves
- B microwaves
- C radio waves
- D visible light

25 Which statement correctly compares radio waves and X-rays?

- A** Radio waves have a longer wavelength and a greater speed in a vacuum.
- B** Radio waves have a longer wavelength and the same speed in a vacuum.
- C** Radio waves have a shorter wavelength and a greater speed in a vacuum.
- D** Radio waves have a shorter wavelength and the same speed in a vacuum.

26 Point X is near to a bar magnet, as shown.

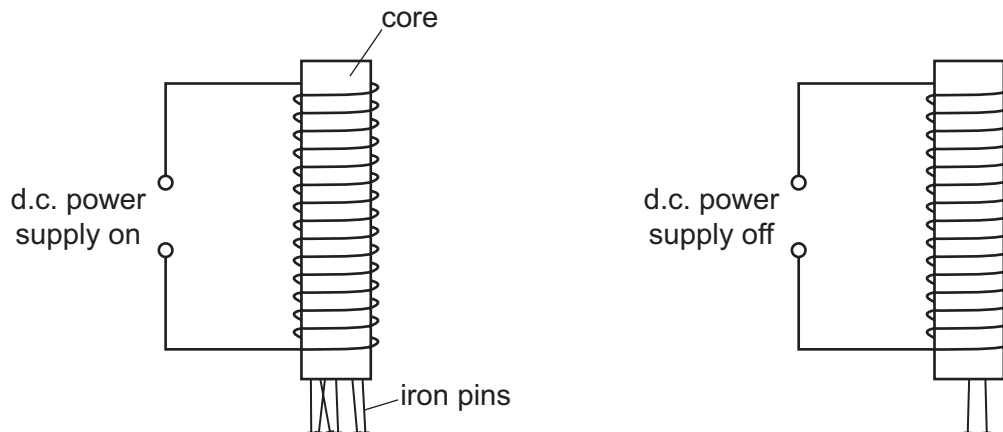
• X



Which arrow indicates the direction of the magnetic field of the bar magnet at point X?



- 27** A student counts how many iron pins an electromagnet picks up when its power supply is switched on. Then, she counts how many pins are picked up when the power supply is switched off.

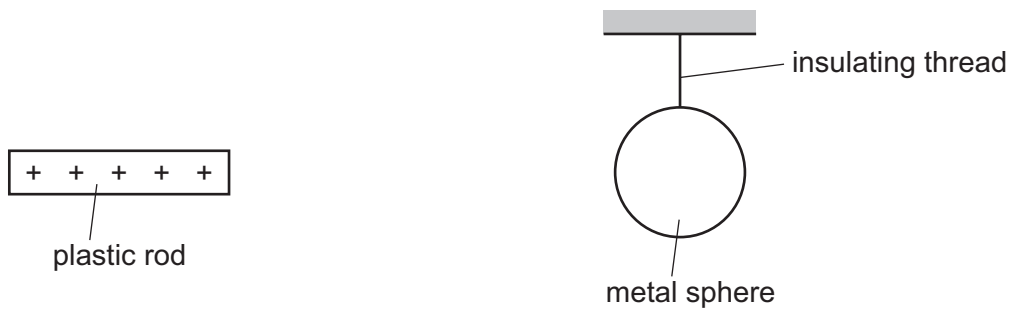


She repeats the experiment using cores made of different materials. The results are shown.

Which core is made out of soft iron?

	pins picked up with the power supply on	pins picked up with the power supply off
A	0	0
B	2	7
C	8	5
D	12	0

- 28 The diagram shows a charged plastic rod and an uncharged metal sphere. The metal sphere is suspended by an insulating thread.



The plastic rod is then moved close to the metal sphere.

Which row is correct?

	observation	the overall state of the metal sphere
A	The rod attracts the sphere.	charged
B	The rod attracts the sphere.	uncharged
C	The rod repels the sphere.	charged
D	The rod repels the sphere.	uncharged

- 29 A large battery is labelled with various items of information about the battery.

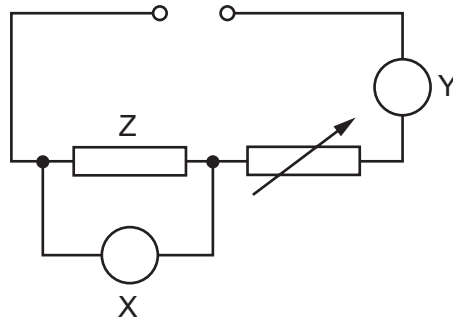
12 V	30 kg	216 kJ	680 A
------	-------	--------	-------

One of these items of information is the electromotive force (e.m.f.) of the battery.

What is the e.m.f. of the battery?

- A** 12 V **B** 30 kg **C** 216 kJ **D** 680 A

- 30 The circuit shown is being used to measure the resistance of resistor Z.



What is the correct combination of meters to determine the resistance of resistor Z?

	meter X	meter Y
A	ammeter	ammeter
B	ammeter	voltmeter
C	voltmeter	ammeter
D	voltmeter	voltmeter

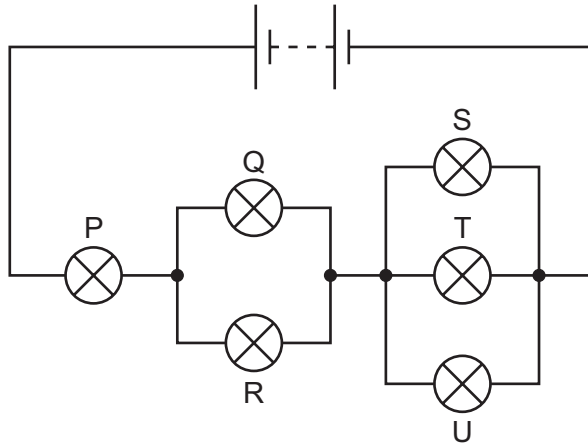
- 31 The diagram shows the circuit diagram symbol of an electrical component.



Which component does the symbol represent?

- A** light-dependent resistor
 - B** relay coil
 - C** thermistor
 - D** variable resistor
- 32 What is the advantage of connecting lamps in parallel in a lighting circuit?
- A** A smaller fuse is needed to protect the lamps.
 - B** If the filament of one lamp breaks, the remaining lamps stay lit.
 - C** The current taken from the supply is less.
 - D** The lamps use less power.

33 The diagram shows a circuit of six identical lamps connected to a battery.



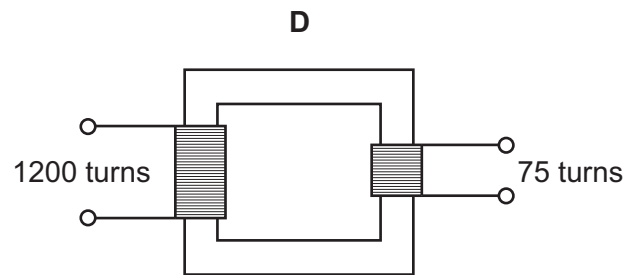
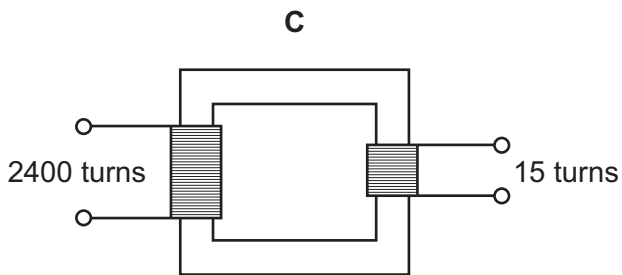
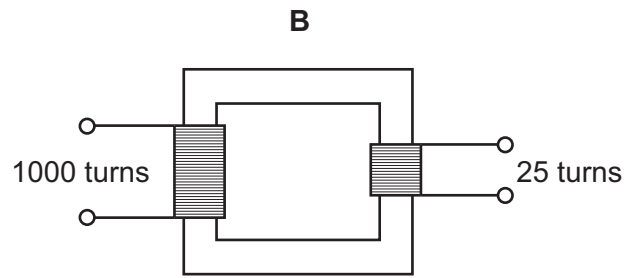
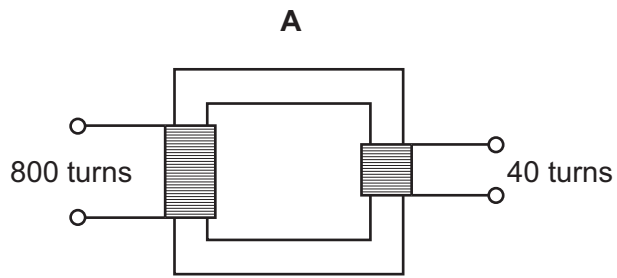
Which lamps are brightest?

- A P only
- B Q and R only
- C S, T and U only
- D P, Q, R, S, T and U are equally bright

34 Why is a fuse used in an electrical circuit?

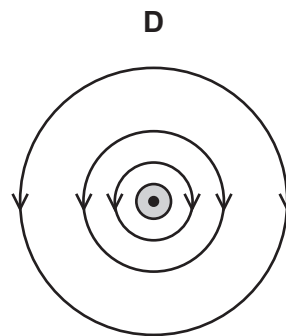
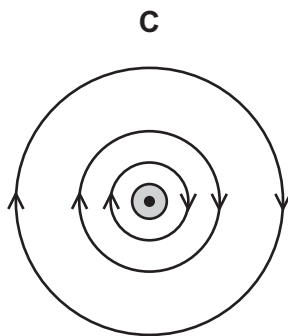
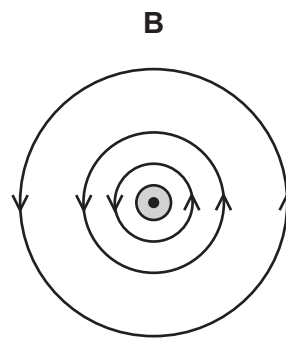
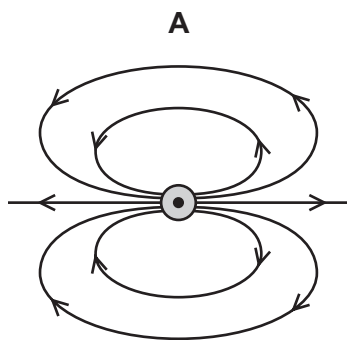
- A so that the current can have only one value
- B to prevent the current becoming too large
- C to provide a path to earth if a fault occurs
- D to save electrical energy

35 Which transformer can change a 240 V a.c. input into a 15 V a.c. output?



36 The diagrams show patterns around a straight wire carrying a current perpendicularly out of the page.

Which pattern represents the magnetic field due to the current in the wire?



- 37** A current-carrying coil in a magnetic field experiences a turning effect.

Three suggestions for increasing the turning effect are given.

- 1 Increase the number of turns on the coil.
- 2 Increase the current in the coil.
- 3 Increase the strength of the magnetic field.

Which suggestions are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 38** A model of an atom consists of small particles orbiting a central nucleus.

Where is the positive charge in an atom?

- A** on the orbiting particles
B in the nucleus
C in the space between the nucleus and the orbiting particles
D spread throughout the atom

- 39** A radioactive source has a half-life of 0.5 hours.

A detector near the source shows a reading of 6000 counts per second.

Background radiation can be ignored.

What is the reading on the detector 1.5 hours later?

- A** 750 counts per second
B 1500 counts per second
C 2000 counts per second
D 3000 counts per second

- 40** Which statement about the radioactive decay of a substance is correct?

- A** It cannot be predicted when a particular nucleus will decay.
B Placing a radioactive substance inside a lead-lined box prevents it from decaying.
C The decay always produces poisonous gases.
D The rate of decay increases if the substance is dissolved in water.

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/11

Paper 1 Multiple Choice (Core)

May/June 2022

MARK SCHEME

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9	A	1
10	C	1
11	D	1
12	A	1
13	A	1
14	B	1
15	C	1
16	B	1
17	A	1
18	C	1
19	B	1
20	B	1
21	A	1
22	C	1
23	A	1
24	A	1
25	B	1
26	D	1
27	D	1
28	B	1

Question	Answer	Marks
29	A	1
30	C	1
31	C	1
32	B	1
33	A	1
34	B	1
35	D	1
36	B	1
37	D	1
38	B	1
39	A	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/12

Paper 1 Multiple Choice (Core)

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
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- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

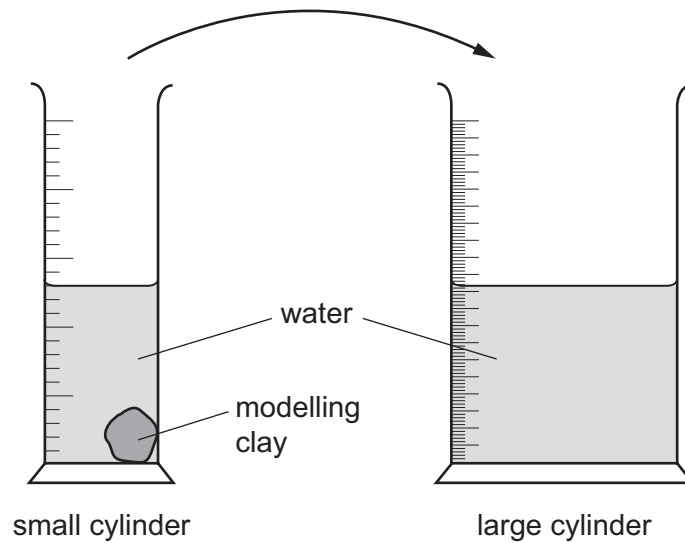
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



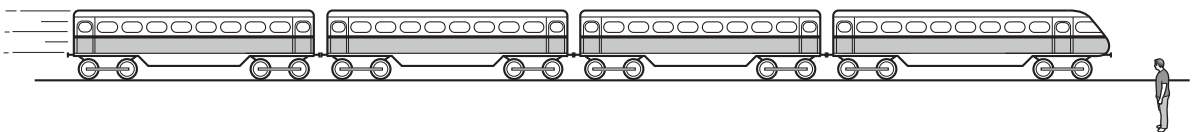
- 1 A lump of modelling clay is moved from a small measuring cylinder to a large measuring cylinder that has twice the diameter.



The reading on the small measuring cylinder goes down by 20 cm^3 .

By how much does the reading on the large cylinder go up?

- A** 10 cm^3 **B** 20 cm^3 **C** 40 cm^3 **D** 80 cm^3
- 2 What is used to determine the distance travelled by an object in motion?
- A** the area under a distance–time graph
B the area under a speed–time graph
C the gradient of a distance–time graph
D the gradient of a speed–time graph
- 3 A man stands next to a railway track.



A train travelling at 40 m/s takes 2.0 s to pass the man.

What is the length of the train?

- A** 20 m **B** 38 m **C** 40 m **D** 80 m

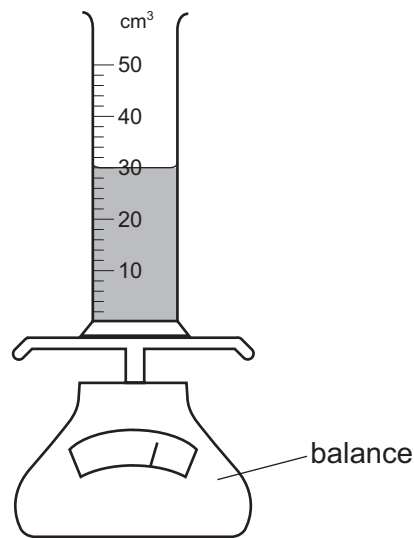
- 4 An object is moved from point X to point Y.

The acceleration of free fall at X is different from that at Y.

Which statement about the object at Y is correct?

- A Both its mass and its weight are different from those at X.
- B Both its mass and its weight are the same as those at X.
- C Its mass is the same as at X but its weight is different.
- D Its weight is the same as at X but its mass is different.

- 5 A measuring cylinder contains 30 cm^3 of a liquid.



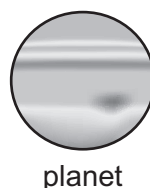
Some more of the liquid is added until the liquid level reaches the 50 cm^3 mark.

The reading on the balance increases by 30 g.

What is the density of the liquid?

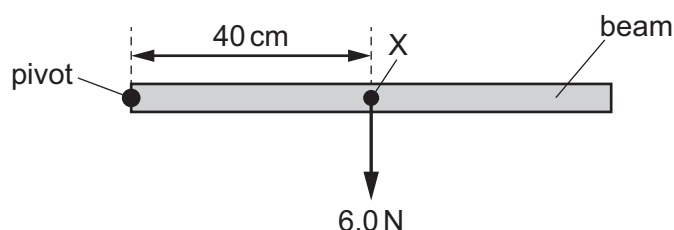
- A 0.60 g/cm^3 B 0.67 g/cm^3 C 1.5 g/cm^3 D 1.7 g/cm^3

- 6 A spaceship approaches and passes a planet.



What can the force of gravity between the spaceship and planet achieve?

- A** It can change the direction of the spaceship, or slow it down, but not speed it up.
 - B** It can change the direction of the spaceship, or speed it up, but not slow it down.
 - C** It can slow down the spaceship, or speed it up, but not change its direction.
 - D** It can change the direction of the spaceship, slow it down, or speed it up.
- 7 A beam is pivoted at one end, as shown.



The beam weighs 6.0 N and its weight acts at a point X 40 cm from the pivot.

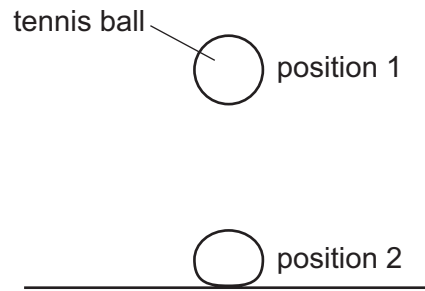
A force of 4.0 N is applied to the beam causing it to balance horizontally.

In which direction and where is the 4.0 N force applied?

- A** vertically downwards at 20 cm to the left of X
 - B** vertically downwards at 20 cm to the right of X
 - C** vertically upwards at 20 cm to the left of X
 - D** vertically upwards at 20 cm to the right of X
- 8 What are the conditions for an object to be in equilibrium?

	forces on object	moment on object
A	no resultant force	no resultant moment
B	no resultant force	resultant moment
C	resultant force	no resultant moment
D	resultant force	resultant moment

- 9 A tennis ball is dropped from position 1. It falls vertically onto a hard surface at position 2.



Which energy changes have taken place between position 1 and position 2?

- A gravitational potential $\rightarrow$ kinetic $\rightarrow$ chemical
 B gravitational potential $\rightarrow$ kinetic $\rightarrow$ elastic (strain)
 C kinetic $\rightarrow$ gravitational potential $\rightarrow$ chemical
 D kinetic $\rightarrow$ gravitational potential $\rightarrow$ elastic (strain)
- 10 A force does work moving an object in the direction of the force.

Which change in the force and distance **always** increases the work done?

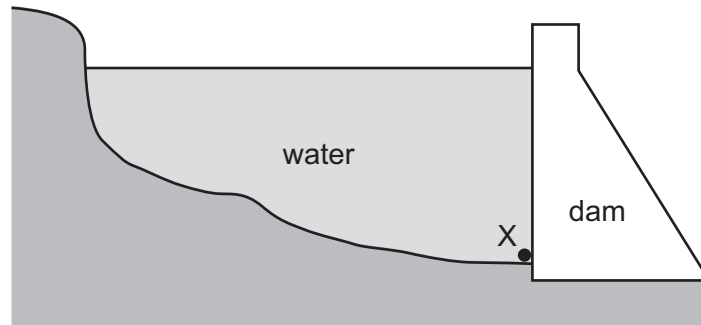
	force	distance
A	greater	same
B	greater	smaller
C	same	smaller
D	smaller	smaller

- 11 The table shows the weights and base areas of four metal blocks.

Which block exerts the greatest pressure on its base?

	weight / N	area of base / m ²
A	3 000	0.20
B	10 000	0.50
C	16 000	2.0
D	20 000	1.5

- 12 The diagram shows a deep reservoir formed by a dam.



On what does the pressure at X depend?

- A the depth of the water at X
 - B the length of the reservoir
 - C the surface area of the water
 - D the thickness of the dam wall
- 13 A particular state of matter consists of molecules that move freely in random directions at high speed. The average speed of the molecules is decreasing.

Which state of matter is being described?

- A a gas cooling
- B a gas being heated
- C a solid cooling
- D a liquid being heated

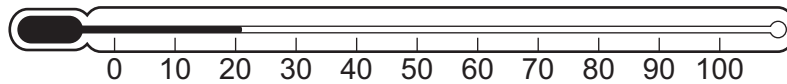
- 14 Small pollen particles are suspended in water.

When viewed with a microscope, the pollen particles can be seen to be moving about irregularly.

What causes this movement?

- A The pollen particles are being bombarded by the heavier particles of the water.
- B The pollen particles are being bombarded by the lighter particles of the water.
- C The temperature of the water is higher than that of the pollen particles.
- D The temperature of the water is lower than that of the pollen particles.

- 15 The diagram shows a liquid-in-glass thermometer.



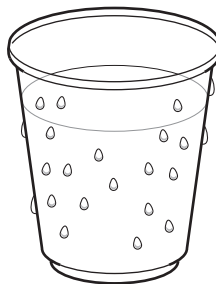
Which physical property of the thermometer is used to measure temperature?

- A** expansion of glass
B expansion of liquid
C mass of glass
D mass of liquid
- 16 An engineer investigates the increase in temperature of the oil in a car engine when it is first switched on.

Which row is correct?

	change in internal energy	explanation
A	increase	The random kinetic energy of the particles increases.
B	increase	The oil evaporates when it is heated.
C	decrease	The potential energy of the particles increases.
D	decrease	The oil changes state to a gas when it is heated.

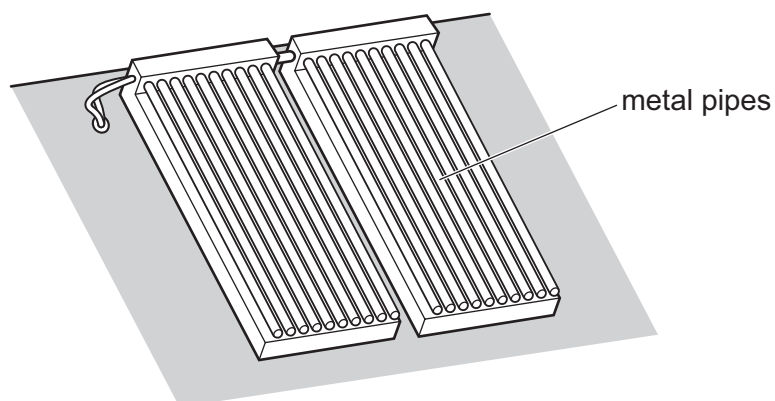
- 17 A glass contains an iced drink on a warm and humid day. Water starts to form on the outside of the glass.



What is the name of the effect by which the water forms?

- A** condensation
B conduction
C convection
D evaporation

- 18** The diagram shows solar water panels on the roof of a house. The panels absorb energy from the Sun to heat up the water in the metal pipes.



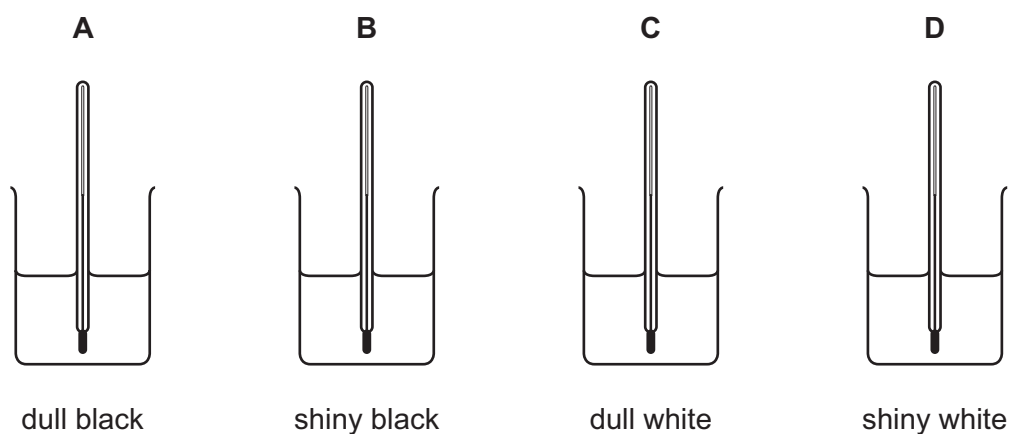
Which row describes how energy is transferred from the Sun and through the metal of the pipes?

	from the Sun	through the metal of the pipes
A	conduction	conduction
B	conduction	convection
C	radiation	conduction
D	radiation	convection

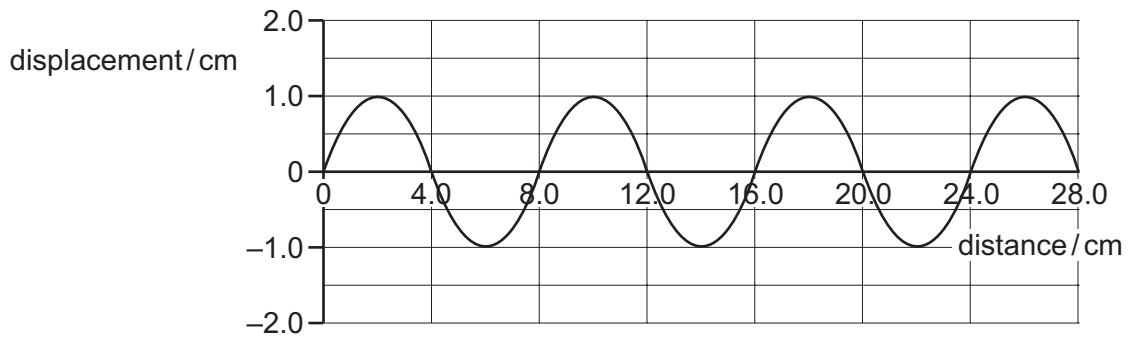
- 19** Four beakers containing equal volumes of water at 10 °C are placed outside in full sunshine on a hot day.

The four beakers are identical except for their surface colour and texture.

Which beaker will heat up the quickest?



20 The diagram shows a wave.



Which row is correct?

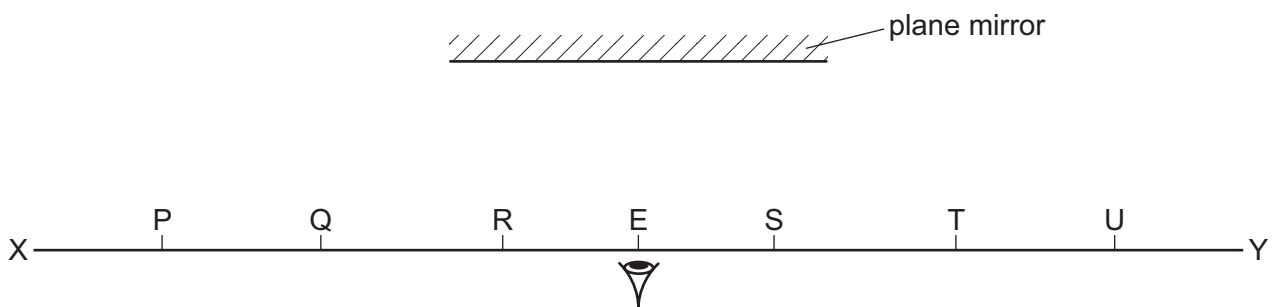
	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

21 A girl is sitting on a rock in the sea looking at passing waves. She notices that five complete wavelengths pass her in 20 s.

What is the frequency of this wave?

- A** 0.25 Hz **B** 4.0 Hz **C** 15 Hz **D** 100 Hz

22 A student uses one eye to look at images in a plane mirror.



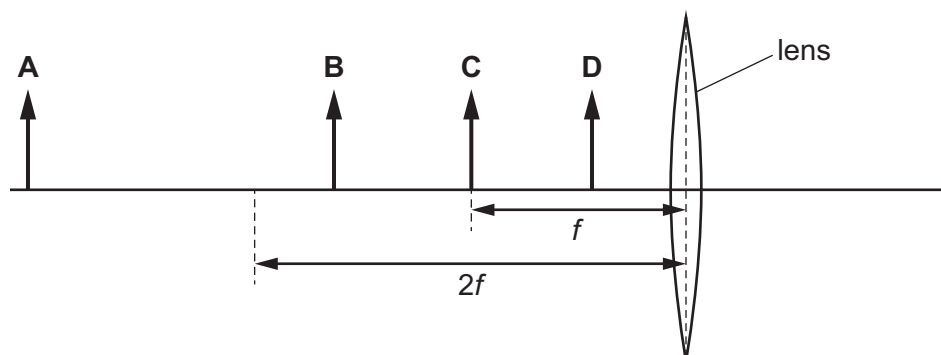
Objects are placed on the line XY.

Which objects give rise to images that can be seen by the eye at E?

- A** P, Q, R, S, T and U
B Q, R, S and T only
C P and U only
D R and S only

- 23** An object is placed in front of a converging lens. The lens has a focal length f .

In which labelled position should the object be placed in order to produce a real image that is smaller than the object?

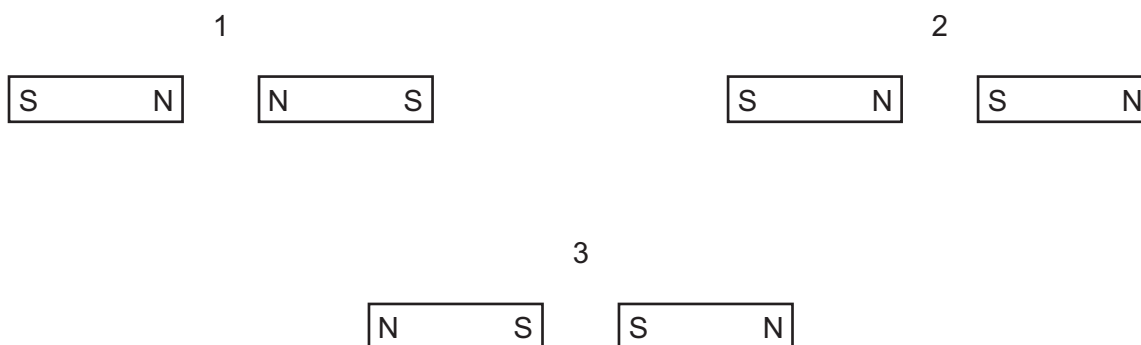


- 24** Microwaves, green light and infrared are three types of electromagnetic radiation.

What is their order when listed by wavelength from the shortest wavelength to the longest?

- A** green light $\rightarrow$ infrared $\rightarrow$ microwaves
B green light $\rightarrow$ microwaves $\rightarrow$ infrared
C infrared $\rightarrow$ green light $\rightarrow$ microwaves
D microwaves $\rightarrow$ infrared $\rightarrow$ green light
- 25** Which statement correctly compares radio waves and X-rays?
- A** Radio waves have a longer wavelength and a greater speed in a vacuum.
B Radio waves have a longer wavelength and the same speed in a vacuum.
C Radio waves have a shorter wavelength and a greater speed in a vacuum.
D Radio waves have a shorter wavelength and the same speed in a vacuum.

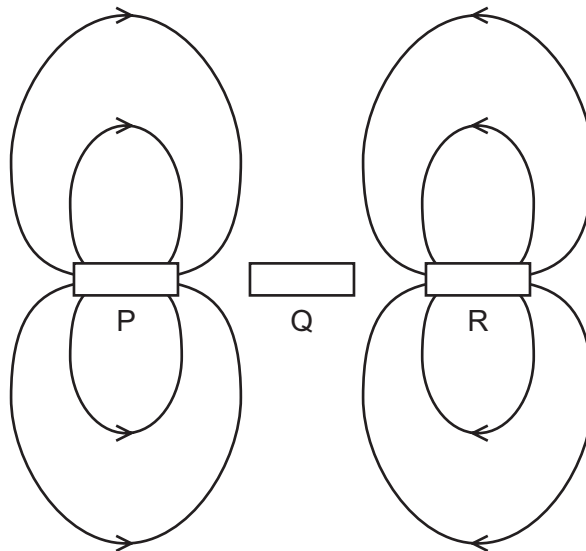
- 26** The three diagrams each show two magnets.



In which diagrams do the two magnets attract each other?

- A** 1 only **B** 1 and 3 **C** 2 only **D** 2 and 3

- 27 The diagram shows the magnetic fields around three objects, P, Q and R, placed close to each other.



Which row shows the nature of each of the objects?

	P	Q	R
A	permanent magnet N S	copper rod	permanent magnet N S
B	permanent magnet N S	copper rod	permanent magnet S N
C	permanent magnet N S	iron rod	permanent magnet N S
D	permanent magnet N S	iron rod	permanent magnet S N

- 28 Iron filings are picked up by an electromagnet.

The current in the electromagnet is switched off and the filings fall on a plastic bench.

A plastic comb is rubbed with a woollen cloth and held just above the iron filings.

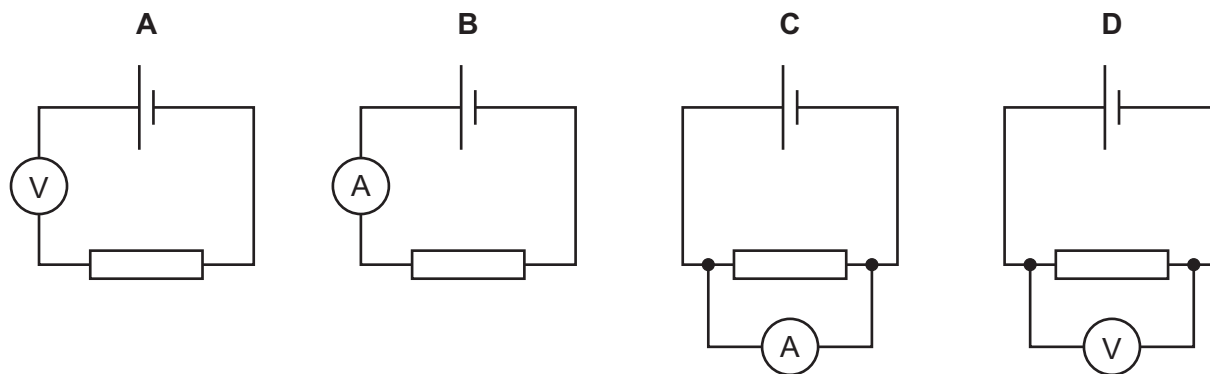
Some of the filings jump and stick to the comb.

Which statement correctly explains the last observation?

- A** The comb has been electrically charged by rubbing with the cloth.
- B** The comb has been magnetised by rubbing with the cloth.
- C** The filings have been electrically charged by the electromagnet.
- D** The filings have been magnetised by the electromagnet.

29 Four circuits are set up.

In which circuit does the meter measure the potential difference (p.d.) across the resistor?

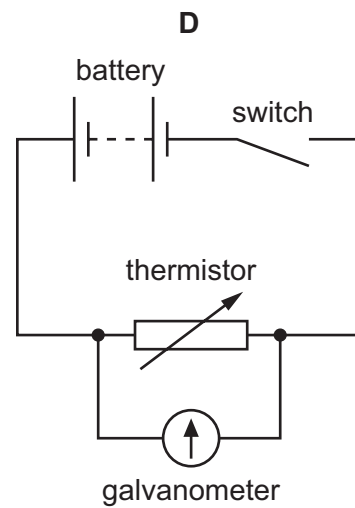
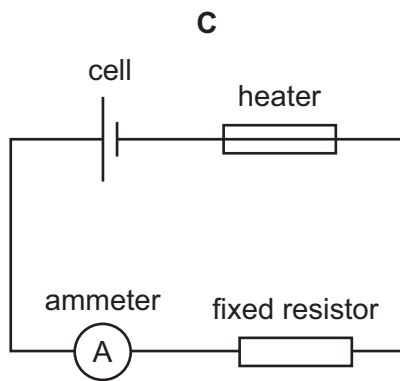
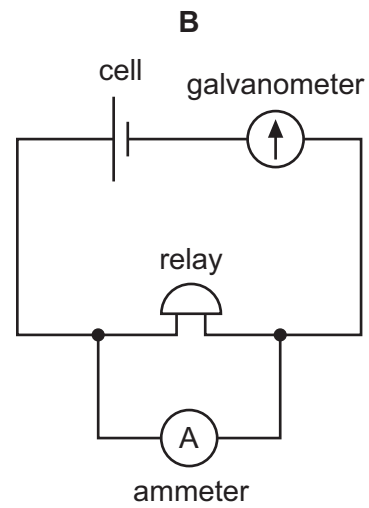
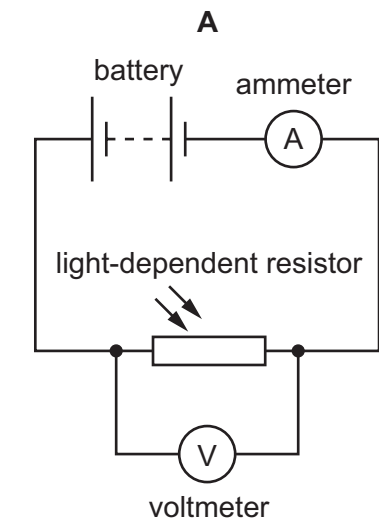


30 A resistor has a potential difference (p.d.) of 12 V across it and a current of 0.60 A in it.

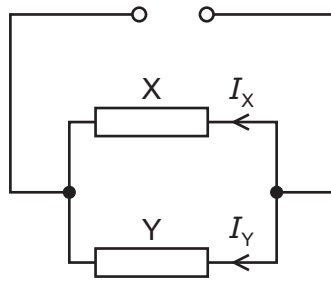
What is the resistance of the resistor?

- A** $0.050\ \Omega$ **B** $2.0\ \Omega$ **C** $7.2\ \Omega$ **D** $20\ \Omega$

31 Which circuit is correctly labelled?



32 The diagram shows an electric circuit.

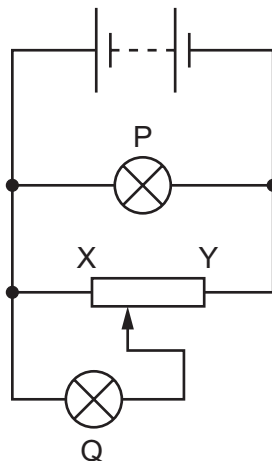


I_X is the current in resistor X. I_Y is the current in resistor Y.

Which statement describes the current from the power supply?

- A greater than I_X and greater than I_Y
- B greater than I_X and less than I_Y
- C less than I_X and greater than I_Y
- D less than I_X and less than I_Y

33 The diagram shows a battery connected to a potential divider and to two lamps, P and Q.



The slider on the potential divider is moved from end X to end Y of the resistor.

Which row shows the effect on the brightness of each lamp?

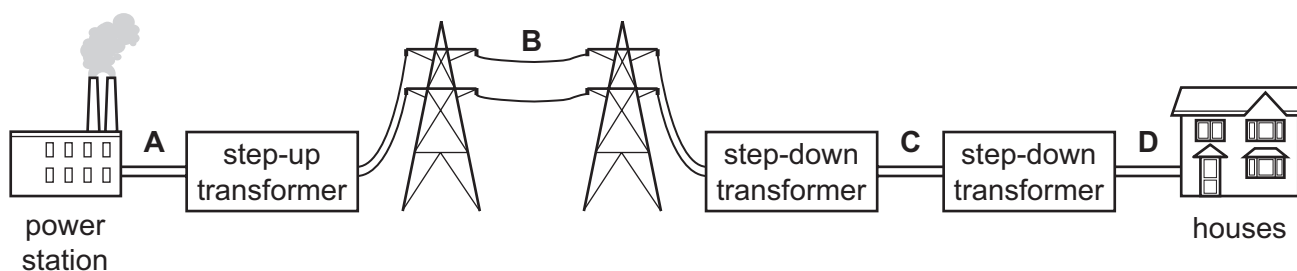
	brightness of P	brightness of Q
A	brighter	brighter
B	brighter	dimmer
C	unchanged	brighter
D	unchanged	dimmer

34 Why is a fuse used in an electrical circuit?

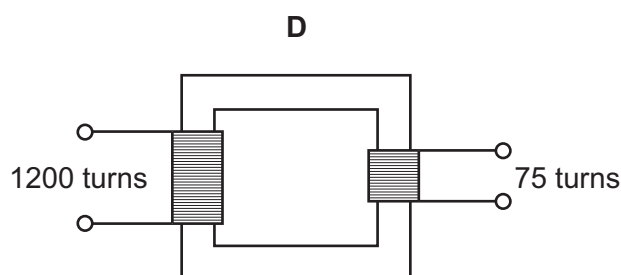
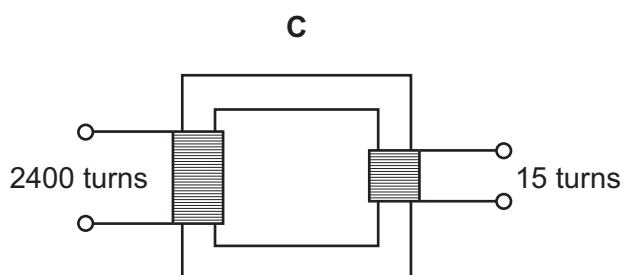
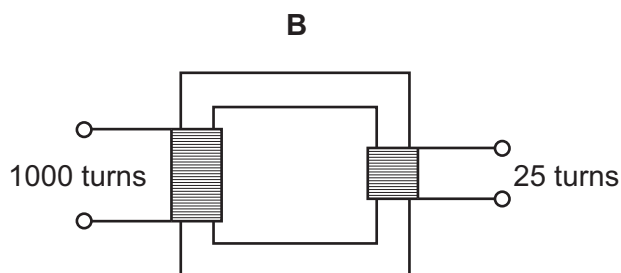
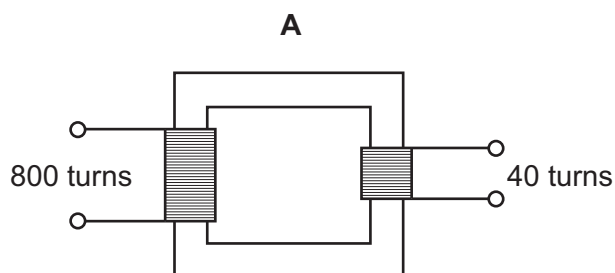
- A so that the current can have only one value
- B to prevent the current becoming too large
- C to provide a path to earth if a fault occurs
- D to save electrical energy

35 Electrical power is transmitted from power stations to homes using the National Grid.

In which part of the National Grid is the voltage highest?



36 Which transformer can change a 240 V a.c. input into a 15 V a.c. output?



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- 37** The coils in two electric motors are identical in size, but motor 1 is observed to spin more quickly than motor 2.

Three suggestions are made to explain this observation.

- 1 The current in the coil of motor 1 is greater than the current in the coil of motor 2.
- 2 The number of turns on the coil of motor 1 is greater than on the coil of motor 2.
- 3 The magnets in motor 1 are stronger than the magnets in motor 2.

Which suggestions give a possible explanation for this observation?

- A** 1 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 38** Which statement describes two atoms of different isotopes of an element?

- A** two atoms with the same nucleon number but different proton number
B two atoms with a different nucleon number but the same proton number
C two atoms with the same nucleon number and the same proton number
D two atoms with a different nucleon number and different proton number

- 39** A radioactive source has a half-life of 0.5 hours.

A detector near the source shows a reading of 6000 counts per second.

Background radiation can be ignored.

What is the reading on the detector 1.5 hours later?

- A** 750 counts per second
B 1500 counts per second
C 2000 counts per second
D 3000 counts per second

- 40** Some nuclei are unstable. They emit radiation and change into nuclei of a different element.

What is this process called?

- A** convection
B electromagnetic induction
C radioactive decay
D the motor effect



Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/12

Paper 1 Multiple Choice (Core)

May/June 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	B	1
3	D	1
4	C	1
5	C	1
6	D	1
7	D	1
8	A	1
9	B	1
10	A	1
11	B	1
12	A	1
13	A	1
14	B	1
15	B	1
16	A	1
17	A	1
18	C	1
19	A	1
20	B	1
21	A	1
22	B	1
23	A	1
24	A	1
25	B	1
26	C	1
27	B	1
28	A	1

Question	Answer	Marks
29	D	1
30	D	1
31	A	1
32	A	1
33	C	1
34	B	1
35	B	1
36	D	1
37	D	1
38	B	1
39	A	1
40	C	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

October/November 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 The times for 10 swings of a pendulum are measured.

measurement	time for 10 swings/s
1	10.12
2	10.48
3	10.24

What is the average time for **one** swing?

- A** 1.028 s **B** 1.036 s **C** 1.042 s **D** 10.28 s

- 2 A car starts from rest.

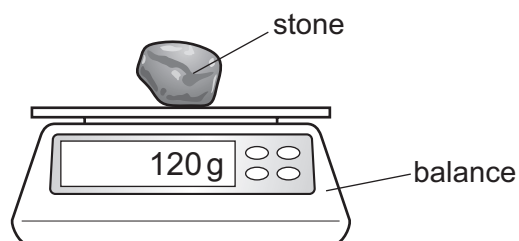
The table shows the readings from its speedometer every 10 s.

time/s	0	10	20	30	40	50	60
$\frac{\text{speed}}{\text{m/s}}$	0	4	8	12	12	12	12

Which row describes the car's motion in the first 30 seconds and in the last 30 seconds?

	motion during first 30 s	motion during last 30 s
A	non-zero acceleration	at rest
B	zero acceleration	constant speed
C	zero acceleration	at rest
D	non-zero acceleration	constant speed

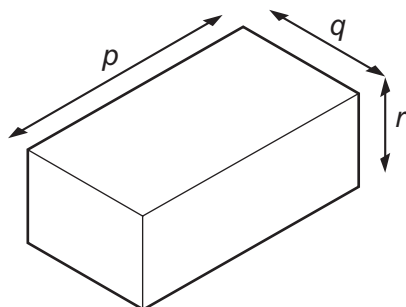
- 3 A stone is placed on a balance as shown.



Which row gives the mass and weight of the stone?

	mass	weight
A	120 g	1.2 N
B	120 g	1200 N
C	1.2 N	120 g
D	1200 N	120 g

- 4 The diagram shows the dimensions of a solid rectangular block of metal of mass m .



Which expression is used to calculate the density of the metal?

- A** $\frac{m}{(p \times q)}$
- B** $\frac{m}{(p \times q \times r)}$
- C** $m \times p \times q$
- D** $m \times p \times q \times r$

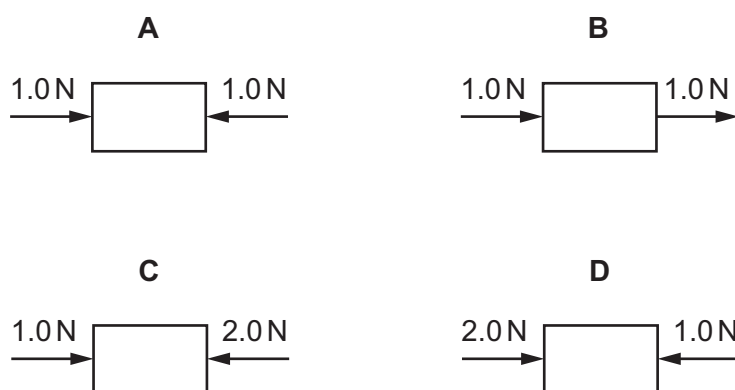
- 5 A force F acts on an object at a distance x from a pivot.

Which two changes both increase the moment of the force about the pivot?

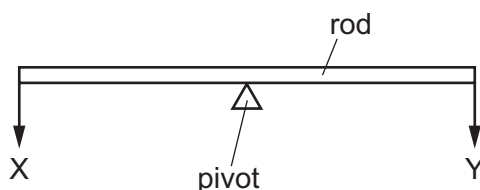
	change 1	change 2
A	decrease F	decrease x
B	decrease F	increase x
C	increase F	decrease x
D	increase F	increase x

- 6 The diagrams represent the only two forces acting on an object.

Which object could be moving to the right at constant speed?



- 7 The diagram shows a uniform rod with its midpoint on a pivot. Two weights X and Y are hung from the rod. The rod is in equilibrium.



Which statement is correct?

- A** The forces at X and Y have different values.
- B** There is a resultant turning effect on the rod.
- C** The resultant force on the rod is zero.
- D** The rod does not have a centre of mass.

- 8 Brakes are used to slow down a moving car.

Into which form of energy is most of the kinetic energy converted as the car slows down?

- A chemical
- B elastic
- C thermal
- D sound

- 9 What is a disadvantage of nuclear fission as a source of energy?

- A Nuclear power stations are expensive to build.
- B Nuclear power stations are unreliable.
- C Nuclear power stations can only provide small quantities of energy.
- D Nuclear power stations release large quantities of carbon dioxide into the atmosphere.

- 10 The statements describe what happens when the power of a machine is increased.

- 1 The work done in a given time decreases.
- 2 The work done in a given time increases.
- 3 The time taken to do a given quantity of work decreases.
- 4 The time taken to do a given quantity of work increases.

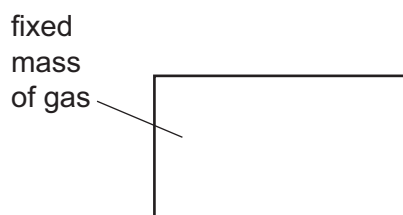
Which statements are correct?

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

- 11 Which quantities, in addition to the acceleration of free fall g , affect the pressure at the bottom of a pond of water?

- A the density of the water and the depth of the pond only
- B the density of the water and the surface area of the pond only
- C the depth of the pond and the surface area of the pond only
- D the depth of the pond, the density of the water and the surface area of the pond

- 12** A fixed mass of gas is trapped in a container. The temperature of the gas is increased but the volume of the gas is kept constant.



How does this change affect the average kinetic energy of the molecules and the pressure on the walls of the container?

	average kinetic energy	pressure
A	increases	increases
B	stays the same	increases
C	increases	decreases
D	decreases	increases

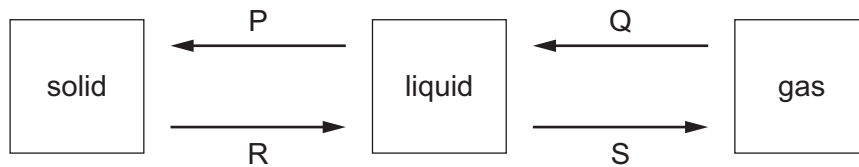
- 13** Which row correctly describes the arrangement and the motion of particles in a solid?

	arrangement of particles	motion of particles
A	far apart	moving randomly from place to place
B	far apart	vibrating about one position
C	tightly packed	moving randomly from place to place
D	tightly packed	vibrating about one position

- 14** What is the temperature difference between the fixed points on the °C temperature scale?

A 10 °C **B** 100 °C **C** 110 °C **D** 120 °C

- 15** In the diagram, each box represents a state of matter and each arrow represents a change of state.



Which row correctly identifies the changes of state?

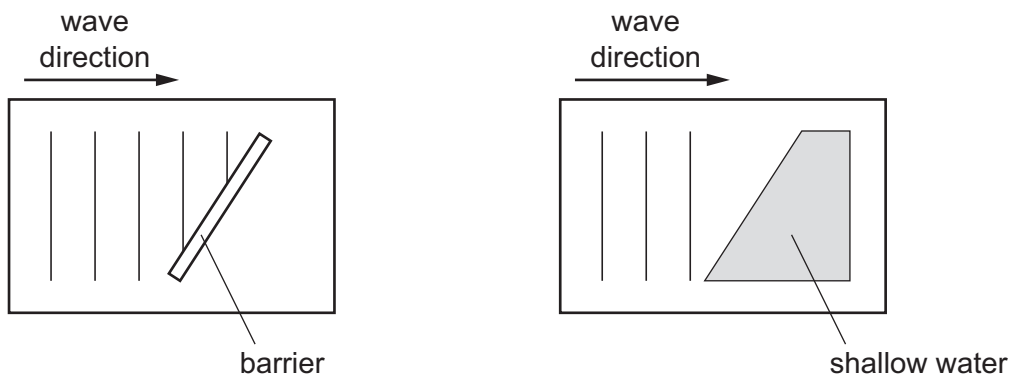
	P	Q	R	S
A	freezing	condensation	boiling	melting
B	boiling	melting	condensation	freezing
C	freezing	condensation	melting	boiling
D	condensation	freezing	melting	boiling

- 16** Four thermometers, with their bulbs painted different colours, are placed at equal distances from a radiant heater.

Which thermometer shows the slowest temperature rise when the heater is first switched on?

- A** dull black
 - B** dull white
 - C** shiny black
 - D** shiny white
- 17** Which method of transfer of thermal energy is caused by changes in density?
- A** conduction
 - B** convection
 - C** evaporation
 - D** radiation

18 The diagrams show two sets of wavefronts in a ripple tank.



A student makes two statements about the waves.

- 1 When the waves reflect from the barrier the direction changes but the wavelength remains the same.
- 2 When the waves refract as they enter the shallow water the direction remains the same, but the wavelength changes.

Which statements are correct?

- A statement 1 and statement 2
- B statement 1 only
- C statement 2 only
- D neither statement 1 nor statement 2

19 What is the correct order of the colours in a spectrum of white light?

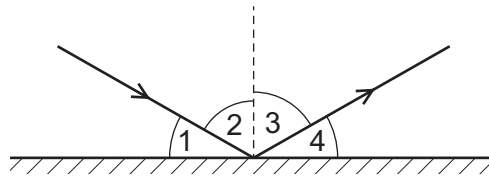
- A blue → green → yellow
- B blue → yellow → green
- C yellow → blue → green
- D green → blue → yellow

20 An object is placed 30 cm in front of a plane mirror.

Which statement describes the image of the object?

- A The image is the same size and 30 cm from the object.
- B The image is the same size and 60 cm from the object.
- C The image is smaller and 30 cm from the object.
- D The image is smaller and 60 cm from the object.

- 21 A ray of light is shone onto the surface of a mirror.



Which two angles represent the angle of incidence and the angle of reflection?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 22 The diagram shows the electromagnetic spectrum. The numbers indicate the approximate wavelength at the boundaries between the various regions of the spectrum.

For a device to be able to make use of electromagnetic radiation, it needs an aerial of approximately the same size as the wavelength of the radiation it is designed to work with.

P	Q	R	S	T	U	V
	1 m	10^{-3} m	7×10^{-7} m	4×10^{-7} m	10^{-8} m	10^{-11} m

Which statement is correct?

- A** A mobile phone uses radiation from region P.
B A television satellite dish uses radiation from region Q.
C The receptor cells in an eye use radiation from region R.
D The remote controller for a television uses radiation from region U.
- 23 Microwaves and X-rays are regions of the electromagnetic spectrum.

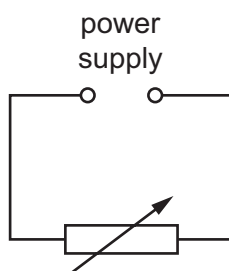
Which statement about microwaves and X-rays is correct?

- A** Microwaves and X-rays have the same frequency.
B Microwaves and X-rays travel at the same speed in a vacuum.
C Microwaves have a shorter wavelength than X-rays.
D Microwaves travel at a lower speed than X-rays in a vacuum.

- 24** An observer stands at the finish line of a 100 m race. He wants to time the winner's run. He starts his stop-watch as soon as he sees the smoke from the starting gun instead of when he hears the bang.

What is the reason for doing this?

- A** Light travels much faster than sound.
B There is a risk he might respond to an echo from a wall.
C Humans react slower to sound than to light.
D Humans react more quickly to sound than to light.
- 25** Which description of a current in a metal is correct?
- A** a flow of electrons
B a flow of molecules
C a flow of positive atoms
D a flow of protons
- 26** The diagram shows a circuit containing a variable resistor connected to a variable power supply.



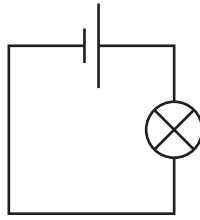
The table shows the currents for different values of the potential difference (p.d.) and the resistance.

p.d./V	resistance/ Ω	current
3.6	12	I_1
1.2	12	I_2
3.6	6	I_3

What is the order of the currents from smallest to largest?

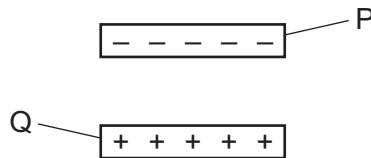
- A** $I_1 \rightarrow I_2 \rightarrow I_3$ **B** $I_1 \rightarrow I_3 \rightarrow I_2$ **C** $I_2 \rightarrow I_1 \rightarrow I_3$ **D** $I_3 \rightarrow I_1 \rightarrow I_2$

- 27 The e.m.f. of the cell in this circuit is 1.5V.



What does e.m.f. stand for?

- A** electromagnetic field
B electromagnetic force
C electromotive field
D electromotive force
- 28 A negatively charged plastic rod P is placed above a positively charged plastic rod Q.

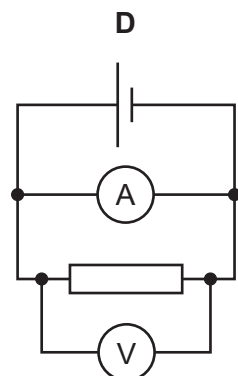
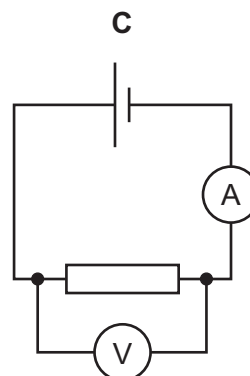
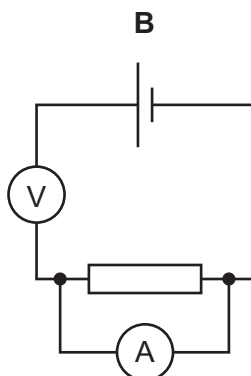
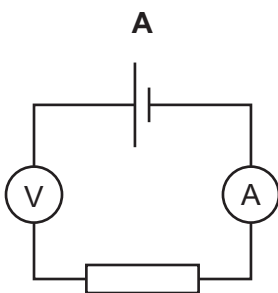


What are the directions of the electrostatic forces on rod P and on rod Q?

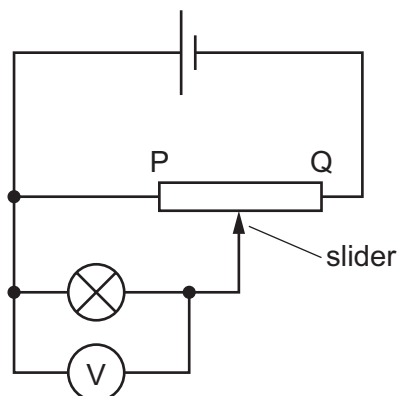
	electrostatic force on rod P	electrostatic force on rod Q
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

- 29 A student finds the resistance of a resistor.

Which circuit is used to measure the potential difference (p.d.) across the resistor and the current in it?



- 30 The circuit diagram shows a variable potential divider.

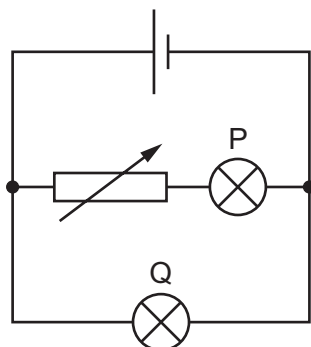


The slider is moved from P towards Q.

What happens to the reading on the voltmeter and to the brightness of the lamp?

	reading on voltmeter	brightness of lamp
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 31 The diagram shows a circuit containing a cell, a variable resistor and two bulbs, P and Q.

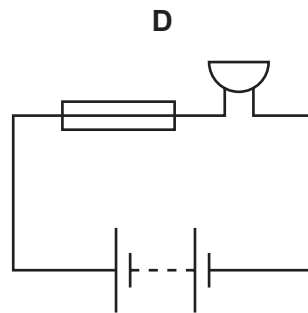
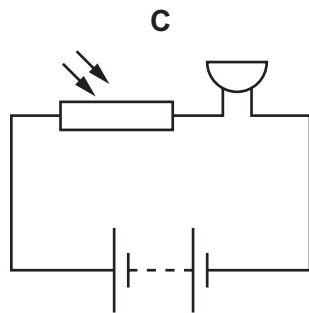
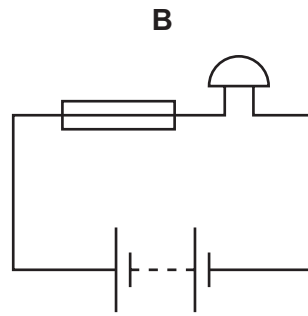
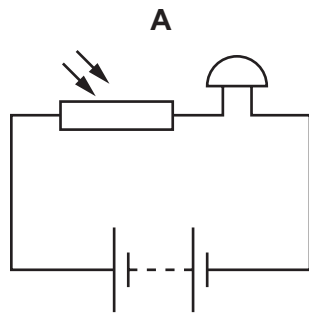


The resistance of the variable resistor is increased.

What happens to the brightness of each bulb?

	brightness of bulb P	brightness of bulb Q
A	dimmer	dimmer
B	dimmer	remains the same
C	remains the same	dimmer
D	remains the same	remains the same

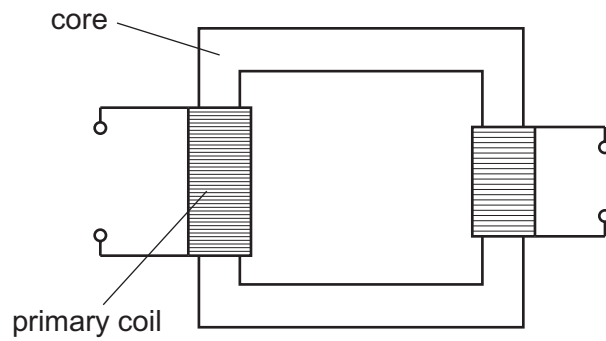
32 Which diagram shows a circuit containing a battery, a fuse and a buzzer?



33 Why is a fuse used in an electric circuit?

- A** to increase the circuit resistance
- B** to prevent short circuits
- C** to reduce the power loss
- D** to stop the cables from overheating

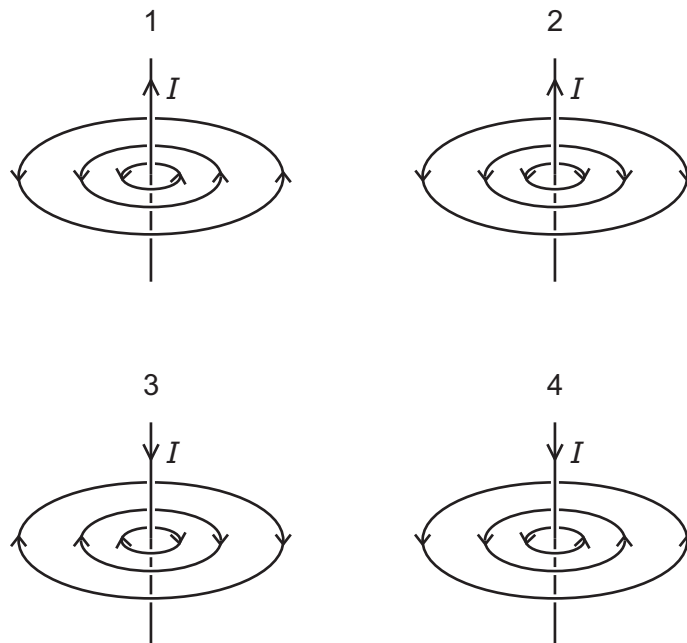
34 The diagram represents a transformer.



Which row shows materials suitable for making the core and the primary coil?

	core	primary coil
A	iron	copper
B	iron	plastic
C	steel	copper
D	steel	plastic

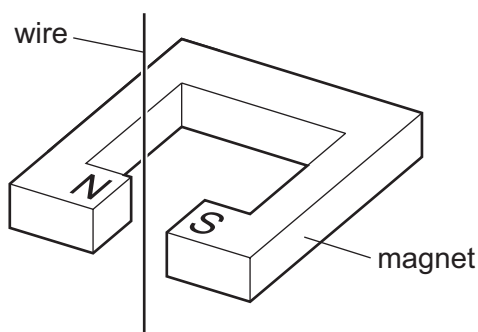
35 The diagrams show the magnetic field lines around a wire carrying a current, I .



Which diagrams are correct?

- A** 1 only **B** 2 and 3 **C** 4 only **D** 1 and 3

- 36 A wire is placed between the poles of a magnet.



Which statements are correct?

- 1 An a.c. current in the wire causes a changing force on it.
- 2 A downward d.c. current in the wire causes a constant force on it.
- 3 An upward d.c. current in the wire causes a constant force on it.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 37 The nuclide notation for the isotope lithium-7 is ${}^7_3\text{Li}$.

How many neutrons are there in an atom of lithium-7?

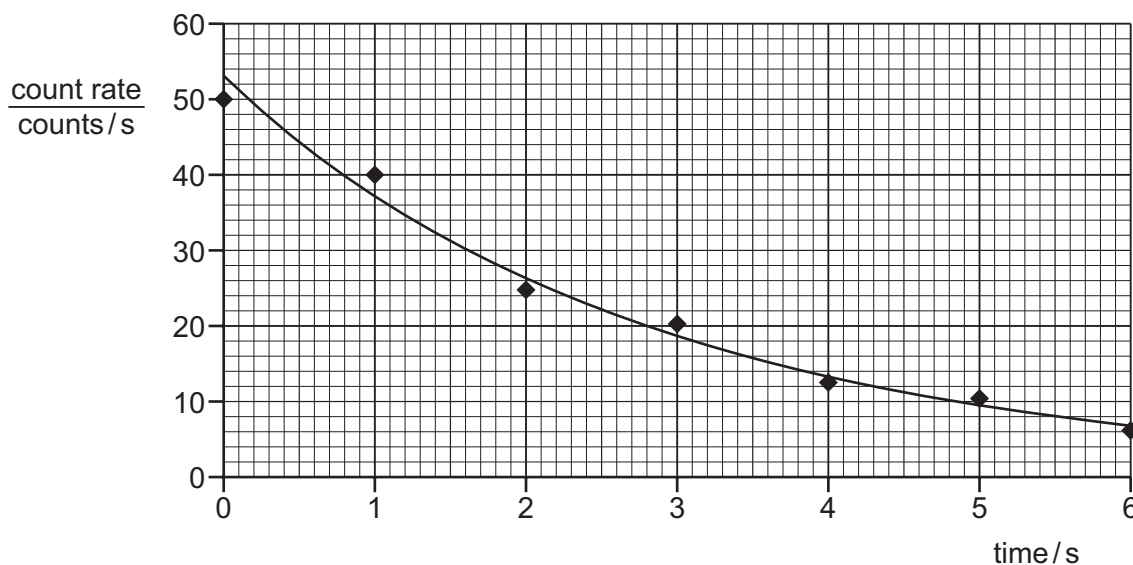
A 3 **B** 4 **C** 7 **D** 10

- 38 A radioactive material is emitting α -particles. The radioactive material is used in a demonstration in a school laboratory experiment.

Which safety precaution must be taken by the person carrying out the experiment?

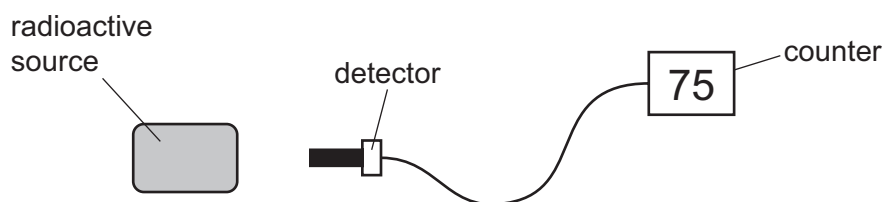
- A** Handle the source with tongs.
- B** Place the source on a heat-proof mat.
- C** Surround the experiment with a lead screen.
- D** Wear goggles.

- 39 The graph shows how the count rate from a radioactive isotope changes with time.



What is the half-life of this isotope?

- A** 2.0 s **B** 6.0 s **C** 12 s **D** 53 s
- 40 A student measures the count rate near a radioactive source using a detector of ionising radiation. The diagram shows the arrangement.



The counter reads 75 counts per minute.

When the source is taken away, the reading on the counter decreases to 5 counts per minute.

What was the rate of emission from the radioactive source when the counter reading is corrected for background radiation?

- A** 5 counts per minute
B 15 counts per minute
C 70 counts per minute
D 80 counts per minute

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Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

October/November 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	D	1
3	A	1
4	B	1
5	D	1
6	A	1
7	C	1
8	C	1
9	A	1
10	C	1
11	A	1
12	A	1
13	D	1
14	B	1
15	C	1
16	D	1
17	B	1
18	B	1
19	A	1
20	B	1
21	C	1
22	B	1
23	B	1
24	A	1
25	A	1
26	C	1
27	D	1
28	B	1

Question	Answer	Marks
29	C	1
30	D	1
31	B	1
32	D	1
33	D	1
34	A	1
35	D	1
36	D	1
37	B	1
38	A	1
39	A	1
40	C	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

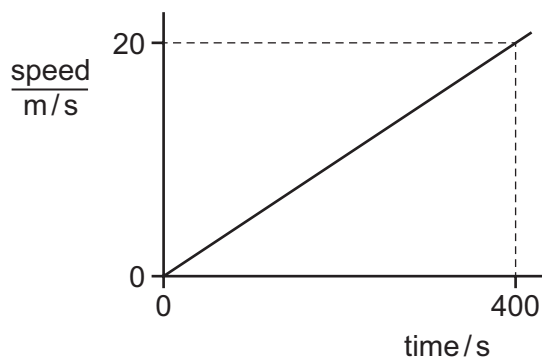
This document has **16** pages.



1 Which unit is a unit of weight?

- A kilogram
- B kilojoule
- C kilometre
- D kilonewton

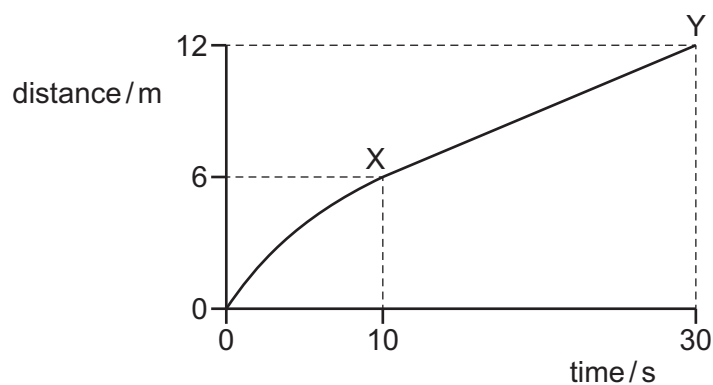
2 The graph represents the motion of a vehicle.



What is the distance travelled by the vehicle in 400 s?

- A 20 m B 400 m C 4000 m D 8000 m

3 The diagram shows a distance–time graph for an object moving in a straight line.



What is the average speed between X and Y?

- A 0.20 m/s B 0.30 m/s C 0.40 m/s D 0.60 m/s

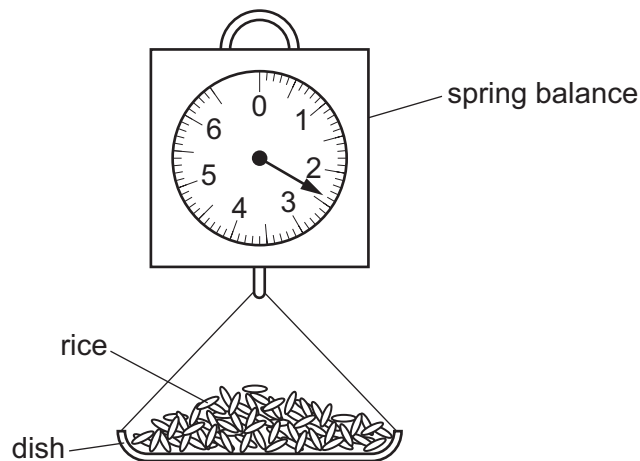
- 4 A space rocket travels to the Moon.

The acceleration of free fall is greater on the Earth than it is on the Moon.

How do the mass and weight of the space rocket on the Moon compare with their values on the Earth?

	mass on the Moon	weight on the Moon
A	less than on the Earth	same as on the Earth
B	less than on the Earth	more than on the Earth
C	same as on the Earth	less than on the Earth
D	same as on the Earth	more than on the Earth

- 5 A shopkeeper pours rice into a dish that hangs from a spring balance. He records the reading.

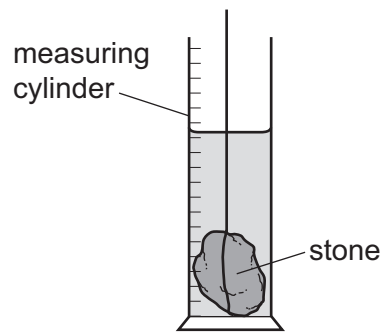


A customer buys some pasta. The shopkeeper notices that the reading on the spring balance, with just pasta in the dish, is the same as it was with just rice in the dish.

Which quantity **must** be the same for the rice and for the pasta?

- A** density
- B** temperature
- C** volume
- D** weight

- 6 A student determines the density of an irregularly shaped stone. The stone is slowly lowered into a measuring cylinder partly filled with water.

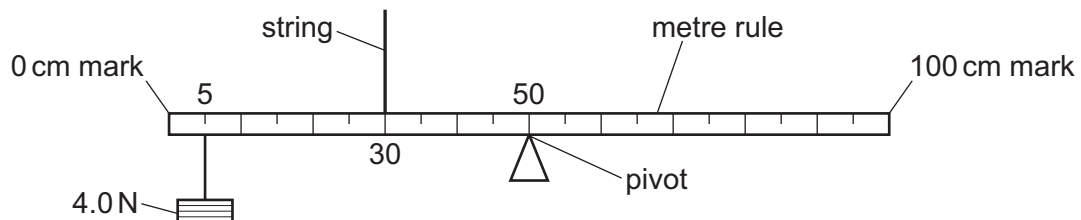


Which other apparatus does the student need to calculate the density of the irregularly shaped stone?

- A a balance
 - B a thermometer
 - C a metre rule
 - D a stop-watch
- 7 Four objects are moving.

Which object has a zero resultant force acting on it?

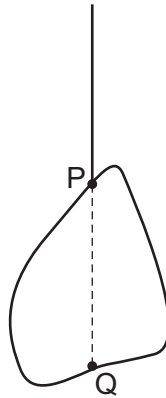
- A the object moving at a decreasing speed
 - B the object moving at an increasing speed
 - C the object moving at a constant speed in a circle
 - D the object moving at a constant speed in a straight line
- 8 The diagram shows a uniform metre rule. The rule is pivoted at its mid-point. A weight of 4.0 N is suspended from the rule at the 5 cm mark. The rule is held by a string at the 30 cm mark. The rule is in equilibrium.



What is the upward force that the string exerts on the rule?

- A 0.67 N
- B 4.0 N
- C 6.0 N
- D 9.0 N

- 9 An irregularly shaped metal plate is freely suspended from a point P and is in equilibrium, as shown. Point Q is vertically below P.



Which statement about the position of the centre of gravity of the metal plate is correct?

- A It is at P.
 B It is at Q.
 C It is halfway between P and Q.
 D Further investigation is needed to determine its position.
- 10 Which row about the change of energy in the energy store must be correct?

	process	energy store	change of energy in store
A	water pumped up to a high-altitude dam	gravitational potential energy of water	increases
B	water pumped up to a high-altitude dam	kinetic energy of water	decreases
C	air passes through a wind turbine	gravitational potential energy of air	increases
D	air passes through a wind turbine	kinetic energy of air	increases

- 11 A rock of weight 50 N falls a vertical distance of 7.0 m from rest.

What is the change in the gravitational potential energy store of the rock?

- A decrease of 7.1 J
 B decrease of 350 J
 C increase of 7.1 J
 D increase of 350 J

- 12** A child runs up a set of stairs four times. The time taken for each run is recorded.

Which time is measured when the child's useful power is greatest?

- A** 10 s **B** 20 s **C** 30 s **D** 40 s

- 13** A woman has a weight of 600 N. She stands on a horizontal floor. The area of her feet in contact with the floor is 0.050 m^2 .

What is the pressure she exerts on the floor?

- A** $1.2 \times 10^3 \text{ N/m}^2$
B $2.4 \times 10^3 \text{ N/m}^2$
C $1.2 \times 10^4 \text{ N/m}^2$
D $2.4 \times 10^4 \text{ N/m}^2$

- 14** Gases, liquids and solids are made up of small particles.

Which row gives the relative separation of the particles?

	gas	liquid	solid
A	far apart	far apart	far apart
B	far apart	far apart	close together
C	far apart	close together	close together
D	close together	close together	far apart

- 15** Which statements about evaporation of water are correct?

- 1 Evaporation causes the remaining liquid to cool.
- 2 During evaporation, the more energetic particles escape from the surface of the liquid.
- 3 Evaporation only happens at 100°C .

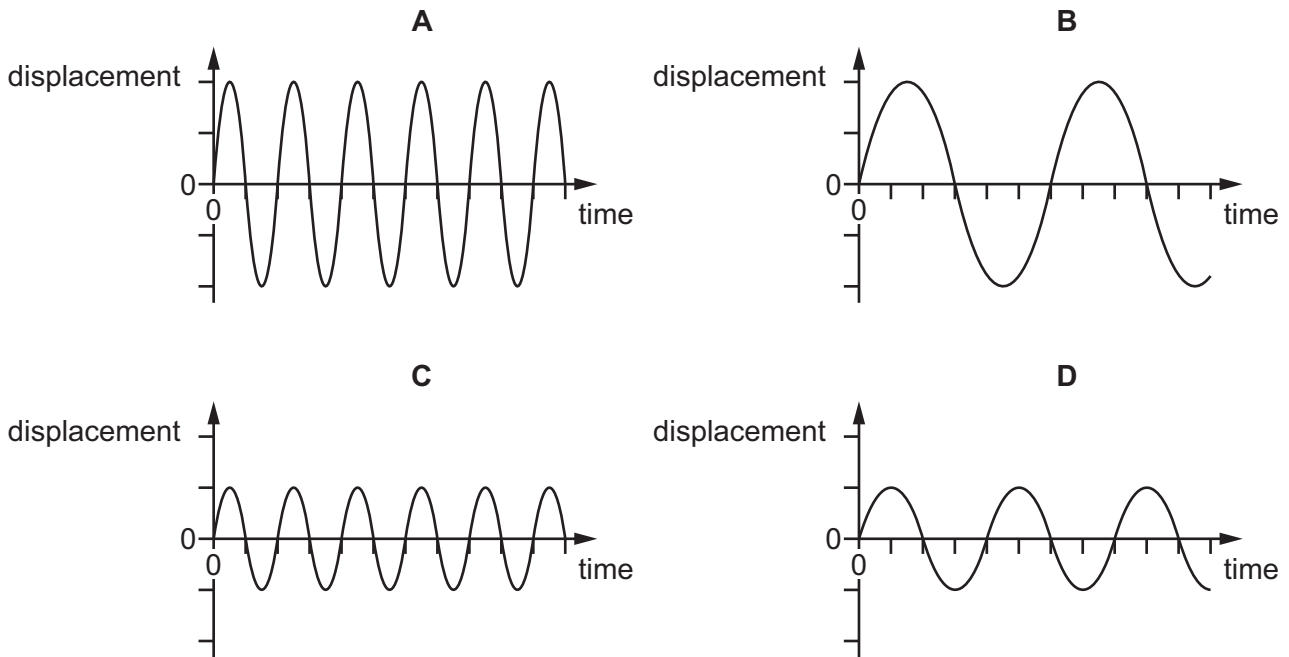
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 16** Which surface is the worst absorber of infrared radiation?

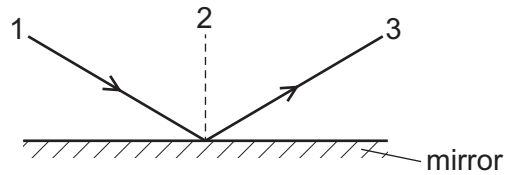
- A** dull black
B dull white
C shiny black
D shiny white

- 17 The diagrams show graphs of displacement against time for four waves. All the graphs are drawn to the same scale.

Which wave has the largest amplitude and the highest frequency?



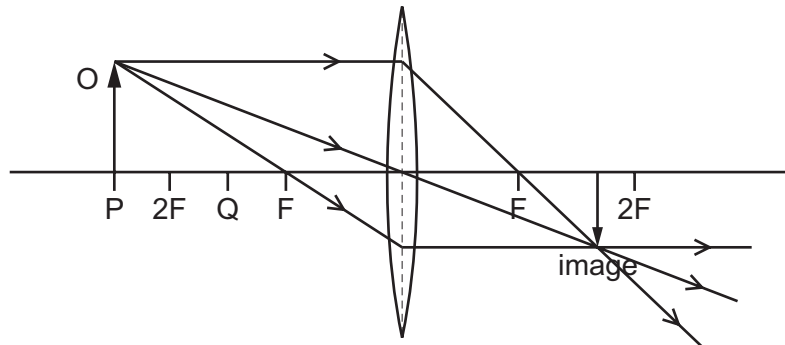
- 18 A student draws a diagram to show the directions of a light ray reflecting off a plane mirror.



What are the correct terms for the lines drawn?

	normal	incident ray	reflected ray
A	1	2	3
B	1	3	2
C	2	1	3
D	2	3	1

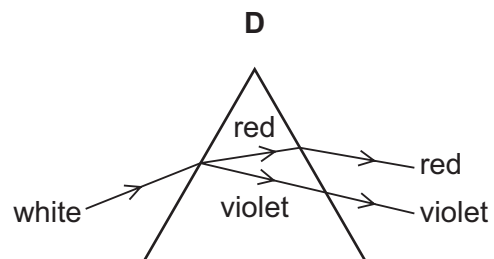
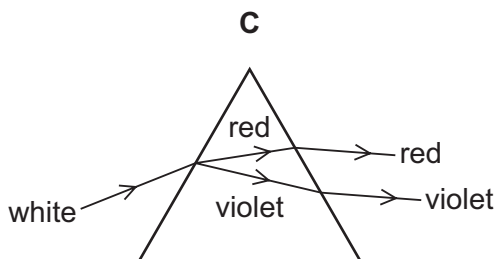
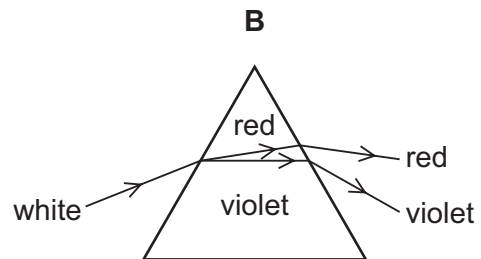
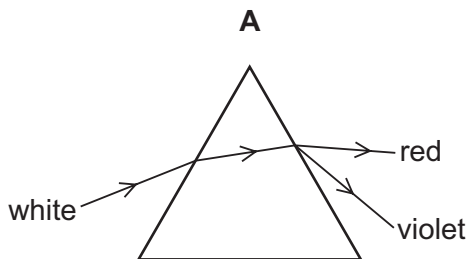
- 19** An object O is placed at point P near to a thin converging lens. The diagram shows three rays from the top of O passing through the lens. Each point F is one focal length from the centre of the lens. Each point $2F$ is two focal lengths from the centre of the lens.



The object O is moved to point Q on the diagram.

Which type of image is produced when the object O is at point Q ?

- A** inverted and the same size as the object
 - B** inverted and enlarged
 - C** upright and the same size as the object
 - D** upright and enlarged
- 20** Which diagram shows the dispersion of white light by a glass prism?



21 Which statement is correct?

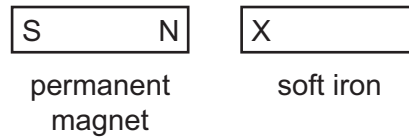
- A A remote controller emits microwave radiation.
- B A remote controller emits infrared radiation.
- C A remote controller emits ultrasound.
- D A remote controller emits ultraviolet radiation.

22 Student X fires a starting pistol which produces smoke and sound. Student Y is standing 100 m away and sees the smoke the instant it is produced. The speed of sound in air is 340 m/s.

What is the time delay between student Y seeing the smoke and hearing the sound?

- A 0.29 s B 0.59 s C 1.7 s D 3.4 s

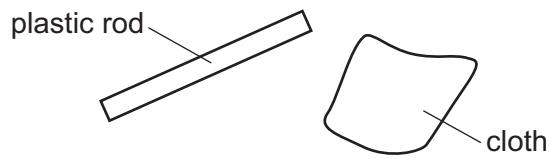
23 An unmagnetised piece of soft iron is placed close to a strong permanent magnet, as shown.



What is the induced polarity of end X of the soft iron and in which direction does the magnetic force act on the soft iron?

	polarity of end X	direction of force on the soft iron
A	N	to the left
B	N	to the right
C	S	to the left
D	S	to the right

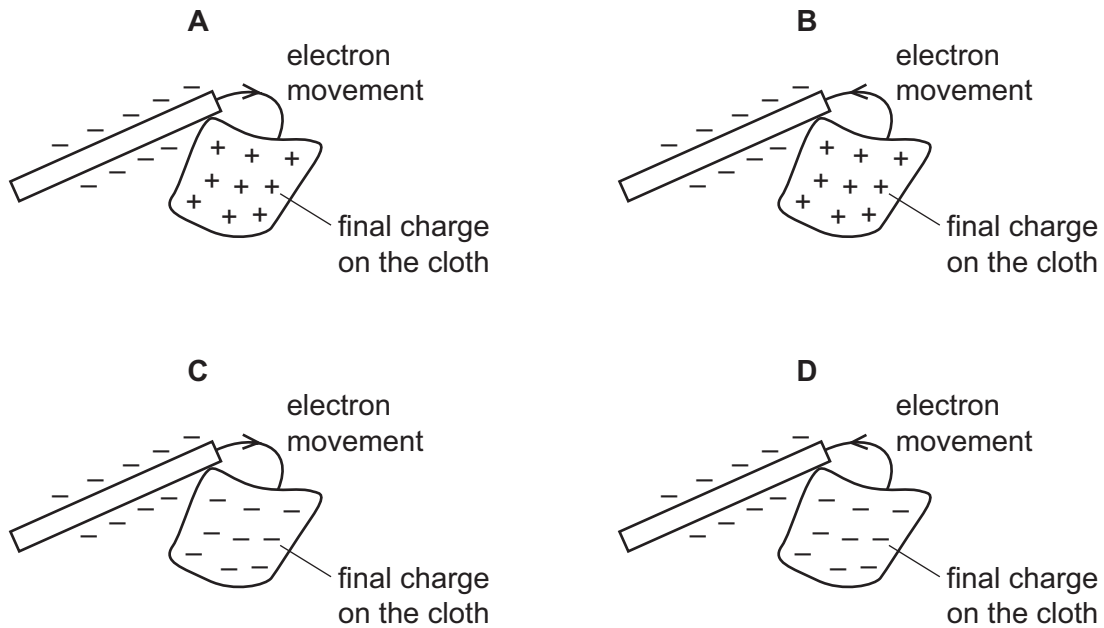
- 24** A plastic rod is rubbed with a cloth.



The rod and the cloth both become charged as electrons move between them.

The rod becomes negatively charged.

Which diagram shows how the rod becomes negatively charged and shows the final charge on the cloth?

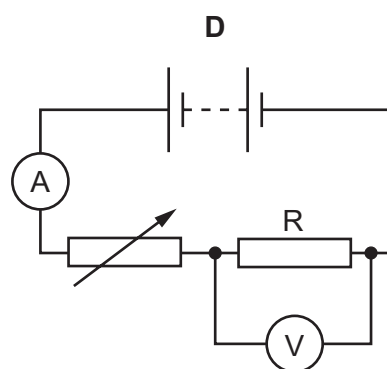
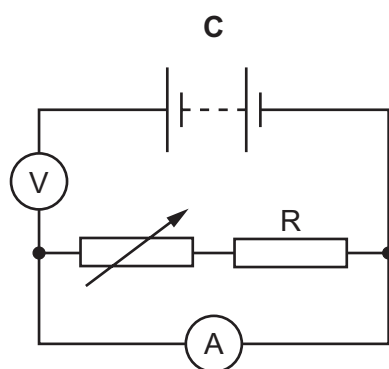
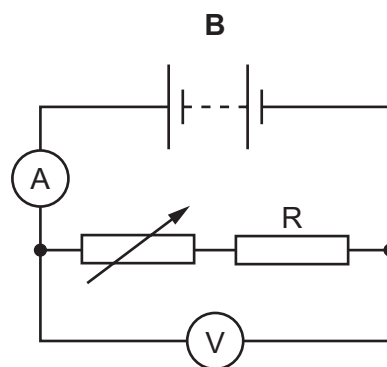
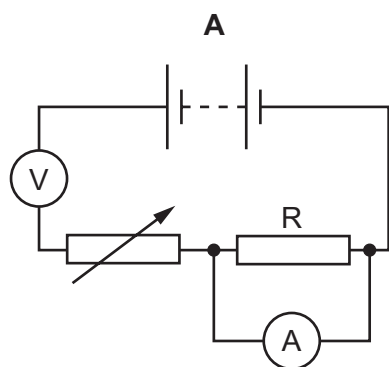


- 25** Which particles move to cause a current in a copper wire?

- A** copper atoms
- B** electrons from the copper atoms
- C** protons from the copper nuclei
- D** neutrons from the copper nuclei

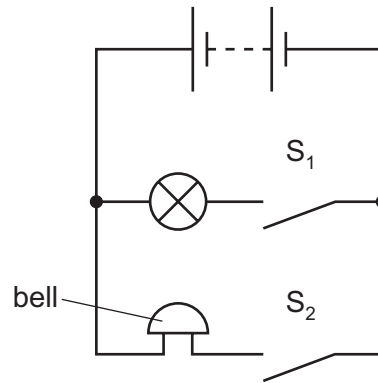
- 26 Four students draw a circuit diagram of the apparatus used to measure the resistance of resistor R.

Which circuit is correct?



27 The battery on a bicycle is connected in parallel to its lamp and bell.

The circuit includes two switches, S_1 and S_2 .



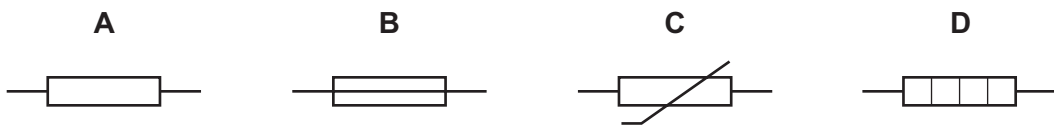
The cyclist closes S_1 to light the lamp.

She then also closes S_2 to sound the bell.

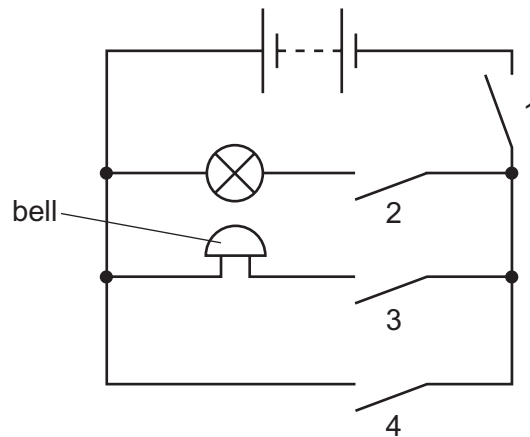
What happens to the current in the battery and the power output from the battery when the cyclist closes S_2 ?

	current in the battery	power output from the battery
A	increases	increases
B	increases	stays the same
C	stays the same	increases
D	stays the same	stays the same

28 Which symbol represents an electric heater?



29 A student connects the circuit shown.



Which switches must be closed for the bell to ring without lighting the lamp?

- A** 1 and 2 only **B** 1 and 3 only **C** 1, 3 and 4 **D** 2, 3 and 4

30 A double-insulated electrical appliance must be connected safely to the electricity supply.

Which statement is correct?

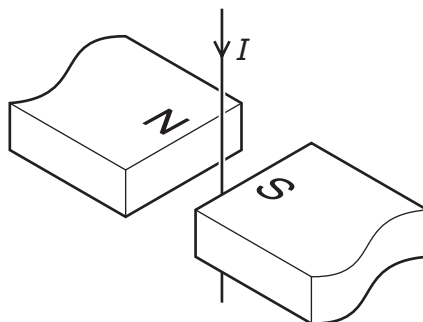
- A** It must be connected with a fuse and an earth wire.
B It can be connected with a fuse only.
C It can be connected with an earth wire only.
D It does not need a fuse or an earth wire.

31 A simple electric generator induces an electromotive force (e.m.f.).

Which modification would increase the induced e.m.f.?

- A** Increase the number of turns in the coil of the generator.
B Increase the distance between the magnetic poles.
C Reduce the strength of the magnetic field around the coil.
D Reverse the direction of the magnetic field.

- 32 The diagram shows a vertical wire carrying a current I placed between the poles of a magnet.



What is the direction of the force on the wire exerted by the magnetic field?

- A** from N to S
B from S to N
C horizontal and at right angles to the direction from N to S
D parallel to the wire
- 33 What is an advantage of transmitting electricity at a high voltage?
- A** It is faster.
B It is safer.
C Less energy is wasted.
D Less equipment is needed.
- 34 How are positive and negative ions formed from atoms?

	positive ion	negative ion
A	add positive charge to the nucleus	add an electron to the atom
B	add positive charge to the nucleus	remove positive charge from the nucleus
C	remove an electron from the atom	add an electron to the atom
D	remove an electron from the atom	remove positive charge from the nucleus

- 35 A nucleus of element X is represented as ${}^{56}_{26}\text{X}$.

Which is an isotope of element X?

- A** ${}^{26}_{56}\text{X}$ **B** ${}^{54}_{26}\text{X}$ **C** ${}^{56}_{24}\text{X}$ **D** ${}^{54}_{28}\text{X}$

36 Which statement about the random decay of the nuclei in a sample of uranium-238 is correct?

- A** The probabilities of an alpha-particle, a beta-particle or a gamma ray being emitted from a nucleus in the sample are equal.
- B** The probability of a nucleus in the sample decaying decreases as time passes.
- C** The probability of a nucleus decaying in any ten minute interval is the same for all the nuclei in the sample.
- D** The probability of a nucleus in the sample decaying increases as time passes.

37 The count rate due to a sample of a radioactive isotope is measured for 80 minutes.

time / minutes	count rate counts/second
0	480
20	380
40	300
60	240
80	190

What is the half-life of the isotope?

- A** 20 minutes **B** 40 minutes **C** 60 minutes **D** 80 minutes

38 Which planet in our Solar System is nearest to the Sun and what is the nature of the planet?

	planet	nature
A	Mercury	rocky
B	Mercury	gaseous
C	Venus	rocky
D	Venus	gaseous

39 The Sun consists mostly of two elements.

What are these two elements?

- A** helium and nitrogen
- B** hydrogen and helium
- C** hydrogen and oxygen
- D** oxygen and nitrogen

40 Which statement about the Milky Way is correct?

- A** It is a galaxy.
- B** It is a group of galaxies.
- C** It is a group of stars outside our own galaxy.
- D** It is a group of stars which are part of our galaxy.

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/11

Paper 1 Multiple Choice (Core)

May/June 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

Question	Answer	Marks
1	D	1
2	C	1
3	B	1
4	C	1
5	D	1
6	A	1
7	D	1
8	D	1
9	D	1
10	A	1
11	B	1
12	A	1
13	C	1
14	C	1
15	B	1
16	D	1
17	A	1
18	C	1
19	B	1
20	B	1
21	B	1
22	A	1
23	C	1
24	B	1
25	B	1
26	D	1
27	A	1
28	D	1

Question	Answer	Marks
29	B	1
30	B	1
31	A	1
32	C	1
33	C	1
34	C	1
35	B	1
36	C	1
37	C	1
38	A	1
39	B	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/12

Paper 1 Multiple Choice (Core)

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

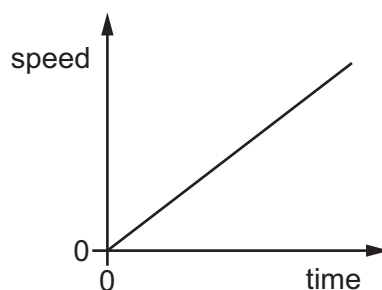
This document has **16** pages. Any blank pages are indicated.



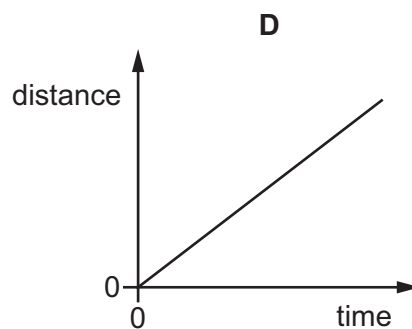
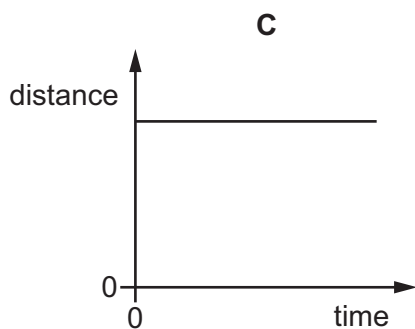
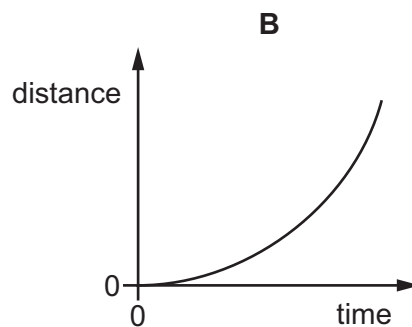
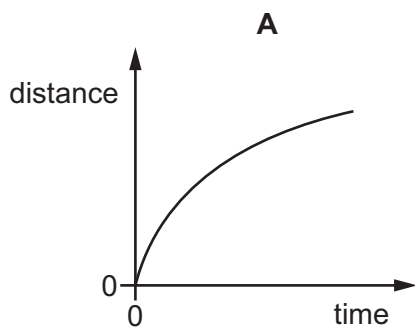
- 1 Which single apparatus is used to find the volume of a solid cube and which single apparatus is used to find the volume of a quantity of liquid?

	volume of solid cube	volume of liquid
A	balance	balance
B	balance	measuring cylinder
C	ruler	balance
D	ruler	measuring cylinder

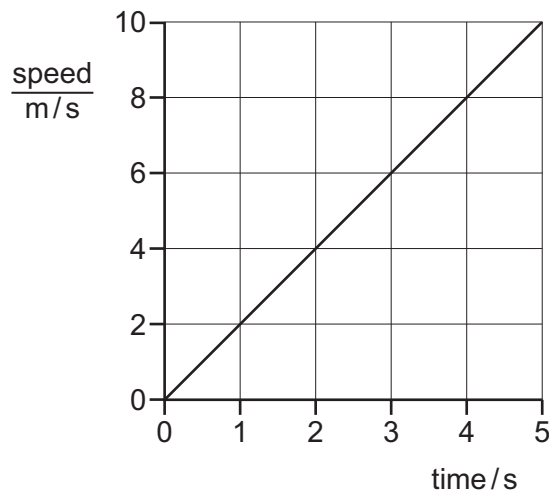
- 2 The speed–time graph represents a short journey.



Which distance–time graph represents the same journey?



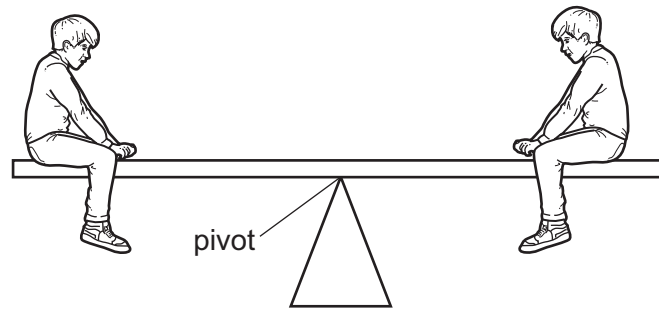
- 3 The graph represents the motion of a car.



How far has the car moved between 0 and 5 s?

- A** 2 m **B** 10 m **C** 25 m **D** 50 m
- 4 Which statement about mass or weight is **not** correct?
- A** Masses can be compared using a balance.
B Mass is a force.
C Weights can be compared using a balance.
D Weight is a force.
- 5 Which two quantities must be known to determine the density of a material?
- A** mass and area
B mass and volume
C weight and area
D weight and volume

- 6 Two boys are sitting on a see-saw. The see-saw is in equilibrium and remains horizontal.

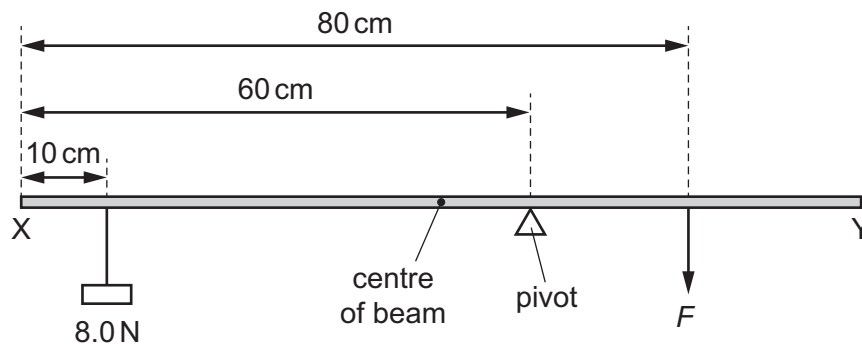


What affects the moment of each boy about the pivot?

- A his distance from the pivot only
 - B his height above the ground and his weight
 - C his weight only
 - D his weight and distance from the pivot
- 7 A uniform metre rule is pivoted in equilibrium at the 50 cm mark. A mass of 25 g is placed at the 30 cm mark on the rule.

What is the smallest mass that can be placed on the rule to restore equilibrium?

- A 5 g
 - B 10 g
 - C 15 g
 - D 25 g
- 8 A uniform beam XY is 100 cm long and weighs 4.0 N.



The beam rests on a pivot 60 cm from end X.

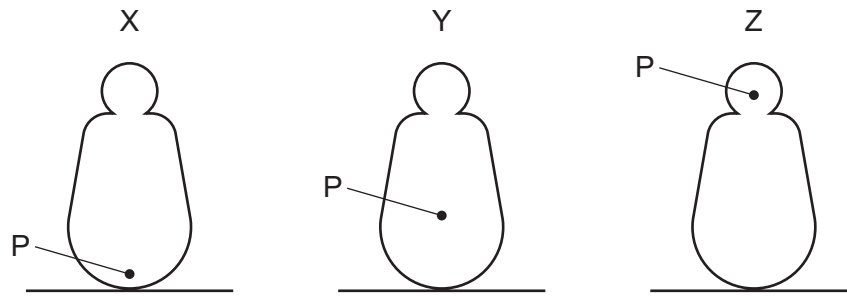
A load of 8.0 N hangs from the beam 10 cm from end X.

The beam is kept balanced by a force F acting on the beam 80 cm from end X.

What is the magnitude of force F ?

- A 8.0 N
- B 18 N
- C 22 N
- D 44 N

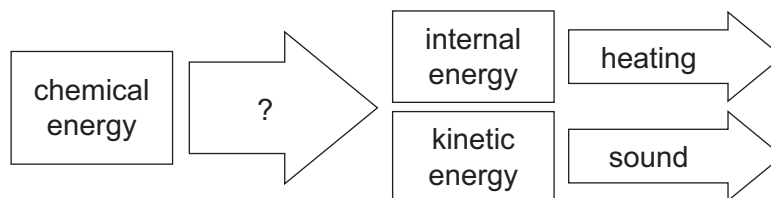
- 9 Three children's toys, X, Y and Z, are the same size and shape. They have weights at different positions inside so that the position of the centre of gravity of each toy is different. Each toy's centre of gravity is marked P.



Which toy is the most stable and which toy is the least stable when balanced in the positions shown?

	most stable	least stable
A	X	Y
B	X	Z
C	Y	X
D	Y	Z

- 10 The diagram shows the energy stores for a mobile (cell) phone and how the energy is transferred between stores.



What describes how the chemical energy is transferred?

- A** electrical work done
 - B** mechanical work done
 - C** electromagnetic waves
 - D** sound waves
- 11 A moving object is brought to rest by a resistive force of 50 N over a distance of 5.0 m.

What is the work done by the force?

- A** 0.10 J
- B** 10 J
- C** 55 J
- D** 250 J

- 12 Which two physical quantities must be used to calculate the power developed by a student running up a flight of steps?
- A force exerted and the vertical height of the steps only
 - B force exerted and the time taken only
 - C work done and the vertical height of the steps only
 - D work done and the time taken only
- 13 A rectangular marble block has dimensions 1 m by 1 m by 5 m and weighs 125 000 N.
The marble block is stored with the long side resting on the ground, as in diagram 1.

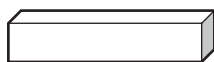


diagram 1



diagram 2

What is the change in the pressure on the ground due to the block when the block is stored as in diagram 2 rather than diagram 1?

- A a decrease of $25\,000\text{ N/m}^2$
 - B an increase of $100\,000\text{ N/m}^2$
 - C an increase of $125\,000\text{ N/m}^2$
 - D no change
- 14 Four students describe the phrase 'absolute zero' during a lesson on the particle model.
Which student is correct?
- A This is the lowest possible temperature.
 - B Particles in a solid start vibrating.
 - C Particles do not have any weight.
 - D Particles have the least gravitational potential energy.

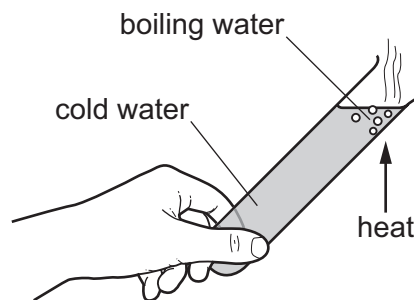
- 15** At the surface of a liquid, the more energetic molecules can escape from the liquid into the atmosphere.

Which name is given to this process?

- A** boiling
- B** condensation
- C** evaporation
- D** melting

- 16** A teacher puts some cold water in a test-tube.

She holds the bottom of the test-tube while heating the top.

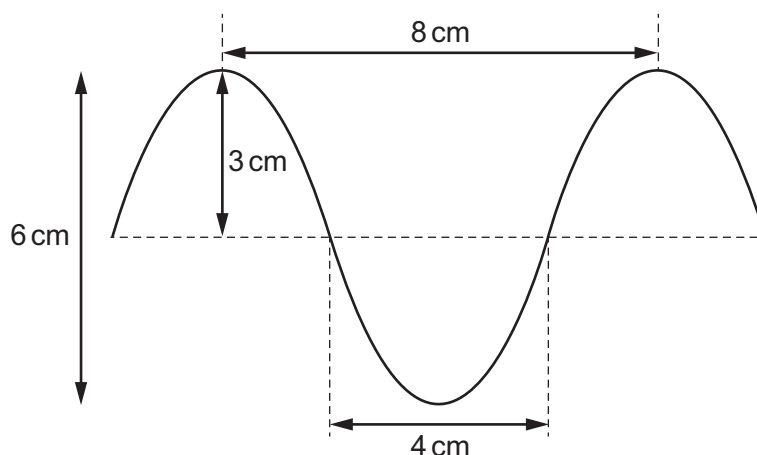


The water at the top boils but she continues to hold the test-tube as the bottom remains cold.

Which conclusion about water is made from this experiment?

- A** Water is a bad conductor.
- B** Water is a bad convector.
- C** Water is a good conductor.
- D** Water is a good convector.

17 The diagram shows a wave.



What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

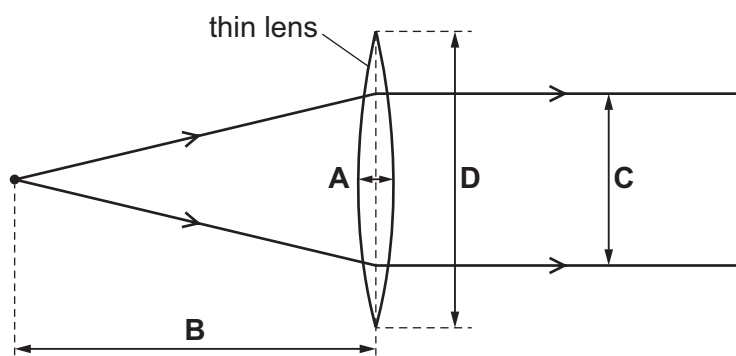
18 A light ray strikes a plane mirror and is reflected.

Which angle is always equal in size to the angle of reflection?

- A** the angle between the incident ray and the mirror
- B** the angle between the incident ray and the normal to the mirror
- C** the angle between the reflected ray and the mirror
- D** the angle between the reflected ray and the incident ray

19 The diagram shows two diverging rays of light passing through a lens and emerging parallel to each other.

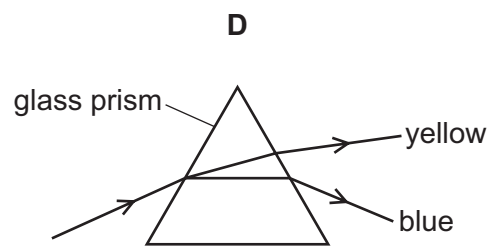
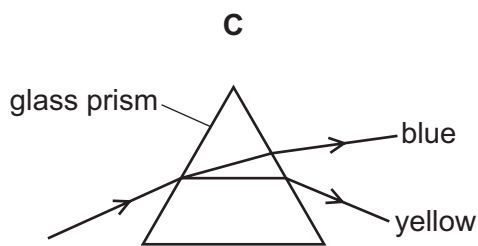
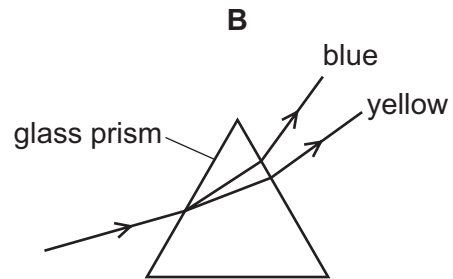
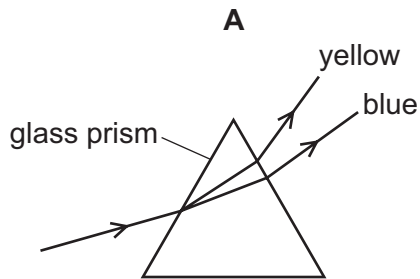
Which labelled distance is the focal length of the lens?



20 A beam of light consists of yellow and blue light.

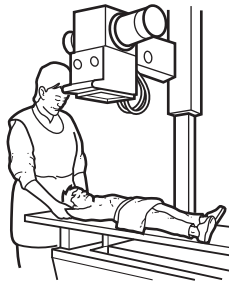
The beam of light is incident on a glass prism.

Which diagram is correct?



21 The two devices shown use different types of electromagnetic waves.

medical scanning



remote controller



Which types of waves are used in these devices?

	medical scanning	remote controller
A	ultraviolet	infrared
B	ultraviolet	microwaves
C	X-rays	infrared
D	X-rays	microwaves

- 22** A boy shouts and hears the echo from a tall building 2.2 s later.

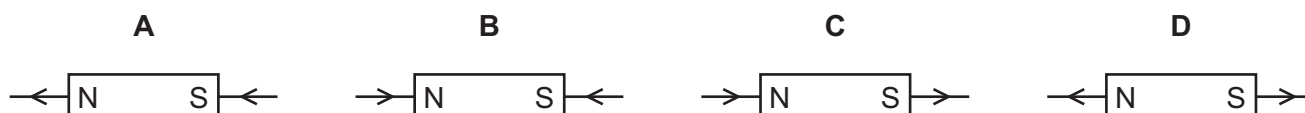
The speed of sound in air is 330 m/s.

How far away from the boy is the building?

- A** 150 m **B** 300 m **C** 360 m **D** 730 m

- 23** The magnetic field of a bar magnet can be represented by magnetic field lines.

Which diagram shows two magnetic field lines correctly?



- 24** A plastic rod is rubbed with a dry cloth. The rod becomes positively charged.

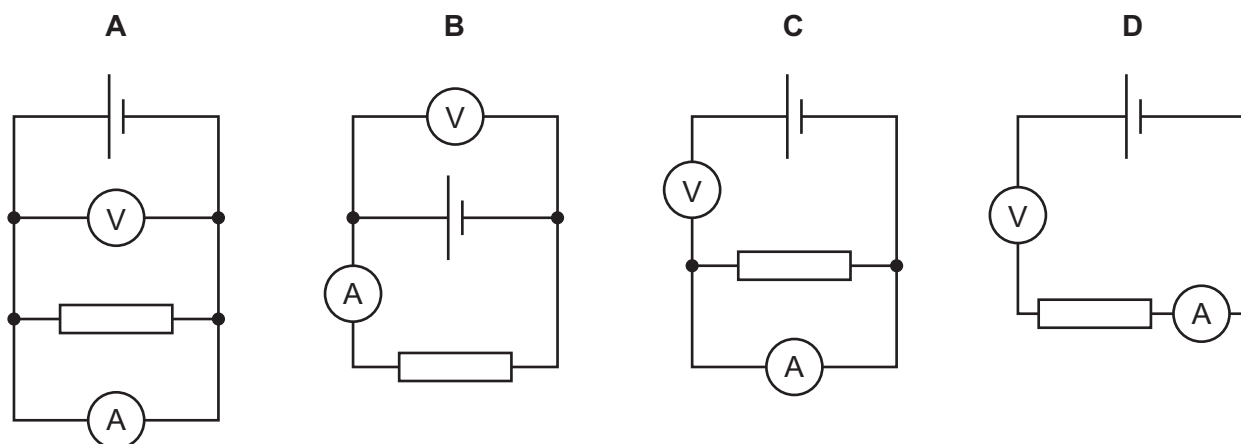
Why has the rod become positively charged?

- A** It has gained electrons.
B It has gained neutrons.
C It has lost electrons.
D It has lost neutrons.

- 25** Which statement about electric current in a conductor is correct?

- A** In a d.c. circuit, the electric current gradually decreases along the conductor.
B In a d.c. circuit, the free electrons flow back and forth.
C In an a.c. circuit, the electric current remains exactly the same all the time.
D In an a.c. circuit, the flow of charge changes direction continually.

- 26** Which circuit can be used to measure the resistance of a resistor?



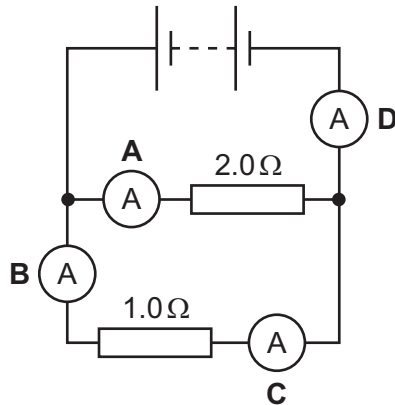
- 27 A lamp rated 12 V, 2.0 A is switched on for 60 s.

How much energy is transferred?

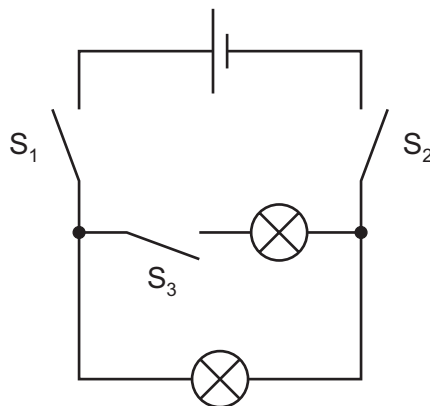
- A 0.40 J B 10 J C 360 J D 1400 J

- 28 The circuit diagram shows a battery connected to two resistors. Four labelled ammeters are connected into the circuit.

Which ammeter shows the largest reading?



- 29 Two lamps are connected in parallel.



Which switches must be closed so that both lamps light?

- A S_1 and S_2 only
 B S_1 and S_3 only
 C S_2 and S_3 only
 D S_1 , S_2 and S_3

30 Which statement about electrical safety is correct?

- A If a device is double insulated, it does not need a fuse.
- B A device that has a normal operating current of 3.0 A must be protected by a 3.0 A fuse.
- C Switches must always be connected into the live supply wire.
- D The metal casing of an electrical device must be connected to the neutral wire.

31 A student investigates the output voltage induced across a coil of wire by a bar magnet.

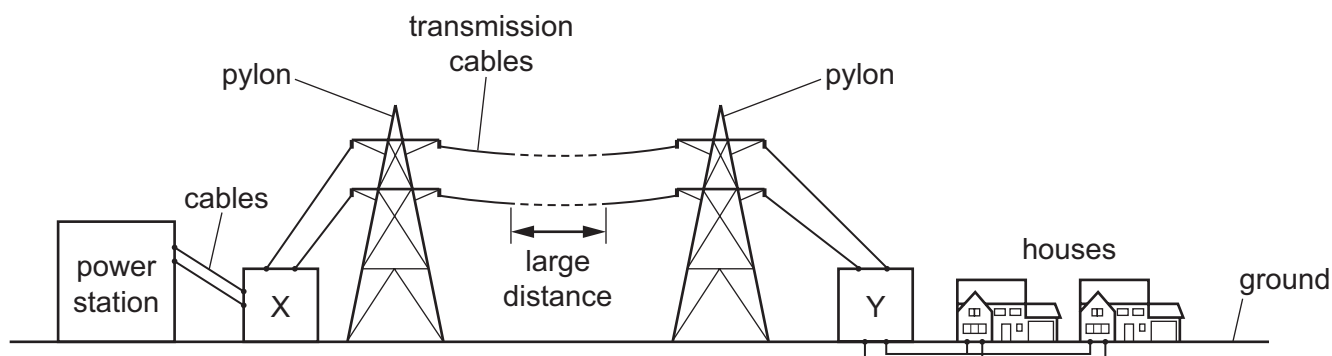
When will the induced voltage have the greatest value?

- A The student slowly moves the bar magnet into the coil of wire.
- B The student leaves the bar magnet stationary in the coil of wire.
- C The student quickly removes the bar magnet from the coil of wire.
- D The student places the bar magnet at rest outside the coil of wire.

32 In which device is the magnetic effect of a current **not** used?

- A electromagnet
- B loudspeaker
- C potential divider
- D relay

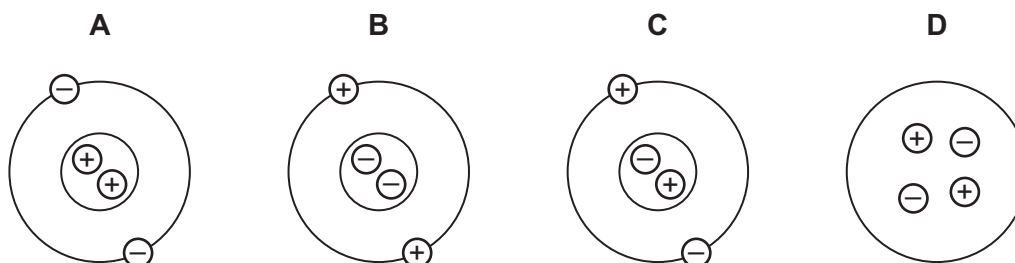
- 33** The diagram represents the transmission of electricity from a power station to homes that are many kilometres away. Two transformers are labelled X and Y.



What type of transformers are X and Y?

	X	Y
A	step-down transformer	step-down transformer
B	step-down transformer	step-up transformer
C	step-up transformer	step-down transformer
D	step-up transformer	step-up transformer

- 34** Which diagram represents the positions of the charged particles of an atom?



- 35** The table shows the composition of three different nuclei.

nucleus	number of protons	number of neutrons
X	3	3
Y	3	4
Z	4	3

Which nuclei are isotopes of the same element?

- A** X, Y and Z **B** X and Y only **C** X and Z only **D** Y and Z only

- 36 A sample of a radioactive isotope emits 9600 α -particles per second.

After 40 hours the rate of emission has fallen to 600 α -particles per second.

What is the half-life of this isotope?

- A** 4.0 hours **B** 8.0 hours **C** 10 hours **D** 20 hours

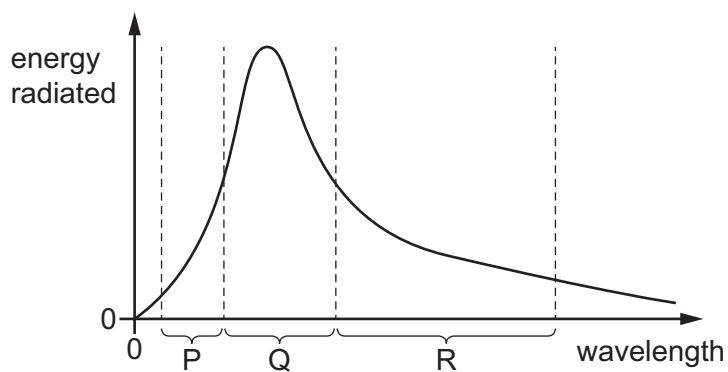
- 37 Which row states a harmful effect and a beneficial effect of ionising radiation on living things?

	harmful effect	beneficial effect
A	kills cancer cells	kills cancer cells
B	kills cancer cells	mutates living cells
C	mutates living cells	kills cancer cells
D	mutates living cells	mutates living cells

- 38 Which statement about the Solar System is correct?

- A** All the planets are rocky.
B Only the Earth has a moon.
C Pluto is a dwarf planet.
D There are many stars in the Solar System.

- 39 The graph shows the energy radiated by the Sun at different wavelengths. Most of the energy is radiated in just three parts of the electromagnetic spectrum, labelled P, Q and R.



Which parts of the electromagnetic spectrum are P and R?

	P	R
A	gamma ray	radio
B	infrared	ultraviolet
C	radio	gamma ray
D	ultraviolet	infrared

40 An astronomer observes redshift in the light from a distant galaxy.

Which statement about redshift is correct?

- A** It is the decrease in the observed wavelength of red light emitted from receding galaxies.
- B** It is evidence that the Universe is contracting and supports the Big Bang Theory.
- C** It is evidence that the Universe is expanding and supports the Big Bang Theory.
- D** Redshift is when light from receding galaxies appears blue.

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/12

Paper 1 Multiple Choice (Core)

May/June 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	D	1
2	B	1
3	C	1
4	B	1
5	B	1
6	D	1
7	B	1
8	C	1
9	B	1
10	A	1
11	D	1
12	D	1
13	B	1
14	A	1
15	C	1
16	A	1
17	B	1
18	B	1
19	B	1
20	D	1
21	C	1
22	C	1
23	A	1
24	C	1
25	D	1
26	B	1
27	D	1
28	D	1

Question	Answer	Marks
29	D	1
30	C	1
31	C	1
32	C	1
33	C	1
34	A	1
35	B	1
36	C	1
37	C	1
38	C	1
39	D	1
40	C	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

October/November 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
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- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m / s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

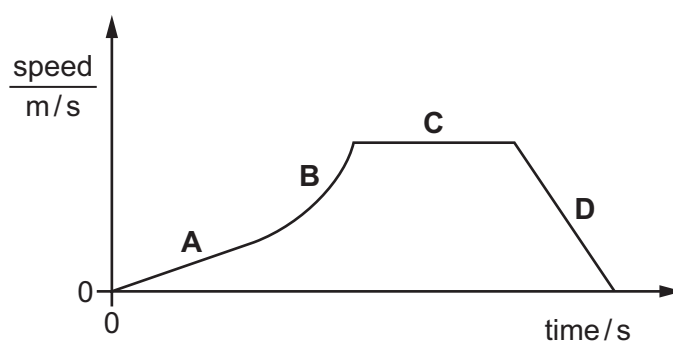


- 1 In order to determine the period of a pendulum, a student times one complete swing of the pendulum using an analogue stop-watch with a second hand.

Which change of method produces the greatest improvement in accuracy?

- A asking a friend with a shorter reaction time to take the measurement
 - B measuring the time for 100 swings of the pendulum and dividing it by 100
 - C measuring the time for a half swing of the pendulum and doubling it
 - D using a digital timer
- 2 The graph shows the speed of a car travelling through a town.

Which section of the graph represents a period when the car is decelerating?

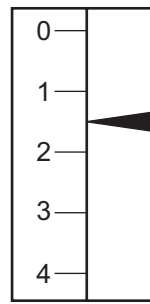


- 3 An object weighs 19 N on a planet where the acceleration of free fall is 3.8 m/s^2 .

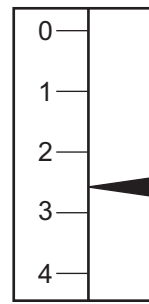
What is the mass of the object?

- A 0.20 kg
- B 1.9 kg
- C 5.0 kg
- D 72 kg

- 4 An object is suspended from a spring balance on the Earth. The same object is suspended from the same spring balance on another planet.



reading on
Earth



reading on
other planet

Which statement explains the difference between the two readings?

- A** Both the mass and the weight of the object are greater on the other planet.
 - B** The mass of the object is greater on the other planet than on the Earth, but the weight is unchanged.
 - C** The spring stretches more easily when on the other planet.
 - D** The weight of the object is greater on the other planet than on the Earth, but the mass is unchanged.
- 5 A student has a bottle of cooking oil.

She determines the density of the cooking oil.

Which apparatus does she need?

	balance	measuring cylinder	ruler	thermometer
A	✓	✓	✓	✓
B	✓	✓	✓	✗
C	✓	✓	✗	✗
D	✓	✗	✗	✗

key

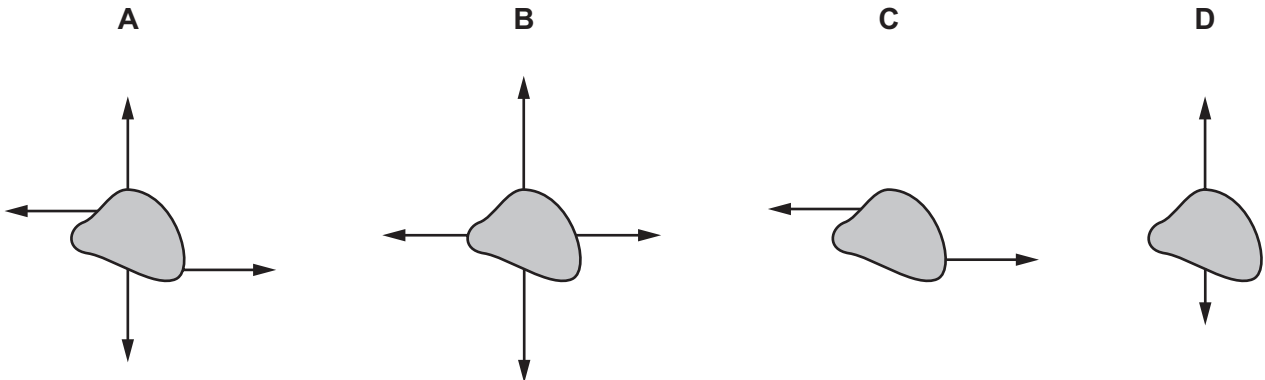
✓ = needed

✗ = not needed

- 6 Forces are applied to four identical objects.

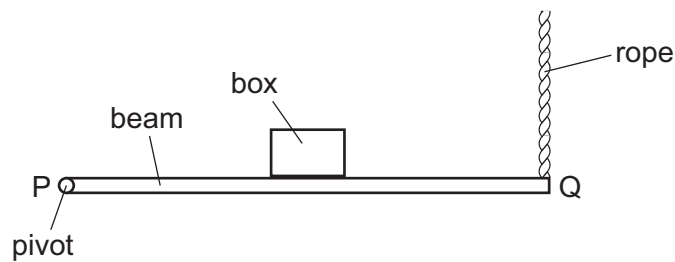
The length of each arrow indicates the magnitude of the force.

Which object is in equilibrium?



- 7 The diagram shows a wooden beam PQ which is attached to a wall by a pivot at P and kept in a horizontal position by a vertical rope attached at Q.

A box has been placed on the beam.

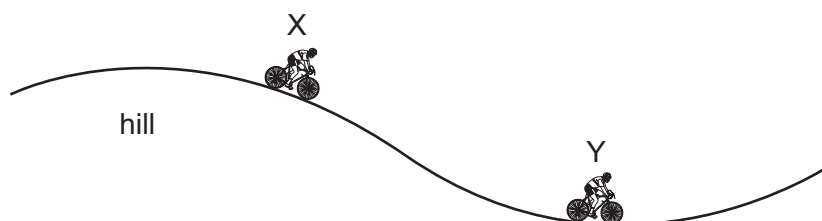


Which changes **must** reduce the tension in the rope at Q?

- A** Decrease the mass of the box and move it towards P.
- B** Decrease the mass of the box and move it towards Q.
- C** Increase the mass of the box and move it towards P.
- D** Increase the mass of the box and move it towards Q.

- 8 A cyclist travels down a hill from rest at point X without pedalling.

The cyclist applies his brakes and the cycle stops at point Y.



Which energy transfers have taken place between X and Y?

- A gravitational potential $\rightarrow$ kinetic $\rightarrow$ internal (thermal)
 - B gravitational potential $\rightarrow$ internal (thermal) $\rightarrow$ kinetic
 - C kinetic $\rightarrow$ gravitational potential $\rightarrow$ internal (thermal)
 - D kinetic $\rightarrow$ internal (thermal) $\rightarrow$ gravitational potential
- 9 Which energy resource is non-renewable?
- A geothermal
 - B natural gas
 - C solar
 - D wind
- 10 A pump does 460 000 J of work to raise water to fill a tank. It takes 7 minutes to fill the tank.
- What is the power of the pump?
- A 1.1 kW B 66 kW C 3200 kW D 190 000 kW
- 11 When a diver swims down from the surface of the water to a depth of 10 m, the pressure experienced increases from $100\,000\text{ N/m}^2$ to $200\,000\text{ N/m}^2$.
- Which statement explains this increase in pressure?
- A The density of the water increases with depth.
 - B The gravitational field strength increases with depth.
 - C The weight of water above the diver increases with depth.
 - D Water cannot be compressed.

12 Why can a gas be compressed easily into a smaller volume?

- A The particles are far apart.
- B The particles do not attract each other.
- C The particles move randomly.
- D The volume of each particle can be reduced.

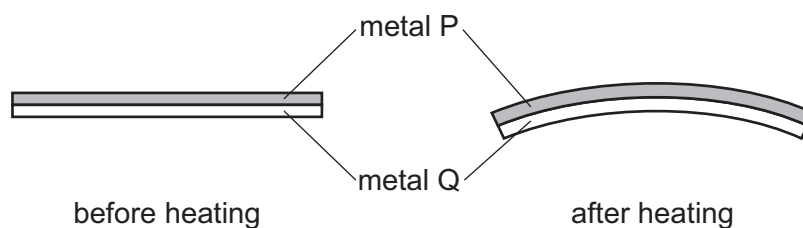
13 The melting point of mercury is -39°C .

What is the melting point of mercury in kelvin?

- A -234 K B 61 K C 234 K D 312 K

14 A bimetallic strip is used to control the temperature of an electrical appliance. It is made of two different metals fixed together.

The diagram shows the shape of the bimetallic strip before and after heating.

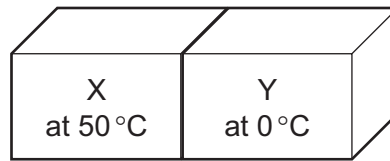


Which statement is correct?

- A Metal P contracts more than metal Q on heating.
- B Metal Q contracts more than metal P on heating.
- C Metal P expands more than metal Q on heating.
- D Metal Q expands more than metal P on heating.

- 15** A student has two blocks of metal, X and Y. The temperature of X is 50°C and the temperature of Y is 0°C .

The two blocks are placed in contact with each other, as shown.

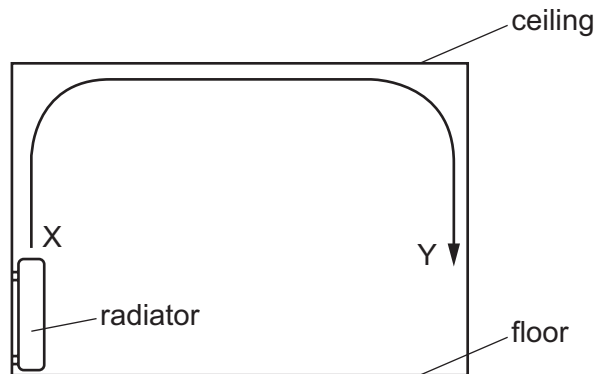


After some time, both blocks have a temperature of 25°C .

What has happened to the internal energy of each block?

	internal energy of X	internal energy of Y
A	decreased	decreased
B	decreased	increased
C	increased	decreased
D	unchanged	unchanged

- 16** The diagram shows the view of a room heated by a radiator. The arrowed line from X to Y is the path of the convection current in the air.



Which row about the air temperature and the air density at X and at Y is correct?

	air temperature	air density
A	higher at X	higher at X
B	higher at X	higher at Y
C	higher at Y	higher at Y
D	higher at Y	higher at X

17 Which statements about waves are correct?

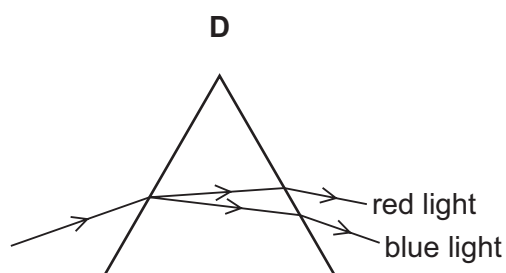
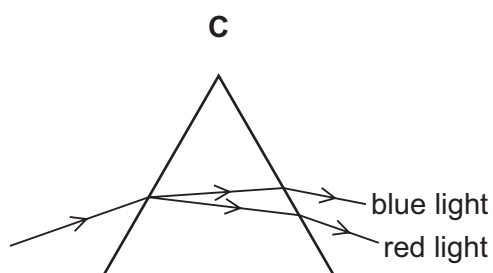
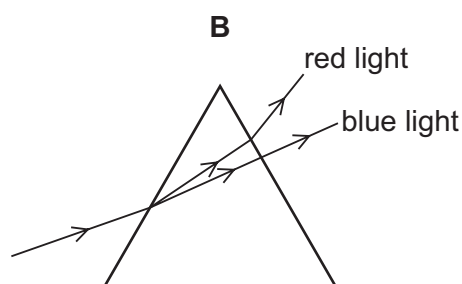
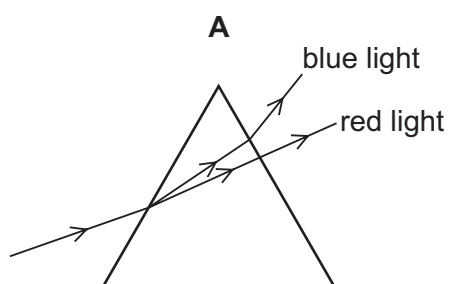
- 1 Only longitudinal waves can undergo diffraction.
- 2 All waves can undergo refraction.
- 3 All waves can undergo reflection.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

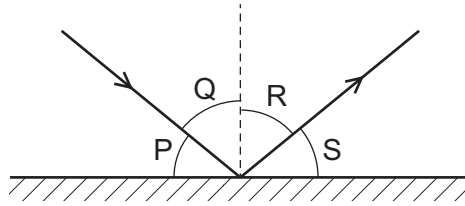
18 Which quantities relating to a wave on the surface of water can **both** be measured in metres?

- A** amplitude and frequency
B amplitude and wavelength
C amplitude and wave speed
D frequency and wavelength

19 Which diagram correctly shows the dispersion of white light through a glass prism?



- 20 A ray of light is reflected by a plane mirror.



Which row shows the angle of incidence and the angle of reflection?

	angle of incidence	angle of reflection
A	P	Q
B	P	S
C	Q	R
D	R	S

- 21 A thin converging lens is used to produce a real image of an object.

Which statement about the real image is always correct?

- A** It is nearer to the lens than the object.
- B** It is on the opposite side of the lens to the object.
- C** It is the same size as the object.
- D** It is upright.

- 22 Where do all types of electromagnetic waves travel at the same speed?

- A** air
- B** a vacuum
- C** glass
- D** water

- 23 Dogs can hear sounds in the range from 100 Hz to 45 kHz.

Which statement is correct?

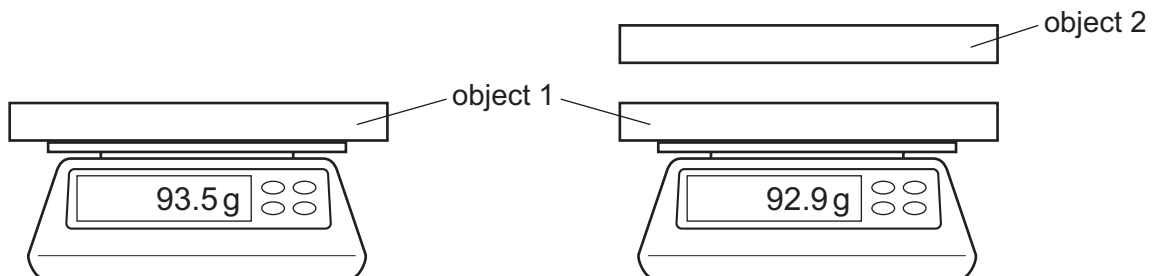
- A** Any sound a dog can hear can also be heard by a human.
- B** Any sound a human can hear can also be heard by a dog.
- C** Dogs can hear some low frequency sounds that are silent for humans.
- D** Dogs can hear some high frequency sounds that are silent for humans.

- 24** Which metal could be used for a permanent magnet and which metal could be used for the core of an electromagnet?

	permanent magnet	core of electromagnet
A	iron	copper
B	iron	steel
C	steel	copper
D	steel	iron

- 25** Object 1 is given a negative charge and placed on a balance.

Object 2, which is also charged, is brought close to object 1 and the reading on the balance changes as shown.



Which action would further decrease the reading on the balance?

- A** Add the same number of electrons to both objects.
 - B** Remove the same number of electrons from both objects.
 - C** Transfer electrons from object 1 to object 2.
 - D** Transfer electrons from object 2 to object 1.
- 26** Which statement about a voltmeter is correct?
- A** It has a scale which is marked in amperes (A).
 - B** It must be connected in series in a circuit.
 - C** It measures potential difference (p.d.).
 - D** It must have three terminals.

27 How does the resistance of a metallic wire change:

- as its length increases
- as its cross-sectional area decreases?

	resistance as length increases	resistance as cross-sectional area decreases
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

28 A plastic rod is rubbed with a cloth. The rod becomes positively charged because of the movement of charged particles.

Which row gives the name of these charged particles and the direction in which they move?

	charged particles	direction of movement
A	electrons	from cloth to rod
B	electrons	from rod to cloth
C	protons	from cloth to rod
D	protons	from rod to cloth

29 Two 2.0Ω resistors are connected in parallel.

What is the combined resistance of the resistors?

- A** less than 2.0Ω
- B** exactly 2.0Ω
- C** more than 2.0Ω , but less than 4.0Ω
- D** exactly 4.0Ω

30 An electric heater is plugged into the mains supply using a fused plug.

The current in the heater is 10 A.

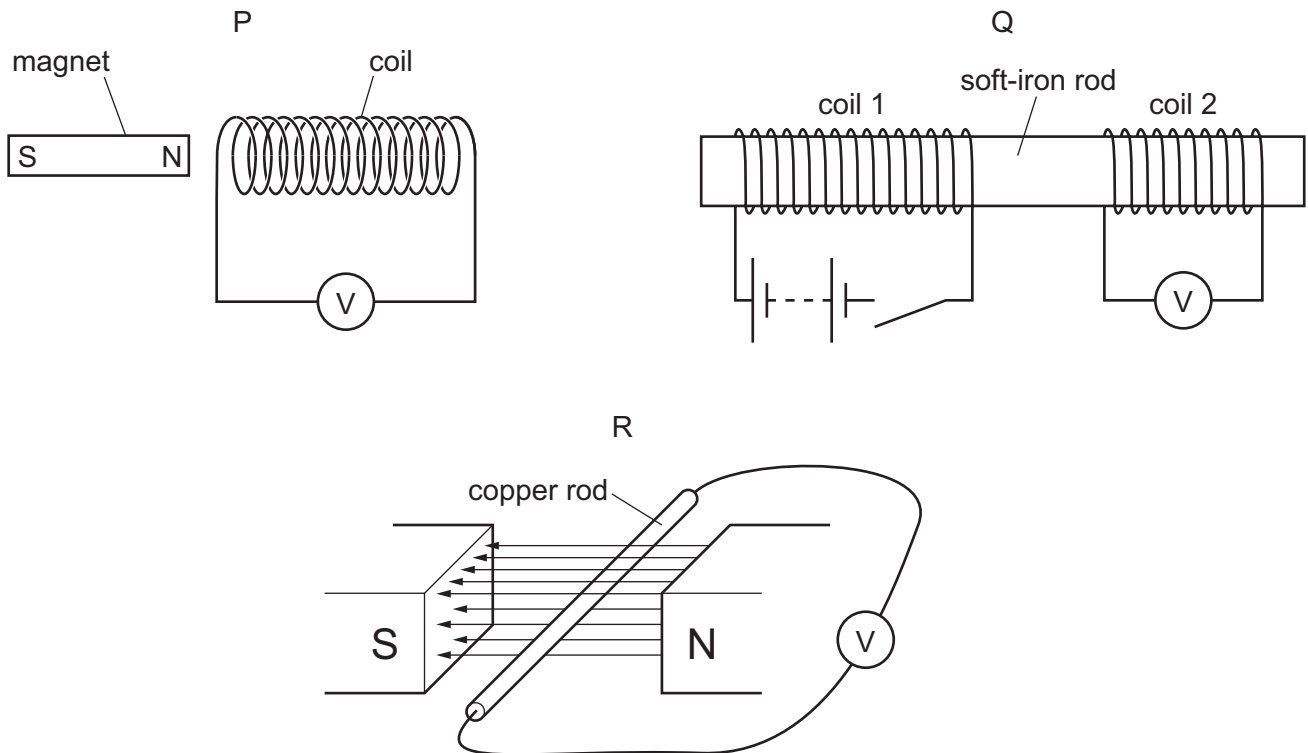
The cable attached to the heater is rated at 15 A.

The fuses available are rated at 1.0 A, 3.0 A, 5.0 A and 13 A.

Which fuse should be used?

- A** 1.0 A **B** 3.0 A **C** 5.0 A **D** 13 A

31 A teacher sets up the equipment for three demonstrations, P, Q and R.



In demonstration P, the magnet is moved to the right.

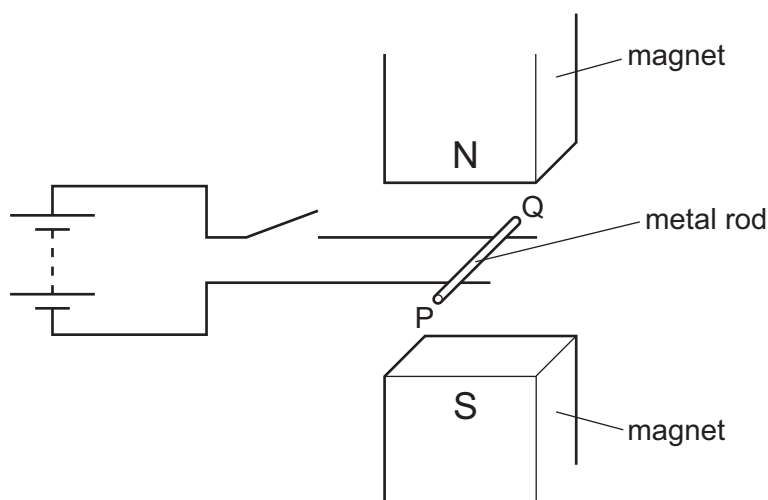
In demonstration Q, the switch is closed.

In demonstration R, the copper rod is moved vertically upwards.

Which demonstrations can be used to demonstrate electromagnetic induction?

- A** P and Q only **B** P and R only **C** Q and R only **D** P, Q and R

- 32** A metal rod PQ rests on two horizontal metal wires that are attached to a battery. The rod lies between the poles of a magnet.



When the switch is closed, the rod moves to the right.

What could be changed so that the rod moves to the left?

- A** Open the switch.
 - B** Reverse the battery terminals and exchange the poles of the magnet.
 - C** Reverse the battery terminals but without exchanging the poles of the magnet.
 - D** Turn the metal rod around (P and Q exchanged).
- 33** A transformer in a computer is used to transform the mains voltage of 240 V to 12 V.

The number of turns on the secondary coil is 2000.

Which statement about the transformer is correct?

- A** It is a step-down transformer and has 100 turns on its primary coil.
- B** It is a step-down transformer and has 40 000 turns on its primary coil.
- C** It is a step-up transformer and has 100 turns on its primary coil.
- D** It is a step-up transformer and has 40 000 turns on its primary coil.

- 34** Atom P has 6 electrons, 6 protons and 6 neutrons.

Atom Q has 6 electrons, 6 protons and 7 neutrons.

Which statement is correct?

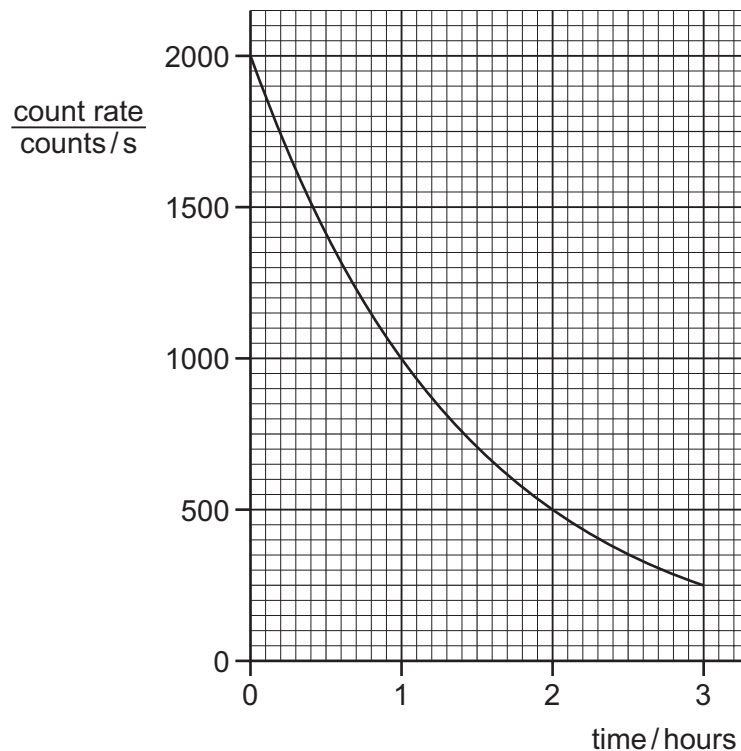
- A** Atom P has atomic number 12.
- B** Atom P has nucleon number 6.
- C** Atom Q has nucleon number 13.
- D** Atoms P and Q are different chemical elements.

- 35 α -particles, β -particles and γ -rays are emitted by radioactive nuclei when they decay.

Which emissions can be deflected by an electric field?

- A α -particles, β -particles and γ -rays
- B α -particles and β -particles only
- C β -particles and γ -rays only
- D γ -rays and α -particles only

- 36 The graph shows the count rate from a radioactive source over a period of time.



What is the half-life of the source?

- A 0.5 hour
 - B 1.0 hour
 - C 1.5 hours
 - D 3.0 hours
- 37 A scientist carries out an experiment using a sealed source which emits β -particles. The range of the β -particles in the air is about 30 cm.

Which precaution is the most effective to protect the scientist from the radiation?

- A handling the source with long tongs
- B keeping the temperature of the source low
- C opening all windows in the laboratory
- D washing his hands before leaving the laboratory

- 38** Which planet orbits the Sun between Mars and Saturn?
- A** Earth
 - B** Jupiter
 - C** Mercury
 - D** Neptune
- 39** Which statement about the Sun is correct?
- A** The Sun is a dwarf star consisting mostly of hydrogen and oxygen.
 - B** The Sun is a giant star consisting mostly of helium and carbon dioxide.
 - C** The Sun is a medium-sized star consisting mostly of hydrogen and helium.
 - D** The Sun is a medium-sized star consisting mostly of nitrogen and oxygen.
- 40** Which quantity does a light-year measure?
- A** an angle
 - B** a distance
 - C** a speed
 - D** a time

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17	D	1
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19	D	1
20	C	1
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25	D	1
26	C	1
27	D	1
28	B	1

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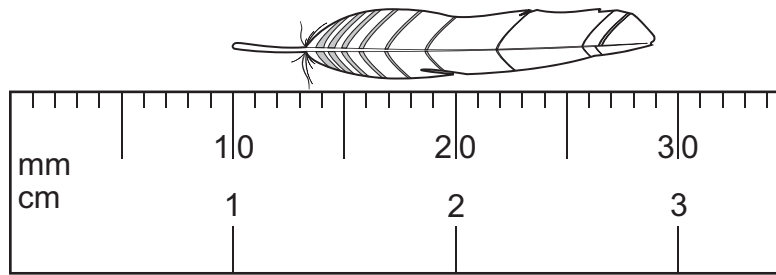
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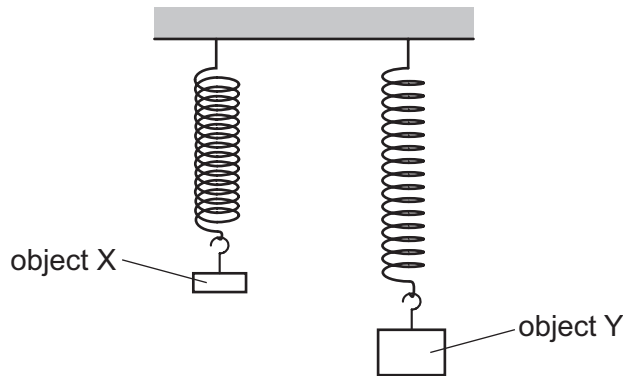


- 1 The diagram shows an enlarged drawing of the end of a metre ruler. It is being used to measure the length of a small feather.



What is the length of the feather?

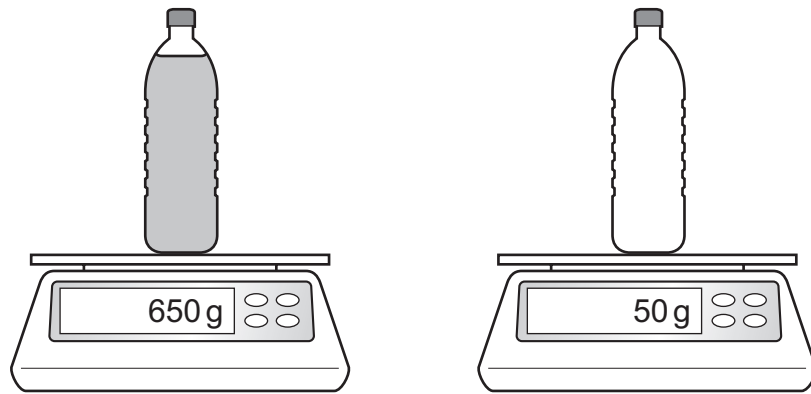
- A** 19 mm **B** 29 mm **C** 19 cm **D** 29 cm
- 2 The diagram shows two objects, X and Y, suspended from identical springs. The extension of each spring is different.



Which row compares the mass and the weight of the two objects?

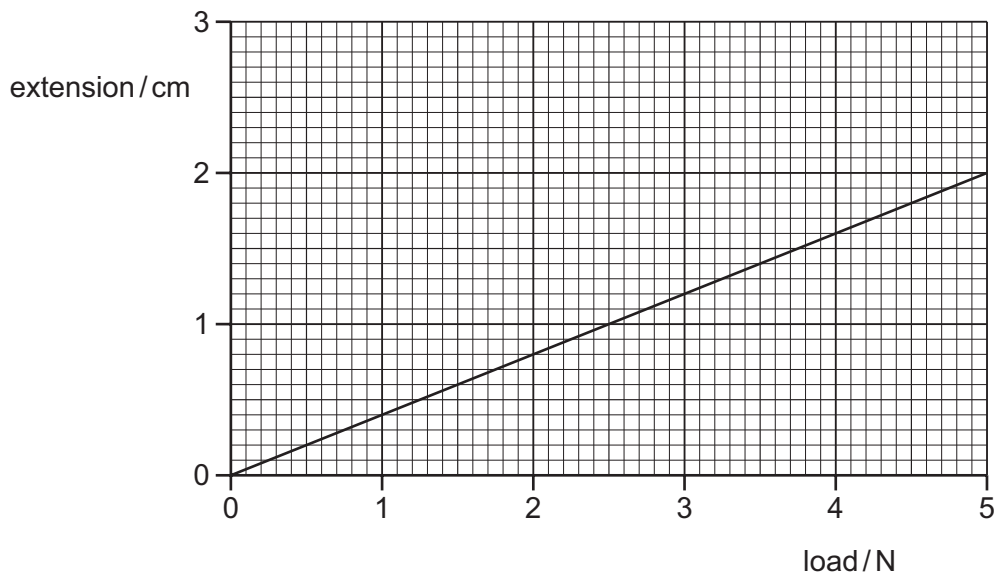
	mass	weight
A	both objects have the same mass	both objects have the same weight
B	both objects have the same mass	the weight of X is less than the weight of Y
C	the mass of X is less than the mass of Y	both objects have the same weight
D	the mass of X is less than the mass of Y	the weight of X is less than the weight of Y

- 3 A plastic bottle contains 750 cm^3 of oil. The diagram shows the mass of the bottle being measured when it is full and then when it is empty.



What is the density of the oil?

- A** 0.80 g/cm^3 **B** 0.93 g/cm^3 **C** 1.1 g/cm^3 **D** 1.3 g/cm^3
- 4 The extension–load graph for a spring is shown. The unstretched length of the spring is 15.0 cm .

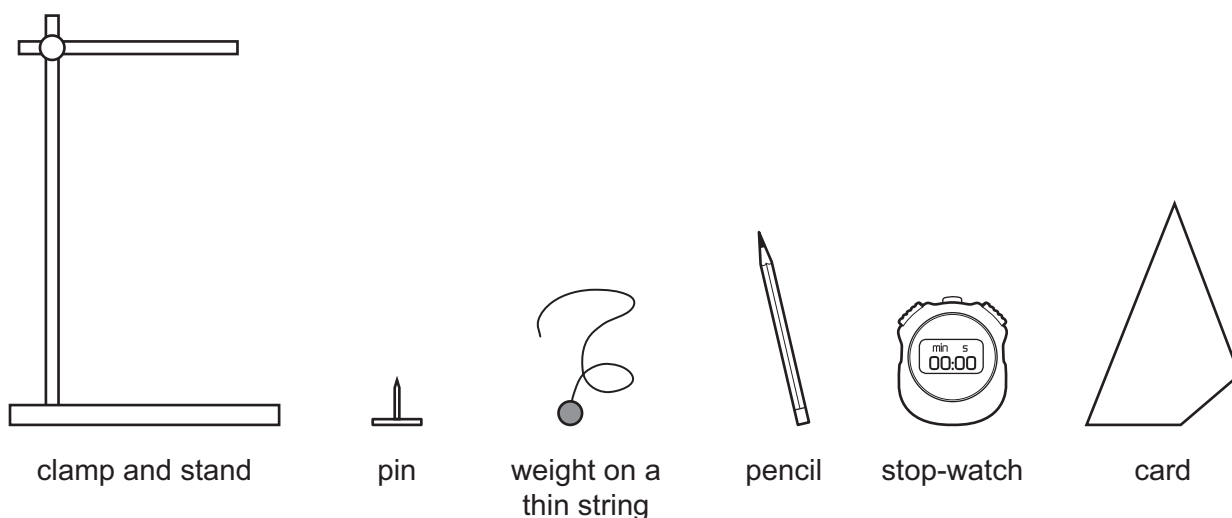


When an object of unknown weight is suspended on the spring, the length of the spring is 16.4 cm .

What is the weight of the object?

- A** 0.55 N **B** 0.67 N **C** 3.5 N **D** 4.1 N

- 5 A student determines the position of the centre of gravity of a piece of card. The diagram shows the equipment that is available. The equipment is not drawn to scale.



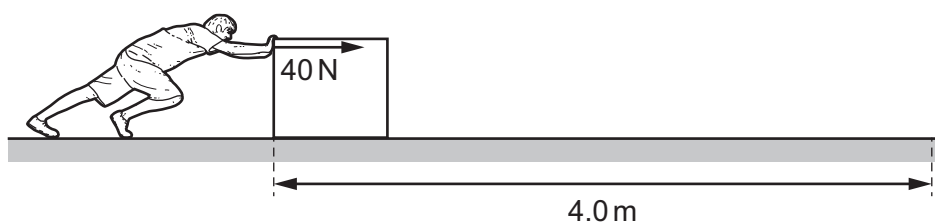
Which piece of equipment is **not** needed?

- A pencil
 - B pin
 - C weight on a thin string
 - D stop-watch
- 6 A cyclist stops his bicycle using the brakes.

Which row about energy stores is correct?

	decreases	increases
A	kinetic energy store of bicycle	chemical energy store of brakes
B	kinetic energy store of bicycle	thermal energy store of brakes
C	thermal energy store of brakes	kinetic energy store of bicycle
D	thermal energy store of brakes	chemical energy store of bicycle

- 7 A man applies a force of 40 N to push a box along the floor.

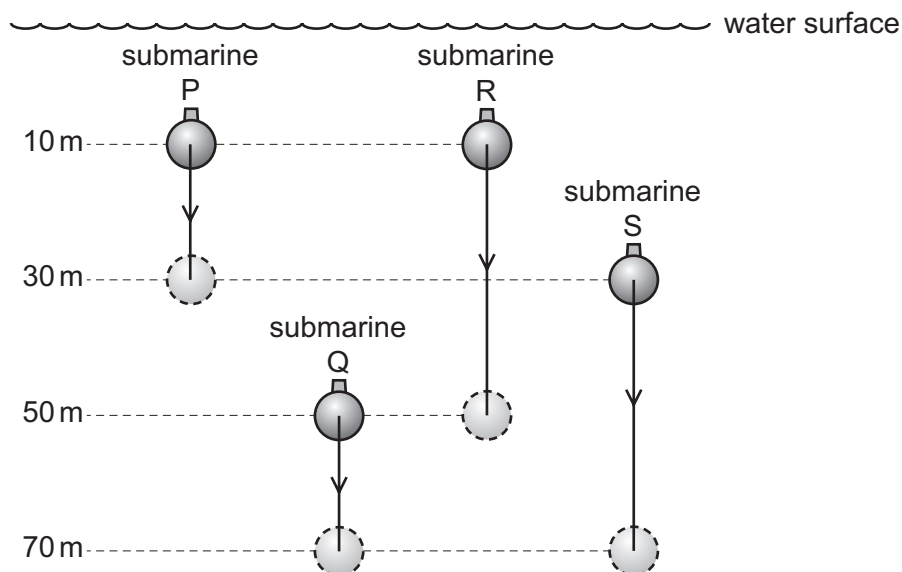


How much power is required to push the box 4.0 m in 3.0 seconds?

- A 3.3 W
- B 30 W
- C 53 W
- D 480 W

- 8 Four identical submarines, P, Q, R and S, are lowered from one depth to another in water of a constant density.

The initial and final depths of each submarine are shown.



Which row is correct for the difference in pressure from the initial to the final depth of each submarine?

	least change in pressure	greatest change in pressure
A	submarines P and Q	submarines R and S
B	submarines P and Q	submarine R only
C	submarine P only	submarines R and S
D	submarine P only	submarine R only

- 9 Some terms describing changes of state are listed.

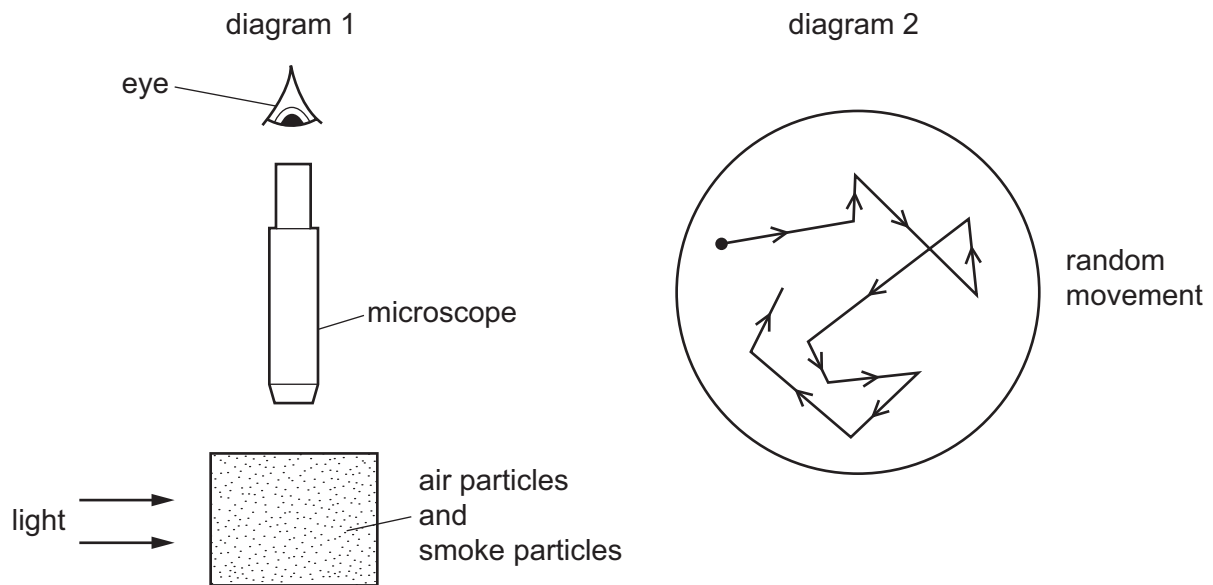
- 1 boiling
- 2 solidification
- 3 condensation
- 4 evaporation

Which two terms identify the same change of state?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

10 Diagram 1 shows apparatus being used to observe smoke particles.

Diagram 2 shows how a smoke particle moves randomly.

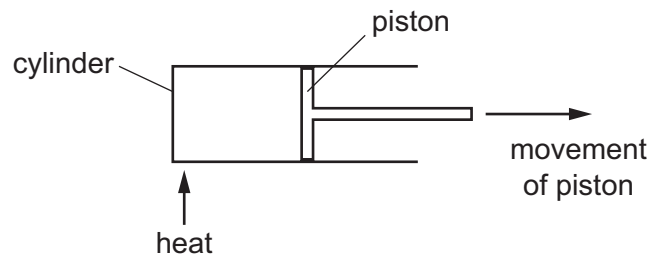


Why do the smoke particles move randomly?

- A** They are hit by air particles.
 - B** They are less dense than air.
 - C** They are moved by convection currents.
 - D** They gain energy from the light.
- 11** How is a temperature in degrees Celsius ($^{\circ}\text{C}$) converted to a temperature in kelvin (K)?
- A** Add 273 to the temperature in $^{\circ}\text{C}$.
 - B** Divide the temperature in $^{\circ}\text{C}$ by 273.
 - C** Multiply the temperature in $^{\circ}\text{C}$ by 273.
 - D** Subtract 273 from the temperature in $^{\circ}\text{C}$.

- 12 The diagram shows a frictionless piston inside a cylinder.

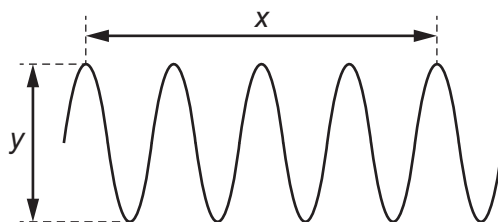
The air inside the cylinder is heated. The piston moves in the direction shown.



Which statement about the air inside the cylinder is correct?

- A The temperature and volume increase at constant pressure.
 - B The temperature and pressure increase at constant volume.
 - C The temperature, volume and pressure all increase.
 - D The volume and pressure increase at constant temperature.
- 13 What increases when the temperature of a copper block increases?
- A the number of copper atoms in the block
 - B the melting point of the block
 - C the internal energy of the block
 - D the density of the block
- 14 A textbook gives the description of a thermal process as 'more-energetic molecules escape from the surface of a liquid which causes the liquid to cool'.
- Which process is being described?
- A boiling
 - B Brownian motion
 - C condensation
 - D evaporation
- 15 In which situation is radiation the **main** method by which energy is transferred?
- A heating a pan of water using a gas camping stove
 - B energy reaching the Earth from the Sun
 - C heating the air in a room with a convection heater
 - D giving gravitational potential energy to a glider when it is lifted by thermal currents

16 The diagram represents a wave.



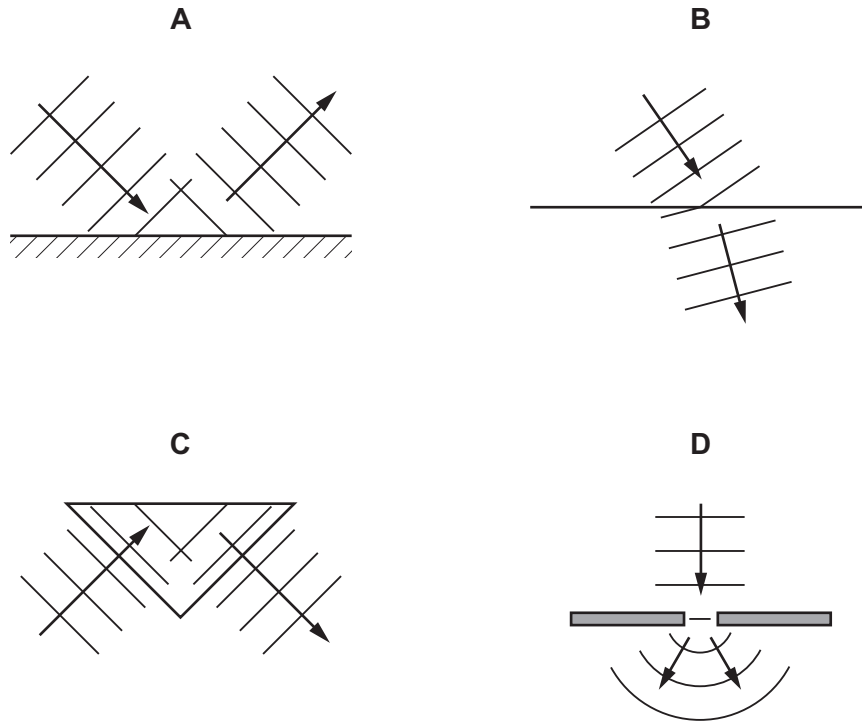
Which row gives the wavelength and the amplitude of the wave?

	wavelength	amplitude
A	x	y
B	y	x
C	x	$\frac{y}{2}$
D	$\frac{x}{4}$	$\frac{y}{2}$

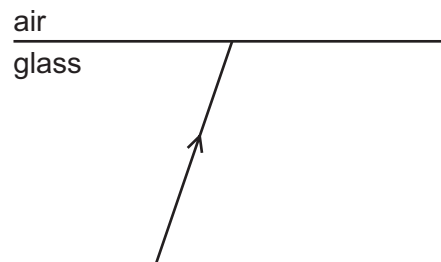
17 Which row describes a seismic P-wave?

	type of wave	direction of vibration
A	longitudinal	parallel to the direction of propagation of the wave
B	longitudinal	perpendicular to the direction of propagation of the wave
C	transverse	parallel to the direction of propagation of the wave
D	transverse	perpendicular to the direction of propagation of the wave

18 Which diagram shows waves diffracting?



19 The diagram shows a ray of light in glass incident on the surface between the glass and air.



What happens if the angle of incidence is made larger than the critical angle for the glass?

- A The angle of refraction becomes equal to 90° .
- B There is a refracted ray and a ray reflected inside the glass.
- C There is a refracted ray only.
- D There is only a ray reflected inside the glass.

20 Red light of frequency 430×10^{12} Hz has a wavelength of 700×10^{-9} m.

What is possible for blue light?

- A a frequency of 190×10^{12} Hz and a wavelength of 450×10^{-9} m
- B a frequency of 190×10^{12} Hz and a wavelength of 950×10^{-9} m
- C a frequency of 670×10^{12} Hz and a wavelength of 450×10^{-9} m
- D a frequency of 670×10^{12} Hz and a wavelength of 950×10^{-9} m

- 21** Infrared is a useful type of electromagnetic radiation. However, excessive exposure to infrared can cause harmful effects.

Which row gives a use and a harmful effect for infrared?

	use	harmful effect
A	detection of cancer	mutations
B	intruder alarms	skin burns
C	detection of fake bank notes	cell damage
D	thermal imaging	internal heating of body cells

- 22** The table shows data for the hearing ranges of different animals.

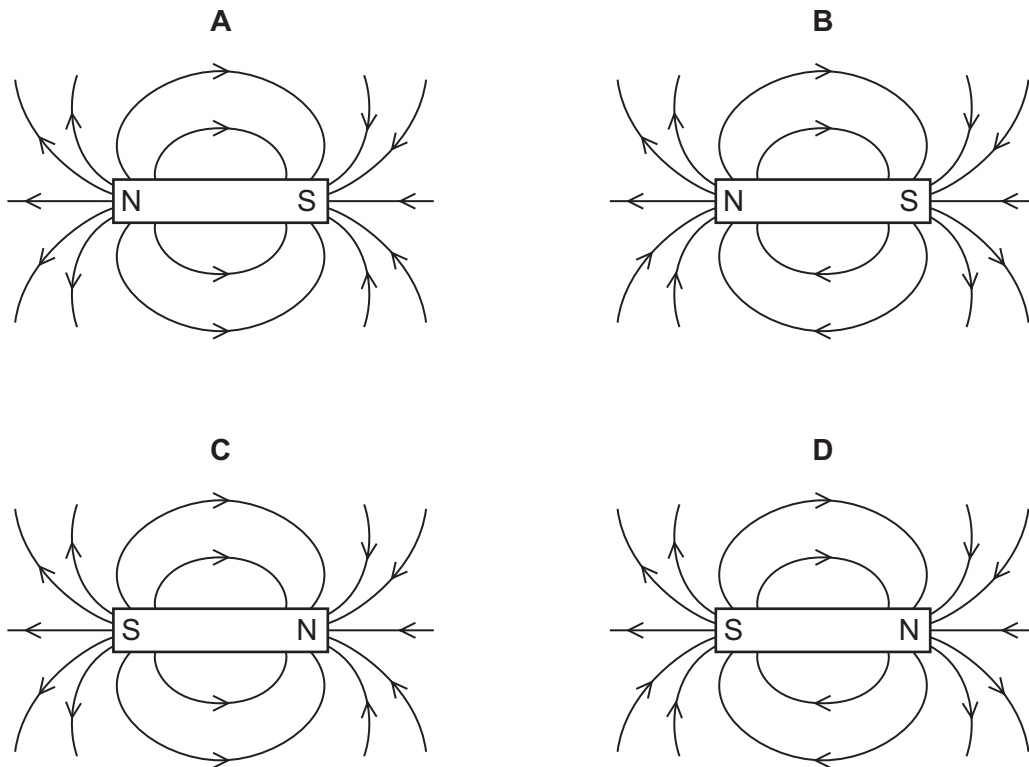
Which animal has the most similar hearing range to a human?

	animal	hearing range / Hz
A	turtle	20–1000
B	mouse	1000–1 000 000
C	moth	20 000–300 000
D	elephant	16–12 000

- 23** Which description of ultrasound is correct?

- A** longitudinal waves with a frequency greater than 20 000 Hz
- B** longitudinal waves with a frequency less than 20 Hz
- C** transverse waves with a frequency greater than 20 000 Hz
- D** transverse waves with a frequency less than 20 Hz

- 24** Which diagram shows the pattern and the direction of the magnetic field lines around a bar magnet?



- 25** Which material is a good electrical conductor?

- A** glass
- B** copper
- C** rubber
- D** wood

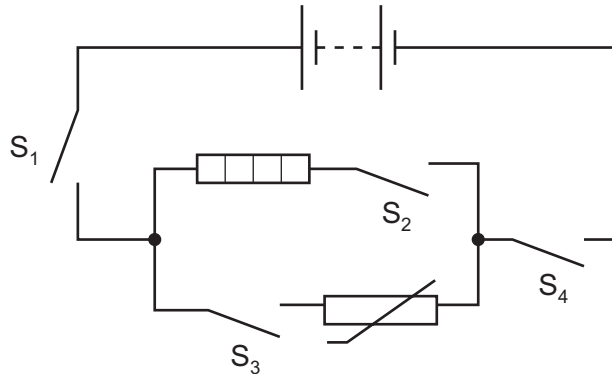
- 26** What is an electric current and how is an ammeter positioned in a circuit to measure the current in a resistor?

	what is an electric current?	positioning of an ammeter
A	a flow of charge	in parallel with the resistor
B	a flow of charge	in series with the resistor
C	a flow of voltage	in parallel with the resistor
D	a flow of voltage	in series with the resistor

27 Which unit is used for the potential difference (p.d.) between two points in an electrical circuit?

- A ampere (A)
- B joule (J)
- C ohm (Ω)
- D volt (V)

28 The circuit in the diagram contains four switches, S_1 , S_2 , S_3 and S_4 .



Which three switches must be closed for the heater to work?

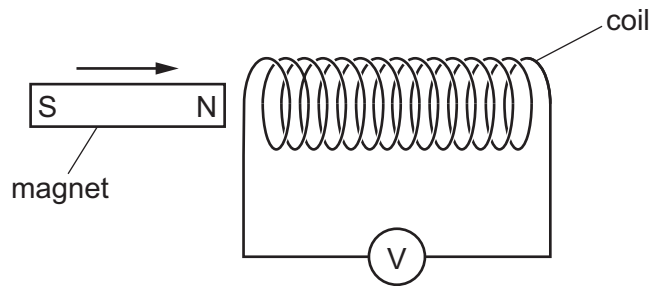
- A S_1 , S_2 and S_3
- B S_1 , S_2 and S_4
- C S_1 , S_3 and S_4
- D S_2 , S_3 and S_4

29 The metal case of an electrical appliance is fitted with an earth connection.

What is the purpose of this?

- A to complete the circuit supplying the appliance so that it works
- B to ensure that the cables supplying the appliance do not become damp
- C to prevent the cables supplying the appliance from overheating
- D to protect a user of the appliance from electric shock

- 30 A student moves a magnet into a coil, as shown. The voltmeter measures the magnitude of the electromotive force (e.m.f.) induced in the coil.



Which quantity does **not** affect the magnitude of the induced e.m.f.?

- A the number of turns per unit length in the coil
 - B the polarity of the magnet
 - C the speed of the magnet
 - D the strength of the magnet
- 31 A student makes a small d.c. motor. The teacher supplies the following equipment.
- battery
 - ammeter
 - voltmeter
 - coil of wire
 - magnets
 - resistor

The student is also able to use other equipment if he wishes.

Which equipment **must** be used?

- A ammeter, battery and resistor
- B ammeter, coil of wire and voltmeter
- C battery, coil of wire and magnets
- D magnets, resistor and voltmeter

- 32** A simple model of the atom consists of small particles orbiting a central nucleus.

Which row is correct?

	charge on nucleus	charge on orbiting particles
A	negative	negative
B	negative	positive
C	positive	negative
D	positive	positive

- 33** What is the charge on a proton and what is the charge on a neutron?

	proton	neutron
A	+1	-1
B	+1	0
C	-1	+1
D	-1	0

- 34** An isotope of radon is radioactive. It decays by emitting an α -particle.

What happens to the nucleus of a radon atom during the emission of the α -particle?

- A** It becomes the nucleus of a different isotope of radon with fewer neutrons.
- B** It becomes the nucleus of a different isotope of radon with more neutrons.
- C** It becomes the nucleus of an element with a higher proton number.
- D** It becomes the nucleus of an element with a lower proton number.

- 35** The nuclide notation of the isotope strontium-90 is $^{90}_{38}\text{Sr}$.

Which statement is correct?

- A** A nucleus of strontium-90 has 38 neutrons.
- B** A nucleus of strontium-90 has 52 neutrons.
- C** A nucleus of strontium-90 has 90 electrons.
- D** A nucleus of strontium-90 has 90 neutrons.

- 36 A box is used for storing radioactive sources.

What is the best material to use for lining the box to prevent radiation from escaping?

- A aluminium
- B lead
- C paper
- D plastic

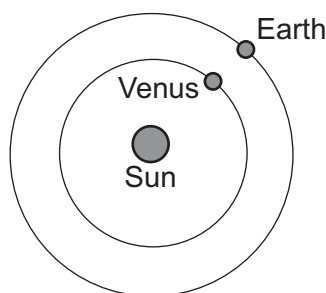
- 37 The Earth is a planet that orbits the Sun once in approximately 365 days.

What does this enable us to explain?

- A the cycle of day and night
- B the cycle of phases of the moon
- C the periodic nature of the seasons
- D the apparent daily motion of the Sun

- 38 The radius of the orbit of Venus around the Sun is 110×10^6 km.

The radius of the orbit of the Earth around the Sun is 150×10^6 km.



The speed of light is 0.30×10^6 km/s.

How long does light take to travel from Venus to the Earth when these planets are closest to each other?

- A 130 s
 - B 370 s
 - C 500 s
 - D 870 s
- 39 What are the three main types of radiation emitted by the Sun?
- A infrared, radio waves and ultraviolet
 - B infrared, visible light and ultraviolet
 - C microwaves, visible light and X-rays
 - D radio waves, X-rays and ultraviolet

40 What is a light-year?

- A** the time for light to travel 1 km in space
- B** the time for light to travel from the Sun to the Earth
- C** the distance travelled in space by light in one year
- D** the distance travelled by light from the Sun to the Earth

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)**0972/11**

Paper 1 Multiple Choice (Core)

May/June 2024

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	D	1
3	A	1
4	C	1
5	D	1
6	B	1
7	C	1
8	A	1
9	B	1
10	A	1
11	A	1
12	A	1
13	C	1
14	D	1
15	B	1
16	D	1
17	A	1
18	D	1
19	D	1
20	C	1
21	B	1
22	D	1
23	A	1
24	A	1
25	B	1
26	B	1
27	D	1
28	B	1

Question	Answer	Marks
29	D	1
30	B	1
31	C	1
32	C	1
33	B	1
34	D	1
35	B	1
36	B	1
37	C	1
38	A	1
39	B	1
40	C	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/12

Paper 1 Multiple Choice (Core)

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Four athletes run twice around a track. The table shows their times at the end of each lap.

Which athlete runs the second lap the fastest?

	time at end of first lap / s	time at end of second lap / s
A	22.99	47.04
B	23.04	47.00
C	23.16	47.18
D	23.39	47.24

- 2 A racing car is fitted with an on-board computer. Every time the car passes the starting line, the computer records the distance travelled in the following two seconds.

Which set of data shows that the car is increasing in speed during the two seconds?

A

time / s	distance travelled / m
0	0
1	100
2	200

B

time / s	distance travelled / m
0	0
1	90
2	180

C

time / s	distance travelled / m
0	0
1	80
2	190

D

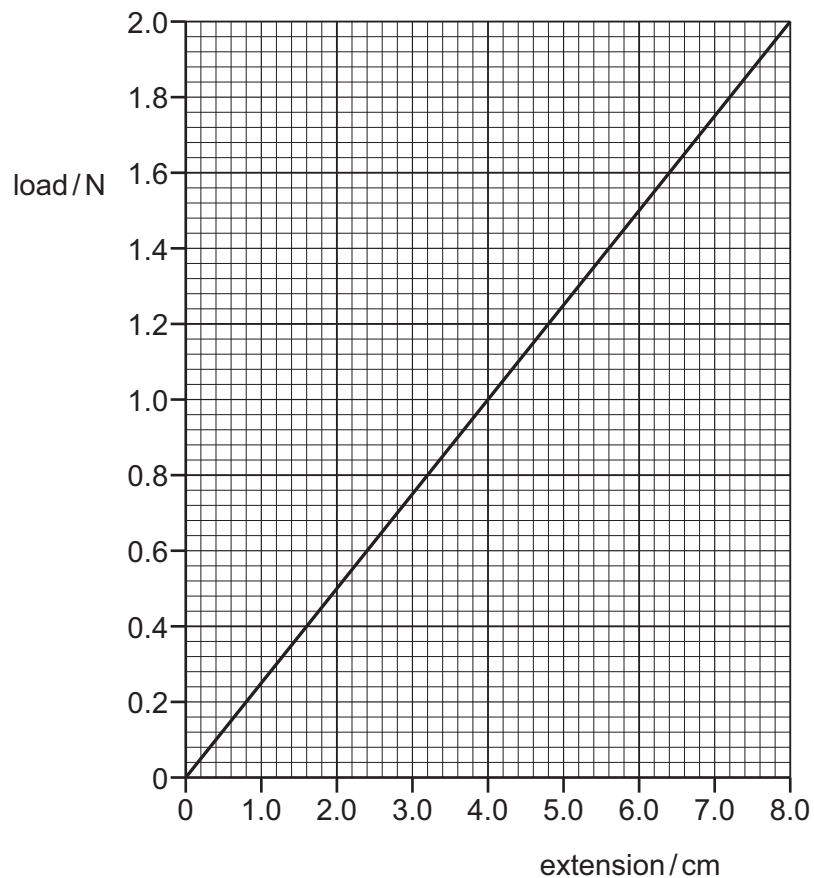
time / s	distance travelled / m
0	0
1	100
2	180

- 3 Which row shows the gravitational field strength on the Earth and the definition of velocity?

	gravitational field strength on the Earth	definition of velocity
A	9.8 kg / N	the change in the speed
B	9.8 N / kg	the change in the speed
C	9.8 kg / N	the speed in a given direction
D	9.8 N / kg	the speed in a given direction

- 4 Which statement about the weight of an object is correct?
- A The weight of an object is the gravitational force per unit mass.
 - B The weight of an object is the gravitational force on an object.
 - C The weight is zero when the object is falling at constant speed in the Earth's atmosphere.
 - D The weight is zero when the object is in orbit around the Earth.
- 5 The diagram shows a load–extension graph for a steel spring.

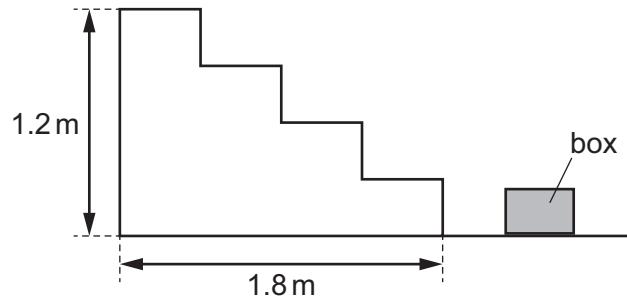
The length of the spring with no load attached is 2.0 cm.



Which load increases the length of the spring to 6.0 cm?

- A 0.5 N B 1.0 N C 1.5 N D 2.0 N
- 6 Which force between two solid surfaces opposes motion?
- A friction
 - B gravity
 - C kinetic
 - D weight

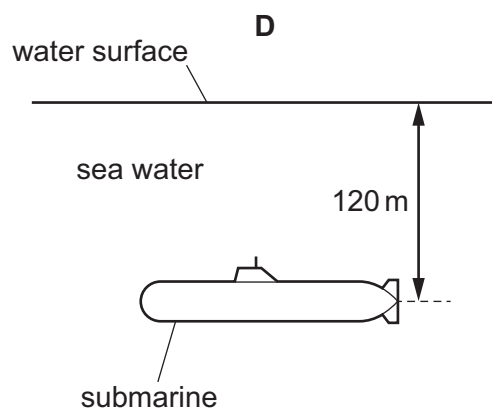
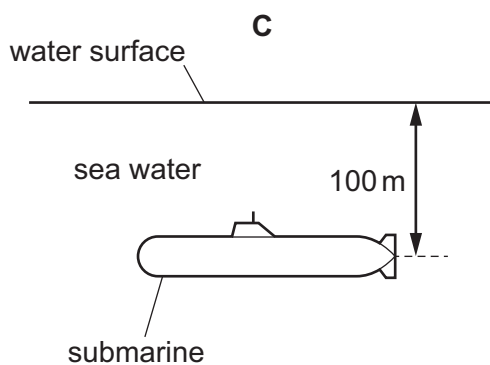
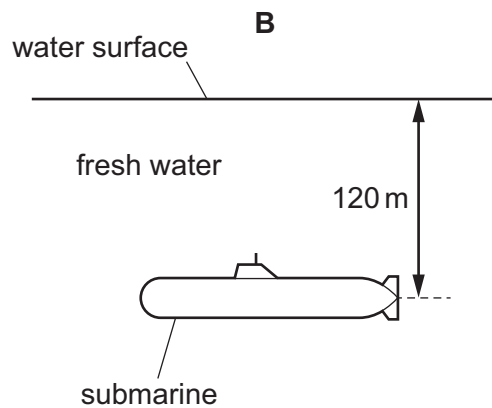
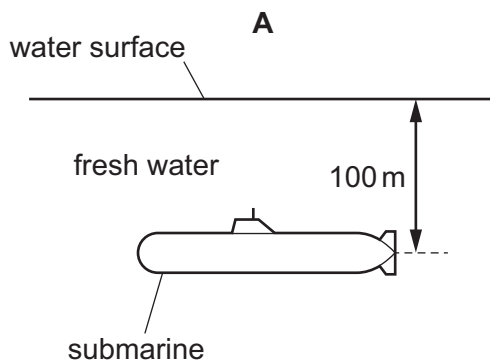
- 7 A box of mass 2.0 kg and weight 20 N is carried to the top of some stairs.



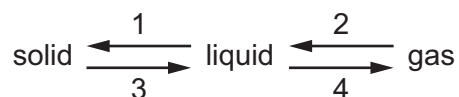
How is the work done against gravity on the box calculated?

- A** 2.0×1.8 **B** 20×1.2 **C** 2.0×1.2 **D** $20 \times 1.2 \times 1.8$
- 8 Four submarines are submerged. The density of fresh water is 1000 kg/m^3 and the density of sea water is 1020 kg/m^3 .

Which submarine experiences the greatest pressure due to the water?



- 9 The diagram shows the changes in state, 1, 2, 3 and 4, between solids, liquids and gases.



What are processes 2 and 3?

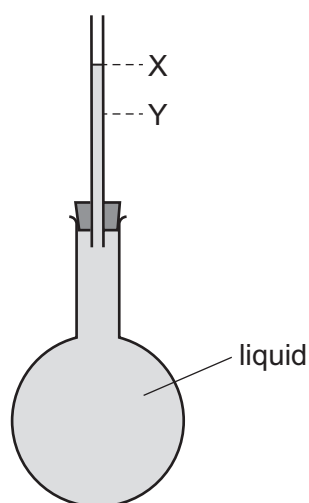
	2	3
A	condensation	melting
B	condensation	freezing
C	evaporation	melting
D	evaporation	freezing

- 10 Copper has a melting point of 1100 °C.

What is the melting point in kelvin?

- A** 727 K **B** 827 K **C** 1373 K **D** 1473 K

- 11 A liquid at room temperature fills a flask and a glass tube to level X.



The flask is placed in ice and the liquid level in the tube changes to level Y.

Why does the level change to level Y?

- A** The flask contracts.
B The flask expands.
C The liquid contracts.
D The liquid expands.

12 Which row correctly describes the state of water at the temperatures shown?

	250 K	300 K	350 K	400 K
A	liquid	liquid	gas	gas
B	solid	solid	liquid	liquid
C	solid	liquid	liquid	gas
D	solid	liquid	gas	gas

13 A liquid is evaporating. The liquid is **not** boiling.

Which statement about the liquid at an instant in time is correct?

- A** Any molecule can escape and from any part of the liquid.
- B** Any molecule can escape, but only from the liquid's surface.
- C** Only molecules with enough energy can escape and only from the liquid's surface.
- D** Only molecules with enough energy can escape, but from any part of the liquid.

14 A student stirs a hot liquid in a pan with a spoon.

Which row explains which material the spoon should be made from so that the student does **not** burn their hand?

	material of spoon	explanation
A	metal	it is a good conductor
B	metal	it is a good insulator
C	wood	it is a good conductor
D	wood	it is a good insulator

15 Thermal radiation is emitted from all objects.

Which mediums can thermal radiation travel through?

	glass	water	air	vacuum
A	✓	✓	✓	✓
B	✓	✓	✓	✗
C	✗	✓	✓	✓
D	✗	✗	✗	✓

key

✓ = can travel through this medium

✗ = cannot travel through this medium

- 16** The radius of a circular pond is 3.0 m.

Circular waves are made at the centre of the pond.

There are 10 whole waves between the centre and the edge of the pond and it takes 5.0 s for all 10 waves to reach the edge.

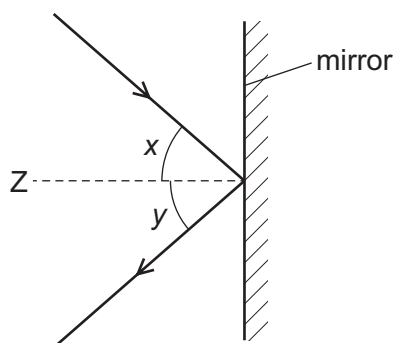
What is the wavelength and the frequency of the waves?

	wavelength / m	frequency / Hz
A	0.30	0.50
B	0.30	2.0
C	0.60	0.50
D	0.60	2.0

- 17** How does the direction of vibration of a longitudinal wave compare to its direction of propagation and which wave example is longitudinal?

	direction of vibration compared to direction of propagation	wave example
A	at right angles	ultrasound waves
B	at right angles	ultraviolet waves
C	parallel	ultrasound waves
D	parallel	ultraviolet waves

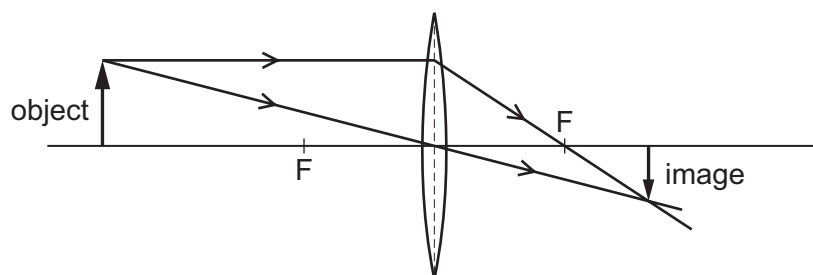
- 18 The diagram shows a ray of light reflecting from a mirror.



Which row shows the correct names for x , y and Z ?

	x	y	Z
A	angle of incidence	angle of reflection	normal
B	angle of incidence	angle of reflection	principal focus
C	angle of reflection	angle of refraction	normal
D	angle of reflection	angle of refraction	principal focus

- 19 A converging lens forms an image of an object placed in front of it.



What are the characteristics of the image?

- A** real, inverted, diminished
- B** real, upright, enlarged
- C** virtual, inverted, diminished
- D** virtual, upright, enlarged

- 20** White light can be split into different colours by passing it through a prism.

What is the name of this process?

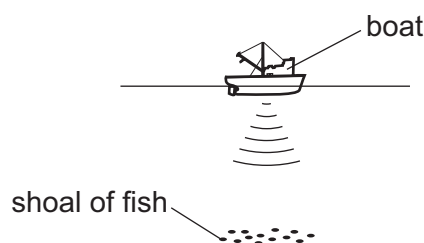
- A** diffraction
- B** dispersion
- C** reflection
- D** refraction

- 21** Excessive exposure to some regions of the electromagnetic spectrum can cause harmful effects to people.

Which regions cause mutation of cells in the body, skin burns and internal heating of body cells?

	mutation of cells	skin burns	internal heating of body cells
A	radio waves	infrared	visible light
B	radio waves	ultraviolet	microwaves
C	X-rays	infrared	microwaves
D	X-rays	ultraviolet	visible light

- 22** A pulse of sound is produced at the bottom of a boat. The sound travels through the water and is reflected from a shoal of fish. The sound reaches the boat again 1.2 s after it is produced. The speed of sound in the water is 1500 m/s.



How far below the bottom of the boat is the shoal of fish?

- A** 450 m
- B** 900 m
- C** 1800 m
- D** 3600 m

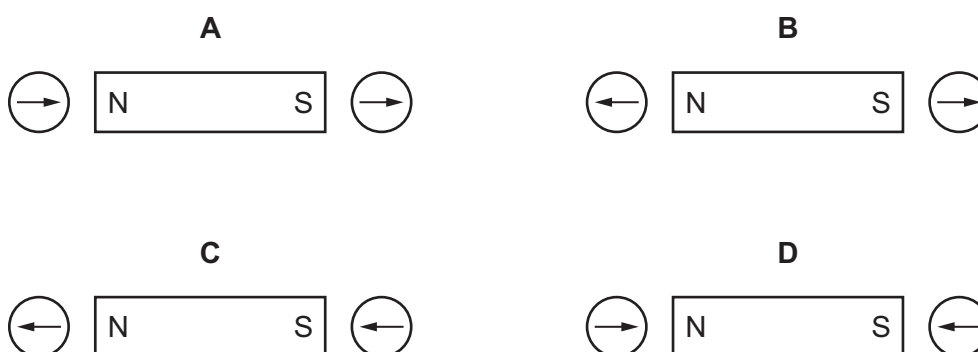
- 23** The frequency of a sound wave increases from 31 000 Hz to 32 000 Hz.

What is the nature of the wave and how does it sound to a human with normal hearing?

- A** It is a longitudinal wave with a decreasing pitch.
- B** It is a transverse wave with an increasing pitch.
- C** It is a longitudinal wave that cannot be heard by a human with normal hearing.
- D** It is a transverse wave that cannot be heard by a human with normal hearing.

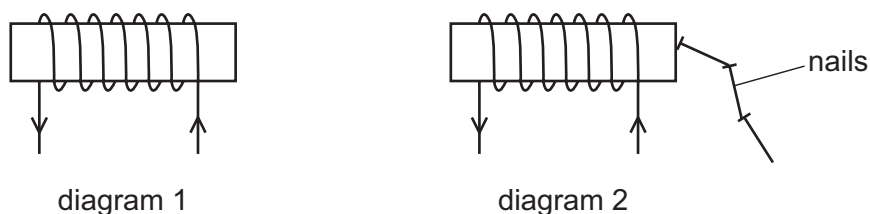
- 24** Small compasses are used to determine the direction of a magnetic field.

Which diagram shows the direction that the compass needles point when placed near the N and S poles of a bar magnet?



- 25** A metal bar is placed inside a current-carrying coil, as shown in diagram 1.

There is a small current in the coil. The bar holds a few nails, as shown in diagram 2.



When there is no current in the coil, the nails drop off.

Which row is correct?

	metal from which the bar is made	effect of a larger current in the coil
A	soft iron	it makes no difference
B	soft iron	the bar holds more nails
C	steel	it makes no difference
D	steel	the bar holds more nails

- 26** There is a current I in a resistor of resistance R . The potential difference (p.d.) across the resistor is V .

Which equation gives the power P dissipated by the resistor?

A $P = IR$ **B** $P = IV$ **C** $P = \frac{I}{R}$ **D** $P = \frac{I}{V}$

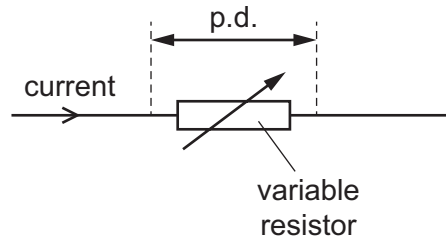
27 A student tests the electrical conduction of four materials.

- aluminium
- iron
- plastic
- silver

Which materials are good conductors of electricity?

- A** aluminium, iron and silver
- B** aluminium and silver only
- C** iron, silver and plastic
- D** plastic only

28 This question is about the potential difference (p.d.) across, and the current in, a variable resistor.



The resistance of the variable resistor is increased.

Which statement is correct?

- A** The current can remain constant if the p.d. is decreased.
- B** The current can remain constant if the p.d. is increased.
- C** The p.d. can remain constant if the current is kept constant.
- D** The p.d. can remain constant if the current is increased.

- 29** Diagram 1 shows a resistor connected in a circuit. Diagram 2 shows an identical resistor connected in parallel with the first resistor.

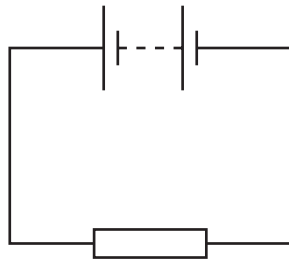


diagram 1

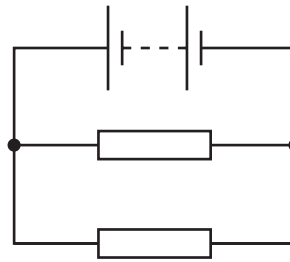
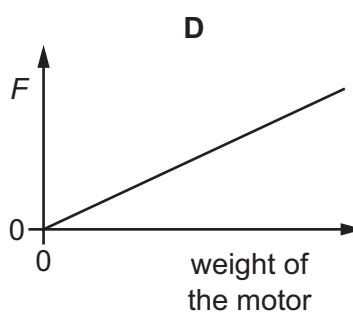
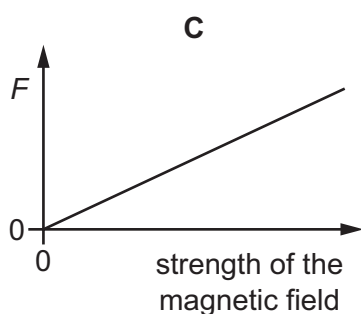
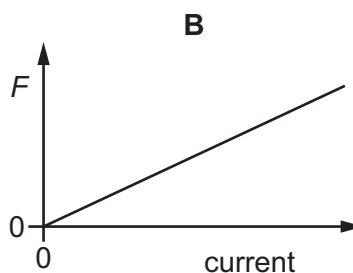
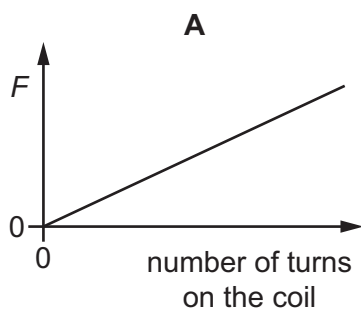


diagram 2

What is the combined resistance of the two resistors in diagram 2?

- A** greater than in the circuit of diagram 1
 - B** less than in the circuit of diagram 1, but greater than zero
 - C** the same as in the circuit of diagram 1
 - D** zero
- 30** Where must a fuse be connected in a mains electric circuit?
- A** the earth wire only
 - B** the live wire only
 - C** the neutral wire only
 - D** the live wire and the earth wire

- 31 Which graph does **not** show the correct trend for the force F causing the turning effect on the coil in a d.c. motor?



- 32 Which statement about the structure of an atom is correct?

- A** Negative electrons surround a neutral nucleus.
- B** Negative electrons surround a positive nucleus.
- C** Positive electrons surround a neutral nucleus.
- D** Positive electrons surround a negative nucleus.

- 33 Four statements about isotopes of a certain element are listed.

Which statement about the isotopes **must** be correct?

- A** They are radioactive.
- B** They are unstable.
- C** They have the same number of neutrons.
- D** They have the same number of protons.

- 34 What makes a contribution to a person's annual dose of ionising radiation?

- A** food and drink
- B** greenhouse gases
- C** mobile phone masts
- D** radio and TV transmissions

- 35** The emissions from a radioactive source pass through a sheet of lead that is 10 mm thick.

Which type of radiation is emitted from the source and how is the radiation affected by an electric field?

	type of radiation	effect of electric field on radiation
A	α	radiation deflected
B	α	radiation not deflected
C	γ	radiation deflected
D	γ	radiation not deflected

- 36** Radioactive waste is often mixed with liquid glass. This mixture solidifies and is stored in steel drums.

Why is this process used to store radioactive waste?

- A** It reduces the decay rate.
 - B** It reduces the half-life of the waste.
 - C** The waste is contained in a solid block so it cannot leak out.
 - D** The glass is transparent so the waste can be seen.
- 37** Which planet is classed as a rocky planet?
- A** Jupiter
 - B** Saturn
 - C** Uranus
 - D** Venus
- 38** What is the cause of the phases of the Moon?
- A** the movement of the Earth around the Sun
 - B** the movement of the Moon around the Sun
 - C** the movement of the Moon around the Earth
 - D** the movement of the Sun around the Moon
- 39** What makes up the Universe?
- A** many billions of galaxies
 - B** Mercury, Venus, the Earth, Mars, Jupiter, Saturn, Uranus and Neptune only
 - C** the Sun, the Earth and the Moon only
 - D** the Solar System only

40 The Sun mostly consists of which gases?

- A** helium and hydrogen
- B** helium and nitrogen
- C** hydrogen and oxygen
- D** oxygen and nitrogen

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)**0972/12**

Paper 1 Multiple Choice (Core)

May/June 2024**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	C	1
3	D	1
4	B	1
5	B	1
6	A	1
7	B	1
8	D	1
9	A	1
10	C	1
11	C	1
12	C	1
13	C	1
14	D	1
15	A	1
16	B	1
17	C	1
18	A	1
19	A	1
20	B	1
21	C	1
22	B	1
23	C	1
24	C	1
25	B	1
26	B	1
27	A	1
28	B	1

Question	Answer	Marks
29	B	1
30	B	1
31	D	1
32	B	1
33	D	1
34	A	1
35	D	1
36	C	1
37	D	1
38	C	1
39	A	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

October/November 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

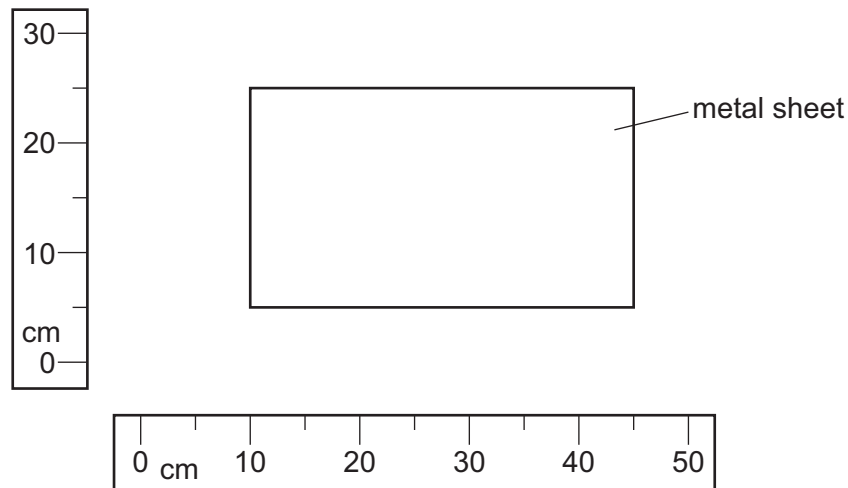
INFORMATION

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- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

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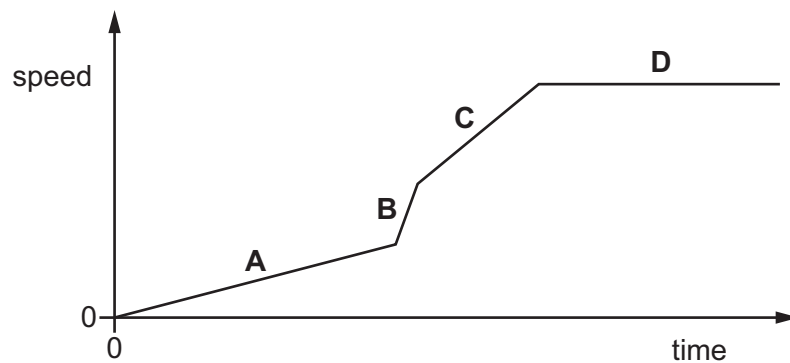
- 1 The diagram shows a rectangular metal sheet close to two rulers.



What is the area of the metal sheet?

- A** 700 cm^2 **B** 875 cm^2 **C** 900 cm^2 **D** 1125 cm^2
- 2 The diagram shows the speed–time graph of an object.

In which section does the object have the largest acceleration?



- 3 An object begins to fall close to the Earth's surface. Air resistance can be ignored.

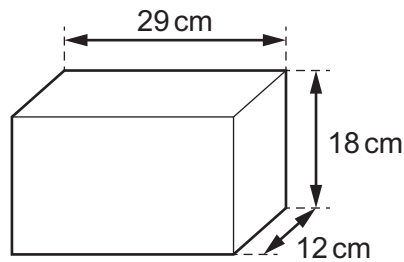
Which statement about the object's acceleration is correct?

- A** The acceleration is constant.
B The acceleration decreases as the body falls.
C The acceleration increases as the body falls.
D The acceleration is zero.

4 Which quantity is equal to gravitational force?

- A gravitational field strength $\times$ mass
- B gravitational field strength $\times$ weight
- C mass per unit weight
- D weight per unit mass

5 A concrete building block has the dimensions shown.



The mass of the block is 15 000 g.

What is the density of the block?

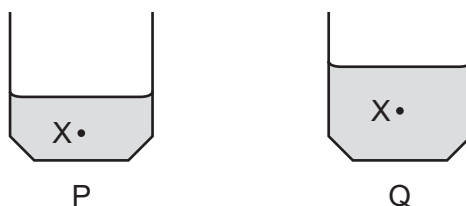
- A 43 g/cm^3 B 2.4 g/cm^3 C 0.42 g/cm^3 D 0.023 g/cm^3

6 What is measured by the moment of a force?

- A the acceleration produced by the force
- B the turning effect of the force
- C the time for which the force acts
- D the increase in energy caused by the force

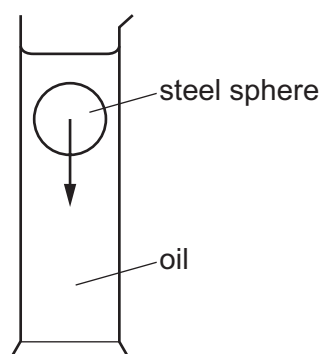
- 7 Two identical containers, P and Q, are partly filled with different quantities of sand.

The position of the centre of gravity for each container is labelled X.



Which container is more stable and what is the reason why it is more stable?

- A** P is more stable because it has a lower centre of gravity.
B P is more stable because it has a smaller mass.
C Q is more stable because it has a greater depth of sand.
D Q is more stable because it has a higher centre of gravity.
- 8 The diagram shows a steel sphere falling through a cylinder of oil.

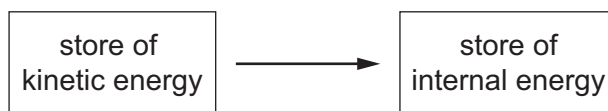


Which row indicates what happens to the steel sphere and what happens to the oil as the steel sphere falls?

	the steel sphere becomes warmer	the oil becomes warmer
A	yes	yes
B	yes	no
C	no	yes
D	no	no

- 9 A moving object is brought to rest by a frictional force of 30 N over a distance of 5.0 m.

The diagram shows the energy transferred between stores.



How much energy is transferred by this force and how is the energy transferred?

	energy transferred / J	how the energy is transferred
A	6.0	by electrical working
B	6.0	by mechanical working
C	150	by electrical working
D	150	by mechanical working

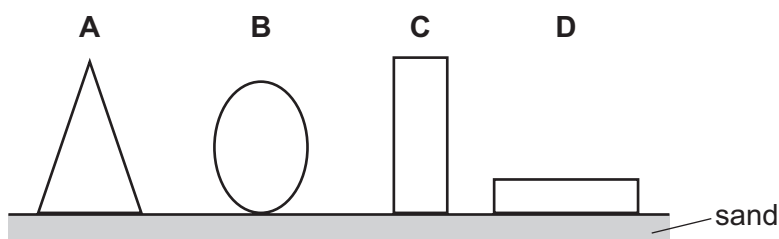
- 10 A motor transfers 24 J of energy in 60 seconds.

What is the power output of the motor?

- A** 0.40 W **B** 2.5 W **C** 24 W **D** 1400 W

- 11 An artist makes four sculptures with circular bases and places them on sand. All the sculptures are of equal weight and volume.

Which sculpture is least likely to sink into the sand?



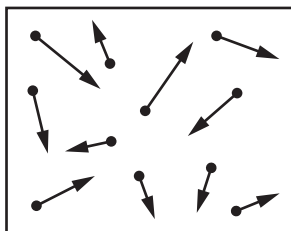
- 12 A quantity of water is boiled to form the same mass of steam.

Which row shows how the volume and density of the water change?

	volume	density
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 13 The diagram represents gas particles moving around in a sealed container.

The gas particles collide with the walls of the container.



The temperature of the gas is increased.

What happens to the average speed of the gas particles and what happens to the number of collisions by the gas particles with the walls of the container?

	average speed of gas particles	the number of collisions with the walls of the container
A	increases	less frequent collisions
B	increases	more frequent collisions
C	stays the same	less frequent collisions
D	stays the same	more frequent collisions

- 14 A metal object is heated strongly in an oven.

What happens to the volume of the object and the internal energy stored in the object?

	volume	internal energy
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 15 Which row describes the process of melting?

	initial state	final state	change in temperature
A	liquid	gas	yes
B	liquid	solid	no
C	solid	gas	yes
D	solid	liquid	no

- 16** A beaker of water is heated and thermal energy travels through the water by convection.

What happens to the density of the water when it is heated and how does the water move?

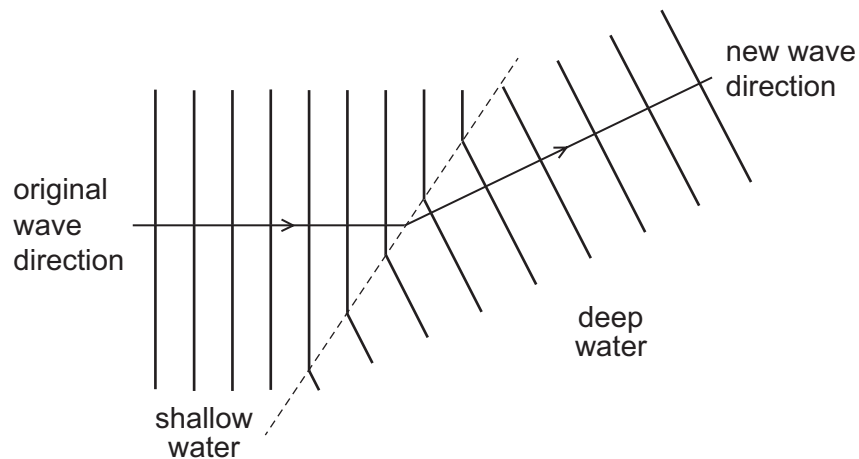
- A** The density decreases and the heated water moves downwards.
- B** The density decreases and the heated water moves upwards.
- C** The density increases and the heated water moves downwards.
- D** The density increases and the heated water moves upwards.

- 17** Waves travel across the surface of water.

What is meant by the amplitude of the wave?

- A** the maximum distance of a water particle from its mean position
- B** how far the wave travels every second
- C** the number of waves passing a point every second
- D** the distance between the top of consecutive waves

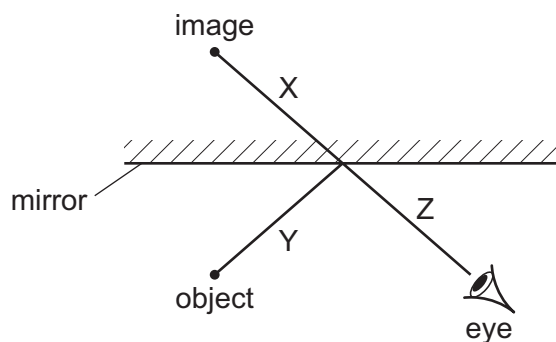
- 18** Water waves change direction when they move from shallow water to deep water.



What is the name of this effect?

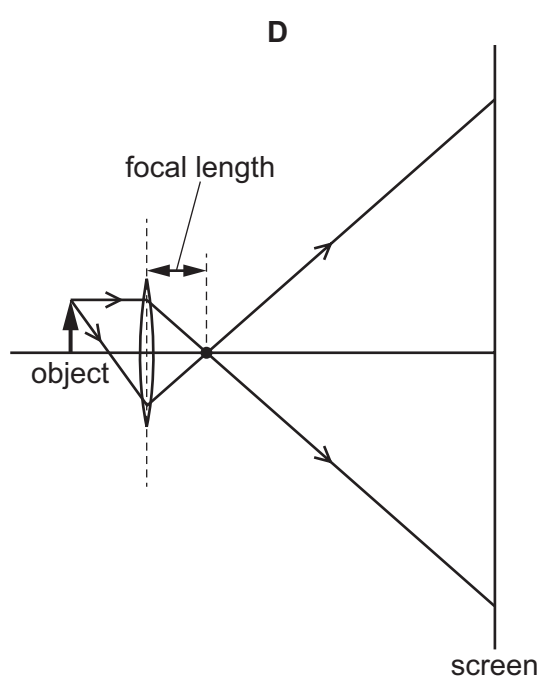
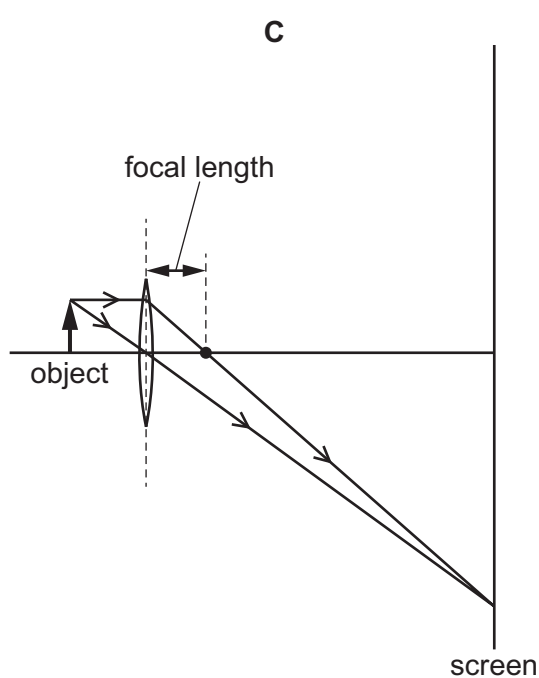
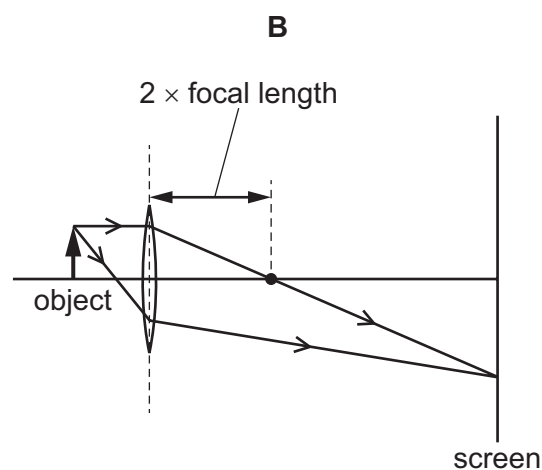
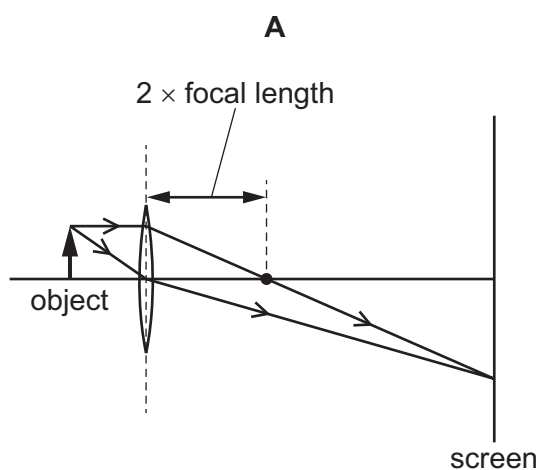
- A** diffraction
- B** dispersion
- C** reflection
- D** refraction

- 19 The ray diagram shows the formation of an image when light is reflected by a plane mirror.

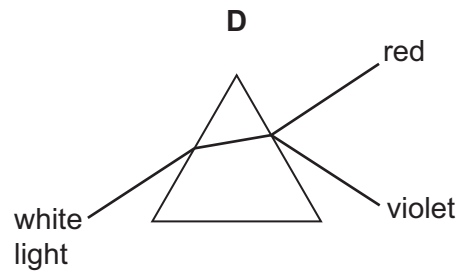
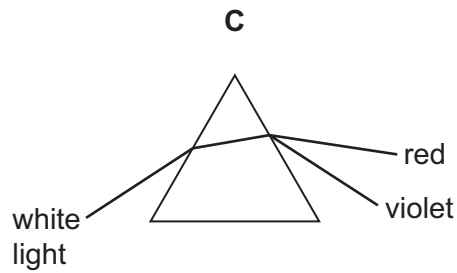
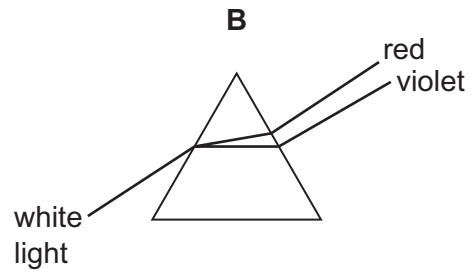
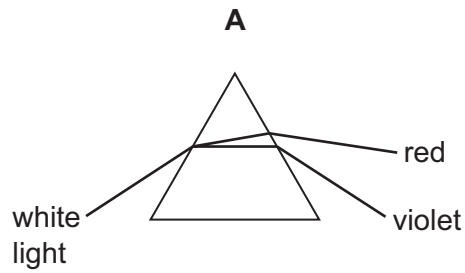


Which lines represent light rays?

- A** X, Y and Z **B** X and Y only **C** X and Z only **D** Y and Z only
- 20 Which diagram shows how an image of an object is formed on a screen by a converging lens?



21 Which diagram shows the dispersion of white light by a glass prism?



22 Which regions of the electromagnetic spectrum are used for satellite television and for security marking?

	satellite television	security marking
A	microwaves	infrared
B	microwaves	ultraviolet
C	X-rays	infrared
D	X-rays	ultraviolet

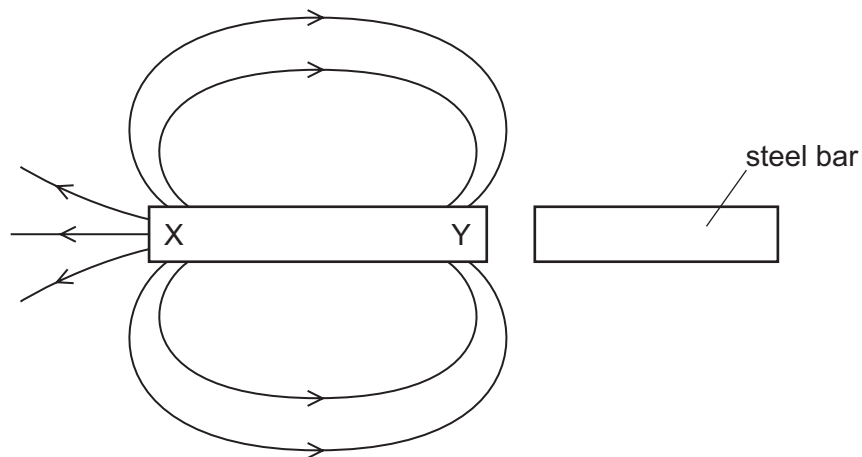
23 The horn on a ship makes a sound. The captain on the ship hears an echo from a cliff 4.0 s later.

The speed of sound is 340 m/s.

How far away is the cliff from the ship?

- A** 170 m **B** 340 m **C** 680 m **D** 1400 m

- 24** A bar magnet is next to a steel bar. Some of the magnetic field lines due to the magnet are shown.



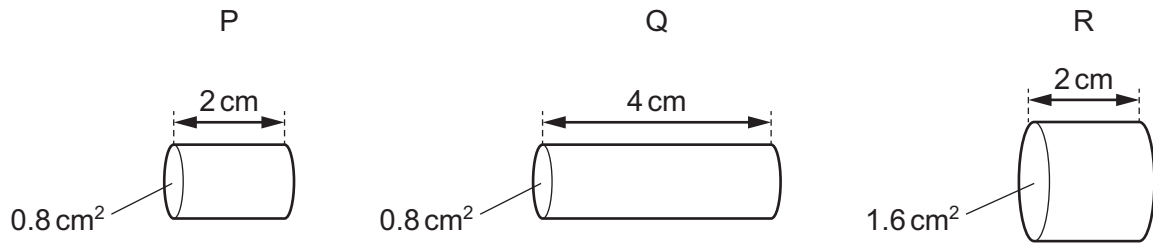
What are the correct poles of the magnet and would the steel bar attract or repel the bar magnet?

	X	Y	force
A	N pole	S pole	attraction
B	N pole	S pole	repulsion
C	S pole	N pole	attraction
D	S pole	N pole	repulsion

- 25** Which row about electrical conduction in metals is correct?

	statement 1	statement 2
A	It is the movement of free electrons.	Electric charge is a flow of current.
B	It is the movement of positive ions.	Electric charge is a flow of current.
C	It is the movement of free electrons.	Electric current is a flow of charge.
D	It is the movement of positive ions.	Electric current is a flow of charge.

- 26 The diagram shows three wires, P, Q and R. They are all made from the same metal.



Which list gives the wires in order of resistance, from lowest to highest?

- A P → Q → R
 - B Q → R → P
 - C R → P → Q
 - D R → Q → P
- 27 A 60 W lamp operates at normal brightness for 2.0 minutes.

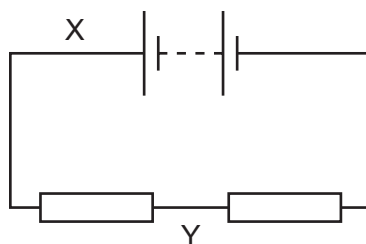
How much energy is transferred by the lamp?

- A 0.50 J
 - B 30 J
 - C 120 J
 - D 7200 J
- 28 What is a thermistor?

- A a container to keep a hot liquid warm
- B a circuit to control room temperature
- C a temperature-dependent resistor
- D a heater for a room

- 29** The diagram shows a circuit containing two identical resistors connected to a battery.

The current is measured at X and at Y.



Which row is correct?

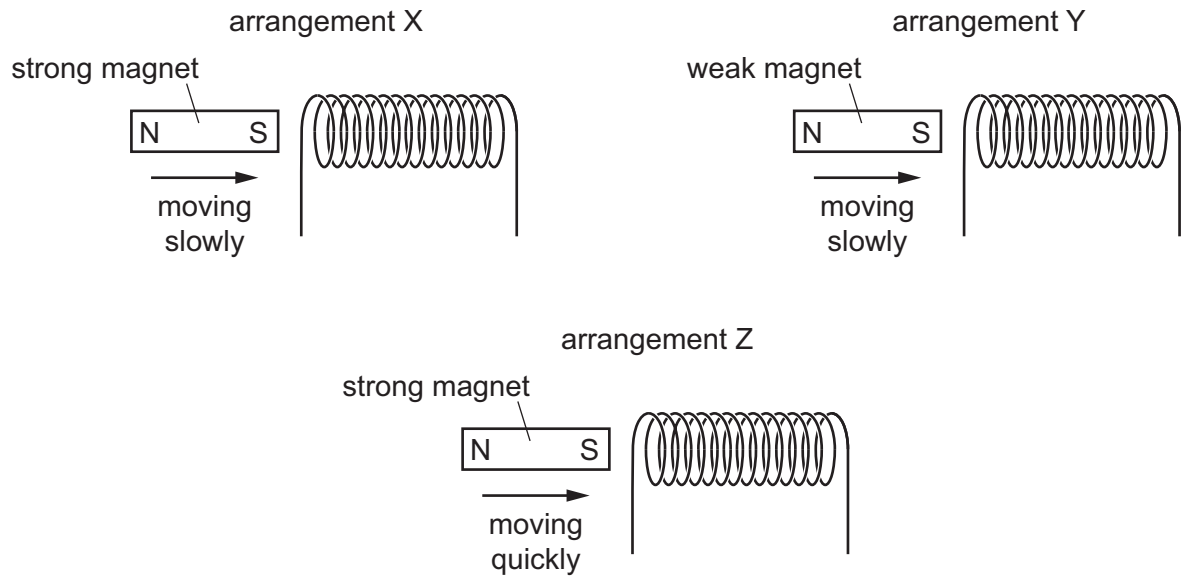
	arrangement of resistors	current at Y compared with current at X
A	parallel	less than X
B	parallel	equal to X
C	series	less than X
D	series	equal to X

- 30** The mains voltage is 120 V.

Which fuse should be fitted to an electric kettle with a power of 1.5 kW?

- A** 3 A **B** 5 A **C** 10 A **D** 13 A

- 31 The diagrams show a strong magnet and a weak magnet moving into the same coil of wire at different speeds.

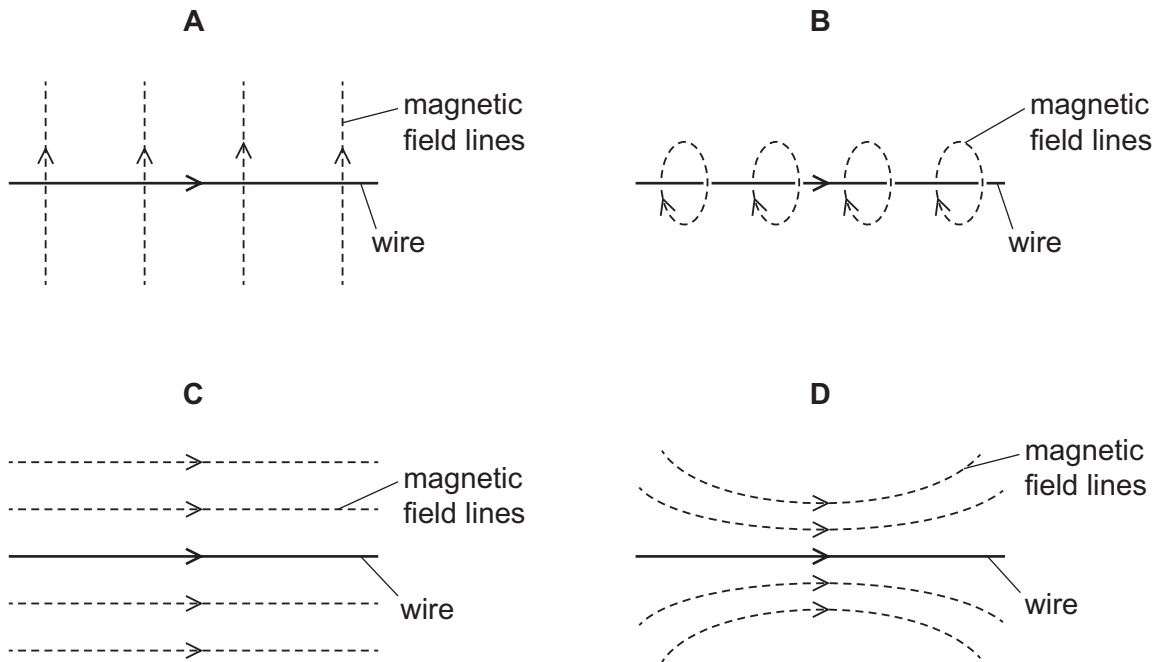


Which arrangement induces the largest electromotive force (e.m.f.) and which arrangement induces the smallest e.m.f.?

	largest e.m.f.	smallest e.m.f.
A	X	Y
B	Y	Z
C	Z	X
D	Z	Y

32 The diagrams show a current-carrying wire with an arrow in the direction of the current.

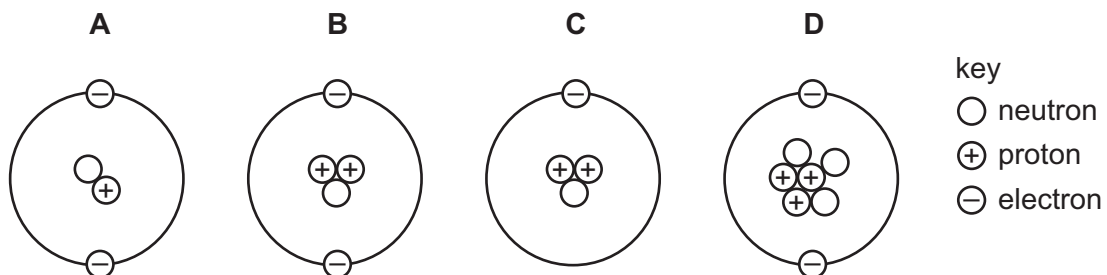
Which diagram shows the magnetic field produced by the current?



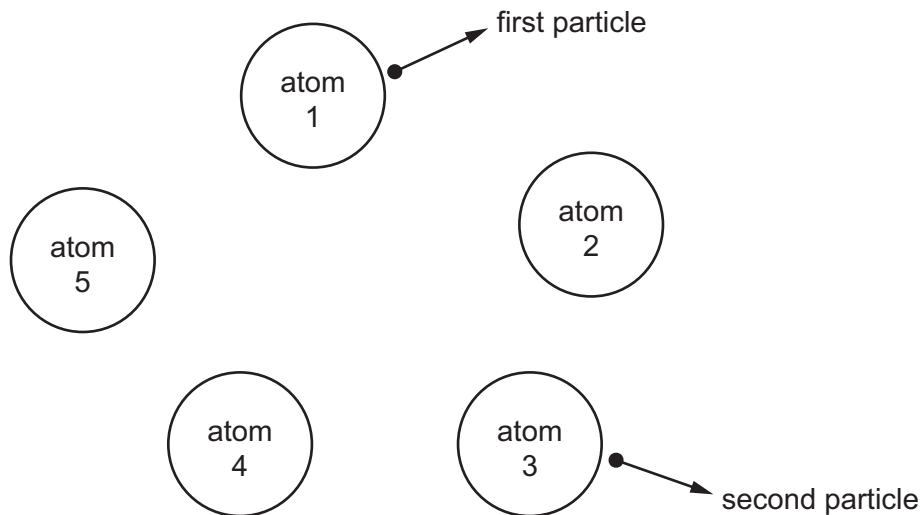
33 What are the best materials to use for the construction of a transformer?

- A** copper for the core and steel for the coils wound around it
- B** copper for the core and copper for the coils wound around it
- C** soft iron for the core and copper for the coils wound around it
- D** steel for the core and copper for the coils wound around it

34 Which diagram represents the structure of a neutral atom?



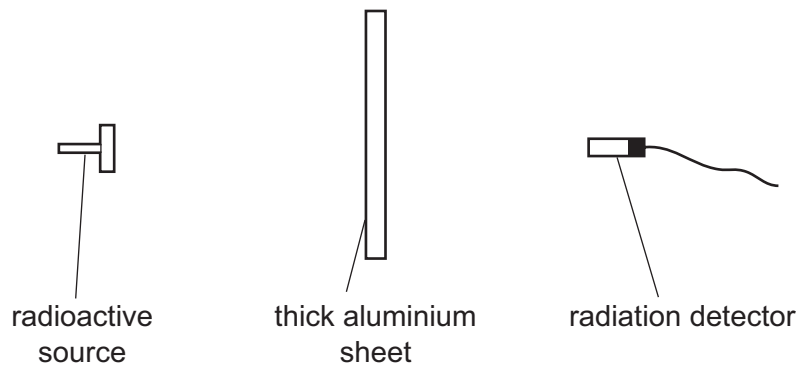
- 35 The diagram shows five atoms in a radioactive substance. The atoms each give out an α -particle.



Atom 1 is the first to give out a particle. Atom 3 is the second to give out a particle.

Which atom will give out the next particle?

- A atom 2
 - B atom 4
 - C atom 5
 - D impossible to tell
- 36 The diagram shows a radioactive source, a thick aluminium sheet and a radiation detector.



The radiation detector shows a reading greater than the background reading.

Which type of radiation is being emitted by the source and detected by the detector?

- A α -radiation
- B β -radiation
- C γ -radiation
- D infrared radiation

- 37 The half-life of the radioactive isotope caesium $^{137}_{55}\text{Cs}$ is 30 years.

Starting with 30 g of the isotope, which mass of the isotope remains after 90 years?

- A 10.0 g B 7.50 g C 3.75 g D 1.25 g

- 38 Which list gives the names of the planets in the correct order?

- A Mercury, Mars, Earth, Venus, Jupiter, Saturn, Uranus, Neptune
 B Mercury, Venus, Earth, Mars, Saturn, Uranus, Jupiter, Neptune
 C Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune
 D Venus, Mars, Earth, Mercury, Jupiter, Saturn, Uranus, Neptune

- 39 Which row describes the Sun and the Milky Way?

	the Sun	the Milky Way
A	a galaxy	a galaxy
B	a galaxy	a star
C	a star	a galaxy
D	a star	a star

- 40 What is the approximate diameter of the Milky Way?

- A 100 light-years
 B 1000 light-years
 C 10 000 light-years
 D 100 000 light-years

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Cambridge IGCSE™ (9–1)

PHYSICS**0972/11**

Paper 1 Multiple Choice (Core)

October/November 2024

MARK SCHEME

Maximum Mark: 40

Published

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3	A	1
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5	B	1
6	B	1
7	A	1
8	A	1
9	D	1
10	A	1
11	D	1
12	C	1
13	B	1
14	D	1
15	D	1
16	B	1
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19	D	1
20	C	1
21	A	1
22	B	1
23	C	1
24	A	1
25	C	1
26	C	1
27	D	1
28	C	1

Question	Answer	Marks
29	D	1
30	D	1
31	D	1
32	B	1
33	C	1
34	B	1
35	D	1
36	C	1
37	C	1
38	C	1
39	C	1
40	D	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
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- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

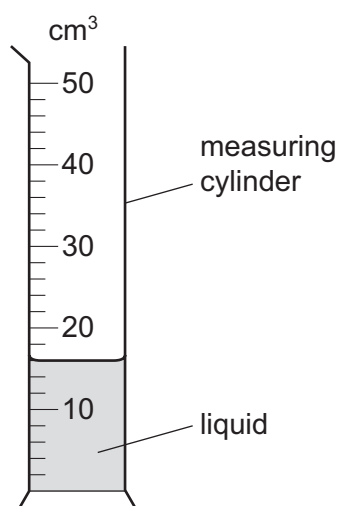
INFORMATION

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This document has **16** pages. Any blank pages are indicated.



- 1 The diagram shows a measuring cylinder containing liquid.



What is the volume of liquid in the measuring cylinder?

- A** 13.0 cm³ **B** 13.5 cm³ **C** 16.0 cm³ **D** 17.0 cm³
- 2 Which statement about the speed and velocity of an object is correct?
- A** The speed and velocity of an object are always greater than zero.
- B** The speed of an object is always greater than its velocity.
- C** The velocity of an object is equal to its speed in a given direction.
- D** Speed can be measured in m/s but velocity can only be measured in km/h.
- 3 The table shows a student's times in three races.

race	distance / m	time / s
1	100	12
2	150	17
3	200	25

What is the order of the student's average speeds in each race, from lowest to highest?

- A** 1 → 2 → 3 **B** 1 → 3 → 2 **C** 2 → 1 → 3 **D** 3 → 1 → 2

- 4 Which row gives the correct weight for the mass shown when it is at the Earth's surface?

	mass / kg	weight / N
A	2.00	19.6
B	10.0	1.02
C	10.0	9.80
D	20.0	2.04

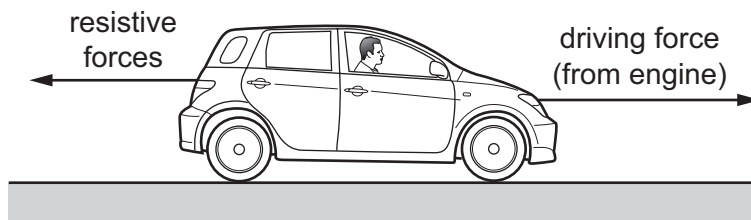
- 5 The masses and volumes of four objects are shown.

Which object floats in a liquid of density 1.4 g/cm^3 ?

	mass of object / g	volume of object / cm^3
A	1.0	0.5
B	2.0	3.0
C	6.0	4.0
D	9.0	6.0

- 6 A car is driven from rest on a long straight road. The car engine exerts a constant driving force.

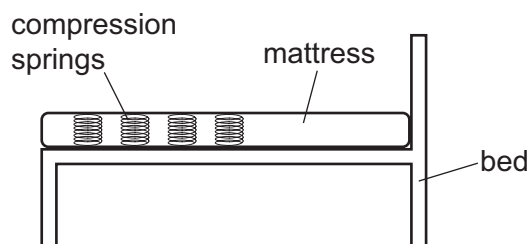
The diagram shows the horizontal forces acting on the car. The resistive forces are proportional to the speed of the car.



Why does the car eventually reach a maximum speed?

- A** The resistive forces decrease to make the acceleration of the car negative.
- B** The resistive forces decrease to make the acceleration of the car zero.
- C** The resistive forces increase to make the acceleration of the car negative.
- D** The resistive forces increase to make the acceleration of the car zero.

- 7 A mattress placed on a bed contains many compression springs.



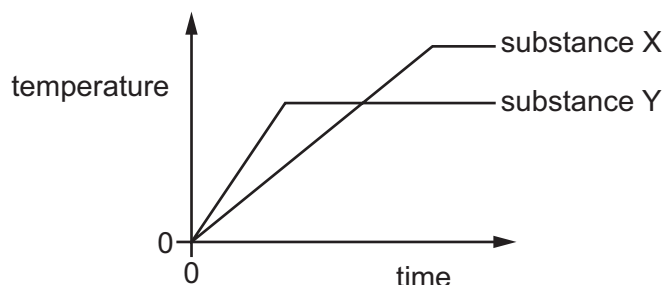
Which energy store of the mattress **increases** when a person lies on the bed?

- A gravitational potential
 B chemical
 C elastic
 D kinetic
- 8 Which energy resource does **not** contain a store of chemical energy?
- A biofuel
 B coal
 C nuclear fuel
 D oil
- 9 A person carries 20 tiles from the ground to the roof of a house. Each tile has a mass of 1.2 kg. The roof of the house is 15 m above the ground.
- How much work does the person do against gravity on the tiles in carrying them to the roof?
- A 37 J B 180 J C 360 J D 3500 J
- 10 Which pair of changes, applied at the same time, always increases the pressure that a brick exerts on the ground?

	weight of brick	area of brick in contact with ground
A	increase	increase
B	increase	decrease
C	decrease	increase
D	decrease	decrease

- 11 Two different pure substances, X and Y, are heated. Both substance X and substance Y are initially in the solid state.

The graph shows how the temperature of each substance changes with time.



What does the graph tell us about the substances?

- A Substance X has a lower boiling point than substance Y.
- B Substance X has a lower melting point than substance Y.
- C Substance Y has a lower boiling point than substance X.
- D Substance Y has a lower melting point than substance X.

- 12 A liquid substance changes into a gas.

Which property of the substance remains the same?

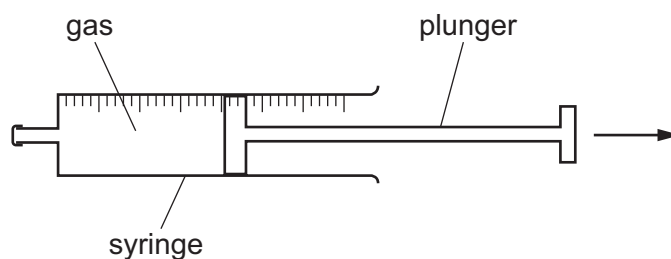
- A ability to flow
 - B compressibility
 - C shape
 - D volume
- 13 At room temperature, iron is difficult to compress.

At the same temperature, oxygen is much easier to compress.

Which comparison of the particles in iron and the particles in oxygen explains this?

- A The iron particles are closer together.
- B The iron particles have a greater mass.
- C The iron particles can be magnetised.
- D The iron particles have less average kinetic energy.

- 14 A teacher has a gas in a sealed syringe with a movable plunger.



The teacher slowly moves the plunger out of the syringe. The temperature of the gas remains constant.

Which row explains what happens to the pressure of the gas in the syringe?

	pressure of gas	explanation
A	decreases	gas particles collide with the syringe wall less frequently
B	decreases	gas particles collide with the syringe wall more frequently
C	increases	gas particles collide with the syringe wall less frequently
D	increases	gas particles collide with the syringe wall more frequently

- 15 Evaporation occurs from the surface of a pool of water.

Which statement describes this change of state?

- A** Electrons move from the liquid and become a gas.
- B** Molecules that move from the liquid have the same energy as the molecules that stay in the liquid.
- C** The more-energetic molecules escape the liquid.
- D** The more-energetic molecules remain in the liquid.

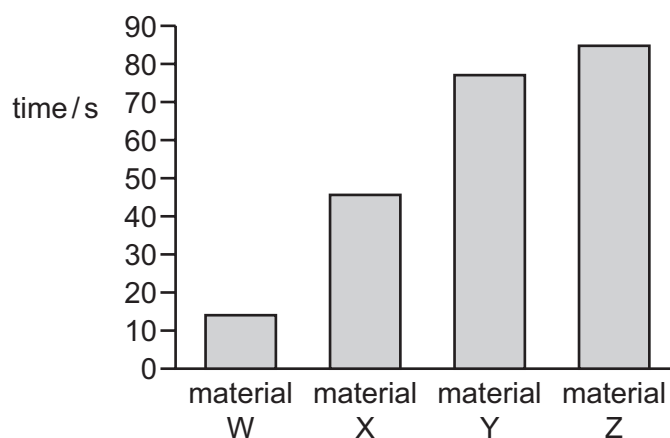
- 16** Students test four rods of different materials, W, X, Y and Z, to compare how well the materials conduct thermal energy.

The rods are all the same size and shape.

The students use wax to stick a pin to one end of the first rod. The wax melts when it is warm. The other end of the rod is heated and the time taken for the pin to fall is recorded.

The experiment is repeated for each rod.

The students plot a chart of the time taken for each pin to fall.



Which statement is correct?

- A** X is a better thermal conductor than Y.
 - B** Y is a better thermal conductor than W.
 - C** Z is a better thermal conductor than W.
 - D** Z is a better thermal conductor than X.
- 17** A student has four metal containers filled with the same volume of water at a temperature of 80 °C.

The metal containers are put in a room at a temperature of 20 °C.

The metal containers are identical except for the texture and colour of their surfaces.

Which surface will emit energy at the greatest rate?

- A** dull black
- B** shiny black
- C** dull white
- D** shiny white

- 18 A water wave has a speed of 2.0 m/s.

4.0 complete waves pass a point every 10 seconds.

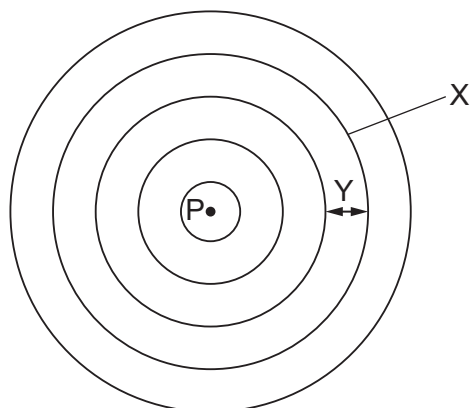
What is the wavelength of the wave?

- A** 0.50 m **B** 0.80 m **C** 5.0 m **D** 8.0 m

- 19 Which row correctly describes the vibrations of a transverse wave and also gives a correct example of a transverse wave?

	description of vibrations	example of a transverse wave
A	at right angles to the wave direction	sound
B	at right angles to the wave direction	water wave
C	parallel to the wave direction	sound
D	parallel to the wave direction	water wave

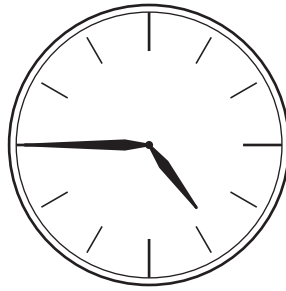
- 20 The diagram shows a view from above of waves on the surface of a pond after the water has been disturbed at point P.



What are the correct labels for X and Y?

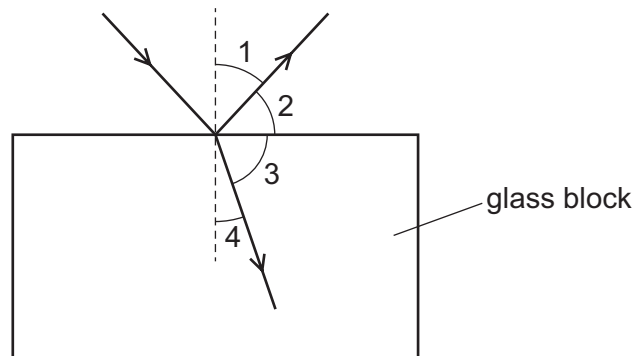
	X	Y
A	amplitude	wavelength
B	wavefront	wavelength
C	amplitude	wavefront
D	wavefront	amplitude

- 21 The diagram shows the image of a clock as seen by a person standing in front of a plane mirror.



What is the actual time?

- A 4:15 B 4:45 C 7:15 D 7:45
- 22 The diagram shows a ray of light incident on the surface of a glass block.



The ray of light is partially reflected back into the air and partially refracted into the glass block.

Which row correctly identifies the angle of reflection and the angle of refraction?

	angle of reflection	angle of refraction
A	1	3
B	1	4
C	2	3
D	2	4

- 23** The amplitude and the frequency of a sound wave both increase.

What happens to the loudness and what happens to the pitch of the sound?

	loudness	pitch
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 24** Which statements about sound and ultrasound are correct?

	sound	ultrasound
A	humans can hear frequencies between 20 Hz and 20 000 Hz	the speed in air is 330 m/s
B	humans can hear frequencies less than 20 000 Hz	the speed in air is 100 m/s
C	the speed in air is 100 m/s	has a frequency greater than 20 000 Hz
D	the speed in air is 330 m/s	has a frequency less than 20 Hz

- 25** Which row describes the magnetic properties of soft iron and steel?

	soft iron	steel
A	easy to demagnetise	easy to demagnetise
B	easy to demagnetise	hard to demagnetise
C	hard to demagnetise	easy to demagnetise
D	hard to demagnetise	hard to demagnetise

- 26 Two charged rods, X and Y, are brought, in turn, near to a negatively charged rod.

X is attracted by the negatively charged rod, and Y is repelled.

What are the charges on X and Y?

	charge on X	charge on Y
A	negative	negative
B	negative	positive
C	positive	negative
D	positive	positive

- 27 The table gives data for four different electrical devices.

Which device develops the greatest power?

	device	voltage	current
A	car headlight	12 V	3.0 A
B	cooling fan	110 V	0.40 A
C	electric spark generator	400 kV	0.10 mA
D	mains lamp	240 V	0.20 A

- 28 A mains-operated radio is a double-insulated appliance.

Which statement is correct?

- A** The radio does **not** need a neutral connection.
- B** The radio **must** be earthed.
- C** The radio circuit is safe **without** a fuse or trip switch.
- D** The outer casing of the radio **cannot** become live.

- 29 Which statement describes a magnetic field?

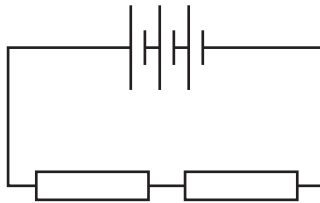
- A** It is a region where all objects experience a force.
- B** It is a region where all electric charges experience a force.
- C** It is a region where all magnetic poles experience a force.
- D** It is a region where all metals experience a force.

- 30 A plastic rod is charged by friction. The plastic rod becomes positively charged.

Which particles are transferred in this process?

- A atoms
- B electrons
- C protons
- D neutrons

- 31 A teacher makes a circuit using three 1.5 V cells and two $20\ \Omega$ resistors connected in series.



Which row gives the combined electromotive force (e.m.f.) of the cells and the combined resistance of the resistors?

	combined e.m.f. / V	combined resistance / Ω
A	1.5	10
B	1.5	40
C	4.5	10
D	4.5	40

- 32 A student investigates the turning effect on a current-carrying coil placed in a magnetic field.

The student changes the current in the coil and the number of turns on the coil.

Which two changes together **must** increase the turning effect?

- A decreasing the current and decreasing the number of turns
- B decreasing the current and increasing the number of turns
- C increasing the current and decreasing the number of turns
- D increasing the current and increasing the number of turns

- 33** There is a current of 3.0 A in a conductor.

The current is changed to 6.0 A in the opposite direction.

Which effect does this have on the magnetic field around the conductor?

- A** It decreases and it acts in the opposite direction.
- B** It decreases and it acts in the same direction.
- C** It increases and it acts in the opposite direction.
- D** It increases and it acts in the same direction.

- 34** The table shows the contribution to background radiation from the four most common sources.

source	percentage of total background radiation / %
radon gas (in the air)	48
rocks and buildings	13
food and drink	10
cosmic rays	11

Which percentage of background radiation comes from all other sources?

- A** 14% **B** 18% **C** 21% **D** 82%

- 35** The unstable nucleus of an element undergoes radioactive decay. The proton number remains unchanged.

Which description of this radioactive decay event is correct?

- A** The nucleus emits an α -particle.
- B** The nucleus emits a β -particle.
- C** The nucleus emits γ -radiation.
- D** The nucleus emits an α -particle, a β -particle and γ -radiation together.

- 36 When they lose electrons,1..... form2..... charged3..... .

Which words correctly complete the sentence?

	1	2	3
A	atoms	negatively	ions
B	ions	negatively	atoms
C	atoms	positively	ions
D	ions	positively	atoms

- 37 An isotope of krypton has an atomic number of 36 and a mass number of 78.

What is the number of protons in a krypton nucleus?

- A** 36 **B** 42 **C** 78 **D** 114

- 38 What is the time taken for the Earth to spin once on its axis and the time taken for the Earth to orbit the Sun?

	spin on its axis	orbit the Sun
A	24 hours	1 month
B	24 hours	365 days
C	1 month	1 month
D	1 month	365 days

- 39 Electromagnetic radiation from distant galaxies is redshifted when observed from the Earth.

This is evidence that supports the Big Bang Theory.

Which statement is correct?

- A** Light from galaxies that are moving towards the Earth is redshifted.
B Redshift is an increase in the wavelength of electromagnetic radiation.
C The Big Bang Theory says that the Solar System is expanding out into the Universe.
D The Big Bang Theory says that, billions of years ago, the Universe expanded to its current size in a few seconds.

40 Which statement about the Milky Way is correct?

- A** It is a galaxy made up of billions of stars.
- B** It is a group of galaxies in our Solar System.
- C** It is all the stars that we can see in the night sky.
- D** It is the path taken by a comet when it gets close to the Sun.

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)**0972/11**

Paper 1 Multiple Choice (Core)

May/June 2025**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	C	1
3	D	1
4	A	1
5	B	1
6	D	1
7	C	1
8	C	1
9	D	1
10	B	1
11	D	1
12	A	1
13	A	1
14	A	1
15	C	1
16	A	1
17	A	1
18	C	1
19	B	1
20	B	1
21	C	1
22	B	1
23	D	1
24	A	1
25	B	1
26	C	1
27	D	1
28	D	1

Question	Answer	Marks
29	C	1
30	B	1
31	D	1
32	D	1
33	C	1
34	B	1
35	C	1
36	C	1
37	A	1
38	B	1
39	B	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/12

Paper 1 Multiple Choice (Core)

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 A pendulum starts swinging and its motion is timed. The time measured for 20 complete swings is 30 s.

What is the time for **one** complete swing of the pendulum?

- A** 0.67 s **B** 0.75 s **C** 1.5 s **D** 3.0 s

- 2 Which piece of apparatus is used to determine the volume of an irregularly shaped solid?

- A** balance
B measuring cylinder
C ruler
D thermometer

- 3 A speed–time graph is plotted for the motion of a car.

Which quantity does the area under the speed–time graph represent?

- A** acceleration of the car
B average speed of the car
C distance travelled by the car
D final velocity of the car

- 4 Which quantity is an example of a force?

- A** density
B mass
C volume
D weight

- 5 Four hollow glass balls, P, Q, R and S, each have a volume of 30 cm^3 .

The mass of each ball is shown.

ball	mass / g
P	15
Q	25
R	35
S	45

Which balls will float in a liquid of density 0.95 g/cm^3 ?

- A** P and Q **B** P only **C** R and S **D** S only

- 6 A car travels at constant speed along a straight road that has a slight downwards slope.

Which statement is correct?

- A There is a resultant force on the car down the slope.
- B There is a resultant force on the car up the slope.
- C There is a resultant force on the car vertically downwards.
- D There is no resultant force on the car.

- 7 A torque wrench is a tool that ensures nuts or bolts are tightened to the required amount.

A mechanic sets the torque wrench to produce a moment of 50 N m.

The torque wrench has a length from the centre of the bolt to the handle of 600 mm.

What is the smallest force the mechanic needs to apply to tighten the bolt correctly?

- A 0.080 N B 30 N C 83 N D 30 kN

- 8 The work done W by a force is related to the magnitude F of the force and the distance d moved in the direction of the force.

Which equation for W is correct?

- A $W = d \div F$ B $W = d + F$ C $W = F \div d$ D $W = F \times d$

- 9 Which energy resource requires a power station with a boiler to produce steam when generating electricity?

- A coal
- B tides
- C water behind a dam
- D wind

- 10 A scientist heats a solid substance until it turns into a liquid, and then into a gas.

Which row describes the two changes of state of the substance as it is heated?

	change of state 1	change of state 2
A	freezing	condensing
B	freezing	boiling
C	melting	condensing
D	melting	boiling

- 11** Gases can be compressed, but liquids cannot be compressed.

Which statement explains this difference?

- A** Each particle in a gas is more compressible than each particle in a liquid.
- B** Particles in a gas are further apart than particles in a liquid.
- C** Particles in a gas attract each other more strongly than particles in a liquid.
- D** Particles in a gas move more slowly than particles in a liquid.

- 12** A fixed mass of gas at a constant temperature is trapped in a cylinder which has a movable piston.

Which row states and explains what happens to the pressure of the gas in the cylinder when the volume of the gas increases?

	pressure	explanation
A	decreases	gas particles collide with the cylinder at a lower speed
B	decreases	gas particles collide with the cylinder less frequently
C	increases	gas particles collide with the cylinder at a higher speed
D	increases	gas particles collide with the cylinder more frequently

- 13** Which quantity does **not** change when there is an increase in temperature?

- A** the density of a steel block
- B** the diameter of the hole in a metal nut
- C** the length of an iron rod
- D** the mass of a metal coin

- 14** A person holds an empty glass beaker and pours hot water into it.

Why does it take a few seconds before his hand starts to feel hot?

- A** Glass is a poor thermal conductor.
- B** Water is a poor thermal conductor.
- C** Glass is a better thermal conductor than water.
- D** Water is a better thermal conductor than glass.

- 15 A liquid is heated and it expands.

How does this lead to the formation of a convection current?

- A The density of the heated liquid decreases.
- B The density of the heated liquid increases.
- C The mass of the heated liquid particles decreases.
- D The mass of the heated liquid particles increases.

- 16 Some electric hotplates are metal and do **not** glow when hot.

Other hotplates are glass and have a heater underneath that glows red. The heater makes the glass hot.

With both hotplates, there is no air gap between the hotplate and a metal pan placed on the hotplate.

Which statement is correct?

- A The glass hotplate transfers less energy by radiation to the base of the pan than the metal hotplate.
- B The glass hotplate transfers energy by convection to the base of the pan.
- C The metal hotplate transfers energy by conduction to the base of the pan.
- D The metal hotplate transfers energy by convection to the base of the pan.

- 17 Twelve crests of a water wave pass a boat anchored at sea in a time of 180 s.

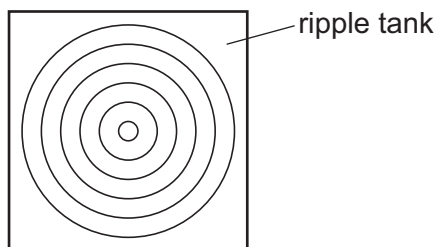
The distance between consecutive crests of the wave is 10 m.

What is the speed of the wave?

- A 0.056 m/s B 0.67 m/s C 15 m/s D 18 m/s

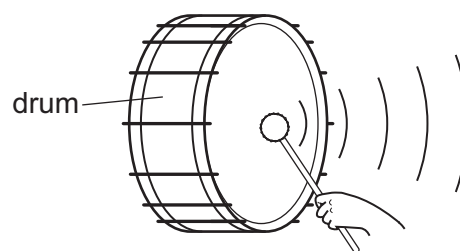
18 The diagrams show examples of wave motion.

waves on water



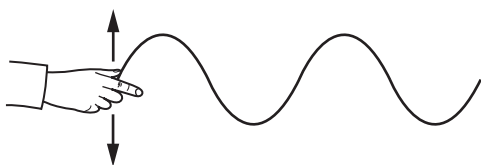
1

waves in air



2

waves on a rope



3

waves in a spring



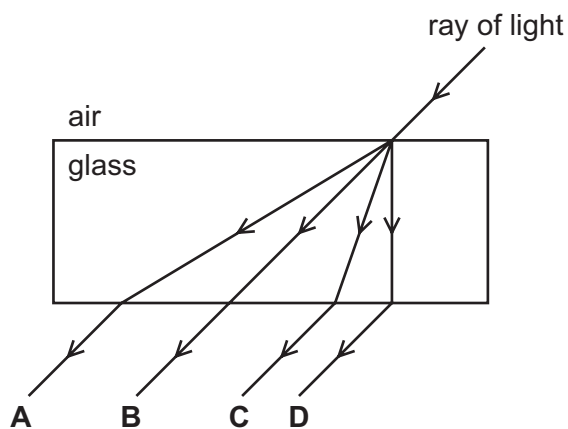
4

Which waves are longitudinal?

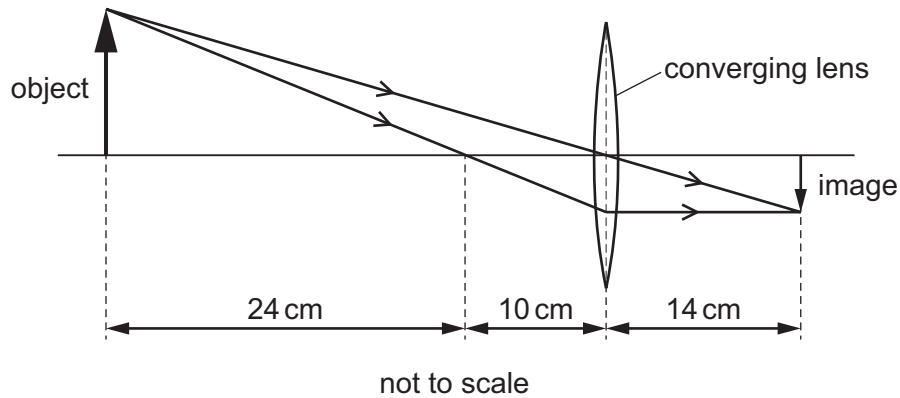
- A** 1 only **B** 2, 3 and 4 **C** 2 and 3 only **D** 2 and 4 only

19 A ray of light passes through a glass block.

What is the path of the light?



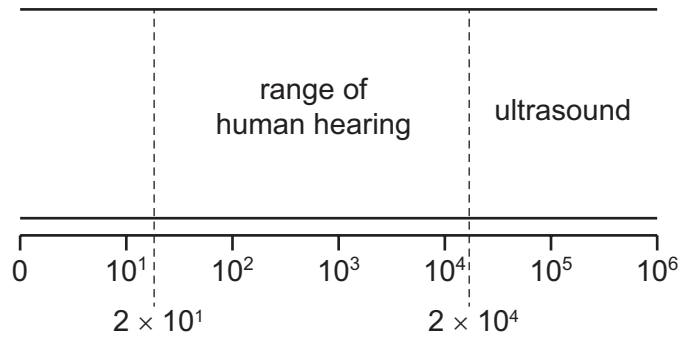
- 20 The diagram shows a converging lens forming an image.



What is the focal length of the converging lens?

- A** 10 cm **B** 14 cm **C** 24 cm **D** 34 cm
- 21 Which type of radiation does a remote controller use to send its instructions to a television (TV)?
- A** gamma rays
B infrared radiation
C ultraviolet radiation
D X-rays
- 22 A person exposed to excessive radiation from certain parts of the electromagnetic spectrum is likely to suffer cell damage.
- Which radiations cause cell damage?
- A** gamma rays and ultraviolet
B gamma rays only
C green light and red light
D radio waves and green light

23 The diagram shows the ranges of human hearing and of ultrasound waves.



Which characteristic of sound waves do the numbers on the diagram refer to?

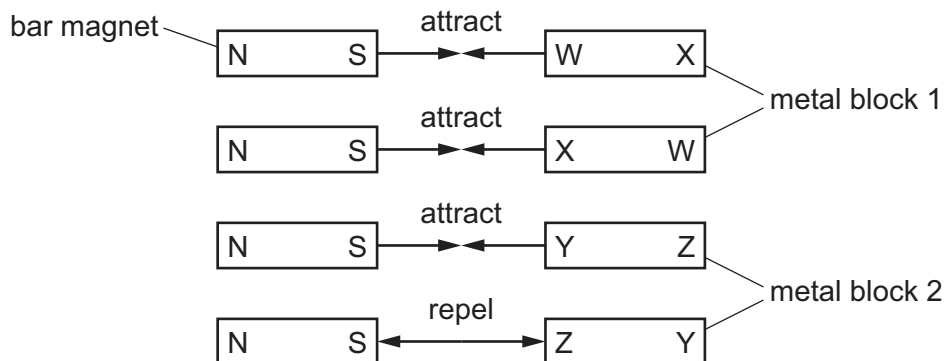
- A amplitude in cm
- B frequency in Hz
- C speed in metres/second
- D wavelength in metres

24 A student has a bar magnet and two metal blocks.

The student places the bar magnet next to end W and end X of metal block 1.

The student then places the bar magnet next to end Y and end Z of metal block 2.

The diagram shows which ends of the metal blocks are attracted or repelled by the bar magnet.



Which metal blocks are bar magnets?

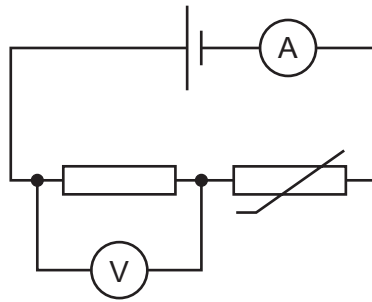
- A both metal block 1 and metal block 2
- B metal block 1 only
- C metal block 2 only
- D neither metal block 1 nor metal block 2

- 25** A glass rod is rubbed with a silk cloth. The glass rod becomes positively charged.

Which statement explains how this happens?

- A** Negative charges move from the glass rod to the silk cloth.
 - B** Negative charges move from the silk cloth to the glass rod.
 - C** Positive charges move from the glass rod to the silk cloth.
 - D** Positive charges move from the silk cloth to the glass rod.
- 26** Which statement describes what is happening when there is an electric current in a metal wire?
- A** Electrons in the wire vibrate.
 - B** Electrons move along the wire.
 - C** Positive charges in the wire vibrate.
 - D** Positive charges move along the wire.
- 27** Which **two** electrical quantities have the same unit?
- A** current and charge
 - B** electromotive force (e.m.f.) and potential difference (p.d.)
 - C** potential difference (p.d.) and current
 - D** resistance and electromotive force (e.m.f.)
- 28** A 10 V d.c. power supply is connected across a resistor.
- The current in the resistor is 0.050 A.
- What is the resistance of the resistor?
- A** $0.0050\ \Omega$ **B** $0.50\ \Omega$ **C** $2.0\ \Omega$ **D** $200\ \Omega$

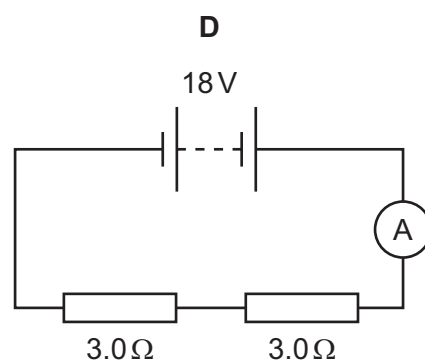
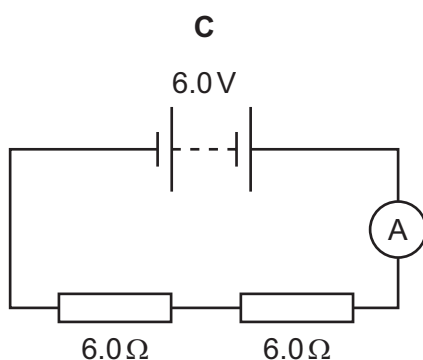
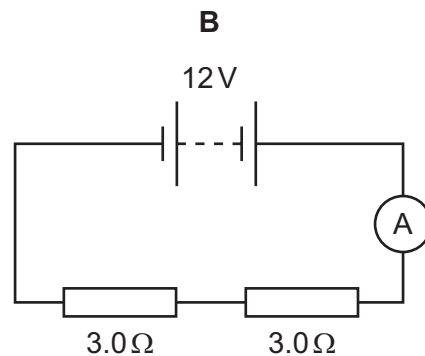
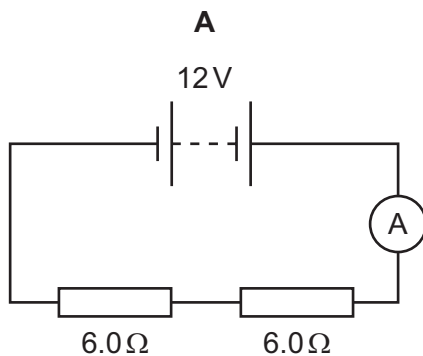
- 29 A resistor and a thermistor are connected in a series circuit, as shown.



What happens to the readings on the ammeter and the voltmeter when the thermistor is heated?

	ammeter reading	voltmeter reading
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 30 In which circuit does the ammeter read 2.0 A?



- 31 A 240 V mains-powered vacuum cleaner has a power output of 2700 W.

Which fuse is suitable for use in the plug of the vacuum cleaner?

- A** 3 A **B** 5 A **C** 10 A **D** 13 A

- 32 The transformer in a laptop power supply is used to change mains voltage from 240 V to 20 V.

The transformer has 600 turns on the secondary coil.

Which type of transformer is it and how many turns does the transformer have on the primary coil?

	transformer type	number of turns on primary coil
A	step-down	50
B	step-down	7200
C	step-up	50
D	step-up	7200

- 33 An ion of the element oxygen is negatively charged.

Which change causes the ion to become a neutral oxygen atom?

- A** gain of electrons
- B** gain of protons
- C** loss of electrons
- D** loss of protons

- 34 A nuclide of nitrogen is represented as $^{15}_7\text{N}$.

What is the number of neutrons in this nuclide?

- A** 7 **B** 8 **C** 15 **D** 22

- 35 Which list **only** includes sources of naturally occurring background radiation?

- A** cosmic rays, radon gas, nuclear power stations
- B** food and drink, mobile phones, cosmic rays
- C** rocks and buildings, food and drink, radon gas
- D** rocks and buildings, mobile phones, nuclear power stations

- 36** A student compares alpha emissions, beta emissions and gamma emissions.

Which statement is correct?

- A** Alpha emissions are the most ionising.
- B** Beta emissions are the most penetrating.
- C** Beta emissions are part of the electromagnetic spectrum.
- D** Gamma emissions are streams of charged particles.

- 37** In school laboratories, radioactive sources are stored in lead-lined boxes.

Why is lead used?

- A** Lead neutralises radiation.
- B** Lead absorbs radiation.
- C** Lead repels radiation.
- D** Lead dissolves radiation.

- 38** The Sun, the Earth and the Moon are all bodies in the Solar System.

The Earth has a periodic cycle of day and night and a periodic cycle of seasons.

Which row relates the periods of these cycles to the motions of these bodies?

	period of day and night	period of the seasons
A	the Earth orbits the Sun	the Earth rotates about its axis
B	the Earth orbits the Sun	the Moon orbits the Earth
C	the Earth rotates about its axis	the Earth orbits the Sun
D	the Earth rotates about its axis	the Moon orbits the Earth

- 39** The strength of the gravitational field at the surface of a planet depends on the1..... .

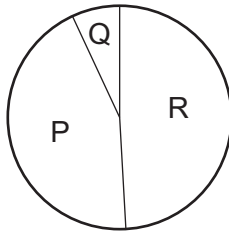
The strength of the gravitational field2..... as the distance from the planet increases.

Which words correctly complete the sentences?

	1	2
A	atmospheric pressure	decreases
B	mass of the planet	decreases
C	atmospheric pressure	increases
D	mass of the planet	increases

- 40** The Sun radiates most of its energy in three regions of the electromagnetic spectrum.

The diagram shows the proportion of the total energy radiated in each of these regions.



Region P is visible light and region Q is ultraviolet.

What is region R?

- A** infrared
- B** microwaves
- C** radio waves
- D** X-rays

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)**0972/12**

Paper 1 Multiple Choice (Core)

May/June 2025

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	B	1
3	C	1
4	D	1
5	A	1
6	D	1
7	C	1
8	D	1
9	A	1
10	D	1
11	B	1
12	B	1
13	D	1
14	A	1
15	A	1
16	C	1
17	B	1
18	D	1
19	C	1
20	A	1
21	B	1
22	A	1
23	B	1
24	C	1
25	A	1
26	B	1
27	B	1
28	D	1

Question	Answer	Marks
29	D	1
30	B	1
31	D	1
32	B	1
33	C	1
34	B	1
35	C	1
36	A	1
37	B	1
38	C	1
39	B	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/11

Paper 1 Multiple Choice (Core)

October/November 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

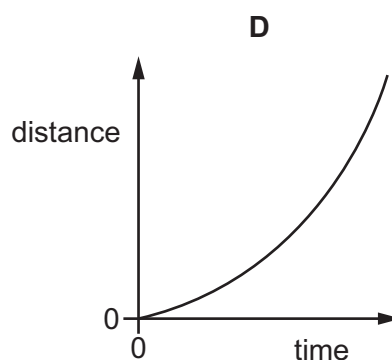
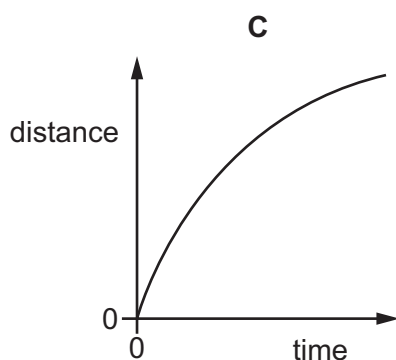
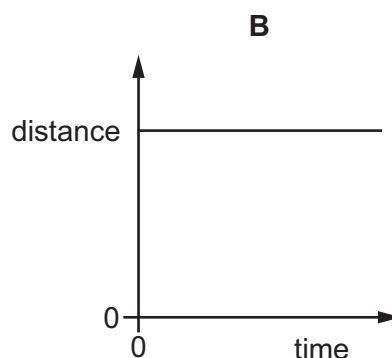
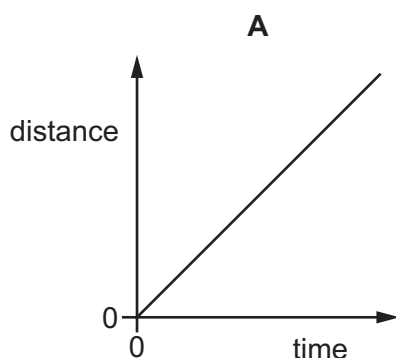


1 Which measuring instrument is used to determine the volume of water in a cup?

- A clock
- B measuring cylinder
- C ruler
- D thermometer

2 The diagrams show distance–time graphs for four objects.

Which object is moving with increasing speed?



3 A space station orbits above the Earth. There is an astronaut in the space station.

In the space station, the acceleration of free fall is 7.5 m/s^2 .

On Earth, the acceleration of free fall is 9.8 m/s^2 .

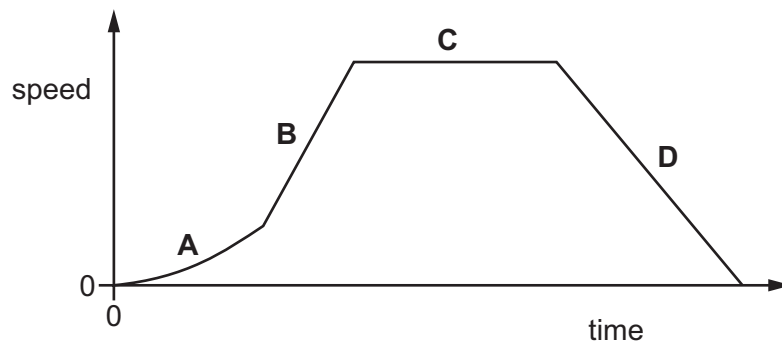
Which row about the astronaut's mass and weight in the space station is correct?

	mass of astronaut	weight of astronaut
A	same as on the Earth	less than on the Earth
B	same as on the Earth	greater than on the Earth
C	lower than on the Earth	less than on the Earth
D	lower than on the Earth	greater than on the Earth

- 4 A student is asked to predict whether a solid floats in a liquid.

Which information does the student require?

- A the density of the liquid and the mass of the solid
 - B the density of the solid and the density of the liquid
 - C the density of the solid and the mass of the liquid
 - D the mass of the solid and the mass of the liquid
- 5 Which property of an object **never** changes when a force acts on the object?
- A mass
 - B shape
 - C size
 - D speed
- 6 A car is travelling along a straight horizontal road. The speed–time graph is shown.
- In which labelled part of the journey is the resultant force on the car zero?



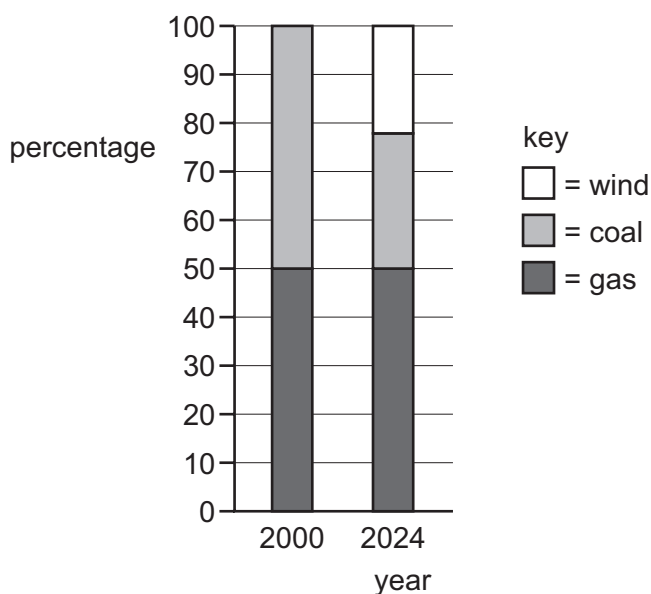
- 7 Which list contains only energy stores?
- A electrostatic, kinetic, light
 - B electrostatic, kinetic, nuclear
 - C electrostatic, light, nuclear
 - D kinetic, light, nuclear

- 8 What needs to be known to calculate the work done by a force acting on an object?

	the size of the force	the distance the force moves the object	the time for which the force acts	
A	✓	✓	✓	key ✓ = needed x = not needed
B	✓	✓	x	
C	✓	x	✓	
D	✓	x	x	

- 9 Different energy resources are used to produce electricity on a small island.

The chart shows the percentages of each energy resource used in 2000 and in 2024.



Which row correctly describes any changes in the percentage for 2024 compared with 2000?

	percentage of energy resource used	
	coal	renewable
A	decrease	no change
B	no change	increase
C	decrease	increase
D	no change	no change

10 Which situation is **not** explained by the equation $P = \frac{F}{A}$?

- A using a longer spanner than normal to undo a tight nut
- B hammering a nail into a piece of wood
- C tractors using wide tyres in a muddy field
- D a sharp kitchen knife cutting vegetables more easily than a blunt one

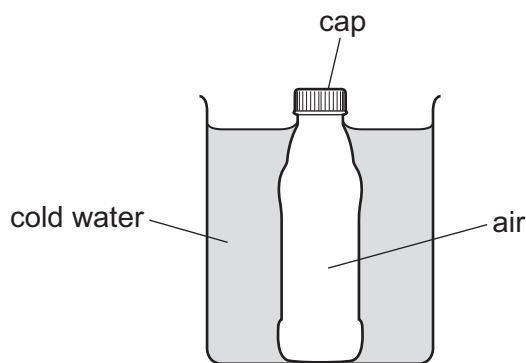
11 A substance is heated. The substance changes state from a liquid to a gas.

When the gas is cooled, it first becomes a liquid and finally a solid.

What is the order of changes of state for this substance?

- A condensation → melting → freezing
- B vaporisation → condensation → melting
- C vaporisation → condensation → freezing
- D melting → vaporisation → freezing

12 A glass bottle containing warm air is sealed with a screw cap and then cooled in cold water.



The contraction of the glass bottle can be ignored.

What remains the same during the cooling?

- A the air pressure inside the bottle
- B the energy of the air particles in the bottle
- C the force on the cap made by the air particles in the bottle
- D the volume of air in the bottle

- 13** A gas is stored in a container of constant volume.

The temperature of the gas is increased.

What happens to the particles of the gas?

- A** The distance between the particles increases.
- B** The mass of the particles increases.
- C** The number of particles increases.
- D** The speed of the particles increases.

- 14** One end of a shiny metal rod is heated, and the other end quickly gets hot.

Which statement describes why the other end quickly gets hot?

- A** Metals are good thermal conductors.
- B** Metals are poor thermal conductors.
- C** Shiny surfaces are good emitters of infrared radiation.
- D** Shiny surfaces are poor emitters of infrared radiation.

- 15** Vacuum flasks usually have silvered walls that help to keep the contents of the flask hot.

Why are the walls silvered?

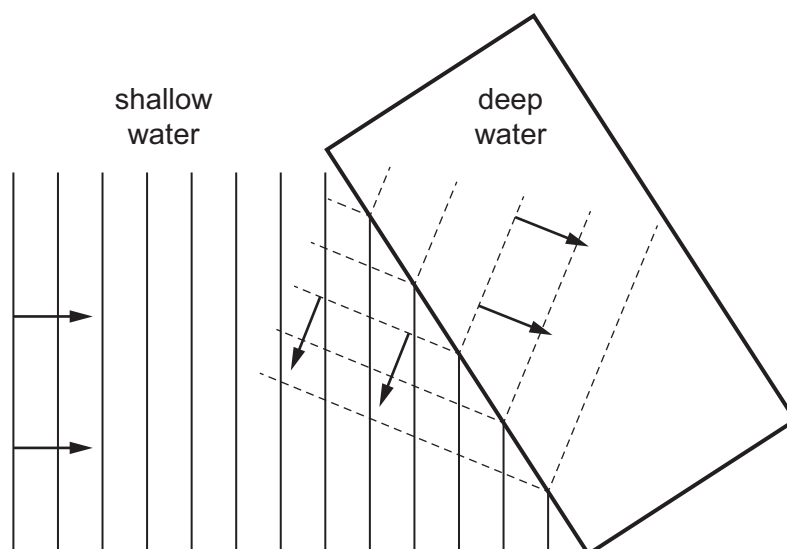
- A** to absorb thermal energy from the air around the flask
- B** to increase the rate of convection inside the flask
- C** to reduce energy loss to the surroundings by conduction
- D** to reflect thermal radiation back into the flask

- 16** The temperature of the air in the atmosphere around us increases.

Which row describes the change in the density of the air and which form of thermal energy transfer occurs as a result of this change?

	what happens to the density of the air	thermal energy transfer that occurs as a result
A	decreases	conduction
B	decreases	convection
C	increases	conduction
D	increases	convection

- 17 A student draws a diagram to show two different properties of a water wave. The arrows show the wave directions.



Which two wave properties does the diagram show?

- A** refraction and diffraction
B reflection and dispersion
C reflection and diffraction
D reflection and refraction
- 18 A seismic wave consists of vibrations that are parallel to the direction of propagation.

What is the type of wave and what is the name of the seismic wave?

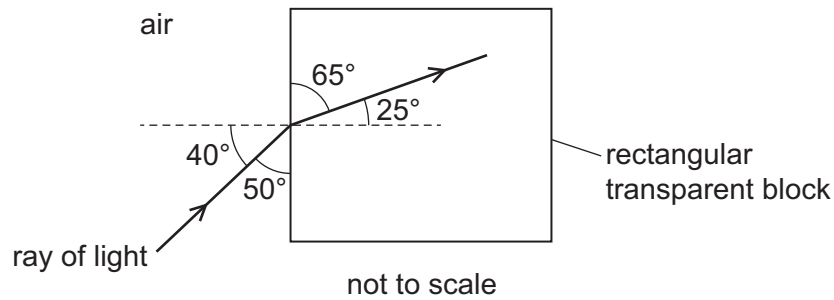
	type of wave	name of seismic wave
A	longitudinal	P-wave
B	longitudinal	S-wave
C	transverse	P-wave
D	transverse	S-wave

- 19 A person stands 1.0 m in front of a plane mirror. The mirror is moved away from the person at a speed of 1.0 m/s.

Which statement is correct?

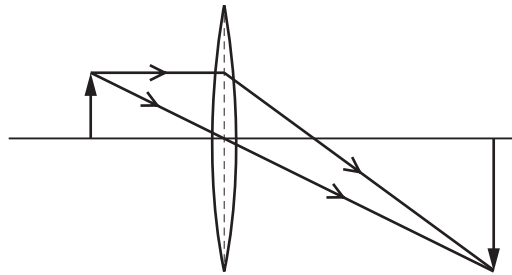
- A** The image moves away from the person at a speed of 1.0 m/s.
B The image moves away from the person at a speed of 2.0 m/s.
C The image moves towards the person at a speed of 1.0 m/s.
D The image moves towards the person at a speed of 2.0 m/s.

- 20 A ray of light passes from air into a rectangular transparent block.



What is the angle of refraction?

- A 25° B 40° C 50° D 65°
- 21 The diagram shows the formation of an image by a thin converging lens.



Which terms describe the image?

- A diminished and upright
- B real and inverted
- C virtual and diminished
- D virtual and inverted
- 22 Valuable objects are often marked with a special ink. In visible light, the ink **cannot** be seen.

The ink can be seen when exposed to the right type of electromagnetic waves.

Which type of electromagnetic waves are used for this type of security marking?

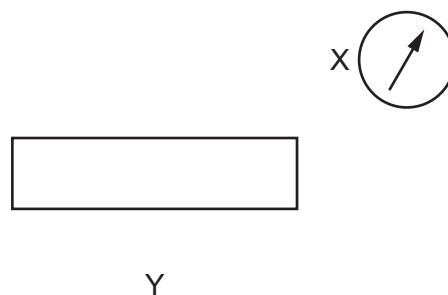
- A infrared
- B microwaves
- C ultraviolet
- D X-rays

- 23** A siren is emitting a sound. As time passes, the sound becomes louder and higher pitched.

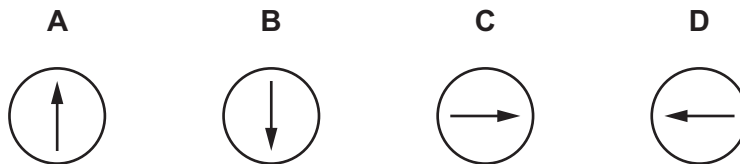
What is happening to the amplitude and to the frequency of the emitted sound wave?

	amplitude	frequency
A	decreasing	decreasing
B	decreasing	increasing
C	increasing	decreasing
D	increasing	increasing

- 24** A plotting compass is placed at position X near to a bar magnet. The diagram shows the direction in which the compass points.



Which diagram shows the direction in which the plotting compass placed at position Y points?



- 25** A student rubs a plastic rod with a cloth.

Electrons move from the cloth to the rod.

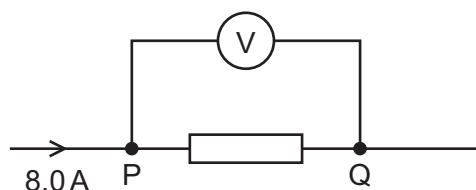
Which statement about the rod is correct?

- A** It is negatively charged and is attracted by a negatively charged object.
- B** It is negatively charged and is attracted by a positively charged object.
- C** It is positively charged and is attracted by a negatively charged object.
- D** It is positively charged and is attracted by a positively charged object.

26 Which expression is used to calculate electrical resistance?

- A charge $\div$ potential difference
- B current $\div$ potential difference
- C potential difference $\div$ charge
- D potential difference $\div$ current

27 The diagram shows part of an electric circuit. The reading on the voltmeter is 16 V. The current in the resistor is 8.0 A.



One coulomb of charge flows from P to Q through the resistor.

How much energy is transferred in the resistor?

- A 2.0 J
- B 8.0 J
- C 16 J
- D 128 J

28 What is the combined resistance of a $3.0\ \Omega$ resistor and a $6.0\ \Omega$ resistor connected in parallel?

- A $2.0\ \Omega$
- B $4.5\ \Omega$
- C $9.0\ \Omega$
- D $18\ \Omega$

29 A $5.0\ \Omega$ resistor is connected in series with a $10\ \Omega$ resistor.

There is a constant current in the $5.0\ \Omega$ resistor.

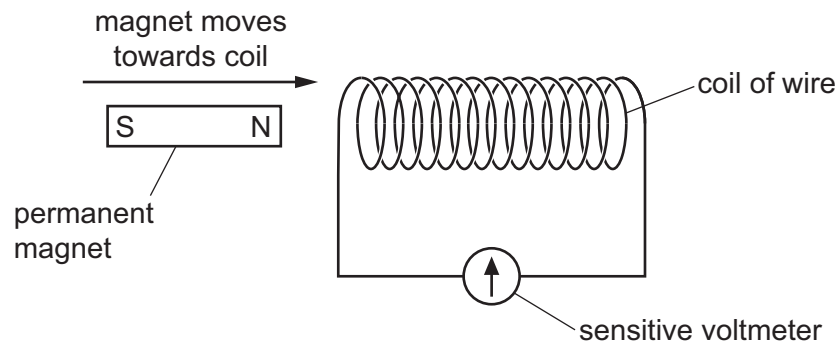
Which statement about the two resistors is correct?

- A The current is greater in the $5.0\ \Omega$ resistor.
- B The current is greater in the $10\ \Omega$ resistor.
- C The potential difference (p.d.) is greater across the $5.0\ \Omega$ resistor.
- D The potential difference (p.d.) is greater across the $10\ \Omega$ resistor.

30 Which statement explains where the switch for an electric lamp is connected?

- A The switch is in **either** the live wire **or** in the neutral wire as both wires carry the current.
- B The switch is in the earth wire to prevent the fuse from blowing when the lamp is in use.
- C The switch is in the live wire so that the lamp is disconnected from the mains supply when the switch is off.
- D The switch is in the neutral wire to prevent current leakage when the lamp is switched off.

- 31 The diagram shows a magnet moving towards a coil of wire.



When the magnet moves towards the coil of wire, an electromotive force (e.m.f.) is induced.

Students are asked for one change that will increase the magnitude of the induced e.m.f.

Three changes are proposed.

- 1 Move the magnet more quickly.
- 2 Use a stronger magnet.
- 3 Increase the number of turns on the coil.

Which changes, on their own, will increase the magnitude of the induced e.m.f.?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only
- 32 What is used to identify the pattern and direction of the magnetic field due to the current in a straight wire?
- A** an ammeter
B a compass
C iron filings
D a voltmeter
- 33 Which description of the structure of an atom is correct?
- A** negatively charged electrons surrounding an uncharged nucleus
B negatively charged electrons surrounding a positively charged nucleus
C positively charged electrons surrounding an uncharged nucleus
D positively charged electrons surrounding a negatively charged nucleus

- 34** A radioactive source is placed 20 mm in front of a detector.

The count rate from the source is 2000 counts/minute.

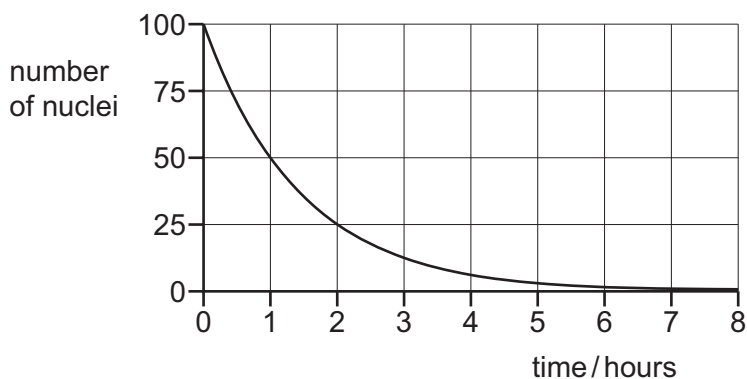
With a thin sheet of paper between the source and the detector, the count rate is 2000 counts/minute.

With a sheet of aluminium 5 mm thick between the source and the detector, the count rate is 1200 counts/minute.

Which radioactive emissions are coming from the source?

- A** alpha, beta and gamma
 - B** alpha only
 - C** beta and gamma only
 - D** beta only
- 35** Which type of radiation is measured using a detector connected to a counter?
- A** visible light
 - B** ionising nuclear
 - C** microwave
 - D** thermal
- 36** A sample of a radioactive isotope contains 100 nuclei of the isotope.

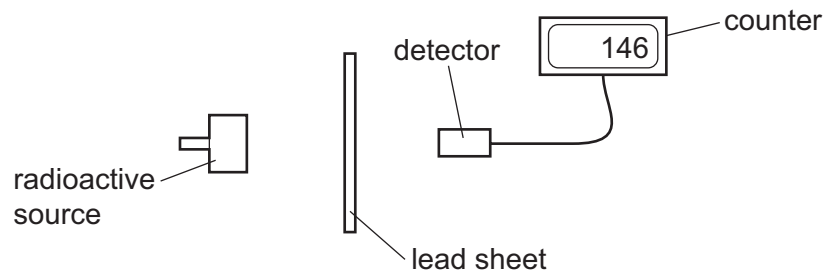
The graph shows the number of nuclei of that isotope that remain in the sample as time passes.



What is the half-life of the radioactive isotope?

- A** 1 hour
- B** 4 hours
- C** 6 nuclei
- D** 50 nuclei

37 A teacher does an experiment to measure radiation passing through a lead sheet.



Three students suggest how the teacher can reduce the risk to her health.

student 1 Move the radioactive source with her bare hands.

student 2 Stay as close as possible to the radioactive source during the experiment.

student 3 Minimise the time of the experiment.

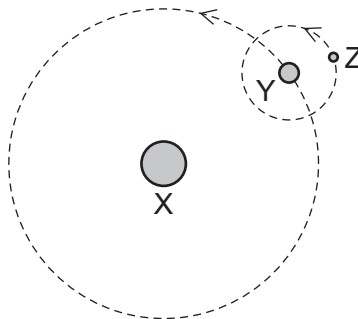
Which students are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 2 only **D** 3 only

38 The diagram represents three objects, X, Y and Z, in space.

Object Y orbits object X.

Object Z orbits object Y.



Which types of object are X, Y and Z?

	object X	object Y	object Z
A	planet	moon	star
B	planet	star	moon
C	star	planet	moon
D	star	moon	planet

39 Which statement about the Sun is correct?

- A** The Sun consists of mainly carbon and nitrogen.
- B** The Sun is the largest planet in the Solar System.
- C** The Sun is a medium-sized star.
- D** The Sun radiates most of its energy in the gamma-ray region of the electromagnetic spectrum.

40 What is the name of the galaxy that contains the Sun?

- A** Andromeda
- B** Milky Way
- C** Red Shift
- D** Universe

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Cambridge IGCSE™ (9–1)

PHYSICS**0972/11**

Paper 1 Multiple Choice (Core)

October/November 2025

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6	C	1
7	B	1
8	B	1
9	C	1
10	A	1
11	C	1
12	D	1
13	D	1
14	A	1
15	D	1
16	B	1
17	D	1
18	A	1
19	B	1
20	A	1
21	B	1
22	C	1
23	D	1
24	D	1
25	B	1
26	D	1
27	C	1
28	A	1

Question	Answer	Marks
29	D	1
30	C	1
31	A	1
32	B	1
33	B	1
34	C	1
35	B	1
36	A	1
37	D	1
38	C	1
39	C	1
40	B	1